

NEURAL BELLMAN OPERATORS: COMPLETE RESEARCH COMPENDIUM, REVISION R14

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This compendium preserves the current main manuscript and proof supplement together with the unchanged earlier main paper, research compendium, and supplemental Sections S.1–S.15. Historical pages retain their original revision and target identities. Their inclusion preserves the full theoretical and experimental record; it does not turn every historical claim into evidence for a current theorem. The R14 response letter is a separate document.

READING THE CURRENT REVISION AND THE PRESERVED RECORD

The current main paper presents finite-query response identification, procurement, enforcement-capacity choice, and a continuous-fee certificate in one stochastic settlement economy. Its current supplement proves the new results and states the arithmetic and model-transfer semantics. Read these two documents first. The following historical papers then preserve the original operator theory, dynamic games, recursive utility, older numerical experiments, and prior proofs without abridgment. Their original assumptions and target identities apply to those pages. The release manifest and preservation inventory identify the source of each unchanged PDF.

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We study procurement when operating service is an endogenous response to a dynamic contract. Finite value queries characterize a sharp outer response set for affine reward perturbations, including ties and near-optimal behavior. Rational support bounds convert this information into participation and purchaser-surplus certificates. An enforcement-capacity theorem and interval response relaxations extend the comparison from a contract menu to continuous fees. In a stochastic settlement economy, menu refinement changes the compulsory-term ranking, and a continuous-instrument certificate identifies executable contracts with explicit global regret bounds. Joint service–surrender queries strengthen coordinate secants; incidence and quality counterfactuals identify the institutional sources of the comparison. A primitive opportunity-ordering result and an adverse parameter map distinguish adjustment gains from financial exposure. Neural proposal, policy evaluation, and verification remain separate, and matched benchmarks retain unfavorable neural outcomes. The arithmetic and open-mandate semantics are integrated into the theorem chain. A model-to-decision transfer theorem separates fixed-array verification from continuous-state approximation, whose constructor discrepancy is audited directly.

KEYWORDS: Dynamic contracts, Response identification, Neural Bellman operators, Verified dynamic programming, Preference adjustment.

1. INTRODUCTION

An operating contract is evaluated through the decisions it induces. A financial guarantee can discourage surrender while reducing the agent’s private value; a compulsory rating term can replace a guarantee while imposing a real enforcement cost. When assumption, portfolio choice, preference adjustment, and surrender are jointly chosen, comparison of private values does not establish a comparison of purchased service. Moreover, an accurate approximation to an optimal policy need not approximate the price of every optimal policy. At an indifference point, an arbitrarily small value error conceal an economically important change in output.

This paper develops a response-set approach to this problem. The object passed on the dynamic program to the purchaser is not a selected policy or a numerical derivative. It is a set of jointly feasible service and surrender moments constrained by verified value queries. A finite family of such queries gives a linear description, in an enlarged space, of every response consistent with the value information. Its support bounds can be checked by rational weak duality. The purchaser then compares the most favorable response to an executable offer with the most favorable response to each competing offer, including nonprocurement. The same construction admits continuous enforcement instruments: interval bounds cover the instruments between evaluated contracts rather than treating the evaluation grid as the feasible set.

There are three parts to the argument. First, Theorem 1 characterizes the sharp response set relative to a finite value oracle for affine reward perturbations. The charac-

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This paper develops the numerical framework accompanying Qi (2024). The author thanks Darrell Duffie for discussions.

optimal behavior. Its sharpness is an information statement: every point in the generated by an affine-response economy consistent with the queries. Additional edge of a particular dynamic economy can, and in our application does, tighten Second, Theorems 2 and 4 convert these response restrictions into procurement continuous-instrument certificates. An enforcement-capacity result separates the imposed on surrender from the capacity required to support it. Third, the application evaluates these comparisons in one stochastic settlement economy, unchanged collection of rewards, transitions, financial interpolation rules, and al actions.

application makes the distinction between institutional restrictions and economic ves consequential. On the original six-point enforcement menu, the optimal sory terms are three intervals with preference adjustment and four without it. g the charge menu to increments of 0.05 changes this comparison to two versus tending the charge to a continuum changes the optimization problem again. ort an executable contract, an upper bound over every charge in the continuum, eparate upper bound for each compulsory term. Thus the coarse-menu result ed as a theorem about its stated institution, while the continuous calculation ther than assumes, its robustness. Neither a separate analytical example nor a le local perturbation substitutes for that test.

service measures also enter the result substantively. Discounted operating dura- he output bought in the baseline contract. It is observed as continued operation, rred from the agent's reported utility. Its coefficient in the agent's flow payoff s a computational instrument for identifying its response. Discounted elective er and wealth-qualified duration are additional moments. A purchaser who s the surrender fee values duration and surrender jointly. Queries in the joint n yield strictly tighter bounds than coordinate-wise restrictions alone; the settle- xperiment demonstrates this reduction within the actual application. Changes incidence, capacity-cost curvature, compulsory-term costs, outside value, and quality then reveal which parts of the procurement comparison are institutional ich are incentive effects.

al Bellman Operators provide the proposal, evaluation, improvement, and cer- n architecture producing the value information. The neural component and ificate have different jobs. A proposal can reduce the search for good feasible policies. It cannot establish its own approximation error or the infeasibility of response. The verification layer therefore remains valid for a poorly trained k and for a nonneural proposal. This separation preserves the operator and mation results developed in the accompanying compendium. It also permits an tive unfavorable benchmark: on the deposited settlement target, the fitted net- less accurate than the nearest-anchor policy, and exact upper-value calculation tes their online comparison. We report these costs rather than attributing the ate's validity to neural prediction accuracy.

adjustment mechanism is examined separately from the purchaser's optimiza- signed opportunity-exposure identity separates the current adjustment gain, re to future adjustment opportunities, and the replacement of the financial action. ing downward drift does not by itself rank the two initial financial mandates. vide primitive sufficient conditions for an ordered opportunity process in a d stopping subclass, and an explicit 54-point map in the settlement economy, ng reversals. The map is not promoted into a uniform theorem between its

tered from a single favorable calibration.

ally, numerical error must be distinguished from model error. We integrate the a-dependent binary64 error derivation and the strictly positive initial-policy connection into the statements used by the paper. We also give a model-to-decision transfer theorem with separate entries for time, state, action, boundary, terminal, and nmetic errors. A direct moment audit of the stored transition exposes a material ection effect. It prevents the error for evaluating fixed arrays from being mistaken he error of approximating the antecedent diffusion. The continuous-state theory is ined, but a numerical conclusion about that diffusion requires the corresponding roximation entries, not a relabeling of the finite-state certificate.

related literature. The affine dependence of rewards on service moments connects analysis to reward transfer and successor features; see [Alegre, Bazzan, and da Silva \(2021\)](#). The envelope theorem of [Milgrom and Segal \(2002\)](#) applies to arbitrary choice at differentiability points and supplies a broader foundation for comparative statics. A finite-query result instead asks what can be established without differentiability n finitely many interval observations, and retains the joint response set needed a purchaser. Regional verification of parametric Markov models is an established ection. Parameter lifting replaces fixed but unknown transition parameters by state-e nondeterministic choices; [Quatmann, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) develop this approach, and [Junges et al. \(2019\)](#) study coverage of parameter regions. We compare an implementation of that relaxation with the common-law count and ed-chord bounds; we do not describe our implementation as an execution of their ware. The full discussion of neural policy iteration, nonlinear operator approxima-, recursive utility, and dynamic equilibrium is retained in the complete historical pendium.

organization. Section 2 specifies the institution and the settlement economy. Section 3 establishes response identification and procurement. Section 4 treats continuous rcement. Section 5 presents the economic evidence. Sections 6 and 7 examine the hanism and computational architecture. Section 8 gives the model-to-decision error ount. The current supplement supplies the proofs and verification details; the histor-compendium and supplements retain all earlier material without silently changing arget identity.

2. THE ECONOMIC ENVIRONMENT

2.1. Operating decisions and affine response moments

Consider a finite-horizon controlled process with live state x at dates $n = 0, \dots, N$, ninal payoff G , and feasible history-dependent operating and stopping policies $\mathcal{P}_j^r$. The public regime r specifies whether deliberate preference adjustment is ilable. The instrument j includes the initial financial mandate and the first date hich elective surrender is permitted. It does not specify the agent’s subsequent sumption, financial position, or stopping rule. Policies may randomize. The finite-e application below has finite later-date action menus and a continuous, piecewise-e first-date financial operator.

For a fixed feasible policy, let $A(p)$ be expected discounted operating duration, $H(p)$ ounted elective-surrender incidence, and $Q(p)$ discounted duration during which

$Q \leq A$. Forced discharge is not elective surrender and does not enter H . At long-benefit parameter d and surrender charge F , private payoff is

$$J_j^r(p; d, F) = B_j^r(p) + dA_j^r(p) - FH_j^r(p), \quad W_j^r(d, F) = \sup_{p \in \mathcal{P}_j^r} J_j^r(p; d, F). \quad (1)$$

feasible policies and their laws are unchanged when only (d, F) is perturbed. This variation is essential: a perturbation that changes feasible actions, wealth dynamics, or surrender technology does not identify service through the same affine equation. We generally write $J(p; t) = \alpha(p) + t \cdot z(p)$, where $t \in \mathbb{R}^k$ is a reward perturbation and $p \in K$, a known compact polytope. The point $t = 0$ denotes the contract under the baseline. For duration and surrender, $t = (\Delta d, -\Delta F)$ and $z = (A, H)$. A quality premium can be included by appending its reward feature and perturbation. The results require neither a smooth optimizer nor a differentiable value function.

2.2. Enforcement, transfers, and participation

The purchaser chooses a mandate σ , compulsory term m , charge F , enforcement technology E , and signing transfer $q \geq 0$, or does not procure. The enforcement technology

$$0 \leq F \leq \zeta E, \quad F \leq \bar{F}, \quad E \geq 0, \quad (2)$$

where $\zeta > 0$ measures effective charge capacity and $\bar{F}$ is the administrative charge. The baseline and incidence counterfactuals use $\bar{F} = 1$; capacity itself can exceed $\bar{F}$ when enforcement is inefficient. Capacity costs $C(E)$, a nondecreasing function; mandatory terms cost $D(m)$. A fraction $\alpha_c \in [0, 1]$ of capacity cost is borne by the agent's private utility numeraire, and the remainder by the purchaser. The purchaser's fraction $\rho_F \in [0, 1]$ of elective surrender fees. The agent's outside value is G_0 . Signing transfer and capacity expenditure do not alter managed wealth or the agent's feasible set in the baseline institution. With an attained exact best response, the minimum transfer inducing weak participation is

$$q_j = [G_0 - W_j(d, F) + \alpha_c C(E)]_+. \quad (3)$$

Participation is resolved by acceptance. A strict participation premium must be paid when that convention is inappropriate; our surplus margins provide the budget for a premium. The purchaser's payoff under response p is

$$\Pi_j(p, q) = bA(p) + \rho_F FH(p) + \nu Q(p) - q - (1 - \alpha_c)C(E) - D(m), \quad (4)$$

where $\nu \geq 0$ and a stated quality valuation ν . All comparisons retain the positive part in the transfer-timing change is neutral only when the discount factors and numeraire are specified; a payment that enters managed wealth is a different dynamic problem and must be re-solved.

Equation (4) is not an unrestricted mechanism-design formulation. The purchaser cannot elicit a private type, select a contingent reporting mechanism, or contract on preference adjustment. It is a complete-information choice of enforceable instruments with endogenous response. Its economic content comes from comparing these instruments before selecting an allocation and from respecting their participation effects.

the quantitative economy uses $N = 8$ intervals of length $1/8$, state (u, X) , and rating action (c, θ, π) . The preference state u lies on 33 nodes in $[1.2, 2.8]$ and wealth on 19 nodes in $[0.5, 2]$, giving 1,617 states. The initial state is $(2, 1.25)$. The cardinal rating primitive and terminal payoff are

$$f(c, u, \theta; d) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 + d, \quad G(u, X) = -0.02(u-2)^2 + 0.1 \log X. \quad (5)$$

adjustment-cost coefficient is $k = 2$. The canonical constructor fixes discounting, volatility, and discharge rewards. The rate of time preference is 0.04; the financial parameters are 0.02 and 0.06, and volatility scales are 0.05 and 0.2 for preferences and wealth. The two endpoint correlation laws are -0.25 and 0.25 .

For endpoint $i \in \{0, 1\}$ let $R_{n,i}$ be the stored reward and $K_{n,i}$ the nonnegative discounted live-state kernel. A common law parameter λ is used at all dates:

$$R_n^\lambda = (1 - \lambda)R_{n,0} + \lambda R_{n,1}, \quad K_n^\lambda = (1 - \lambda)K_{n,0} + \lambda K_{n,1}. \quad (6)$$

The current action is chosen before the endpoint-law draw. Discharge rewards are added once in R ; lost live-state mass is not assigned an additional terminal reward. Pre-date controls consist of a union of two fixed meshes, with 1,565 common actions, the frozen date- and state-dependent proposals, and surrender when permitted. The assumption is in $[0.05, 0.8]$, deliberate drift in $[-0.2, 0.2]$, and common financial rates in $[-0.5, 0.8]$. In the no-adjustment regime, only zero-drift actions are admissible; stochastic preference variation remains.

At the first date, consumption and adjustment pairs are finite but the risky share is optimized over the stored piecewise-affine interpolation of its breakpoints. The restrictive mandate requires $0 < \pi_0 \leq L$ throughout the first interval; the other requires $\pi_0 \leq 0$. The application fixes $L = 0.8$ and $S = 0.5$. Subsequent shares are not restricted by the initial sign. A closure value at $\pi_0 = 0$ supplies an upper supremum for the positive class, not an attained positive policy. Section 7.3 gives the corresponding super-witness construction.

Selective surrender at a nonboundary live state pays $G - F$ and is available only at dates $n \geq m$. Forced discharge pays G without the fee. The contract $m = 8$ has no selective surrender within the operating horizon. We compare contracts at $\lambda = 0.125$ and $d = 0.42425$. Baseline purchaser primitives are $b = 1$, $C(E) = 0.02E^2$, $D(m) = (m-1)/8$, $\rho_F = \nu = 0$, $\alpha_C = \zeta = 1$, and $G_0 = G(2, 1.25)$. These are normalized institutional parameters, not estimates from a dataset. Their alternatives are evaluated in Section 5.

3. RESPONSE SETS AND PROCUREMENT

3.1. A sharp finite-query characterization

Suppose values are enclosed at $t_0 = 0, t_1, \dots, t_M$:

$$L_i \leq W(t_i) \leq U_i, \quad i = 0, \dots, M. \quad (7)$$

An η -response satisfies $J(p; 0) \geq W(0) - \eta$. Exact responses correspond to $\eta = 0$ when the supremum is attained. Define a lifted set with variables v, a, z and one pair (β_i, z_i)

an affine response supporting query i . Impose

$$L_0 \leq v \leq U_0, \quad v - \eta \leq a \leq v, \quad z, z_i \in K, \quad (8)$$

$$a + t_j \cdot z \leq U_j, \quad \beta_i \leq v, \quad (9)$$

$$\beta_i + t_j \cdot z_i \leq U_j, \quad \beta_i + t_i \cdot z_i \geq L_i, \quad \beta_0 = v. \quad (10)$$

(L, U) be its projection onto z . All the restrictions are linear when K is a polytope. Exact feature bounds also give finite intercept bounds: $\beta_i \geq L_i - h_K(t_i)$, where h_K is support function of K .

THEOREM 1—Finite-query response identification: *Assume K is a nonempty compact set, the query intervals are mutually consistent with an affine-response economy, and the desired supremum is attained in that economy or its stated closure. Then every attained supremum has feature vector in $\mathcal{Z}_\eta(L, U)$. Conversely, each feasible lifted point defines a finite-response economy satisfying all the query intervals and having the candidate as an η -optimal point. Hence support bounds over $\mathcal{Z}_\eta(L, U)$ are sharp relative to the query information and the candidate.*

The necessity argument uses the candidate's affine payoff and an optimal response to the query. For sufficiency, take those same affine functions as the entire response menu. The maximum at zero is v ; each query is at least its designated lower bound and no function exceeds its upper bound. The construction proves sharpness without requiring that every feasible lifted point is implementable in the already specified response economy. That distinction allows the theorem to be used as a valid outer approximation while reserving the equality-graph restrictions for stronger fixed-economy variants.

One-dimensional secants are a special case. With duration in $[0, 1]$, queries at $d - h$ and $d + h$ give

$$\max\left\{0, \frac{L(d) - U(d - h) - \eta}{h}\right\} \leq A(p) \leq \min\left\{1, \frac{U(d + h) - L(d) + \eta}{h}\right\}. \quad (11)$$

Increasing h reduces the span of economic curvature but magnifies value error and η . There is no assertion that an arbitrarily small secant is better. With two features, separate queries at $\pm h$ do not rule out a response with both feature coordinates equal to h . Adding (h, h) rules it out and bounds their sum by one in the exact oracle. The theorem gives the finite-menu witnesses and rational certificates for this example and its interval variants.

3.2. Checking support bounds rather than trusting an optimizer

Formulate the bounded linear program as $\max\{c^\top x : Ax \leq b, \ell \leq x \leq u\}$. For any nonnegative vector μ , weak duality gives

$$c^\top x \leq \mu^\top b + \sum_j \max\{(c - A^\top \mu)_j \ell_j, (c - A^\top \mu)_j u_j\}. \quad (12)$$

se: the residual is charged through the box support. In the implementation, a
 ting-point optimizer proposes μ , while rational arithmetic checks nonnegativity,
 ry coefficient, and the complete residual term. A solver's success flag is therefore
 her a feasibility proof nor a certificate of the reported upper bound. The same
 ument applied to $-c$ gives a lower support bound.

3.3. From response identification to institutional choice

or each contract, let $[L_j, U_j]$ enclose private value and let $\mathcal{Z}_j$ contain all η -response
 nents. Define $s_j(z) = bA + \rho_F FH + vQ$ and

$$\underline{q}_j = [G_0 - U_j + \alpha_C C(E)]_+, \quad \bar{q}_j = [G_0 - L_j + \eta + \alpha_C C(E)]_+, \quad (13)$$

$$\underline{\Pi}_j = \inf_{z \in \mathcal{Z}_j} s_j(z) - \bar{q}_j - (1 - \alpha_C)C(E) - D(m), \quad (14)$$

$$\bar{\Pi}_j = \sup_{z \in \mathcal{Z}_j} s_j(z) - \underline{q}_j - (1 - \alpha_C)C(E) - D(m). \quad (15)$$

grant $\bar{q}_j$ induces participation for every feasible η -response. The upper bound
 berately allows a rival its most favorable response and its smallest possibly sufficient
 nt, even when these are not jointly attainable. Retaining their dependence in a lifted
 gram can sharpen this upper bound but is not necessary for its validity.

THEOREM 2—Robust procurement: *Suppose the value and response enclosures above are
 d, and the candidate admits an attained exact response when $\eta = 0$, or feasible η -responses
 n $\eta > 0$. If*

$$\underline{\Pi}_{j_*} > \max\{0, \sup_{j \neq j_*} \bar{\Pi}_j\}, \quad (16)$$

offering j_* with grant $\bar{q}_{j_*}$ strictly dominates nonprocurement and every other participating
 , for every admitted response of the candidate and every admitted response of every rival.
 e generally, if the right side is at most $\underline{\Pi}_{j_*} + \varepsilon$, the candidate is executable with regret at
 t ε relative to the most favorable competing response.

this is a result about contracts with endogenous response, not merely about the out-
 of a policy optimizer. Exact behavior is the economic benchmark. The η statement
 sures robustness to approximate optimization, and is not obtained by substituting
 exact-policy support calculation into an η -response claim. An outside option of zero
 the purchaser is explicitly included.

PROPOSITION 3—Efficient capacity and incidence: *Suppose capacity affects neither
 ating payoffs nor feasible policies except through (2), and C is nondecreasing. For every
 ble (F, E) there is a weakly better purchaser contract with the same F and $E = F/\zeta$. This
 s with binding or slack participation. Under binding participation, changing α_C leaves
 chaser surplus unchanged. Changing ρ_F instead adds $\Delta\rho_F FH$ and generally changes the
 ring.*

Indeed, total capacity and participation expenditure is $[G_0 - W + \alpha_C C]_+ + (1 - \alpha_C)C$,
 nondecreasing function of C . Under binding participation it equals $G_0 - W + C$.

ment technology rather than a restriction imposed on two otherwise independent instruments. This proposition does not apply to a capacity payment that reduces wealth. That intervention changes W and is not an incidence relabeling.

CONTINUOUS ENFORCEMENT WITHOUT AN UNCHECKED MENU APPROXIMATION

Proposition 3, the baseline continuous instrument is $F \in [0, 1]$, with $E = F$. Fix a term and mandate, and write a policy's payoff at the central benefit as $B - FH$, and dA into B . A value query at $(d + \delta, F_i)$ implies

$$B + \delta A - F_i H \leq U(d + \delta, F_i). \quad (17)$$

Inequalities constrain the *same* policy moments across fee cells. At an attained optimum at fee F_i , let $H_i \in [\underline{H}_i, \bar{H}_i]$. Its affine payoff supplies a lower bound at fee:

$$W(d, F) \geq L(d, F_i) - (F - F_i)h_i, \quad h_i = \begin{cases} \bar{H}_i, & F \geq F_i, \\ \underline{H}_i, & F \leq F_i. \end{cases} \quad (18)$$

Thus that an η -response at F must satisfy $B - FH \geq L(d, F_i) - (F - F_i)h_i - \eta$. In cell $F \in [a, b]$, comparison with an exact response at a strictly exterior fee gives monotonicity bounds:

$$H \leq \bar{H}_i + \frac{\eta}{a - F_i} \quad (F_i < a), \quad H \geq \underline{H}_i - \frac{\eta}{F_i - b} \quad (F_i > b). \quad (19)$$

At responses, endpoint bounds apply by the same two-policy argument. The inequalities follow from adding the optimality inequalities at the two fees; no derivative is taken. Intersect them with $[0, 1]$ to obtain $H \in [\underline{h}, \bar{h}]$. Reduce $T = FH$. Its graph over this rectangle is contained in the four affine lines:

$$\begin{aligned} T &\geq aH + \underline{h}F - a\underline{h}, & T &\geq bH + \bar{h}F - b\bar{h}, \\ T &\leq bH + \underline{h}F - b\underline{h}, & T &\leq aH + \bar{h}F - a\bar{h}. \end{aligned} \quad (20)$$

Sorting affine function of $C(F/\zeta)$ gives a lower bound on capacity cost through cell. With $C(F) = cF^2$ and midpoint v , it is $c(2vF - v^2)$. Combining (17)–(20), the lower bound, and moment bounds yields a linear program whose objective is the user's payoff. The upper program may admit a fictitious response or a relaxed response. That enlarges its feasible set and preserves the direction of the bound.

THEOREM 4—A continuum procurement certificate: *For each term and mandate, let $\mathcal{F}$ be a family of many closed fee cells cover the entire feasible interval. On every cell impose valid query, C , product, capacity-cost, and participation relaxations as above. Let U_i be a rationally supported upper objective and let L_* be the lower payoff of an executable candidate. Then*

$$\sup_{j, p, q \text{ participating}} \Pi_j(p, q) - L_* \leq \max_i \{0, \max U_i\} - L_*. \quad (21)$$

Step	Regime	(F, m, σ)	Surplus lower	Rival	Margin
0.20	Adjustment	0.60, 3, +	0.0960214	0.80, 1, +	0.00059979
0.20	No adjustment	0.60, 4, +	0.0580640	0.80, 1, −	0.00022405
0.10	Adjustment	0.60, 3, +	0.0960214	0.70, 2, +	0.00009979
0.10	No adjustment	0.30, 6, +	0.0583927	0.60, 4, +	0.00032847
0.05	Adjustment	0.65, 2, +	0.0972714	0.55, 3, +	0.00009979
0.05	No adjustment	0.55, 4, +	0.0590227	0.75, 2, +	0.00000842

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). Exact responses; value
 vance 10^{-7} and support allowance 10^{-9} . Lower bounds and margins are rounded down. Each margin
 ompares the candidate lower bound against every other offer upper bound and nonprocurement.

articular, if the candidate lower payoff exceeds the maximum upper bound on every cell with
 ferent term, its compulsory term is identified even if the optimal fee is not attained or the
 ining same-term regret is positive.

he last qualification matters at a surrender indifference. The least fee eliminating
 nder may leave a low-service best response at the boundary. A slightly larger fee
 implement high service, while the infimum cost itself is not achieved under adverse
 election. Our reported object is consequently an executable contract with a bound
 inst the global supremum, not an asserted maximizer at an untested kink. Every
 ined cell appears in the witness; a collection of favorable evaluation points cannot
 s as a continuum certificate.

he algorithm begins with the deposited fee queries, adds benefit perturbations
 ere the response upper bound is loose, and subdivides the cell with the largest
 fied purchaser upper bound. It stops at a specified regret tolerance or an explicit
 putational budget. No convergence flag is interpreted as economic separation.
 eported tolerance is met only when the final cover and its inequalities establish
 . Wider benefit perturbations can be valuable because they constrain implausible
 n-duration responses that remain compatible with very local value queries. The
 ice of query is thus directed by the purchaser’s unresolved comparison.

5. CONTRACT CHOICE IN THE SETTLEMENT ECONOMY

5.1. Nested menus and continuous fees

able I reports exact-response comparisons on three nested menus. The six-point
 u and the two refinements use the same 1,617-state economy, the same later-date
 ons, the same first-date interpolation, and the same public permissions. Only the set
 urchasable fees changes. All ties within a contract are enclosed by an independently
 nstructed Bellman equality graph, rather than resolved in favor of the purchaser.
 displayed rival is the contract furnishing the largest competing upper payoff.
 he original three-versus-four term comparison is therefore correct for its institu-
 but not invariant to menu refinement. This is economically informative: a fee
 ntly above a surrender threshold can substitute for a compulsory interval. A coarser
 u makes that substitution more expensive because it requires excess capacity. A
 m about commitment length that omits the enforceability menu can conflate this
 retization effect with the preference-adjustment mechanism.

Regime	Fee	Term	Surplus lower	Global upper	Regret upper
Adjustment	0.72880859	1	0.097598279	0.097599163	0.000000883
No adjustment	0.80087891	1	0.059936055	0.059936149	0.000000093

nonical manifest 54adb353c5b85210d (full hash in the release manifest). Both mandates, all eight and every $F \in [0, 1]$ are covered; $E = F$ is derived from the capacity theorem. Behavioral tolerance Regret and upper bounds are rounded up. Exact input fees and outward grants: Adjustment: 0.80859375, $q = 0.792634995286$; No adjustment: $F = 0.80087890625$, $q = 0.830297219351$.

continuous experiment applies Theorem 4 on $F \in [0, 1]$ after separately optimizing capacity under (2). Its final candidates, global upper bounds, and term-specific pricing bounds appear in Table II. Both identified compulsory terms are one interval. Adjusted contract requires a smaller charge than the no-adjustment contract, so the continuously optimized comparison concerns the price of enforcement rather than the difference in term length. These calculations establish the stated regret guarantee over the continuous fee interval; they do not approximate continuous consumption or settlement choices. The latter remain part of the unchanged settlement target. Numerical comparisons have distinct logical forms. A positive finite-menu margin excludes every other menu item. A continuous-fee certificate bounds a supremum over a covered interval and can identify a term without identifying a unique fee. An equality graph is not automatically an η -response graph. For the original menu we retain the separate secant certificate with $\eta = 10^{-8}$ and the service-price rectangle. Its smallest deposited margin is 7.6880857×10^{-5} . For the continuous comparison, the behavioral tolerance is reported in the table, not inherited from the original menu.

5.2. Joint service and surrender information

For a fee-receiving purchaser the relevant operating output is $A + FH$. Table III fixes the settlement contracts near surrender transitions, evaluates coordinate reward and then adds the two joint directions. In each case the lifted response set uses the feature box and value-error allowance. The upper bound falls strictly; it is merely a smaller error bar on a separately estimated derivative. The gain arises because a single policy cannot simultaneously select the most favorable duration from a response and the most favorable fee receipt from another without satisfying the value inequalities.

The experiment gives the finite-query theorem a direct economic use beyond the one-dimensional secant argument. It does not claim that the oracle set equals the feasible response set of the settlement dynamic program. Exact equality-graph supports can be derived when the full Bellman solution is available. Finite-query supports are useful for value certificates, rather than complete feasible-response descriptions, are the tractable objects, and when the purchaser changes the weights on several service items.

5.3. Institutional sensitivity

Table IV records counterfactuals on the 0.05 fee menu. Each row uses actual settlement contract moments. Capacity cost is evaluated at the efficient level $E = F/\zeta$; changes in

Regime	Mandate	η	Coordinate upper	Joint upper	Reduction
Adjustment	+	0.0000	0.955820	0.882345	0.073475
Adjustment	+	0.0001	0.963945	0.887345	0.076600
Adjustment	−	0.0000	0.915092	0.882145	0.032947
Adjustment	−	0.0001	0.923217	0.887145	0.036072
No adjustment	+	0.0000	0.891508	0.890449	0.001059
No adjustment	+	0.0001	0.899133	0.895449	0.003684
No adjustment	−	0.0000	0.888871	0.888088	0.000783
No adjustment	−	0.0001	0.896496	0.893088	0.003408

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). The objective is $A + FH$. Adjustment uses $(m, F) = (2, 0.625)$ and no adjustment $(4, 0.525)$. The coordinate oracle uses $h = 0.02$ and queries; the joint oracle adds (h, Fh) and its negative. Query error is 10^{-10} . Bounds concern the same y feature vector, not independent coordinate selections.

TABLE IV

INSTITUTIONAL COUNTERFACTUALS IN THE SAME SETTLEMENT ECONOMY

Institution	Adjusted offer	Margin	Fixed offer	Margin
Baseline	0.65, 2, +	0.0000997	0.55, 4, +	0.0000084
All fees to purchaser	0.65, 2, +	0.0000997	0.60, 3, +	0.0002224
Half fees to purchaser	0.65, 2, +	0.0000997	0.55, 4, +	0.0001202
Capacity paid by purchaser	0.65, 2, +	0.0000997	0.55, 4, +	0.0000084
Capacity cost shared	0.65, 2, +	0.0000997	0.55, 4, +	0.0000084
Efficiency 0.8	0.35, 5, + †	-0.0000003	0.00, 7, +	0.0001960
Efficiency 1.2	0.75, 1, +	0.0005553	0.85, 1, +	0.0002775
Capacity coefficient 0.01	0.75, 1, +	0.0007747	0.85, 1, +	0.0008747
Capacity coefficient 0.04	0.35, 5, +	0.0000998	0.00, 7, +	0.0007947
Cubic capacity cost	0.65, 2, +	0.0003347	0.75, 2, +	0.0000811
Quadratic term cost	0.45, 4, +	0.0004372	0.55, 4, +	0.0000210
Quality premium +0.05	0.65, 2, +	0.0000997	0.55, 4, +	0.0000082
Quality premium −0.05	0.65, 2, +	0.0000997	0.80, 1, −	0.0006062
Fees and quality premium	0.65, 2, +	0.0000997	0.60, 3, +	0.0002050
Outside value +0.05	0.65, 2, +	0.0000997	0.55, 4, +	0.0000084
Outside value −0.05	0.65, 2, +	0.0000997	0.55, 4, +	0.0000084
Service price 0.9	0.65, 2, +	0.0000997	0.00, 1, +	0.0004559
Service price 1.1	0.65, 2, +	0.0000997	0.75, 2, +	0.0000110

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). All rows use the 0.05 menu and exact-response graph supports. Offers are (F, m, σ) . A dagger denotes an unresolved strict ring, not a certified winner. Capacity is F/ζ , the charge ceiling remains one, and participation is equated with its positive part. These are finite-menu sensitivities; they are not continuous-fee claims.

recipient use joint duration–surrender supports; quality changes use the wealth-lified duration moment. These are not substitutions from the separate analytical tract example in the supplement.

three distinctions organize the results. First, shifting a numeraire capacity payment between parties leaves the allocation unchanged when participation binds, as Proposition 3 predicts. The outside-value shifts in the table change required compensation not the operating incentives. Second, transferring surrender fees to the purchaser changes the social value of a response with elective stopping and can change the chosen n . This is not a transfer-incidence invariance. Third, the shape of enforcement and

and enforcement efficiency can eliminate the baseline term difference even while changes private continuation values.

ity valuation is equally substantive. Rewarding duration at wealth at least not the same service as undifferentiated operating time. A negative premium nge the preferred initial financial mandate in the no-adjustment regime. The definition is therefore not innocuous. The method handles the alternative by g the joint moment, not by declaring all service measures equivalent. A signing ntering wealth, a different discharge rule, or a different action set requires new c values and is outside these incidence-only calculations.

remaining sensitivity concerns the numerical query scale. For the original g offers, the deposited h checks at 0.0025 , 0.005 , and 0.01 retain positive no- nent margins of approximately 3.04×10^{-5} , 1.14×10^{-4} , and 1.56×10^{-4} under fified conservative construction. These are secant/error tradeoffs, not proof continuous model has negligible approximation error. The complete queries, nts, and error conventions are supplied with the tables.

6. ADJUSTMENT OPPORTUNITIES AND FINANCIAL EXPOSURE

6.1. *An exact decomposition*

V^a and V^0 denote continuation values with and without deliberate adjustment, $O = V^a - V^0$. At an adjusted optimal first action in class σ , keep consumption estment fixed and replace its drift by zero. Suppose this counterfactual action is . Let K_σ be its discounted continuation row. Define g_σ as the gain from replacing o-drift action by the adjusted action evaluated with V^a , and B_σ as the zero-drift ctual's payoff evaluated with V^0 minus the optimal no-adjustment payoff. ≤ 0 and

$$= g_\sigma + K_\sigma O + B_\sigma, \quad \Omega_+ - \Omega_- = g_+ - g_- + (K_+ - K_-)O + B_+ - B_-. \quad (22)$$

entity neither asserts that the signed exposure term is positive nor identifies a e economic ordering. For $D = K_+ - K_- = D^+ - D^-$ and an enclosure $\ell \leq O \leq u$, p box bound is

$$D^+ \ell - D^- u \leq DO \leq D^+ u - D^- \ell. \quad (23)$$

an absolute norm before this multiplication loses the direction of exposure. se, showing that negative drift is dominated does not order the two financial or the distribution of future opportunities.

6.2. *Primitive conditions in an ordered stopping subclass*

eful sufficient condition can be stated for a restricted but explicit economy. Let follow an exogenous order-preserving Markov transition, with common state- ndent survival and discount factors. Before an irreversible preference upgrade, nt may choose its exercise date. Consumption is an exogenous nondecreasing n of wealth taking values in $(0, 1]$. An upgrade raises the preference index to $u' > u$ and entails a wealth-independent cost. The terminal increment from ing is nondecreasing in wealth (it may be zero), and the upgrade does not change

Component	Positive	Negative	Near zero	Minimum	Maximum
Current-gain difference	16	11	27	-0.0019671	0.0000544
Future signed exposure	15	12	27	-0.0036131	0.0003893
Financial replacement	0	0	54	0.0000000	0.0000000
Total option difference	15	12	27	-0.0055802	0.0004436

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). All 54 combinations of three laws, three benefits, three fees, and two terms are retained. Counts use a reporting tolerance of 10^{-9} ; full signed intervals and values are deposited. The map is not a proof of a sign between grid cells.

wealth transition or consumption rule. There is no additional wealth-dependent surrender option in this subclass. The incremental flow is

$$\Delta f(c) = \frac{c^{1-u'}}{1-u'} - \frac{c^{1-u}}{1-u}. \quad (24)$$

ough its level can have either sign, its derivative is $c^{-u'} - c^{-u} \geq 0$. Coupled ordered wealth paths therefore order the present value of exercising at any common stopping time.

PROPOSITION 5—Ordered adjustment opportunities: *Under the preceding primitives, the value of the irreversible upgrade option is nondecreasing in wealth at every date. If the future initial financial transition first-order stochastically dominates the nonpositive transition at the same total discounted mass, then its future opportunity exposure is weakly larger. With equal current gains and equal replacement terms, (22) gives $\Omega_+ \geq \Omega_-$. Strict inequality follows if the stochastic dominance is strict on a set where the opportunity value is strictly increasing and that set receives positive exposure.*

The assumptions locate the mechanism rather than hiding it in a value-function ordering. Monotone consumption, a single upgrade, common survival, and the absence of wealth-dependent surrender are doing work. The full settlement economy has endogenous consumption, continuous preference adjustment on its mesh, and elective surrender. It need not satisfy this proposition. The decomposition remains valid there, but its sign must be checked rather than imported from the subclass.

6.3. A map including adverse cases

The settlement map uses $\lambda \in \{0, 0.125, 1\}$, $d \in \{0.35, 0.42425, 0.5\}$, $F \in \{0, 0.4, 0.8\}$, and $m \in \{1, 4\}$. All 54 points are retained. Each entry recomputes the adjusted and unadjusted problems, the zero-drift counterfactual rows, the signed exposure, and the wealth-bin contributions. Table V summarizes the sign frequencies and ranges; the complete rows make the reversals visible.

The map shows why a local sign is not a transferable primitive theorem. Changing the time at which surrender is permitted changes the opportunity process reached after each initial financial action. The current drift gain can be positive while the future signed contribution reverses the class difference. The replacement term is zero in the ordered map; this is an outcome of these first-action choices, not a general cancellation

mentary roles: one identifies sufficient structure, and the other records where restricted application departs from it.

7. NEURAL BELLMAN OPERATORS AND THE COMPUTATIONAL COMPARISON

7.1. *Proposal, evaluation, improvement, and certification*

discounted finite-horizon problem, let T_n be the Bellman operator and T_n^p its policy counterpart. A neural operator proposes a value or policy indexed by policy parameters. Policy evaluation yields an executable lower payoff; Bellman improvement searches the admissible actions; an independent upper construction bounds that search or proposal might have missed. These are separate mathematical objects. A proposed policy with evaluated value L and a valid optimal upper bound U certifies at most $U - L$, regardless of how it was proposed.

This compendium proves the continuous operator, approximation, recursive-utility, equilibrium versions with their own regularity and contraction assumptions. Here a bounding-law certificate uses the mixture structure in (6). A fixed policy's value is a polynomial in the common law parameter. Its coefficients can be evaluated using the deposited policy and kernels. Restricting the polynomial to a subinterval of the parameter space gives a lower witness throughout that interval. An upper construction on counts of endpoint draws while preserving a common current witness before the next draw. Signed chord corrections provide a compressed alternative. A parameter-dependent arithmetic account applies separately to policy coefficients, upper bounds, restrictions, and signed comparisons.

These constructions do not assume that the common law parameter can be selected at every state. A rectangular parameter-lifting bound permits such extra choices and consequently an informative, potentially looser comparator. Neither signed chord nor neural approximation is guaranteed to dominate an exact finite-state solution. Their costs and certificate widths must be compared on a common target and the same lower witnesses and the same error convention.

7.2. *Matched evidence and its limits*

Table VI includes teacher construction, fitting, online evaluation, and the final policy comparison of the actual deposited neural, polynomial, and nearest-anchor proposals. The parameter laws are 0.12 and 0.13; tests include 0, 0.125, 0.25, 0.75, and 1. The network is a standard two-hidden-layer, 32-unit model, not a relabeled interpolation scheme. The results use the same 1,617-state, eight-interval settlement economy used by the economic

analysis. On this scale the nearest-anchor proposal is the strongest of the three in achieved value. The network's larger gaps are charged rather than excused by its fit. The comparison includes an exact upper solve and policy verification, which limits the benefit from a cheap proposal. The measurements consequently do not establish a clear advantage in state dimension, action complexity, or end-to-end accuracy per se. The title denotes the maintained operator architecture and its theory; the theoretical procurement theorem relies on verified response and value inequalities, not on the claim of neural superiority.

The supplemental computational table reports common-law chord and count results over $[0.12, 0.13]$, $[0, 0.25]$, and $[0, 1]$, together with the independently specified rectangular-lifting implementation. Endpoint solves and lower-witness work are

Regime	Proposal	Teacher	Fit	Online max.	Gap max.	KiB
Adjustment	Neural	1.576	1.125	1.742	0.806	22.1
Adjustment	Polynomial	1.576	0.005	1.762	0.0902	5.8
Adjustment	Nearest	1.576	0.000	1.710	2.13×10^{-07}	101.1
No adjustment	Neural	1.798	0.750	1.897	0.568	13.4
No adjustment	Polynomial	1.798	0.003	1.870	0.0147	1.8
No adjustment	Nearest	1.798	0.000	1.795	2×10^{-07}	101.1

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). Seconds are from the completed R13 execution, not a timing rerun in R14. Online work includes proposal, policy evaluation, DP upper reference, and independent verification. Maxima cover five test laws and both mandates. Reports stored proposal parameters or anchor policies, not process peak memory. No fitting step is used for nearest-anchor lookup; teacher cost remains charged. The experiment uses one target size and recorded neural seed.

ged to every method using them. The comparison with parameter lifting is a comparison of algorithms on the same arrays, not an external-package speed ranking. Peak memory and hardware identities are attached to their execution records. Repeated sampling distributions, high-dimensional scaling, and failure probabilities across random sampling runs are not inferred from a single recorded fit. A missing measurement is recorded as such rather than entered as zero.

7.3. Arithmetic and an open financial mandate

All binary64 entries of the canonical target are treated as exact rational inputs to its mathematical Bellman problem. Evaluation error is then derived under a stated arithmetic model: correctly rounded basic operations, round-to-nearest, finite nonnegative sparse data, no overflow, gradual underflow, and reductions within the documented precision-count envelope. The derivation is not a formal proof of every possible implementation of a sparse numerical library.

Let s bound nonzero terms per sparse row, M discounted row mass, and B_{n+1} a bound on the magnitude of continuation values. With unit roundoff u and $\gamma_k = ku/(1 - ku)$, row and reward calculation has a forward-error envelope of the form

$$e_n \leq Me_{n+1} + \gamma_{k_n}(\bar{R}_n + MB_{n+1}) + k_n \text{sub}, \quad (25)$$

where k_n also covers mixture, reward, and comparison operations and sub charges normal absolute error. Maximization is nonexpansive in the sup norm. The actual fit derives s , M , B_n , and each operation envelope from all stored rows. Its largest error and across the inherited certificate operations is below 7.057×10^{-11} . The inherited economic records charge 10^{-7} per value; new value queries charge 10^{-10} only after audit verifies that this allowance covers their operations. Equality-graph support calculations charge their separately derived support allowance.

For a positive first mandate, let $\bar{W}_+$ be the optimum on $[0, L]$. If its optimizing financial fit is strictly positive, the associated policy is an attained feasible witness. If it is zero, choose a positive share ϵ in the adjacent affine segment while keeping the same first assumption-adjustment pair and continuation policy. The value loss is the segment of times ϵ and is charged explicitly. This produces a feasible lower payoff, while $\bar{W}_+$ remains an upper supremum. An exact-response procurement statement may use the

responses and the charged closure gap. The equality of an open supremum closure value never makes the zero action feasible.

8. FROM A NUMERICAL TARGET TO AN ECONOMIC DECISION

8.1. A complete conditional error chain

, denote the Bellman operator of an intended model and $\hat{T}_n$ that of a numerical. An interpolation or lifting operator I_n places numerical functions on the model's space. Suppose, for a stated envelope of admissible continuation functions,

$$\|T_n I_{n+1} v - I_n \hat{T}_n v\| \leq \delta_n, \quad \|I_n v - I_n w\| \leq \Lambda_n \|v - w\|. \quad (26)$$

rm, domain, boundary treatment, and continuation envelope are part of this. A residual evaluated only at selected nodes does not establish it. Write the allowance as

$$\delta_n \leq \delta_n^{\text{time}} + \delta_n^{\text{trunc}} + \delta_n^{\text{state}} + \delta_n^{\text{action}} + \delta_n^{\text{first}} + \delta_n^{\text{kernel}} + \delta_n^{\text{boundary}} + \delta_n^{\text{arithmetic}}, \quad (27)$$

$\delta_N^{\text{terminal}}$ bound the terminal discrepancy. Each entry refers to a specified interpolation operator, so errors are not omitted or counted twice. In particular, first-date interpolation and later-date consumption-adjustment discretization are separables.

THEOREM 6—Model-to-decision transfer: *Suppose the model Bellman operators are η -close in continuation value with factors M_n , the lifting and residual bounds (26) hold, and the terminal discrepancy is at most δ_N . Then the value discrepancy is at most Δ_n , where*

$$\Delta_N = \delta_N, \quad \Delta_n = \delta_n + M_n \Delta_{n+1}. \quad (28)$$

required reward query enlarge $[L_i, U_i]$ to $[L_i - \Delta_i, U_i + \Delta_i]$. If the same affine response and feasible-policy interpretation hold in the intended model, Theorems 1 and 2 apply to enlarged intervals. If policies are transferred instead, a uniform two-sided policy-payoff discrepancy Δ maps an intended-model η -response to a target $(\eta + 2\Delta)$ -response, provided the intended policy correspondence is available. A service-moment discrepancy must be charged only when the transferred feature itself changes.

first route avoids a policy-selection error by querying values in the intended model with valid intervals. The second is convenient for policy witnesses but requires intended correspondence; transferring only a favorable numerical policy does not disqualify all intended-model competitors. For a one-dimensional secant, enlarging both endpoints by Δ increases each service error by at most $2\Delta/h$. Participation adds Δ per contract. Hence a simple two-contract sufficient decision-error charge $(2\Delta/h + \Delta)$, before quality, fee, or feature-transfer errors. The smallest purchaser budget at the magnitude of private values, determines whether this budget is adequate.

8.2. What the deposited constructor establishes

canonical arrays are produced from stochastic-settlement primitives by a finite-transition construction and state projection. They are not exact transition probabilities of the original continuous diffusion. The binary64 audit starts after that construction.

rior first action $(c, \theta, \pi) = (0.05, 0, 0)$ at $u = 2$ and computes the normalized stored preference variance with exact rational arithmetic on its row. Both endpoint laws give approximately

$$\text{Var}_{\text{stored}}(u_1) = 0.0008838834764831865, \quad \sigma_u^2 \Delta t = 0.0003125. \quad (29)$$

a drift-free increment smaller than a preference grid cell, linear projection to the adjacent nodes contributes variance $\Delta u \sigma_u \sqrt{\Delta t}$, rather than $\sigma_u^2 \Delta t$. Here their ratio is 8427.... This is a local moment comparison, not a lower or upper bound on value error. It nevertheless rules out identifying array-evaluation accuracy with constructor accuracy.

A raw total-variation comparison between an atomless diffusion transition and a σ -supported law is also inappropriate: they are mutually singular before a common section or lifting is specified. The relevant operator estimate must use an explicit interpolation and a continuation regularity class, or compare consistently projected kernels. The supplement states the corresponding Lipschitz and smooth-test-function alternatives and keeps the boundary and stopping terms separate.

The inherited stress calculation supplies a sufficient neighborhood under hypothetical reward and kernel errors. It does not establish that the diffusion lies in that neighborhood. Accordingly, the paper's executed strict inequalities are about the hash-identified settlement array economy and the continuous *contract* extension within it. The general model-to-decision theorem specifies what is required for the antecedent continuous-state model; its quantitative inclusion has not been established by the posited constructor. This distinction preserves, rather than replaces, the continuous-state theory and provides a testable approximation requirement for it.

9. CONCLUSION

A dynamic value certificate becomes an economic contract certificate only after the reduced response, participation payment, and enforcement cost are treated together. The reward queries can identify a joint outer set of all responses without differentiation or a favorable tie-breaking rule. Rational support bounds make that information available by a purchaser. A complete cover of the enforcement interval then replaces an unchecked menu approximation with a bound against every continuous instrument. The settlement application illustrates why each step matters. Refining enforcement changes the compulsory-term comparison in the original institution; fee incidence and quality valuation alter which response the purchaser values; and signed future exposure can reverse a local adjustment advantage. These findings do not disappear when a neural proposal is inaccurate, because proposal quality and certificate validity are separate. Nor does a verified finite-array calculation by itself establish a diffusion approximation. The contribution is the explicit chain from dynamic value information to response restrictions and then to an executable institutional choice, with each numerical modeling error charged at the comparison where it matters.

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This supplement proves the finite-query response characterization, robust procurement comparison, efficient-capacity result, continuous-fee certificate, primitive opportunity ordering, and model-to-decision transfer theorem. It supplies the all-response graph argument, exact rational dual semantics, complete fee-cover requirements, arithmetic operation account, positive-mandate witness construction, and direct constructor diagnostic. Computational accounts retain unfavorable controls. A separate unchanged historical compendium and supplement preserve every earlier section, result, reference, and adverse experiment.

S.16. FINITE VALUE QUERIES AND JOINT RESPONSE INFORMATION

The section numbering continues the historical supplement: S.1–S.15 are retained in accompanying R12 supplement and complete compendium. The present sections give the current manuscript's additional results. Their numerical target is the R13 conical settlement array system identified in the replication appendix, not a reassignment of the identity of any earlier computation. References prefixed "paper" point to current main manuscript.

S.16.1. *Proof of the finite-query characterization*

PROOF OF THEOREM 1: Let an affine-response economy have payoffs $J(p; t) = a + t \cdot z(p)$ with $z(p) \in K$. For an attained η -response p , put $v = W(0)$, $a = \alpha(p)$, and $z = z(p)$. The inequalities $v - \eta \leq a \leq v$ follow from the definition of response tolerance. For every fixed policy is feasible at every queried perturbation, $a + t_j \cdot z \leq W(t_j) \leq U_j$. At each query i , choose a maximizing response p_i in the stated economy or its closure, put $(\beta_i, z_i) = (\alpha(p_i), z(p_i))$. Its center payoff cannot exceed v , and none of its queried payoffs can exceed the respective U_j . At its own query it attains $W(t_i) \geq L_i$. At zero choose $\beta_0 = v$. These choices establish every lifted inequality. The intercept does not truncate them: from $\beta_i + t_i \cdot z_i \geq L_i$ and $t_i \cdot z_i \leq h_K(t_i)$, it follows that $\beta_i \geq L_i - h_K(t_i)$.

Conversely, take any feasible lifted point. Form a finite economy whose response menu consists of the candidate affine payoff $a + t \cdot z$ and the query-supporting payoffs $\beta_i + t_i \cdot z_i$, for $i = 0, \dots, M$. All these feature vectors lie in K . At zero their maximum is v because all intercepts are at most v and $\beta_0 = v$. The candidate's intercept is at least $v - \eta$, so it is an η -response in this constructed economy. At query j , every menu payoff is at most U_j , while its designated support payoff is at least L_j . The constructed value therefore satisfies every original interval observation.

The construction proves attainability of each feasible lifted feature vector across compatible finite economies. It establishes the asserted sharpness, not a claim that a dynamic transition system can implement every affine menu. If an application supplies additional linear restrictions on feature vectors, they must be imposed on the

genuinely open policy set, necessity can be interpreted on a specified compact of achievable payoff–feature pairs. If no such closure or attainment argument is available, approximate query maximizers can be used after an additional lower-query followed by a limit when justified. The main theorem states its attainment or assumption explicitly. In the settlement application the first financial operator is piecewise affine on a compact closure, later-date menus are finite, and limits of its feature pairs are available. A closure support function is valid as an outer bound on a positive mandate, but is not an executable positive policy.

S.16.2. Secants, a sharp example, and rational duals

For a fixed duration, a fixed η -response obeys

$$W(d+h) \geq J(p; d+h) = J(p; d) + hA(p) \geq W(d) - \eta + hA(p).$$

A symmetric inequality at $d-h$ gives the lower bound. Substitution of the adverse endpoints and intersection with $[0,1]$ gives equation (11). Randomization is none of these affine identities. Neither the proof nor the implemented bound limit as h tends to zero.

Example $W(t) = \max\{t_1, t_2\}$ illustrates the importance of a joint query. At zero the four coordinate points $\pm he_1, \pm he_2$, the candidate affine response $t_1 + t_2$ never meets the observed values. Adding it to the two original responses therefore yields a viable economy with feature vector $(1,1)$ at zero. At (h,h) , however, that candidate and the original value is h . The additional query excludes it. In the exact oracle, the date optimal at zero has intercept zero, so this diagonal query gives $z_1 + z_2 \leq 1$. The visible vector $(1,0)$ attains that bound. The deposited rational examples extend the construction to interval errors and response tolerances; the verifier checks both the visible affine menus and the primal and dual inequalities.

For completeness, let x satisfy $Ax \leq b$ and $\ell \leq x \leq u$, and take any $\mu \geq 0$. Then

$$c^\top x = \mu^\top Ax + (c - A^\top \mu)^\top x \leq \mu^\top b + \sum_j \max\{r_j \ell_j, r_j u_j\}, \quad r = c - A^\top \mu.$$

This proves equation (12). The residual term cannot be dropped merely because a solver reports approximate stationarity. In the deposited certificate all input elements, proposed multipliers, products, and sums in this inequality are interpreted as rational numbers. The displayed floating upper endpoint is rounded outward to the resulting fraction. Zero-multiplier rows can be removed exactly without affecting the bound or the residual; this is the basis of the compact continuum witness. No rows are never removed by a numerical threshold.

S.16.3. Fixed-economy response supports

A lifted query set does not use all of the dynamic economy's restrictions. When an exact Bellman value is available, we also compute support bounds on an enlarged query graph. Let $V_n(x)$ be the exact value, $Q_n(x, a)$ its exact action value, and suppose the exact values and action values have audited errors. An action used with positive

$v(x, a) = V_n(x)$. Otherwise replacing it by an optimal continuation would strictly improve the initial policy on a positive-probability event. The implementation retains every feasible action whose computed action value is within eight audited Bellman-error bounds of the computed value. Under the stated symmetric envelope this contains every exact optimal action. Retaining extra actions does not invalidate an upper support bound. For a direction w in the space of moments, run a separate dynamic program maximizing $w \cdot z$ over the retained actions. The corresponding program for $-w$ supplies a lower bound. The reward is the desired moment increment, not the agent's utility reward; surrender contributes to H and waiting intervals contribute to A and Q . Discounted live-state kernels propagate the current moments, with no additional fee for forced discharge. The first-date financial rule is affine within each stored segment. An optimal interior financial point lies in a segment whose endpoint values support its optimum; feature values for a fixed continuation are affine on that segment as well. Endpoint feature reports consequently enclose every interior exact response. On the positive open set the graph is an outer graph on the closure. Exact-response lower statements additionally establish that a positive feasible maximizer exists; otherwise only near-optimal positive witnesses are admitted. This procedure encloses all exact history-independent and randomized optimal responses, not merely the selected deterministic maximizer. It does not enclose all η -responses by default. An initially η -optimal policy can incur a much larger conditional Bellman loss at a rarely visited state. Expanding every η mask by a single unqualified η would not prove a global statement. The paper therefore uses the finite-query theorem or the continuous-cell incentive inequalities for explicit near-optimal response results, and labels the exact equality-graph results separately.

S.17. PROCUREMENT, ENFORCEMENT CAPACITY, AND TRANSFER INCIDENCE

PROOF OF THEOREM 2: A feasible η -response has private value at least $W_j - \eta \geq \eta$. With grant $\bar{q}_j$, its participation payoff is at least $\bar{G}_0$ after its capacity expenditure. $\bar{q}_j$ is a common executable grant for all admitted responses. Every participating response to an arbitrary offer has private value at most $W_j \leq U_j$ and must consequently be at least $\underline{q}_j$. Both conclusions retain the positive part and apply when participation is weak. By the response-set inclusion, every admitted candidate response supplies operating output worth at least $\inf_{z \in Z_j} s_j(z)$, and every rival response supplies at most its corresponding supremum. Subtract the appropriate grant and remaining costs to obtain the lower and upper purchaser bounds. The upper calculation can give a rival two advantages which no single response attains together; that only makes it more conservative. Comparing the candidate lower bound against all rival upper bounds and zero proves optimal selection. Replacing strict separation by an upper difference of ε proves the regret bound. The argument does not require a particular optimal-policy selection or a differentiable private value. Q.E.D.

When acceptance at indifference is not the institution's convention, add a positive premium ξ to the candidate grant. Its guaranteed surplus falls by exactly ξ . Any ξ

exact response cannot be made exact merely by increasing the transfer, because a private grant does not change operating incentives. That case requires an explicit η -response or an attainment argument.

OF PROPOSITION 3: Fix F , the term, and the mandate. Under the assumption that the private value and the response set are independent of E once $F \leq \zeta E$. Write $c = W$ and $c = C(E)$. Total participation and capacity expenditure is

$$\Phi(c) = [a + \alpha_C c]_+ + (1 - \alpha_C)c.$$

if $\alpha_C c < 0$, its slope where defined is $1 - \alpha_C \geq 0$; if $a + \alpha_C c > 0$, the slope is one. Continuity at the kink proves that Φ is nondecreasing on its entire domain, including where $\alpha_C = 1$. Reducing E to F/ζ therefore weakly increases purchaser payoff by increasing the operating response. Under binding participation, $\Phi(c) = a + c$, so changing E changes total expenditure unchanged. A change in fee recipient instead changes the user's operating payoff by $\Delta \rho_F FH$ and need not preserve a response ranking. *Q.E.D.*

A charge ceiling in the application is $\bar{F} = 1$. Under an efficiency intervention the charge interval is held fixed and the efficient capacity is F/ζ ; for $\zeta < 1$ it may be lower. A fixed capacity ceiling would create a different feasible charge interval that is not silently imposed in the reported counterfactuals. Similarly, moving a capacity ceiling into the wealth state invalidates the hypothesis that the operating problem is independent of its incidence. That experiment must change the initial state or transition probabilities and repeat the value calculations.

Since price and compulsory-term cost vary over a rectangle while operating costs and response sets are unchanged, every displayed purchaser lower or upper bound is affine in those prices apart from fixed, already evaluated participation terms. The differences are affine. The minimum of each difference over the rectangle occurs at a vertex, so checking every candidate-rival comparison at all vertices establishes the price rectangle. This argument underlies the retained original-menu regional analysis. It does not apply to a primitive which changes the policy response itself. New response bounds cover that change.

S.18. THE CONTINUOUS-FEE CERTIFICATE

S.18.1. *Incentive inequalities across fees*

term and mandate. Write a fixed policy's central-benefit payoff as $B - FH$. Every query satisfies $B + \delta A - F_i H \leq W(d + \delta, F_i) \leq U_i$, proving the query rows used in the upper program. Let p_i be an exact response at a center-benefit query fee F_i , with H_i . Its payoff at F is $W(d, F_i) - (F - F_i)H_i$. This is a lower bound on $W(d, F)$, replacing $W(d, F_i)$ by L_i and H_i by the adverse endpoint proves equation (18). An exact response at F has payoff at least this quantity minus η . In the no-moment restriction, let p be an η -response at F . Optimality at F_i and near-optimality at F give

$$\begin{aligned} J(p; F_i) &\leq J(p_i; F_i), \\ J(p; F) &\geq J(p_i; F) - \eta. \end{aligned}$$

F , this gives $H(p) \leq \bar{H}_i + \eta / (a - F_i)$; if $F \leq b < F_i$, it gives $H(p) \geq \underline{H}_i - \eta / (F_i - b)$. If $\eta = 0$, the weak endpoint inequalities follow by taking the exact-response inequality at an endpoint and the graph's bounds there. At positive η , dividing by a zero endpoint difference would be invalid, which is why the general formula uses strictly exterior inequalities.

All these comparisons involve one fixed policy's moments. They do not replace a common fee by a separately chosen fee at each state, and they do not use the coordinate-wise extremes of different responses as if jointly attained. For an open mandate the closure policy class has the same continuous affine payoff limit; its supporting planes remain valid outer information. An executable exact candidate is checked separately for positive attainment.

S.18.2. Product, cost, and participation relaxations

Let $F \in [a, b]$ and $H \in [\ell, h]$. The nonnegative products $(F - a)(H - \ell)$ and $(b - F)(h - H)$ yield the two lower inequalities in equation (20). The products $(b - F)(H - \ell)$ and $(F - a)(h - H)$ yield its two upper inequalities. Every true product $T = FH$ satisfies them, even though a relaxed point need not. The box $a\ell \leq T \leq bh$ is also valid because the variables are nonnegative. For quadratic capacity cost, the identity

$$cF^2 - c(2vF - v^2) = c(F - v)^2 \geq 0$$

gives the cell-wide tangent bound. Lowering capacity expenditure in the participation constraint can only increase the optimistic purchaser objective.

The implementation uses variables (B, A, H, F, T, q) and the baseline objective $A - D(m)$. It imposes the queried payoff upper inequalities, the central response lower bounds with $B - T$, the four product inequalities, and

$$q \geq G_0 - B + T + c(2vF - v^2), \quad q \geq 0.$$

A fee-receiving generalization adds $\rho_F T$ to the objective. The deposited baseline continuum calculation has $\rho_F = 0$; fee-recipient sensitivity on the finite refined menu is separately executed and is not called a continuous-fee sensitivity result.

Every true response has $0 \leq A, H \leq 1$. The all-horizon norm audit gives a payoff envelope below 200, so the box $-200 \leq B \leq 200$ covers every policy's central-benefit, no-fee payoff. The grant box deserves a separate argument. An arbitrarily large grant is feasible but never improves purchaser surplus. Every participating offer is weakly dominated, at its same response, by a minimally sufficient grant. Since $W(d, F) \geq L(d, 1)$ for $F \leq 1$, an η -response's minimal grant is no greater than $[G_0 - L(d, 1) + \eta]_+$ in the baseline. A fee-one value query for every term and mandate checks that this is below 20. Thus restricting the optimistic search to $0 \leq q \leq 20$ loses no purchaser supremum; it is not an assertion that the institution forbids larger transfers. Every payoff, moment, fee, product, and grant box has a stated economic justification.

PROOF OF THEOREM 4: Take any feasible fee and the corresponding term and mandate. At least one deposited cell contains it by the exact rational coverage check. Every admitted response satisfies all that cell's query and response inequalities. Its product satisfies the product relaxation and its capacity cost exceeds the tangent. Replace any excess grant by the minimally sufficient grant as just explained. This

original purchaser payoff. Rational weak duality bounds that objective by the
 ed U_I . Taking the supremum over fees, mandates, terms, and responses proves
 bal upper bound. Subtracting the executable candidate lower bound yields
 n (21). Applying the same argument only to cells with a different term proves
 entification. Q.E.D.

proof is unaffected by a failure of fee-optimum attainment. It bounds a supre-
 the candidate is a concrete binary64 fee interpreted as its exact rational number,
 attained positive first mandate, a separately enclosed all-response service level,
 outward sufficient grant. Its displayed regret is rounded upward. The proof also
 clear why a finite set of successful grid evaluations cannot replace the cell cover.

S.18.3. *Search and independent verification*

search starts from the deposited R13 queries and may add central fees and benefit
 ations. Local benefit shifts constrain service near a contract; wider shifts can
 fictitious high-duration responses left in the local oracle set. The query policy
 nputational choice, not an economic restriction. The final compact certificate
 every nonzero dual constraint and every query defining a variable bound. Its
 d constraint system produces exactly the same rational weak-duality bound
 e removed rows have zero multiplier.

checker does not call the search's cell-construction function. It reconstructs each
 d row from its economic role and query identifier, verifies every box source,
 utes the dual residual, and checks all fee intervals as exact fractions. For each
 ight terms and both mandates, sorted intervals must start at zero, end at one,
 n without a gap. Candidate lower payoffs are recomputed from private value,
 onse duration, actual charge, and the positive-part grant. Every table's global
 er-term bounds are reductions of those verified cell inequalities.

successful search is not itself the independent check. Conversely, the checker need
 n the adaptive search sequence to establish the final inequalities. It reads the
 deposited witness and independently reconstructs its mathematical content.
 timings, including exploratory queries that do not survive compaction, are
 d separately from check-only timings. The initial loose search is not presented
 cessful regret certificate.

S.19. OPPORTUNITY EXPOSURE AND PRIMITIVE ORDERING

S.19.1. *The exact decomposition and its signed enclosure*

and subtract the zero-drift counterfactual payoff with adjusted continuation,
 en the same payoff with no-adjustment continuation. The first difference is
 immediate reward of the fixed counterfactual is unchanged between the two
 s, so the second difference is exactly $K_\sigma(V^a - V^0)$. The third difference is B_σ .
 positivity follows from feasibility of the counterfactual in the no-adjustment
 n. Subtracting the two classes proves equation (22).

ne signed row $D = D^+ - D^-$, positivity gives $D^+\ell \leq D^+O \leq D^+u$ and $D^-\ell \leq$
 D^-u . Subtract the adverse endpoints to obtain equation (23). These bounds
 rp for a box restriction on O : choose its lower or upper endpoint at each state
 ng to the sign of D . With two independently enclosed continuation values

Regime	Term	Upper surplus	Incumbent minus upper
Adjustment	1	0.097599163	-0.000000883
Adjustment	2	0.097496242	0.000102038
Adjustment	3	0.097290205	0.000308075
Adjustment	4	0.097346496	0.000251784
Adjustment	5	0.096625464	0.000972815
Adjustment	6	0.095684008	0.001914272
Adjustment	7	0.096891057	0.000707223
Adjustment	8	0.090734023	0.006864257
No adjustment	1	0.059936149	-0.000000093
No adjustment	2	0.059312494	0.000623561
No adjustment	3	0.059695575	0.000240481
No adjustment	4	0.059878982	0.000057074
No adjustment	5	0.059852451	0.000083605
No adjustment	6	0.058787186	0.001148870
No adjustment	7	0.059714479	0.000221577
No adjustment	8	0.055276700	0.004659356

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). Each upper bound
 ers both mandates and the entire fee interval for its term. A negative same-term difference is the
 ining regret bound, not a failure of term identification.

adius ϵ , the opportunity radius is 2ϵ and the signed exposure uncertainty is at
 st $2\epsilon\|D\|_1$, plus a separate arithmetic charge. The reported map adds 8ϵ for the row
 tures, opportunity subtraction, two sparse products, and final difference; each is
 minated by the audited operation envelope. The charge is additional to, and not a
 placement for, the value-error term.

PROOF OF PROPOSITION 5: For $u' > u$ and $0 < c \leq 1$,

$$\frac{\partial}{\partial c} \left(\frac{c^{1-u'}}{1-u'} - \frac{c^{1-u}}{1-u} \right) = c^{-u'} - c^{-u} \geq 0.$$

position with nondecreasing consumption makes the upgrade's flow increment
 decreasing in wealth. Its terminal increment is nondecreasing by assumption. The
 lth transition preserves the order of increasing functions, and survival and dis-
 nting multiply expectations by common nonnegative scalars. Backward induction
 efore makes the payoff of exercising the irreversible upgrade immediately nonde-
 s in wealth.

ne option to wait has continuation equal to the expectation of next period's option
 e, multiplied by the common survival and discount factor. Starting from the non-
 ecreasing terminal exercise payoff (and the option not to exercise), induction shows
 both immediate exercise and waiting have nondecreasing values. Their maximum
 nondecreasing. The wealth-independent exercise cost does not change this ordering.
 s proves the opportunity-value statement.

rst-order stochastic dominance with equal total discounted mass orders the expec-
 on of every nondecreasing function, including this opportunity value. Hence the
 re exposure under the positive initial transition is at least that under the nonpositive
 sition. Equality of the current gains and replacement terms then makes equation (22)

ch the opportunity value strictly increases, the expectation inequality is strict,
the final assertion. Q.E.D.

proposition is deliberately a primitive result for a specified subclass, not a
n that O increases with wealth in every endogenous-preference model. State-
ent killing can reverse the expectation ordering; endogenous consumption can
the assumed monotone consumption rule; a wealth-dependent surrender payoff
different stopping option; and repeated preference adjustment changes the
structure. The 54-point settlement map keeps these richer features and records
tual signed consequences. All points, including negative differences and zero
ment terms, appear in the deposited table and are independently recomputed
eir full continuation arrays.

S.20. ARITHMETIC, OPEN-CLASS WITNESSES, AND TARGET TRANSFER

S.20.1. *The data-dependent arithmetic account*

numerical target is the exact rational interpretation of stored binary64 entries.
struction history is separately identified. The arithmetic model assumes round-
est correctly rounded basic operations, gradual underflow, no overflow, and
e reduction respecting the recorded maximum operation count. Let $u = 2^{-53}$
 $\phi = 2^{-1074}$. For $ku < 1$, put $\gamma_k = ku / (1 - ku)$. Bounds below treat every resulting
4 scalar in the audit as an exact rational number before the outward conversion
r display.

γ nonnegative sparse row with at most s entries has true mass bounded above

$$\frac{\hat{m} + (s + 1)\text{sub}}{1 - \gamma_{\max(1,s)}},$$

$\hat{m}$ is its computed row sum. The audit scans all rows, checks nonnegativity and
ss, and uses the largest computed row sum together with the largest row length.
bounds the largest true row mass without assuming that the floating maximum
es the exact maximizing row. Stored duplicate summation has already been
rated into the target entries. The largest bound is below one. First-date matrices
uded, not treated as an uncharged initialization.

bound the absolute reward over $d \in [-2, 2]$, $F \in [0, 2]$, and adjustment-cost co-
t at most two. Let M bound discounted mass. With terminal bound $B_0 = \|G\|_\infty$,
 $B_h = \max\{\|G\|_\infty + 2, R + MB_{h-1}\}$. The operation counts for Bellman evalua-
lity Bernstein coefficients, count-information upper recursion, signed-majorant
ation, and polynomial restriction are respectively

$$2s + 20, \quad 2s + 28, \quad 2s + 40, \quad 2s + 30, \quad 6N^2 + 12.$$

deposited target $s = 16$ and $N = 8$, giving 52, 60, 72, 62, and 396. Maximiza-
ects an operand and is nonexpansive in its input error. Multiplication by a
ative kernel propagates continuation error by at most M . Reward formation,
e weights, and the recorded sums are bounded by the corresponding γ_k term
normal absolute charge. This gives the Bellman, policy, and count recurrences
orm in equation (25).

Operation	Error bound
Bellman value	2.79×10^{-12}
Count information upper	3.87×10^{-12}
Fixed policy bernstein	3.22×10^{-12}
Restricted policy	4.1×10^{-11}
Restricted signed chord	7.06×10^{-11}
Signed chord upper	3.28×10^{-11}
Signed transport	1.44×10^{-11}

et: canonical manifest 54adb353c5b85210d (full hash in the release manifest). All rows and the date operator are included. These values bound arithmetic under the stated operation model, not the el-to-array approximation. The full rational bound, row-mass envelopes, and operation counts are listed.

or the signed-majorant calculation, $\|(K_b - K_a)(V_b - V_a)\| \leq 4M\|V\|$. Its error relation also charges the two endpoint-value errors before propagating the previous orant error. The chord combination and its Bernstein restriction have additional ration envelopes. Signed transport has a separate charge from the width of the tinuation enclosure. Taking the maximum over these derived recurrences gives the ertised-operation ceiling test; the allowance is a number to be checked, not an input nise from which the bound is inferred.

he global value norm reaches approximately 108.03 under the deliberately broad it parameter box. It is not replaced by a convenient but unjustified bound such en. The largest derived arithmetic bound is less than 7.057×10^{-11} . Original R13 curement and extension records charge 10^{-7} per value. New R14 value queries rge 10^{-10} after checking the same operation envelope. Graph support arithmetic is ved separately and enclosed with 10^{-9} . Parameter additions used to describe a joint ry are charged when their binary64 result differs from the nominal perturbation; the nd on the resulting reward discrepancy is added before accepting the value-query wance.

he audit is conditional on its arithmetic and reduction conventions. It is not a proof ne sparse-library binary, the processor, or arbitrary compiler transformations. An lementation using a different reduction graph, flush-to-zero, an unbounded approx- te primitive, or overflow must supply a compatible bound or fail the check. The ependent reconstruction tests consistency with the recorded graph and primitives; es not turn an untested software convention into a machine-checked theorem.

S.20.2. Strictly positive lower policies

t a fixed first consumption–adjustment pair, suppose the positive closure optimum urs at the zero-share knot i , with adjacent strictly positive knot $k \leq L$. Let q_i, q_k be r action values under the same continuation policy. For $t \in (0, 1)$, the interpolated on has positive share $t\pi_k$ and payoff $(1 - t)q_i + tq_k$. Its closure loss is $t(q_i - q_k)$ when difference is nonnegative. Choosing t so that this loss is no more than a specified k gives a feasible positive lower policy, while the closure optimum remains an er bound on the positive supremum.

he record includes the two knot identifiers, nonnegative weights summing to the interpolated action, its positive share, its value, and the charged closure gap.

permissions, and value reconstruction. If the closure optimizer is already strictly ϵ , a one-knot witness establishes attainment. A zero action is never accepted because its computed value equals the positive supremum. In the current exact-se incumbent certificates, the independently computed positive optimum is separated from the zero boundary, so the grant is attached to an attained class.

S.20.3. Proof of model-to-decision transfer

OF THEOREM 6: Suppose at date $n + 1$ the intended value differs from the numerical value by at most Δ_{n+1} . Add and subtract the intended operator applied to the lifted value. Lipschitz continuity of T_n bounds the first difference by $M_n \Delta_{n+1}$; the intended operator residual bounds the second by δ_n . Terminal initialization and the induction yield equation (28). The lifting norm is part of the residual condition and any transition between numerical and model norms; it is not silently set to zero when an interpolation can amplify error.

By this value bound at each reward query. The enlarged intervals contain the true values. If the reward perturbation changes the same affine features without changing feasible policies, the necessity argument for the response-set theorem applies directly to intended-model responses. The procurement theorem then uses their common set. No mapping of a selected numerical optimizer is needed for this route. On the policy-correspondence route, suppose all corresponding policy payoffs differ by at most Δ in both directions and that the correspondence covers all competitors for the two suprema. The two optimal values differ by at most Δ . If intended value is η -optimal, its target counterpart has target payoff at least $W - \eta - \Delta$, and its target optimum is at most $W + \Delta$. The target regret is therefore at most $\eta + 2\Delta$. A feature discrepancy changes purchaser output and must be added to the regret bound. This proves the stated alternative and explains the need for two-sided error.
Q.E.D.

On the one-dimensional secant, increasing the adverse value difference by 2Δ widens the interval by $2\Delta/h$. The positive-part transfer map is one-Lipschitz in private value, so it changes by at most Δ . A candidate-rival difference can therefore require the sum of both parties' charges, $2(2b\Delta/h + \Delta)$, before any fee, quality, or direct feature error. Multi-query supports should instead be recomputed with the widened oracle intervals; using separately optimized coordinates may be unnecessarily conservative.

S.20.4. The constructor diagnostic and a usable approximation norm

On the drift-free preference action at an interior node, the unprojected one-step increment in the constructor is $\pm a$, where $a = \sigma_u \sqrt{\Delta t} < \Delta u$. Linear interpolation onto the preference grid leaves mass at the current node and moves total mass $a/\Delta u$ to its neighbors. Its mean increment is zero and its variance is $(a/\Delta u)(\Delta u)^2 = \Delta u a$. The projected increment and matching drift-free diffusion have variance $a^2 = \sigma_u^2 \Delta t$. At a fixed mesh and time step this gives the two quantities in equation (29). The checker computes the stored row's variance directly as its exact second moment minus the square of its exact first moment after normalization, rather than relying on this derivation.

Moment discrepancy by itself does not give the sign of the value or purchaser error. A usable transfer estimate can instead compare lifted continuation values. For

states gives

$$|\mathbb{E}v(X) - \mathbb{E}v(\hat{X})| \leq L\mathbb{E}\|X - \hat{X}\|.$$

a twice differentiable continuation and centered projection, a Hessian envelope ends the second-order contribution by one half the envelope times the projection's end moment. Neither condition is automatic for a stopping value across exercise forced-discharge boundaries. Boundary hitting, terminal interpolation, action maximization, and the interpolation's domain must each be covered in the residual chain. Bound on an interior smooth test function cannot be extended across a discontinuous stopping boundary without argument.

The numerical diagnostic is therefore deposited as a constructor test, not as a small-order certificate. The inherited operator-neighborhood stress calculation is retained in its exact conditional interpretation. It supplies a sufficient decision budget for a model known to lie in that neighborhood; it does not verify the original diffusion's membership. The current main paper retains the full continuous-state theory while paring this outstanding quantitative inclusion from the completed continuous-fee result on the stored settlement economy.

S.21. COMPUTATION, REPLICATION, AND PRESERVATION

S.21.1. *Work accounts and unfavorable controls*

The R13 proposal experiment actually trains the neural network and deposits its weights, class inventory, polynomial coefficients, anchor policies, and every test policy. Whether solves, fitting, proposal time, fixed-policy evaluation, exact upper reference, policy verification are recorded separately. The network's unfavorable outcomes are retained. A near-zero time for a nearest-anchor lookup is not a zero-cost end-to-end certificate; its teacher and upper-reference work remain charged. The memory does distinguish model parameters from process peak resident memory. A cumulative process maximum is not a per-query allocation measurement.

The common-law and rectangular-lifting comparisons use the same immutable arrays. The latter is this repository's implementation of a standard optimistic relaxation, not a benchmark run of an external package. The tables below retain broader law intervals rather than extrapolating the small correction on a narrow economic region.

The new continuous-fee search has a different purpose and is not added to neural training time as though it were neural training. Its query acquisition, rational bound construction, cell subdivision, and independent verification are recorded separately. The search does not exploit a claimed neural error guarantee. No unmeasured high-dimensional scaling advantage, random-seed success rate, or asymptotic superiority is inferred from these small finite-state experiments.

S.21.2. *Source identities and a check-only entry point*

The reviewed report is pinned to the R13 science snapshot it actually assessed. The completed R13 extension deposit is identified separately and is not retroactively ascribed as evidence available to that referee. R14 uses the unchanged canonical manifest, imports the completed extension witnesses, and adds current source, new arrays, response queries, rational continuum cells, mechanism arrays, generated

Regime	Law interval	Bound	Uniform gap	Total seconds
Adjustment	[0.12, 0.13]	Chord	2.01×10^{-07}	1.707
Adjustment	[0.12, 0.13]	Count	2×10^{-07}	3.884
No adjustment	[0.12, 0.13]	Chord	2.06×10^{-07}	1.768
No adjustment	[0.12, 0.13]	Count	2.05×10^{-07}	4.036
Adjustment	[0, 0.25]	Chord	7.78×10^{-07}	1.817
Adjustment	[0, 0.25]	Count	2×10^{-07}	4.057
No adjustment	[0, 0.25]	Chord	2.41×10^{-06}	1.868
No adjustment	[0, 0.25]	Count	5.13×10^{-07}	4.095
Adjustment	[0, 1]	Chord	9.52×10^{-06}	1.724
Adjustment	[0, 1]	Count	2.1×10^{-07}	3.859
No adjustment	[0, 1]	Chord	1.92×10^{-05}	1.809
No adjustment	[0, 1]	Count	9.3×10^{-07}	4.011

nonical manifest 54adb353c5b85210d (full hash in the release manifest). Both methods are their endpoint solves, common feasible-policy lower bank, and own upper calculation. Larger law are retained. The rectangular-lifting execution is recorded separately in the stress file and uses this y’s implementation, not an external solver package.

TABLE S.4
RECTANGULAR PARAMETER-LIFTING CONTROL

Regime	Cells	Lower coefficients	Uniform gap	Total seconds
Adjustment	1	4	0.000891	2.579
Adjustment	2	8	0.000446	5.082
Adjustment	4	16	0.000223	10.220
Adjustment	8	32	0.000112	21.419
No adjustment	1	4	0.00119	2.850
No adjustment	2	8	0.000597	5.801
No adjustment	4	16	0.000299	11.594
No adjustment	8	32	0.000151	23.173

nonical manifest 54adb353c5b85210d (full hash in the release manifest). Every row covers the interval [0, 1]. The number of feasible lower-policy coefficients is reported explicitly. Timings include ded relaxation and matching lower-bound calculation; this is an in-repository implementation of ard algorithm, not an external-package performance claim.

and manuscripts. The release manifest hashes the scientific inputs, the checking tables, and PDFs; the final Git commit identifies the resulting tree without a erential commit hash embedded in its own payload.
primary review command is `python replication/r14/review.py`; its sci-checking component is `python replication/r14/verify.py`. It reads ex-science, reconstructs dynamic values through the independently recoded Engine, ndently reconstructs equality-graph supports, rebuilds economic LP rows and eak-duality bounds, checks fee coverage, and redoes the institutional accounting echanism map. It does not call the producer’s Bellman solver or continuous-cell , and it does not regenerate the numerical target or witness files. An optional path writes only a validation report outside the scientific output directory. Each ipt records hashes of its actual inputs and checks that those files did not change verification. The executable transfer is rounded upward and then charged at ed value in the purchaser lower bound. A replay may reuse nonnegative dual

queries; a floating-point infeasibility or success flag does not replace this calculation. The primary command checks the deposited tables and rebuilds the PDFs in a temporary directory; a newly generated target must never replace the canonical target during a check-only run.

All current validation predicates use explicit exceptions and remain active under optimized Python. Destructive fixtures include a negative rational multiplier, an all-or-nothing constraint, a destroyed upper summary, an out-of-cell fee box, missing interval coverage, a zero-share positive-witness claim, an insufficient participation transfer, and a failure to charge the implemented transfer. These tests specify what the checker rejects. They are not a proof that every software defect has been excluded. The supplementary case receipt identifies the exact checks actually executed and their independence boundaries.

S.21.3. *No-content-loss preservation*

The new manuscript reorganizes the argument around the economic response set of a continuous institution. It does not overwrite the historical R11 or R12 manuscripts, or references, the S.1–S.15 proofs, the original approximation and operator results, the recursive-utility and equilibrium extensions, the negative controls, or the review reports. A preservation inventory records their original and final identities. The complete historical main compendium, R12 main paper, and R12 supplement are included unchanged in the review package, with a cover explaining their target and date. Statements preserved in those historical documents are not silently endorsed as statements about a later numerical target. The new main manuscript and these new proof sections extend the current theorem and evidence chain.

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We study the purchase of operating commitments when an agent controls consumption, investment, preference adjustment, and surrender. A purchaser chooses a financial mandate, guarantee capacity, compulsory term, and participation payment from a specified institutional menu. Value perturbations bound delivered service for every near-optimal response, allowing a contract choice to be certified without selecting a favorable best response. In a stochastic settlement economy, adjustment changes the chosen compulsory term even when the purchaser selects the positive financial mandate in both regimes. A signed opportunity-exposure decomposition distinguishes this procurement comparison from the agent's class ranking. Neural Bellman Operators separate policy proposal from evaluation and certification. Feasible-policy polynomials preserve common transition-law dependence; a signed endpoint correction compresses a count-information upper bound. We certify regional financial and surrender comparisons and report matched broader-law computations. Complete canonical inputs and an independently recoded witness checker connect the economic inequalities to immutable arrays and reject corrupted certificates. Earlier operator, approximation, preference, and equilibrium results are preserved in the accompanying compendium and supplement.

KEYWORDS: Dynamic programming, Procurement, Preference adjustment, Policy iteration, Numerical verification, Computational economics.

1. INTRODUCTION

Operating commitment changes both the payoff to an agent and the service delivered to a purchaser. These two effects need not favor the same financial position. A surrender charge can increase service while reducing private value; a longer compulsory term can replace a financial guarantee, but is not costless to enforce. Preference adjustment changes the continuation values that govern these responses. An economic assessment must therefore compare contracts before selecting the operating allocation, and must distinguish the agent's investment ranking from the purchaser's choice. This paper develops a computational framework for making these comparisons with reliable error bounds. Neural Bellman Operators separate the proposal of a policy from its evaluation, improvement, and certification. The framework accommodates a common parameter that changes the transition law at every date. Feasible-policy values in that common parameter through polynomial representations. On the upper side, count-information relaxation and a signed endpoint correction provide two distinct bounds. Neither an accurate neural fit nor a successful optimizer is itself the numerical certificate.

Our central application is a finite-menu procurement problem in a stochastic settlement economy. A purchaser chooses an initial financial mandate, a surrender-charge capacity, a compulsory term, and a participation payment. Consumption, subsequent investment, deliberate preference adjustment, and elective surrender remain the agent's decisions. Financial capacity and compulsory terms are both priced; not procuring

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This paper develops the numerical framework accompanying Qi (2024). The author thanks Darrell Duffie for discussions.

comparing contracts. This is an institutional-choice problem within an explicit, finite menu, not the implementation of an allocation selected beforehand.

Our results link this economic problem to the computational framework. First, per the agent’s operating benefit bounds the service delivered by *every* sufficiently optimal response, including responses at a tie. This converts value enclosures to procurement-surplus enclosures without differentiability or an assumed selection. Strict interval separation then identifies the winning contract even when rivals choose their most favorable admissible responses. Second, a signed transport complement eliminates downward adjustment on a complete feasible-support tube, while a state opportunity-exposure decomposition explains why this elimination alone does not determine the difference between financial classes. Third, policy-polynomial bootstrapping witnesses establish regional financial and surrender comparisons under a changing law. The procurement and regional calculations share the complete, bounded finite-array economy.

In the central economy, with service price one and the specified enforcement costs, the purchaser selects a positive initial mandate in both adjustment regimes. With no adjustment, the selected contract has capacity 0.6 and three compulsory intervals; with adjustment it has the same capacity and four intervals. These choices are made jointly against all 95 other offers in the respective menu and the outside option. The result remains the same over a rectangle of service prices and term costs. The result does not follow from the agent’s class ranking: the unadjusted agent can prefer the lower class under a common grant even when the purchaser optimally procures the higher mandate. Participation compensation and delivered service account for the difference.

The computational increment is also evaluated on a common target. The signed complement is indispensable to the validity of the endpoint theorem, but need not be large in every application. We therefore report both the narrow economic region and broader intervals, with matched endpoints, lower policies, error levels, and work accounts. Theoretical precision frontiers, neural-free controls, unfavorable permission interventions, alternative initial states, and the recursive-utility and competition extensions are part of the complete research record. They are organized in the accompanying theoretical and computational compendium rather than interleaved with the procurement argument.

1. *Related work.* The reward-transfer component is related to successor features and linear support: [Alegre, Bazzan, and da Silva \(2022\)](#) study policy transfer when rewards are linear in known features. Our fixed-policy reward moments have a specific structure; changing the transition law requires additional arguments. Parametric Markov-model methods already study regional verification and reward properties. [Bonn, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) develop parameter lifting, and [Bonn et al. \(2019\)](#) analyze region construction for parametric chains and decision problems. We do not claim that regional expected-reward control originates here. The upper component is a signed compression of a count-information relaxation that maintains a common action before the next endpoint draw. It is paired with genuinely lower policies. The service comparison below uses elementary convexity of a sum of affine payoffs, rather than an unverified numerical envelope derivative. Envelope arguments are discussed by [Milgrom and Segal \(2002\)](#). The inherited results, policy-improvement, approximation, and equilibrium results and their full discussion are preserved in the compendium and supplement.

certificate, and give the procurement results. Sections 5–6 explain the dynamic mechanism and the changing-law bounds. Section 7 records the target and verification. All new proofs are in Supplement S.15. Supplement S.1–S.14 and the complete compendium retain the earlier results without deleting or rewriting their statements. Their historical numerical targets are distinguished from the newly identified target used here.

2. THE ECONOMY AND CONTRACTIBLE INSTITUTIONS

2.1. Operating payoffs and information

Time is $n = 0, \dots, N$, with $N = 8$ and interval length $1/8$. A live state is $x = (u, X)$, where u is a preference index and X is managed wealth. An operating action is $a = (c, \theta, \pi)$: consumption, deliberate preference drift, and the risky share. Regime $r = \text{adj}$ implements the specified adjustment menu; regime $r = 0$ imposes $\theta = 0$ but does not remove stochastic preference variation. The public regime is not chosen by the purchaser in the experiment.

For each date, state, and feasible action, the canonical economy specifies rewards $R_n(x, a; d)$ and nonnegative discounted live-state kernels $K_{n,k}(x, a, \cdot)$, $k = 0, 1$. Their structure is

$$R_n^\lambda = (1 - \lambda)R_{n,0} + \lambda R_{n,1}, \quad K_n^\lambda = (1 - \lambda)K_{n,0} + \lambda K_{n,1}. \quad (1)$$

The same λ applies at every date. The agent selects the current action before the point-law draw. Boundary discharge rewards are already included in R ; missing live-state mass must not receive those rewards a second time. The coefficient of d in the current reward is discounted operating duration. It is nonnegative. Total discounted duration $A(\pi)$ of a feasible operating and stopping policy satisfies $0 \leq A(\pi) \leq 1$.

The original settlement construction uses the cardinal flow primitive, adjustment rule, boundary rule, and terminal payoff documented in the compendium. The terminal payoff is

$$G(u, X) = -0.02(u - 2)^2 + 0.1 \log X, \quad (2)$$

the initial state is $x_0 = (2, 1.25)$. The finite target has 1,617 states. Its common consumption–adjustment–investment menu and three date- and state-specific frozen proposals are retained. At date zero, conditional on the finite consumption–adjustment menu, the risky share is evaluated by the stored piecewise-affine breakpoint operator. This is continuous optimization of that operator on each permitted share interval, plus optimization over continuous consumption and adjustment controls. Subsequent menus are unchanged. The mathematical object for the new numerical results is the discretized array system, not an unstated diffusion approximation.

The mandate $\sigma = +$ requires $0 < \pi_0 \leq L$ for the whole first interval; $\sigma = -$ requires $-\pi_0 \leq \pi_0 \leq 0$. The closure at zero may be used to bound a supremum for the positive class. A lower policy used to establish an attained positive-class value must have a strictly positive first share. There is no claim that the sign constraint restricts all later financial positions.

The contract specifies a compulsory term $m \in \{1, \dots, 8\}$ and charge $F \geq 0$. At a live, boundary date $n \geq m$, the agent may surrender for $G(x) - F$. Surrender is unavailable before m , and forced discharge still pays $G(x)$. Thus $m = 8$ is noncancellable within

$$J_j^r(\pi; d) = B_j^r(\pi) + dA_j^r(\pi) - FH_j^r(\pi), \quad W_j^r(d) = \sup_{\pi \in \mathcal{P}_j^r} J_j^r(\pi; d), \quad (3)$$

H is discounted elective-surrender incidence. It is not an undiscounted probability-diffusion hitting probability. Adjustment expenditure and discharge rewards. The menu item j collects the enforceable financial mandate and stopping terms. The feasible set is unchanged when d is perturbed for the service calculation.

2.2. Procurement timing and instruments

The purchaser values one unit of discounted service at $b \geq 0$. This service price is independent from the operating-benefit parameter d in the agent's preferences. Public instrumentations are fixed at $L = 0.8$ and $S = 0.5$ in the procurement experiment. They define the ex ante risk standard under which all offers are compared. The purchaser does not optimize over that standard or over an unrestricted mechanism space. The noncontractible menu is

$$\mathcal{J} = \{+, -\} \times \{0, 0.2, 0.4, 0.6, 0.8, 1.0\} \times \{1, \dots, 8\}. \quad (4)$$

The second coordinate is guarantee capacity E , and the surrender charge is $F = E$. The first six capacities are the available standardized enforcement products; the eight coordinates are the available settlement dates. The finite instrument restriction is part of the experimental environment, not an assertion that these points approximate every feasible contract without error. Each publicly known adjustment regime has 96

At time zero the purchaser chooses an offer $j = (\sigma, E, m)$ and a nonnegative signing payment q , or makes no offer. The agent accepts or takes outside value $G(x_0)$. Capacity E is the agent $C(E) = 0.02E^2$ in a separate utility numeraire. The purchaser pays $\kappa(m - 1)/8$ for the compulsory term. Surrender charges accrue to an external enforcement sector and are not purchaser revenue. Neither the signing payment nor the surrender charge enters managed wealth or changes the operating feasible set. After acceptance, the agent chooses all noncontracted controls and stopping decisions to maximize (3). There is no report of private type and no adverse-selection incentive constraint hidden in the complete-information timing.

Given an exact optimal response and value W_j^r , the infimum signing payment is

$$q_j^r = [G(x_0) - W_j^r + C(E)]_+. \quad (5)$$

The positive part is retained even when participation happens to bind in the reported instrument. With a response delivering A , purchaser surplus is $bA - q - D_\kappa(m)$. Weak instrumentation is resolved by acceptance; any required strictly positive acceptance margin is an additional payment and must fit inside the certified surplus gap. When the agent has multiple best responses, the service comparison in the next section covers all of them. This removes the need to select a favorable optimal policy before evaluating the instrument.

For a common enforcement instrument and binding participation, comparing two instruments gives

$$\Pi_+^r(b) - \Pi_-^r(b) = (W_+^r - W_-^r) + b(A_+^r - A_-^r). \quad (6)$$

erence does not certify procurement. When instruments differ, their capacity and costs must also be charged; when participation is slack, (5), rather than the linear simplification, applies.

3. FROM VALUE ENCLOSURES TO A PROCUREMENT DECISION

Let $\eta \geq 0$. A response is η -optimal if its payoff is at least $W_j(d) - \eta$. Exact best responses are included. The following result does not assume uniqueness, smoothness, or coincidence of policies at nearby benefits.

PROPOSITION 1—Service of all near-optimal responses: Suppose $J_j(\pi; d + t) = W_j(d) + tA_j(\pi)$ for every feasible policy and $0 \leq A_j(\pi) \leq \bar{A}$. Let $h > 0$, and suppose $W_j(d + t) \leq u_i$ for $t \in \{-h, 0, h\}$. Every η -optimal response at d satisfies

$$\underline{A}_j = \max \left\{ 0, \frac{\ell_0 - \eta - u_{-h}}{h} \right\} \leq A_j(\pi) \leq \min \left\{ \bar{A}, \frac{u_h - \ell_0 + \eta}{h} \right\} = \bar{A}_j. \quad (7)$$

conclusion holds for randomized responses as well. A positive-class supremum need not be defined for the inequalities to apply to any existing η -optimal feasible response.

smaller h does not automatically sharpen the certificate. It reduces the change in marginal policy but magnifies value-evaluation error by $1/h$. With certified value error and two error contributions in a secant can total $2\epsilon/h$. A larger h can instead include a change in stopping behavior and widen the economic secant. We use $h = 0.01$, $\eta = 10^{-8}$, $\epsilon = 10^{-7}$ per value, and report the resulting intervals rather than interpreting the bounds as exact derivatives.

define an executable participation payment and a lower bound on any feasible payment by

$$\bar{q}_j = [G(x_0) - \ell_0 + C(E) + \eta]_+, \quad \underline{q}_j = [G(x_0) - u_0 + C(E)]_+. \quad (8)$$

first payment secures weak participation for every η -optimal response. The second bounds from below the payment of any participating response, since its operating cost cannot exceed u_0 . For $b \geq 0$, put

$$\underline{\Pi}_j = b\underline{A}_j - \bar{q}_j - D_\kappa(m), \quad \bar{\Pi}_j = b\bar{A}_j - \underline{q}_j - D_\kappa(m). \quad (9)$$

THEOREM 2—Selection from a contractible menu: Suppose the enclosures in Proposition 1 hold for every menu item. If an item j^* satisfies

$$\underline{\Pi}_{j^*} > \max\{0, \max_{j \neq j^*} \bar{\Pi}_j\}, \quad (10)$$

offering $(j^*, \bar{q}_{j^*})$ guarantees strictly greater purchaser surplus than no procurement or any other menu item with any participating η -optimal response. The result allows the selected item to be the most favorable admissible response and each rival its most favorable one. Consequently the selection choice is the same under every selection among exact best responses. The executable payment is not asserted to equal the exact minimum grant.

$D_\kappa(m)$ is affine in κ and the same item satisfies (10) at every vertex of a rectangle in (b, κ) with $b \geq 0$, it satisfies it throughout the rectangle.

Regime	(σ, E, m)	Grant	Service enclosure	Margin
Adjustment	$(+, 0.6, 3)$	0.78921187	$[0.8902122, 0.8902543]$	5.578×10^{-4}
No adjustment	$(+, 0.6, 4)$	0.82466919	$[0.8902122, 0.8902543]$	1.564×10^{-4}

in compares the chosen lower surplus with every rival upper surplus and zero. Service endpoints are displayed outward. Each regime has 96 offers. The minimum strict margin over the eight price-box/regime vertices is 7.688×10^{-5} . Full precision is in the deposited witness record.

rectangle assertion uses affine differences of fixed enclosure endpoints; it does not use the continuum theorem from interior solutions. There are finitely many competitors, including the zero-profit outside option. No differentiation of an optimized policy is required. The result compares allocation and institution jointly, unlike an enforcement mechanism that is only evaluated after fixing a noncancellable allocation.

4. FINANCIAL MANDATES AND THE CHOSEN OPERATING TERM

The procurement experiment uses $\lambda = 0.125$, $d = 0.42425$, and the permissions specified in Table 1. The two extra service evaluations occur at $d \pm 0.01$ on the same complete capacity target. They are not assumed to lie within the narrower regional financial-price box in Section 6. We solve 96 operating/stopping problems per regime at these three benefits, sharing continuation calculations between the two initial offers. Thus there are 192 offers and 288 distinct continuation-parameter queries, with two first-date class comparisons.

Table 1 reports the winning contracts at $(b, \kappa) = (1, 0.02)$. The quoted payment in the value allowance and response tolerance in (8). The nominal service and order moments in the replication record are policy diagnostics; the instrument comparison uses the wider all-response service intervals, not those point values. The listed positive first shares are strictly feasible. All fee levels, terms, and signs, including surrender offers, enter the comparison. Compulsory service is not assumed to be required after the minimum term: subsequent surrender remains an optimized choice.

THEOREM 3—Procurement in the canonical settlement economy: *For the deposited capacity target, menu (4), and the stated value allowances, the unique discrete procurement choice is $(6, 3)$ under adjustment and $(+, 0.6, 4)$ without adjustment, uniformly for*

$$b \in [0.998, 1.02], \quad \kappa \in [0.0198, 0.0202]. \quad (11)$$

Equalities hold for every 10^{-8} -optimal response to the quoted contract and against every competing 10^{-8} -optimal response to every competing item, as well as no procurement. The listed grants follow (8). The certificate does not assert uniformity over additional values of b or over unlisted enforcement products.

The economic effect of adjustment is a change in the chosen commitment term, not an immediate reversal of the purchaser's financial mandate. Improving the agent's continuation opportunity changes the compensation and enforcement needed to deliver service. On the standardized capacity menu, adjustment makes the three-interval contract optimal; without adjustment, four intervals are preferable at the same capacity. This is the only optimized comparison. It is distinct from the earlier continuously chosen

b	Adjustment	Margin	No adjustment	Margin
0.80	No offer	1.090×10^{-2}	No offer	1.090×10^{-2}
0.90	(+, 0.6, 3)	5.620×10^{-4}	(+, 0.0, 1)	4.181×10^{-4}
0.94	(+, 0.6, 3)	5.603×10^{-4}	(+, 0.0, 4)	1.054×10^{-4}
0.98	(+, 0.6, 3)	5.586×10^{-4}	(+, 0.0, 7)	1.623×10^{-4}
1.00	(+, 0.6, 3)	5.578×10^{-4}	(+, 0.6, 4)	1.564×10^{-4}
1.02	(+, 0.6, 3)	5.570×10^{-4}	(+, 0.6, 4)	2.018×10^{-4}
1.05	(+, 0.6, 3)	5.557×10^{-4}	(+, 0.6, 4)	2.698×10^{-4}

positive margin certifies the indicated point choice against the complete menu. A nonpositive margin could be explicitly inconclusive; it would not be labeled a certificate. These point comparisons do not establish behavior between displayed prices.

capacity threshold for implementing a previously selected noncancellable allocation, which remains a valid conditional result in the compendium.

The distinction from private investment ranking is essential. At a common grant, the adjusted agent can prefer the nonpositive class. With individualized participation payments, a purchaser may compensate a lower private value and obtain enough additional service to prefer the positive mandate. Equation (6) identifies that margin. Conversely, at the active-surrender midpoint contract with $E = 0.8$ and $m = 1$, the adjusted positive class can deliver less service; fixing that institution is not innocuous. Allowing the purchaser to choose E and m prevents either conditional comparison from being substituted for the menu optimum.

The price variation in Table II makes the service margin economically visible. Offers that permit more surrender can be attractive when service is less valuable, because they reduce the participation payment. Higher service prices can justify additional compulsory operation. These are certified point comparisons where the reported strict margin is positive; the rectangle in (11) is the separate uniform statement. No interpolation between point choices is used to claim a complete phase diagram. Nor is the selected contract called optimal outside the stated menu.

5. ADJUSTMENT DIRECTION AND CLASS-SPECIFIC OPPORTUNITY EXPOSURE

The result eliminating negative preference drift in both classes does not explain which class gains more from adjustment. We separate these questions. Write W_σ^r for class σ values under a fixed contract and $\Omega_\sigma = W_\sigma^{\text{adj}} - W_\sigma^0$. Let a_σ be an adjusted optimal first period point and let a_σ^0 set its drift to zero while preserving its consumption and financial control. Suppose this counterpart is feasible. Let $O_1 = V_1^{\text{adj}} - V_1^0$, and let K_σ be the one-date kernel of a_σ^0 .

PROPOSITION 4—Opportunity exposure and financial replacement: *With $Q^r(a) = Q(a) + K(a)V_1^r$, define*

$$g_\sigma = Q^{\text{adj}}(a_\sigma) - Q^{\text{adj}}(a_\sigma^0), \quad (12)$$

$$B_\sigma = Q^0(a_\sigma^0) - W_\sigma^0 \leq 0. \quad (13)$$

$$\Omega_+ - \Omega_- = (g_+ - g_-) + (K_+ - K_-)O_1 + (B_+ - B_-). \quad (14)$$

λ	d	(F, m)	Local	Future	Replacement	Total
0.125	0.42425	(0.8, 1)	5.039×10^{-5}	2.952×10^{-4}	0	3.456×10^{-4}
0.125	0.42425	(0.0, 8)	5.039×10^{-5}	3.557×10^{-4}	0	4.061×10^{-4}
0.120	0.42400	(0.0, 8)	5.055×10^{-5}	3.568×10^{-4}	0	4.073×10^{-4}
0.130	0.42450	(0.0, 8)	5.024×10^{-5}	3.546×10^{-4}	0	4.049×10^{-4}

row compares the positive and nonpositive classes. These are pointwise decompositions. The continuation uncertainty radius is approximately 2.1×10^{-7} per future-exposure comparison. The full signed intervals and wealth-bin contributions are deposited and independently reconstructed.

certified $\underline{Q} \leq O_1 \leq \overline{O}$, writing $D = K_+ - K_- = D^+ - D^-$ gives

$$D^+ \underline{Q} - D^- \overline{O} \leq DO_1 \leq D^+ \overline{O} - D^- \underline{Q}. \quad (15)$$

icular, $O_1 \geq 0$ does not determine the sign of DO_1 .

first term is the relative gain from current deliberate drift at the selected financial states. The second measures different exposure to future adjustment opportunities. The third is financial replacement: adjusted financial controls need not solve the unadjusted problem. Dropping that last term without checking its premise can give a false conclusion. Since each B_σ is nonpositive, their difference has no predetermined sign. A interpretable sufficient condition can be stated as an opportunity-tail comparison. Suppose active and cemetery states by a specified opportunity index, and suppose $O_1(x_i) = O_2(x_i)$ is nondecreasing in that order. If the two augmented kernels have equal mass and $\sum_{j=i}^1 D_j \geq 0$ for every $i \geq 2$, then

$$DO_1 = \sum_{i=2}^M T_i(o_i - o_{i-1}) \geq 0. \quad (16)$$

cemetery has zero future option; its inclusion accounts for different live-state tail masses. More generally, bounds on the increments of o_i and signed tail masses follow from lower and upper exposure bounds by assigning adverse endpoints to each signed tail mass. This identifies what a monotone opportunity explanation requires: both a lower bound of future opportunities and an ordering of class exposure. Neither follows from the current drift elimination. The numerical application uses (15); it does not assert an adjusted monotone wealth ordering.

III evaluates the components at the actual benefit midpoint and at two additional benefit points, separating active surrender from a noncancellable contract. The adjusted exposure bounds allow uncertainty in both continuation arrays. Wealth-bin contributions are also deposited so that the continuation contribution can be localized rather than inferred from a favorable aggregate. These are pointwise mechanism decompositions, not a uniform decomposition over the five-dimensional box.

In the direction of drift, let $\mathcal{R}_n$ contain the support of every feasible initial action. The largest permissions and be closed under every feasible subsequent live action. For each negative-drift action a , compare a feasible zero-drift action a^0 with the financial and consumption controls. If $\ell \leq V_{n+1}^\uparrow \leq h$ on this support, a valid

$$R^a - R^{a^0} + (K^a - K^{a^0})\frac{h + \ell}{2} + |K^a - K^{a^0}|\frac{h - \ell}{2} + \varepsilon. \quad (17)$$

ct negativity for every such pair gives, by backward induction, $V^{\text{adj}} = V^\dagger$ throughout support. Frozen proposals must also be feasible and accounted for. Our checker constructs the entire feasible-support tube and the statewise enclosures before testing the inequalities. The support is not selected after observing an optimal policy. This direction result is local to the specified initial state and parameter region. The served alternative-initial-state experiment demonstrates substantial value from outward adjustment near a preference-discharge boundary. That experiment is economically informative: access to discharge changes the continuation tradeoff. It is not pressed to create a global monotonicity claim. Likewise, nearby long- and short-emission interventions can remove the class reversal. These adverse cases delimit the mechanism rather than invalidate a correctly localized theorem.

6. CHANGING-LAW BOUNDS AND REGIONAL ECONOMIC COMPARISONS

6.1. Policy lower bounds and two upper constructions

For a fixed feasible Markov policy, a common law parameter produces a polynomial value over the remaining dates. Let $B_i^H(t) = \binom{H}{i} t^i (1-t)^{H-i}$ denote Bernstein basis functions. On $\lambda \in [a, b]$, $t = (\lambda - a)/(b - a)$, a feasible policy has representation $J^\pi(\lambda) = \sum_i \beta_i^\pi B_i^H(t)$. Its minimum coefficient bounds its value from below. Reward and charge dependence is affine for that fixed policy, so reward corners can be considered jointly while optimizing across policies. In particular, the valid order is

$$\max_{\pi \in \mathcal{B}} \min_{\text{reward corners}, i} \beta_i^\pi, \quad (18)$$

a choice of a different policy inside each coefficient minimum. Each policy must be feasible under the smallest permissions used for the lower bound.

Endpoint optimal values do not in general form an upper chord when a parameter changes the law at every date. The count-information recursion supplies an upper maximization by conditioning on the number of endpoint draws remaining, while preparing one common action before the next draw. Its interior recursion maximizes the conditional mixture of two endpoint backups, not two separately optimized actions selected after the draw. Feasible policies retain the common-parameter dependency on the lower side.

The signed chord compresses that upper information. Put $\delta V_{n+1} = V_{n+1}^b - V_{n+1}^a$. A negative correction can be propagated using

$$M_n = \left[\max_{a_n} \{ (K_n^a - K_n^b) \delta V_{n+1} + \max(K_n^a M_{n+1}, K_n^b M_{n+1}) \} \right]_+. \quad (19)$$

The superscripts a, b on kernels denote parameter endpoints, not actions; the maximization is over feasible controls. Over H remaining dates, the coefficient correction is $(-i)M_n/[H(H-1)]$ for $H \geq 2$, added to the endpoint coefficient chord. The one-period value uses its endpoint chord without that term. Supplement S.10 preserves full ordering and induction proof. The checker independently reconstructs both

Comparison	Certified interval
Adjusted positive minus nonpositive	[0.0001876285, 0.0002023956]
Unadjusted positive minus nonpositive	[-0.0001792987, -0.0001221851]
Unadjusted positive: surrender gain	[0.0000347089, 0.0000864511]

ved endpoints are rounded outward. Both upper constructions independently reproduce these signs. Lower-policy feasibility, corner/degree ordering, upper recursions, and all support-tube transport inequalities are checked from canonical arrays.

recursions on all states and dates. A coefficient array is not accepted because its ne or JSON summary says “upper.”
 signed cross term can have either sign before positive majorization. It cannot be d because a particular numerical example happens to make it small. Conversely, em requiring the term does not establish that it gives a material advantage on workload. Policy proposal can be neural or nonneural without changing the of these witnesses.

6.2. The joint financial and surrender region

ne new canonical target, feasible lower policies and both upper constructions a common box

$$\begin{aligned} \lambda \in [0.12, 0.13], \quad d \in [0.424, 0.4245], \quad F \in [0.7999, 0.8001], \\ L \in [0.7999, 0.8001], \quad S \in [0.4999, 0.5001], \quad m = 1. \end{aligned} \quad (20)$$

sions are handled monotonically: lower policies use the smaller feasible sets, values the larger ones. Reward and charge corners are treated with a fixed policy before taking coefficient minima. Bernstein restriction deals with the continuum. These operations certify a region; finite interior solutions are only necks.

adjusted agent prefers the positive financial class, the unadjusted agent prefers positive class, and the unadjusted positive class has a strictly valuable sur- option relative to its noncancellable value. The third comparison concerns a rfactual class: it does not say that the unconstrained unadjusted agent selects ss and surrenders. Dynamic downward-drift elimination holds on the indepen- reconstructed feasible-support tube. This joint statement is retained alongside, stituted for, the procurement result. The menu optimum chooses a different ion; a single conditional sign comparison cannot settle that choice.

6.3. What broader law variation changes

V compares the two upper components at the same benefit, fee, permissions, nt solutions, and two-endpoint lower-policy bank. It reports the maximum on above an uncorrected endpoint chord, the difference between upper coef- arrays, and a common eight-subcell bound on the per-value upper-lower gap. rader intervals are computational stress tests, not assertions that the financial- l box extends to those intervals. Both methods receive the same endpoint and

Law range	Regime	Correction	Chord gap	Count gap	Work (s)
[0.12, 0.13]	Adj.	1.150×10^{-9}	2.009×10^{-7}	2.000×10^{-7}	0.96/5.03
[0.12, 0.13]	Zero	2.750×10^{-9}	2.058×10^{-7}	2.049×10^{-7}	0.95/5.07
[0, 0.25]	Adj.	7.214×10^{-7}	7.775×10^{-7}	2.000×10^{-7}	0.95/5.07
[0, 0.25]	Zero	1.205×10^{-6}	2.407×10^{-6}	5.128×10^{-7}	0.96/5.10
[0, 1]	Adj.	1.146×10^{-5}	9.520×10^{-6}	2.098×10^{-7}	0.95/5.08
[0, 1]	Zero	1.453×10^{-5}	1.918×10^{-5}	9.298×10^{-7}	0.96/5.02

include 2×10^{-7} evaluation allowance. Work reports chord/count nonendpoint seconds; common point and lower-policy costs are charged to both in the machine-readable account. Complete coefficient differences, endpoint actions and package-level load/validation costs are deposited. The broader gaps are not a financial-sign theorem.

er-policy costs; nonendpoint work is recorded separately. Canonical loading and witness validation are reported at package level and cannot be silently excluded in total reproduction cost.

he narrow economic region is not presented as a demanding demonstration of correction's magnitude. The broader runs identify how law uncertainty changesificate width, the correction, and upper-construction work. Earlier controlled comparisons of precision, policy-bank enrichment, neural-free proposal, and structural drastic cases remain in the compendium with their original target identities. They not falsely reported as freshly retrained experiments. No comparison here is a benchmark against an entire external parameter-synthesis software system, and no claim of universal wall-time superiority follows from a single horizon or machine.

7. TARGET IDENTITY, ARITHMETIC, AND WITNESS VERIFICATION

he new numerical theorems concern one completely deposited array target. Its outputs include every common and frozen-proposal kernel at every date, all reward, validation, adjustment, discharge and exit-discount arrays, terminal and state arrays, and entire first-date breakpoint operator. Large archives are split into hash-identified sets for repository storage. A manifest records archive and array identities. The reader verifies these identities and never invokes the reward or transition constructor. The expected manifest is not overwritten during a check.

his is a newly identified R12 target. It is not described as a bit-identical reproduction of the earlier R11 target merely because numerical differences are small. The previous expected manifest remains unmodified, and a separate comparison lists any differing primitive identities. No unmeasured R11-to-R12 transfer enclosure is used. Instead, new economic and regional witnesses are generated and checked directly on the canonical inputs. The general operator-transfer results remain available for other net comparisons, including diffusion approximation, but their assumptions are not automatically discharged by array agreement.

he arithmetic audit bounds round-to-nearest evaluation of the stored system. Its saved per-value bound is below the charged 10^{-7} allowance; all displayed class-reference bounds charge twice that allowance. Service secants additionally charge the allowances divided by h and the response tolerance. The independent checker is separate binary64 reconstruction of the relevant recursions, not a second real-interval

count. It does not certify the transcendental construction of a different model. Verification has three distinguishable layers. First, immutable identities reject a bad input or witness. Second, semantic checks reconstruct lower-policy feasibility polynomials, all-state upper recursions, signed transport comparisons, the full procurement value and service calculations. Third, reported economic policies must agree with those reconstructed witnesses. The checker does not import the solver's Bellman solver, upper-certificate functions, or region reducer. Shared data and mathematical definitions are explicit; software independence is not independence of the mathematical proof.

Negative tests are executed on disposable fixtures. They replace the referee's upper array by -10^6 , introduce an infeasible policy index, corrupt a transition or row mass, alter a continuation enclosure, supply a stale favorable summary, change the expected target identity. Hash checks reject changed files; semantic checks reject the destroyed certificate and infeasible mathematical witnesses without their original hashes. Passing these tests is evidence about verifier scope, not that every conceivable software defect has been excluded.

The complete execution receipt reports source and deposit identities separately, actual time, peak memory, and the precise checks run. A workflow success at a scientific commit is not silently attributed to a later deposit commit. Reproduction uses the tested canonical inputs by default; creation of a new target is an explicit, separate operation which refuses to overwrite an existing one.

8. CONCLUSION

A computational statement about an agent's preferred financial class is not yet a commitment result. The purchaser values delivered service and must compensate the provider while paying for enforcement. Once these margins are evaluated within a non-contractible menu, preference adjustment changes the chosen compulsory policy even when both regimes favor the same purchased financial mandate. Value comparisons make that comparison robust to ties and near-optimal responses, without the need for a service derivative or an advantageous optimal-policy selection.

The same separation of objects disciplines numerical certification. Policy proposals, policy arrays, lower policies, upper recursions, and economic summaries perform different roles. Signed compression addresses the changing-law upper problem; fixed-polynomials and explicit service enclosures convert its output into decisions. The enforcing canonical target and witness-driven checker connect the reported policies to the mathematical objects they purport to bound. The preserved adverse and broader-law comparisons show where local mechanisms and computational elements are consequential and where they are not. This combination yields an explicit institutional-choice result while retaining the broader operator and economic

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We study how preference adjustment changes financial choice and the price of operating commitments. A dynamic transport test identifies when downward adjustment is dominated on a complete feasible-support tube, without imposing global monotonicity of continuation value. In a stochastic settlement economy, we certify a joint region of transition laws, benefits, surrender charges, and investment permissions in which adjustment reverses the initial position ranking while a counterfactual surrender option remains strictly valuable. The direction of adjustment is explained by continuation gains and their class-specific exposure, rather than inferred from a two-stage example. An explicit technology pricing both guarantee capacity and compulsory terms characterizes the institution that implements a selected operating allocation. Neural Bellman Operators separate policy proposal, evaluation, improvement, and certification. For transition laws that change at every date, a signed endpoint correction compresses a sharper count-information upper value; feasible-policy Bernstein representations supply continuum welfare and decision bounds. Matched comparisons charge common construction, endpoint, and validation costs. Reward-transfer correspondences, cardinal perturbations, adverse permission and surrender cases, structural computational controls, recursive utility, and dynamic competition remain part of the analysis. The numerical claims refer to identified finite-array targets with explicit evaluation and target-transfer accounts.

KEYWORDS: Dynamic programming, Neural policy iteration, Recursive utility, Endogenous preferences, Stochastic control, Computational economics.

1. INTRODUCTION

Economic counterfactuals require more than a numerical solution at a single calibration. A researcher changes an operating benefit, an adjustment cost, or the distribution of a financial shock and asks whether a policy response reflects an economic mechanism rather than approximation error. Repeating an optimization at every parameter value yields a response. Reusing an approximate policy is another, but it is useful only when the welfare loss and the scope of its validity can be evaluated. This paper studies that problem for dynamic economies in which control changes both financial resources and agent's preferences.

Our central question is whether a technology for changing preferences can alter the first financial position once the continuation incentives, surrender right, and cost-enforcing operation are treated jointly. Section 2 gives a constructive answer. Its transport proposition supplies verifiable inequalities under which every negative-drift action is dominated by the same financial action with zero drift on a complete feasible-support tube. It does not assume that preference adjustment is globally upward or that a two-stage direction survives in a stochastic continuation economy. The resulting dynamic explanation separates a local continuation gain from differential exposure to

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This paper develops the numerical framework accompanying Qi (2024). The author thanks Darrell Duffie for discussions.

at initial state near discharge. Joint economic theorem certifies a nonempty five-dimensional region of transition utilities, operating benefits, surrender charges, and long and short permissions. Surrender makes the positive position preferable; without adjustment, the nonpositive position is preferable. Nevertheless, allowing surrender strictly improves the defined positive class without adjustment. It is this active counterfactual option, surrender under every ultimately selected policy, that enters the financial comparison. The result is a continuum statement about an explicitly identified settlement. Its proof uses the changing-law certificate, preserves the same lower policy reward corners, and verifies directional inequalities on all feasible reachable. Separate favorable point calculations cannot supply those conclusions.

Also price the choice of operating institution. Guarantee capacity and compulsory execution have separate resource technologies. The optimal implementing term minimizes the grant required for participation plus the direct cost of enforcing that term. Free, selectable full term recovers the dominance of costly financial enforcement; the term costs can instead select a short financial guarantee or an intermediate compulsory term. This comparison implements a specified continuation allocation, not a restricted mechanism. It distinguishes the economic choice of an instrument from an additional fact that the instrument can implement operation.

Operating mandate also requires an economic account of its obligations. We introduce voluntary participation, an externally funded signing grant, finite enforcement capacity, and a right to surrender after a minimum term. Preference adjustment can match both the compensation required to operate and the enforcement capacity needed to implement continued operation. Its relative value across initial position classes can nevertheless reverse when the surrender margin changes. We derive that distinction between the two investment-permission boundaries using an exact first-date breakdown. These results tie a certified risk comparison to the price of the contract and make it relevant.

Local Bellman Operators (NBO) separate policy proposal, evaluation, improvement, and certification. A critic represents continuation value, an actor proposes a feasible action, an improvement calculation checks that action, and an independent evaluation certifies the induced policy. We use this separation to develop a counterfactual method that separates explicit distinction between changes in rewards and changes in transition laws. The certificate, rather than a training objective or a preferred approximation architecture, defines which economic conclusions are warranted.

First distinction is mathematical. For fixed transitions, the value of a policy is linear in reward coefficients. Policy caches and optimistic linear support exploit this structure. Once a parameter changes the transition law at every date, a fixed policy can have a nonconvex value in that parameter. We identify the operator term that makes an uncorrected endpoint chord fail and construct a backward correction yielding an upper value throughout a parameter interval. Approximation error is second order in interval width under explicit Lipschitz conditions. A refined comparison shows that a localized count-information upper value is sharper, as the correction compresses triangular horizon dependence into a linear number of kernel applications. We give a total coefficient-error bound and a finite-horizon guarantee that remain valid when optimal actions switch. Independently certified policy values have a finite Bernstein representation for an affine mixture of transition laws. Combining their maximum with the corrected upper value and

continuum of transition counterfactuals. The result also admits approximate neural point values after a certified residual correction.

The second distinction is economic. In a portfolio economy with adjustable risk aversion and first-exit liquidation, a constant operating benefit m and a liquidation element ℓ affect behavior through the contrast $d = m - \rho\ell$. The annuity identity explains which contract changes matter, but not whether a portfolio reversal is caused by preference formation. We partition policies by the sign of the risky position held over a positive-length initial rebalancing interval. The slope of a regular indifference locus with respect to adjustment cost equals the difference in adjustment effort between the two risk classes divided by their difference in discounted duration. A separate option-value decomposition compares deliberate preference adjustment with the otherwise optimal economy in which adjustment is unavailable. It identifies the quantity that must change the ranking, but an identity alone does not predict that change. We derive a primitive sufficient condition from consumption exposure and the curvature of CRRA utility with respect to its preference index. It predicts a strictly positive risk-specific adjustment premium over an entire probability-cost family. In the full finite stopped economy, class-specific tensor certificates then establish opposite risk rankings with and without adjustment over a joint region of contract and transition parameters.

The finite implementation retains consumption, risky investment, costly preference adjustment, stochastic preference and wealth shocks, their covariance, and the original settlement rule. It includes the union of both historical action meshes and the frozen proposal proposals. The reward-transfer bank retains its all-state, all-date 10^{-3} welfare certificate and its separate first-action sign exclusions. The new uncertainty experiment changes the probability of two one-step shock laws, each with its own killed transition kernel, and certifies their entire mixture interval. This changes the dynamics; it is not a relabeling of a reward coefficient. The finite-chain mixture is distinguished from a Brownian correlation parameter without a discretization bound.

The economic interpretation is developed further by specifying the decision calendar, settlement lottery, varying its wealth granularity with both adjustment regimes computed, and separating the resulting movement of a risk boundary from the active adjustment option. We retain the original region, including the loss of its no-adjustment sign at a finer-settlement corner. A common nested region is additionally identified for each of three finite protocols. A preference over preference states supplies a cardinal primitive not identified by a CRRA index alone. We prove a uniform option bound under bounded common cardinal tilts and then use full dynamic reoptimization to distinguish that primitive from operating-duration, liquidation, and covariance alternatives. These results concern explicit economic experiments rather than a sign assigned to an unspecified continuous-time limit.

The computational comparisons assign credit to the object doing the work. We compute an exact-oracle scalar optimistic-linear-support comparison, distinguish an exact-distribution objective from simultaneous statewise coverage, and remove all irrelevant actions from the lower policy bank while retaining the original full-menu upper benchmark. In the resource economy, we add an analytically assembled, probability-conditioned parametric quadratic program that reuses the same structure across all queries. It is substantially faster than the learned arm in this short quadratic model. This comparison changes the inference from the earlier crossover: nonlinear continuation can outperform the fitted quadratic approximation without outperforming the exact reusable quadratic program. The transition-law experiments compare

setup and evaluation work to each arm. They also measure autonomous dyadic performance from empty policy caches. The compressed and count constructions occupy different points on a precision–work frontier; the rectangular relaxation is a further methodological comparator. The economic use is a decision-specific region certificate rather than a generic welfare bound placed beside a separately optimized risk bracket. A new comparison charges both upper methods for the same model construction, including endpoint optimization, feasible-policy evaluation, and independent replay at a decision target. It also tests a mesh-only feasible lower bank against the unchanged upper target. This comparison attributes a gain to compression without treating learned models as indispensable.

General error analysis and the broader economic extensions remain part of the future work. A finite-horizon theorem distinguishes evaluation, improvement, boundary, and transition errors. Exact policy improvement provides a conditional benchmark; counterexamples exclude identification of attracting actor stationary points with global optimal policies. Merton and Epstein–Zin examples test actual evaluation and policy updates. Sophisticated temporal selves require a different improvement criterion from ordinary continuation evaluation. Persistent capacity competition tests for structural dynamic deviations at every date and state. These exercises retain their economic purposes rather than becoming an aggregate count of successes.

1.1. Related Literature and the Contribution

Policy improvement and approximate dynamic programming precede neural approximation. Jacka and Mijatović (2015) study continuous-time improvement under explicit hypotheses. The nearest neural approaches also combine continuation approximation and policy improvement: fixed-policy PDE evaluation in Kim, Kim, Kim, and Lee (2025, 2026), operator learning in Lee and Kim (2024), martingale evaluation in Cai and Zhou (2025), and direct control approximation in Han and E (2016). Neural networks, separate parameter blocks, and avoidance of a closed-form maximizer are topics of priority here. Our operator assumptions and error accounts specify when our computations support the economic use of a policy.

Successor features and generalized policy improvement (Barreto et al., 2017), policy caching (Chang, 2021), and policy-cache upper bounds (Nemecek and Parr, 2021) build on the reward-transfer foundations. Alegre, Bazzan, and da Silva (2022) combine linear support with successor features and generalized policy improvement. Our work is a necessary comparator: selecting an unsatisfactory reward weight, switching there, and adding its policy is not a new algorithm merely because the reward is a contract. We prove below that, with exact statewise endpoint optima and one scalar anchor, our adjacent-policy envelope equals the full scalar-OLS policy envelope. We measure the cost of strengthening an initial-distribution guarantee to continuous statewise coverage, but do not present that strengthening alone as a new principle. Our switching rule also differs from full action-value generalized policy improvement; the comparison does not identify the two operators.

An additional transfer result concerns *changing transition laws*, outside the fixed-reward correspondence. Lipschitz dependence of value on rewards and transitions is established in the changing-environment literature (Csáji and Monostori, 2008). Section 13 instead gives an explicit endpoint-supersolution correction involving the difference of a kernel difference and a continuation-value difference. Its implementation

independently computed information-relaxation upper value. The latter is an instance of the general approach of [Brown, Smith, and Sun \(2010\)](#), not a new duality theorem. Conditional verification and parameter lifting ([Quatmann, Dehnert, Jansen, Junges, and van den Broek, 2016](#), [Heck, Quatmann, Spel, Katoen, and Junges, 2025](#)) are also direct comparators for repeated transition parameters; we distinguish their dependence relaxations by implementing a finite-horizon rectangular upper recursion on the same economic model. Section 11 gives the date-augmented reward-MDP correspondence, the order of information and endpoint information, the treatment of killed mass and terminal rewards, and the precise role of signed compression. The economic switching-locus result uses the envelope logic of [Milgrom and Segal \(2002\)](#); the envelope identity organizes the simulation but does not itself establish a duration ordering. The economic results comprise the primitive consumption-exposure condition, the dynamic transport and no-arbitrage contract theorem, and the priced implementation frontier, not a claim to a new envelope theorem.

The broader computational literature includes recursive and projection methods ([Key, Lucas, and Prescott, 1989](#), [Judd, 1998](#)), sparse grids ([Brumm and Scheidegger, 2017](#)), and neural differential-equation solvers ([Raissi, Perdikaris, and Karniadakis, 2019](#), [Han, Jentzen, and E, 2018](#)). Our input-convex critic follows [Amos, Xu, and Kolter, 2017](#). Riccati recursion and a reusable convex quadratic program are the appropriate structural comparators for the corresponding models. We retain both, including results favorable to learned deployment. Thus the paper’s incremental case rests on the controlled compression of transition-law upper information, the resulting precision-risk and anchor-demand comparisons, and a certified preference-adjustment effect on economic choice. The architecture proposes policies; theorems and independent class situations determine which counterfactual conclusions those policies support.

2. DYNAMIC ADJUSTMENT AND THE PRICE OF COMMITMENT

Why does a technology for changing preferences alter the initial financial position, and what does it cost to sustain the activity in which that technology is used? These questions require a common continuation economy. A preference change can improve the value of remaining in operation, accelerate access to discharge, or change the exposure to a future surrender option. A financial guarantee and a compulsory operating plan can implement the same continuation allocation at different resource costs. We establish these distinctions before developing the reusable numerical operators that justify them.

2.1. *The Continuation Economy*

The state is $x = (u, X)$, with preference index u and managed wealth X . An operating plan is $a = (c, \theta, \pi)$: consumption, deliberate preference drift, and the risky share. The eight-date settlement economy, its cardinal preference primitive, and its boundary charge rule are specified in Sections 5 and 9.1. Operating benefits are denoted by v , the probability of the second transition regime by λ , and the elective surrender charge by F . Surrender pays $G(x) - F$ after a compulsory first interval; discharge at the contractual boundary pays $G(x)$. The guarantee and any signing payment use a separate utility numeraire and do not enter managed wealth.

classification lasts for the entire first interval, not an isolated instant. The initial value is optimized continuously subject to long and short permissions L and S , conditional on the inherited finite consumption–adjustment pairs and feasible frozen proposals. Later operating menus are unchanged. Section 10.3 gives the exact break-even representation. Thus the first-date permission experiment and the subsequent adjustment problem are now evaluated as one target, rather than as separate favorable

and W_σ^r for the initial class value in regime $r \in \{\text{adj}, 0\}$, where 0 prohibits deliberate adjustment but retains stochastic preference changes. Let $W_\sigma^\uparrow$ restrict deliberate adjustment to nonnegative values at every date and state. Define

$$\Delta^r = W_+^r - W_-^r, \quad \Omega_\sigma = W_\sigma^{\text{adj}} - W_\sigma^0. \quad (1)$$

The comparative option is $\Omega_+ - \Omega_- = \Delta^{\text{adj}} - \Delta^0$. This identity specifies the comparison; the short inequalities and regional bounds below determine its sign.

2.2. A Dynamic Transport Test

Given economic primitives, write an operating backup as $Q_n^a(v) = R_n^a + K_n^a v$, where R_n^a is the nonnegative discounted live-state kernel. Boundary payments are included in R_n^a . Let $\mathcal{R}_n$ denote a set of reachable states. The sets include the support of every feasible live position under the largest permissions in the parameter region and are closed under every later feasible live transition. They are not the support of a selected optimal

live position. Consider a negative-drift action a and a feasible zero-drift action a^0 with the same permission and risky share. For endpoint regime $k \in \{0, 1\}$, set

$$D_{n,k}^a = K_{n,k}^a - K_{n,k}^{a^0}. \quad (2)$$

Choose $\ell_{n+1} \leq V_{n+1}^\uparrow \leq h_{n+1}$ throughout the parameter region, and put $v = (h + \ell)/2$, $e = (h - \ell)/2$. If the benefit interval is $[d_0, d_1]$, define

$$H_{n,k}^a = \max_{d \in \{d_0, d_1\}} \{R_{n,k,d}^a - R_{n,k,d}^{a^0}\} + D_{n,k}^a v + |D_{n,k}^a| e, \quad (3)$$

$$\bar{g}_n^a = \max_{\lambda \in \{\lambda_0, \lambda_1\}} \{(1 - \lambda)H_{n,0}^a + \lambda H_{n,1}^a\} + \varepsilon_n^a. \quad (4)$$

The value is entrywise, and ε_n^a is an evaluation-error allowance. The surrender enters the continuation bounds, although it is not paid by either current operation.

PROPOSITION 1—Reachable-state elimination of downward adjustment: *Suppose the continuation enclosures and reachable sets just described are valid. Suppose every feasible negative-drift action has the indicated feasible zero-drift counterpart and $\bar{g}_n^a < 0$ at every live state. Then, simultaneously throughout the parameter region and for both initial classes,*

$$W_\sigma^{\text{adj}} = W_\sigma^\uparrow. \quad (5)$$

The conclusion holds at every date on its corresponding reachable set. It makes no assertion about alternative initial states outside those sets. A negative frozen proposal without a feasible counterpart must be checked separately; it cannot be omitted from the test.

ends, not a test on the sign of the reported initial control. The enclosures can be constructed without assuming an optimal selector: apply the upward-restricted Bellman recursion to the lower and upper continuation arrays, minimize or maximize over the point laws, and use the adverse benefit and fee endpoints. Positivity of the kernels serves the enclosure. The signed row difference in (4) retains cancellation that a difference of two unrelated value-error bounds would discard.

The economic content can also be seen directly. When preference-boundary crossing is present and the within-interval stopping fraction α is determined by wealth alone, changing drift from zero to θ shifts the arrival preference index by $\alpha h \theta$. Let $\mathcal{V}$ be interpolated continuation on a live branch and G on a discharged branch. Apart from quadratic adjustment cost, the change in the backup is

$$\mathbb{E} \left[e^{-\rho \alpha h} \int_{u'_0}^{u'_0 + \alpha h \theta} \partial_u \mathcal{V}(z, X') dz \right] - \frac{k \theta^2}{2} \mathbb{E} \left[\frac{1 - e^{-\rho \alpha h}}{\rho} \right]. \quad (6)$$

ewise derivatives suffice for the settlement interpolation. A positive drift is profitable when its discounted continuation gain exceeds its resource cost. A negative drift may instead be valuable close to a preference-discharge boundary. Formula (4) tests these effects without assuming that continuation is globally monotone or differentiable. At the original central contract $(\lambda, d) = (0.125, 0.425)$, both adjusted classes choose $(c) = (0.8, 0.2)$. The local gain from this drift, holding the selected financial control fixed and using adjusted continuation, is 0.01143610 in class + and 0.01138571 in class -. The common current adjustment cost is 0.00498752. The larger part of the difference between the two total adjustment options is not that common cost: it is their different exposure to future adjustment opportunities. The relative local gain is 0.0000503943, the relative future-option exposure is 0.0003556734, summing to 0.0004060676 in this noncancellable economy. Supplement S.14 gives the exact decomposition and its selected-control reconstruction. With $F = 0.80$, the positive unadjusted class gains an active surrender option, reducing the relative future component while leaving the adjusted local comparison unchanged.

These are not the directions in the two-stage primitive example of Section 8.2. That example establishes a separate consumption-exposure mechanism and remains intact. In the dynamic economy, even the baseline cardinal flow can favor a higher preference index: at $c = 0.8$, its derivative in u is positive on $[1.2, 2.8]$. The dynamic conclusion nevertheless requires continuation and discharge, not that flow derivative alone.

The distinction across initial states is substantial. At $(u, X) = (1.25, 1.59375)$ and the central noncancellable contract, full adjustment exceeds upward-only adjustment in the nonpositive class by 1.22113089. Downward-only adjustment exceeds no adjustment by 1.30273111. Holding the full policy's initial financial controls fixed, negative drift increases the discounted next-date mass at the lower preference boundary from 0.589502 to 0.49750624. This is arrival through the stipulated settlement lottery, not a claim about a diffusion hitting probability. The reduction of costly continuing exposure accounts for a one-step gain of approximately 0.43835 before the adjustment and liquidation-benchmark terms. That initial state is outside the support tube certified by the law. There is therefore no global monotonicity claim hidden in (5).

LEM 2—Adjustment, active surrender, and financial permissions: *Consider the array settlement target and the stored piecewise-affine first-date operator just specified, $\lambda = (2, 1.25)$ and one compulsory interval. Let*

$$\begin{aligned} \lambda &\in [0.12, 0.13], & d &\in [0.424, 0.4245], & F &\in [0.7999, 0.8001], \\ L &\in [0.7999, 0.8001], & S &\in [0.4999, 0.5001]. \end{aligned} \quad (7)$$

Throughout this five-dimensional region, (5) holds at all reachable live states and dates. With ϵ -rounded displayed bounds, the initial comparisons satisfy

$$\Delta^{\text{adj}} \in [0.00018762, 0.00020240], \quad (8)$$

$$\Delta^0 \in [-0.00017930, -0.00012218], \quad (9)$$

$$W_+^0(F, 1) - W_+^0(0, 8) \geq 0.000034708. \quad (10)$$

At the first date, the relative adjustment option exceeds 0.0003098. The positive-position comparison shows that the no-adjustment regime has a strictly valuable surrender margin: a policy in that class with a value within 0.000034708 of its supremum cannot avoid elective surrender almost

The last statement concerns the constrained positive class. In this region the unadjusted no-adjustment agent prefers the nonpositive class; the theorem does not guarantee that this selected agent surrenders. An active counterfactual option changes the comparison of financial positions even when it is not exercised under the ultimately optimal position. This distinction is essential to interpreting the contracting experiment. The proof combines two certificates. The first establishes the transport inequalities over the complete feasible-support tube. Its reachable-node counts, including boundary nodes, are 1, 30, 100, 203, 342, 495, 637, 735, 833. Every negative operating drift is strictly dominated by its zero-drift counterpart; the least slack after its arithmetic allowance is 0.000070902. Frozen proposals have nonnegative drift at every reachable live state. The second certificate uses feasible-policy Bernstein coefficients and either the transition-law chord or the count-information upper value. It treats the benefit corners jointly and uses the smallest permissions for feasible lower policies and the largest permissions for upper values. No interpolation of optimized values across the parameter is asserted to be an upper bound without the transition correction. The time interval in (7) is not the earlier implementation interval $[0.85, 0.90]$. The latter allows elective surrender throughout the original finite-menu law-benefit rectangle; the new region implements the noncancellable mandate. The new region instead establishes a strictly valuable surrender option in the indicated class and optimizes financial permissions. Neither result absorbs the other. The unfavorable original corner at $F = 0.80$, with interventions $(L, S) = (0.82, 0.50)$ and $(0.80, 0.51)$, remain in the paper and are included in the new replication package. At the central contract, the first permission makes both regimes prefer a positive position, whereas the second makes them prefer a nonpositive position. These are boundaries of the economic result, not to be removed from its record.

financial guarantee should not be compared with a freely selectable compulsory one whose execution is left unpriced. We therefore specify both technologies. Guarantee capacity E costs the agent $C(E)$ in the separate numeraire, with C continuous and nondecreasing. A compulsory term m costs the principal $D(m)$. These costs pay for different enforcement services. Fees accrue to the external enforcement sector, not to the agent's wealth; they are inactive in the implementation comparison below. No term was included in the original zero-cost procurement table, which remains a useful benchmarking case.

Let x be an adjustment regime and an operating policy selected from the noncancellable allocation set, including a specified selection among ties. Write $W^r = \max_{\sigma} W_{\sigma}^r$ for its agent value and A_r for delivered discounted service. Let $F_r^*(m)$ be the statewise enforcement threshold in Proposition 25, computed on the common feasible-support tube. The principal can choose any of the eight terms and any sufficient capacity, with $F = E$. Above the threshold, continued operation is the stated selection among indifferent best responses. Strictly larger fees give the same implementation without that stopping tie.

PROPOSITION 3—The priced implementation frontier: *Under these primitives, the minimum cost of procuring the selected noncancellable allocation is*

$$P_r = \min_{1 \leq m \leq 8} \{ [G(x_0) - W^r + C(F_r^*(m))]_+ + D(m) \}. \quad (11)$$

Participation is feasible for a principal with service value b if and only if $bA_r \geq P_r$. When participation binds for all eight alternatives, the chosen term minimizes $C(F_r^*(m)) + D(m)$. The capacity frontier for adjustment is weakly below that for no adjustment, but comparisons of principal surplus must still charge the regimes' possibly different delivered service.

For a short-term financial implementation and a costless-capacity hard term, put $z = G(x_0) - W^r$. The short-term implementation is preferred precisely when its saving in direct term cost is at least

$$[z + C(E)]_+ - [z]_+. \quad (12)$$

The expression equals $C(E)$ when participation binds and lies between zero and $C(E)$ otherwise. In particular, a freely selectable, zero-cost full term weakly dominates the positive-cost financial implementation; it strictly dominates when the additional grant is positive.

Equation (11) is an endogenous choice within an explicit enforcement technology, not an assertion of unrestricted optimal mechanism design. It also makes the original selection testable. At zero term cost, the model selects the free full term. With $C(E) = E^2$ and $D(m) = \kappa(m - 1)/8$, the central economy selects $m = 8$ when $\kappa = 0$, $m = 7$ when $\kappa = 0.01$, and $m = 1$ when $\kappa = 0.02$, in both adjustment regimes. The short-term capacities are approximately 0.77675422 with adjustment and 0.81353318 without it. All eight alternatives, grants, service amounts, and break-even service values are reported in the replication record; fees 10^{-5} above each threshold independently reproduce both implemented class values.

Thus preference adjustment has two distinct economic consequences. Its continuation alternatives change the ranking of financial positions on (7). Its improvement of operating efficiency lowers the capacity required to implement a specified continuation allocation. The latter result becomes an institutional comparison only after both capacity and execution are priced. An active surrender contract is not relabeled as an optimal implementation contract merely because it belongs to the same family.

economic conclusion is a joint restriction on direction, financial choice, stopping, and price of enforcing operation. A table of point solutions cannot establish the free on a parameter continuum. Reward-feature transfer alone also cannot do as λ changes the transition law at every date. The signed correction of Theorem 1 and the feasible-policy representation supply the missing law-parameter bounds; the support test then uses statewise continuation enclosures to explain the dynamic decision rather than merely label a relative option.

new chord and count runs use identical action targets, reward corners, and policy corners. Both certify the bounds above. Their work account charges the common kernel construction, endpoint solves, and validation to each method, while recording incremental certificate work separately. Earlier same-target precision frontiers, neural-free comparisons, structural quadratic controls, and unfavorable experiments are retained. Comparison supports a claim about the cost of certifying an economic decision, the necessity of neural training or a universal computational advantage.

Numerical allowance is 10^{-7} per compared value; the difference bounds already exceed twice that amount. It encloses evaluation of the specified stored arrays, not the transcendental construction of another target. To transfer all three displayed value estimates to another model, an additional common error for each compared initial value below 1.7354×10^{-5} would suffice. This is a required budget, not an estimate of model's error. Transferring the directional elimination additionally requires valid base reward and kernel bounds on its reachable sets. The diffusion formulation and approximation analysis retain their separate role.

3. ECONOMIC CONTROL AND SEPARATED BELLMAN OPERATORS

3.1. The Controlled Problem

(u_t, X_t) be a Markov state and let $a_t = (\theta_t, b_t)$ collect internal and external controls. On a specified domain D , the controlled state satisfies

$$dS_t = \mu(t, S_t, a_t) dt + B(t, S_t, a_t) dW_t, \quad a_t \in A(t, S_t), \quad (13)$$

W is a vector of independent Brownian motions and $A(t, s)$ is the feasible correspondence. Correlation is represented in B . More generally, a continuous martingale covariance density $Q(t, s, a)$ gives covariance $\sigma Q \sigma^T$. A predictable covariance not determined by the declared state does not, merely by notation, define the Markov problem; the state must be augmented or the formulation changed. Additive preferences and stopping time $\tau \leq T$, policy evaluation is

$$J^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(v-t)} r(v, S_v, \pi_v) dv + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right]. \quad (14)$$

Stopping rule and payoff are economic primitives. In an exit problem, τ is the first terminal date. Reflection instead requires a regulator and a boundary operator defining its direction; state constraints require viability. Neither interpretation is aided by an unreported numerical projection.

For recursive preferences, evaluation is first defined for each admissible policy by

$$-dY_t^\pi = f(t, S_t, \pi_t, Y_t^\pi) dt - Z_t^\pi dW_t, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (15)$$

son, and any domain restriction on its value argument are imposed before taking premium over policies. Lipschitz drivers on a finite horizon provide a standardmissible class; the Epstein–Zin specification below uses its own positive-domain transversality conditions. Calling Y an argument of utility does not itself establish Y -posedness.

When a classical Markov representation is available, write

$$\mathcal{H}^a[v] = f(t, s, a, v) + \mu(t, s, a)^\top \nabla v + \frac{1}{2} \text{tr}\{B(t, s, a)B(t, s, a)^\top \nabla^2 v\}. \quad (16)$$

additive utility, $f = r - \rho v$. The evaluation equation is $v_t + \mathcal{H}^\pi[v] = 0$ with the specification boundary operator, while the control equation replaces $\mathcal{H}^\pi$ by $\sup_{a \in A} \mathcal{H}^a$. Boundary a , admissibility, and comparison are part of these statements, not consequences of a general parameterization.

3.2. Evaluation, Proposal, and Improvement

Let v_ξ be a critic and a_ω a feasible actor. A differential evaluation step freezes the policy and minimizes

$$L_E(\xi; \pi) = \mathbb{E}_v [(\partial_t v_\xi + \mathcal{H}^\pi[v_\xi])^2] + \lambda_B \mathbb{E}_{v_B} [|Bv_\xi - g_B|^2]. \quad (17)$$

where B is the actual boundary operator. A Dirichlet discrepancy is appropriate for a stopped problem used below, not for an unspecified reflected diffusion. Hard architectures can impose the boundary exactly, although they do not eliminate interior approximation error.

The actor’s Hamiltonian proposal solves, approximately,

$$\max_{\omega} \mathbb{E}_{v_I} [\mathcal{H}^{a_\omega} \{\text{sg}(v_\xi), \text{sg}(\nabla v_\xi), \text{sg}(\nabla^2 v_\xi)\}]. \quad (18)$$

The notation sg freezes the critic’s parameter block. Coefficients depending on the action, including diffusion loadings, retain their action derivatives. Evaluation does not update actor parameters, and actor improvement does not update critic parameters. For example, rival controls are also frozen in a player’s improvement step. These restrictions concern the computational graph, not just the labels attached to two loss functions.

A bounded activation can enforce a box constraint. Shared resource constraints require a different map. For example, $a_i = B_0 \exp(z_i) / (\sum_{j=1}^d \exp(z_j) + \exp(z_0))$ gives $a_i \geq 0$ and $\sum_i a_i \leq B_0$, with z_0 representing slack. General convex sets can use projection; a penalty alone is an approximation whose violations must be measured. An arbitrary policy sampling measure does not turn (18) into the gradient of lifetime welfare. Such interpretation requires occupancy or adjoint weighting and the relevant interchange conditions.

For an implemented transition rule, it is often preferable to use a finite-step operator. For $n = 0, \dots, N-1$, define

$$(\mathcal{T}_n^a v)(s) = R_n(s, a) + \int v(s') K_n(ds' | s, a), \quad \mathcal{T}_n v = \sup_{a \in A_n(s)} \mathcal{T}_n^a v. \quad (19)$$

A nonnegative kernel includes discount and survival. A stopped payoff is included in R_n ; it is not continued at a clipped interior state. The experiments with nonlinear

ent critic to the evaluated target of the selected action. This is a finite-horizon Bellman update, not a simulation of a joint scalar loss. The semigroup implementation, freezing the continuation network’s parameters, does not remove its derivative with respect to the next state. The actor influences the transition, so the composition $v_{n+1}(F(s, a, \epsilon))$ must retain that input derivative. Our tests compare this actor gradient with finite differences and separately with the differential Hamiltonian graph. The distinction prevents a stop-gradient operation appropriate for a Hamiltonian jet from being incorrectly imposed on a step transition.

3.3. The Exact Policy-Improvement Benchmark

The following proposition states the exact object to which neural approximation is applied. Let $\mathcal{V}$ be a classical value space whose norm controls v , v_t , ∇v , $\nabla^2 v$, and the boundary traces required by $\mathcal{B}$. Including the time derivative is essential to pass the continuation equation to a limit.

ASSUMPTION 1—Exact evaluation and improvement: *The policy class Π is compact in the uniform topology. Evaluation $E : \Pi \rightarrow \mathcal{V}$ is uniquely defined, continuous, and has compact image. A selector I attains the pointwise Hamiltonian supremum, maps Π to itself, and is continuous at accumulation points of the policy sequence. Comparison and identification hold for the specified generator, recursive driver, boundary operator, and value class. Evaluated values are bounded above by the optimal value.*

PROPOSITION 4—Exact separated policy improvement: *Under Assumption 1, let $I(\pi_j)$ and $v_j = E(\pi_j)$. The evaluated values increase to a common limit in $\mathcal{V}$. Every evaluation policy p satisfies $E(p) = E(I(p)) = \bar{v}$, and $\bar{v}$ solves the maximized boundary-value problem. If comparison identifies the value, p is optimal.*

The important limit is equality of values, not equality of successive policies. Along with continuity gives $E(p) = \bar{v}$ and $E(I(p)) = \bar{v}$ from the shifted sequence. The continuation equation for $I(p)$ at that same value is the maximized equation. The supplement improves this step in detail. Compactness and closure under an exact selector are essential hypotheses; they are not automatically properties of a finite neural class. Proposition 4 is not an optimizer convergence theorem.

4. APPROXIMATION ERROR AND FEASIBLE IMPROVEMENT

4.1. An Error Account for Policy Value

$\bar{T}_n^a$ be the target evaluation operator and $\tilde{T}_n^a$ the implemented operator. For recursive preferences, these have the form (19). For recursive preferences, the following also permits nonlinear evaluation maps, provided they are order preserving and continuous in the continuation value. We impose those properties on the relevant value rather than presume them for every recursive aggregator.

ASSUMPTION 2—Finite-horizon operators: *For every feasible a , the target operator $\bar{T}_n^a$ is order preserving and satisfies $\|\bar{T}_n^a v - \bar{T}_n^a w\|_\infty \leq \gamma_n \|v - w\|_\infty$. The same bound holds for $\tilde{T}_n^a$; the supremum is well defined. Terminal values and the selected policy values*

ed one for a finite-horizon recursive problem.

or learned values $\widehat{v}_n$ and a feasible policy π_n , define

$$\epsilon_n = \left\| \widehat{v}_n - \widetilde{\mathcal{T}}_n^{\pi_n} \widehat{v}_{n+1} \right\|_{\infty}, \quad (20)$$

$$\delta_n = \left\| \widetilde{\mathcal{T}}_n \widehat{v}_{n+1} - \widetilde{\mathcal{T}}_n^{\pi_n} \widehat{v}_{n+1} \right\|_{\infty}, \quad (21)$$

$$\kappa_n = \sup_{s,a} \left| \widetilde{\mathcal{T}}_n^a \widehat{v}_{n+1}(s) - \mathcal{T}_n^a \widehat{v}_{n+1}(s) \right|, \quad \eta = \|\widehat{v}_N - g_N\|_{\infty}. \quad (22)$$

first quantity is evaluation error; the second is a payoff improvement gap, not an on distance. The third compares operators on the actual learned continuation. It udes transition, quadrature, interpolation, and stopped-payoff errors when those present. Let $\Gamma_{n,j} = \prod_{i=n}^{j-1} \gamma_i$, with an empty product equal to one.

THEOREM 5—Policy-value error account: *Under Assumption 2, the target policy value satisfies*

$$\|\widehat{v}_n - J_n^{\pi}\|_{\infty} \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta. \quad (23)$$

the target optimal value V_n^* ,

$$0 \leq V_n^*(s) - J_n^{\pi}(s) \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (2\epsilon_j + \delta_j + 2\kappa_j) + 2\Gamma_{n,N} \eta. \quad (24)$$

ne proof is backward error propagation, but the placement of the errors matters. The cy-evaluation recursion contributes $\epsilon_n + \kappa_n$. The optimized recursion contributes $\delta_n + \kappa_n$. Adding the resulting bounds yields (24); separately inflating an actor by operator error would produce an unnecessarily loose coefficient. For additive pped control with exact discount, $\gamma_n \leq e^{-\rho \Delta_n}$. A boundary-payoff error at an inter- mediate exit belongs in the relevant one-step operator error, not only in the terminal or at N .

here is an analogous differential statement. If a classical candidate v has boundary repency at most η , fixed-policy residual at most $\epsilon(t)$, and Hamiltonian gap at most , Itô's formula and verification give

$$V^*(t,s) - J^{\pi}(t,s) \leq 2\eta + \int_t^T e^{-\rho(q-t)} \{2\epsilon(q) + \delta(q)\} dq. \quad (25)$$

stopped martingale and comparison conditions are stated in Supplement S.1. This quantitative link to a differential critic, not a claim that sampled residuals provide required supremum bounds.

finite test set by itself provides a lower observation of a supremum, not an upper ificate. With a known Lipschitz constant L and an h -cover, a residual's uniform gnitude is bounded by its largest tested magnitude plus Lh . Without such regu- larity and coverage information, we report held-out errors as diagnostics. The same

information of a smooth function does not convert a sample average into a uniform bound.

4.2. Two Improvement Safeguards

$Q_n(s, a) = \tilde{\mathcal{T}}_n^a \hat{v}_{n+1}(s)$. For a finite feasible set $C_n(s)$ and neural proposal $a_\omega(s)$, the state safeguard selects

$$\pi_n(s) \in \arg \max_{a \in C_n(s) \cup \{a_\omega(s)\}} Q_n(s, a). \quad (26)$$

It improves the proposal's evaluated target at every queried state. If $C_n(s)$ is a h_a -cover of the feasible set and Q_n has a verified action-Lipschitz constant L_a , the remaining improvement gap is at most $L_a h_a$. Without that information, a finite improvement is not a continuum optimality certificate. In the portfolio computation, the safeguard and report the remaining independently evaluated feasible actions.

In convex cost minimization, a stronger check is available. Let $F_s(a)$ be differentiable and convex on a compact convex feasible set $A(s)$. Define

$$g_s(a) = \nabla F_s(a)^\top a - \min_{b \in A(s)} \nabla F_s(a)^\top b. \quad (27)$$

This implies $0 \leq F_s(a) - \min_b F_s(b) \leq g_s(a)$. For $A = \{a \geq 0 : \sum_i a_i \leq B_0\}$, the linear minimum is $B_0 \min\{0, \min_i \partial_i F_s(a)\}$. The computation therefore obtains an improvement certificate without a global search over neural parameters. The actor is a proposal; the improvement is stopped according to (27).

In the resource application, the continuation critic is convex by construction, so the period criterion is strongly convex in the action. Its neural representation uses additive coefficients on squared-ReLU features together with a convex quadratic and a free affine term. This architecture justifies the convex check for the *learned* continuation criterion. It does not assume that its continuation value is exact. An independent scenario-tree calculation separately assesses the resulting policy's lifetime

PROPOSITION 6—Finite stochastic-program policy certificate: *Let $C(u)$ be the convex nonanticipative finite scenario-tree control problem on a compact convex feasible set U . Let $\hat{u}$ be a feasible reference point, $G(\hat{u})$ its gap defined as in (27), and u^π the feasible tree policy by an evaluated feedback policy. Then*

$$C(u^\pi) - C(\hat{u}) \leq C(u^\pi) - C^* \leq C(u^\pi) - C(\hat{u}) + G(\hat{u}). \quad (28)$$

This certificate is independent of the training residual. It applies at each initial state where the complete scenario tree is evaluated. It is not a bound for an untested continuum of conditions, nor does it cover a different shock distribution. The supplement to the projected-simplex implementation and the independent tree gradient.

smooth Hamiltonian can have a strictly suboptimal attracting stationary point. On [2], consider

$$H(a) = -(a^2 - 1)^2 + 0.2a. \quad (29)$$

negative local maximizer is approximately -0.973994 , while its global maximizer is approximately 1.024120 . The former is locally attracting under gradient ascent and the latter is an improvement gap of approximately 0.399875 . Exact evaluation of $V^a(t) = (T - t)^a$ does not change that fact. Isolation of stationary points cannot turn an actor-critic limit into an optimal policy. An optimality conclusion needs, for example, a verified global concavity or gradient-dominance condition together with the necessary representation and evaluation assumptions. Constant-step Adam and finite-step L-S are not covered by such a conclusion merely because they update separate parameter blocks.

or does a small differential residual select a viscosity solution. For the exit equation $|v'| = 0$ on $(-1, 1)$ with zero boundary values, the relevant value is $|x| - 1$. The solutions

$$w_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2} \quad (30)$$

have the same boundary values and squared residual tending to zero, but converge to the wrong function $1 - |x|$. At zero, that function fails the viscosity subsolution property. Our regression suite retains this example and the biased composite-loss example in Supplement S.2. A convergent approximation of a nonsmooth control problem needs an appropriate mechanism, such as monotonicity, stability, consistency, boundary treatment, and comparison (Barles and Souganidis, 1991); it is not supplied by smooth activation functions alone.

4.4. The Squared-Trace Objective

For a differential critic, let $A = B^\top \nabla^2 v B$ and let z_k be independent standard Gaussian vectors. Conditional on a fixed state, action, and parameter vector, put $q_k = z_k^\top A z_k$, $K^{-1} \sum_k q_k$, and $r_k = a - q_k/2$. Then

$$\mathbb{E}(a - \bar{q}/2)^2 = (a - \text{tr}(A)/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (31)$$

Using r_k^2 instead has an excess expectation independent of K . These are different objectives, even though each individual trace estimate is unbiased.

PROPOSITION 7—An unbiased squared-residual estimand: For $K \geq 2$, define

$$U_K = \frac{(\sum_{k=1}^K r_k)^2 - \sum_{k=1}^K r_k^2}{K(K-1)}. \quad (32)$$

Then $\mathbb{E}U_K = (a - \text{tr}(A)/2)^2$. If differentiation and expectation can be interchanged, its parameter gradient is unbiased for the gradient of that squared residual.

the stability of its minimization, or an accurate policy. Optimizing over actions with the same noisy probes also introduces a selection issue beyond the fixed-action benchmark. We therefore retain exact derivatives for the differential benchmark checks across the probe estimands with repeated independent batches. The executed dimensional resource solver uses a finite-step operator and a convex critic; its results are not attributed to Hutchinson acceleration.

5. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

5.1. A Stopped Operating Economy

First state the continuous-state diffusion benchmark, including its boundary conditions and control margins. The executed regional decision experiments use its fully specified, eight-date lottery-settlement counterpart in Section 9. A certificate of its counterpart is an economic statement about its declared transition protocol; requiring it to this diffusion additionally requires the target-error account developed in Section 10. We now specify an economy in which the preference-adjustment margin is active under the transition law. Wealth is X , risk aversion is u , and the controls are consumption c , preference adjustment θ , and risky-asset share π . Inside $D = (1.2, 2.8) \times (0.5, 2.0)$, the dynamics evolve as

$$du_t = \theta_t dt + \sigma_u dZ_t^u, \quad (33)$$

$$dX_t = \{(r + \pi_t(\mu - r))X_t - c_t\} dt + \pi_t X_t \sigma_X dZ_t^X, \quad d\langle Z^u, Z^X \rangle_t = \varrho dt. \quad (34)$$

The operating contract terminates at $\tau = T \wedge \inf\{t : S_t \notin D\}$. At that time the agent receives the cardinal payoff

$$G(u, X) = -0.02(u - 2)^2 + 0.1 \log X. \quad (35)$$

The payoff summarizes liquidation or the outside option after leaving the operating economy; it is not a transfer restoring the state to the interior.

The feasible box is $c \in [0.05, 0.8]$, $\theta \in [-0.2, 0.2]$, and $\pi \in [-0.5, 0.8]$. We set

$$1, \quad \rho = 0.04, \quad r = 0.02, \quad \mu = 0.08, \quad \sigma_X = 0.2, \quad \sigma_u = 0.05, \quad \varrho = -0.25. \quad (36)$$

The baseline adjustment-cost coefficient is $k = 2$. For a feasible policy,

$$J_k^\pi(t, u, X) = \mathbb{E}_{t,u,X}^\pi \left[\int_t^\tau e^{-\rho(s-t)} \left\{ \frac{c_s^{1-u_s}}{1-u_s} - \frac{k}{2} \theta_s^2 \right\} ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (37)$$

Consumption is measured in the fixed units appearing in (37). An additive or multiplicative transformation depending on u changes incentives once u is controlled. We therefore treat such transformations as innocuous changes of utility representation.

The boundary contract is a substantive specification. Constant preference volatility remains in a closed interval without an exit or regulator mechanism. Moreover, under the displayed consumption bounds, the drift at $X = 0.5$ is negative even at the best feasible portfolio share: its maximum is -0.016 . An unregulated state-contingent interpretation is consequently unavailable. Stopping repairs that problem by modifying the interior diffusion, imposing an unreported reflection, or injecting a boundary condition. It is not equivalent to the earlier clipped transition.

ul check is that G is a strict supersolution for every feasible control. The bounds imply

$$\frac{c^{1-u}}{1-u} - \frac{k\theta^2}{2} + \mathcal{L}^{c,\theta,\pi}G - \rho G \leq -0.8162 \quad (k \geq 0). \quad (38)$$

supplement derives the bound without a grid maximization. Together with a fixed degenerate admissible diffusion, it supplies a boundary-continuity argument for stopped value. The comparison also has economic content: continued operation a cost relative to the liquidation payoff. Exit incentives are therefore part of this operation and must be reported, rather than interpreted as a generic wealth or risk-ersion effect.

5.2. The Operator and Control Margins

Writing subscripts for derivatives, the HJB equation in the interior is

$$\begin{aligned} -V_t = \sup_{c,\theta,\pi} \{ & \frac{c^{1-u}}{1-u} - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + \pi(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}\pi^2 X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X \pi X V_{uX} \}, \end{aligned} \quad (39)$$

on $V = G$ on the spatial boundary and at T . Both diagonal diffusion terms and the drift term belong to the operator. In particular, the generator of u^2 contains σ_u^2 even if preference drift is zero.

For an interior optimum, the first-order conditions are

$$c^{-u} = V_X, \quad k\theta = V_u, \quad 0 = (\mu - r)XV_X + \pi X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X X V_{uX}. \quad (40)$$

When $V_X > 0$, $V_{XX} < 0$, and all relevant controls are interior, these give

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{(\mu - r)V_X}{X\sigma_X^2 V_{XX}} - \frac{\varrho \sigma_u V_{uX}}{X\sigma_X V_{XX}}. \quad (41)$$

The final expression separates a mean-variance term from a preference-hedging term. The sign is not determined by $\varrho < 0$ alone: the cross derivative and curvature also matter. We need the stopped economy satisfy $V_X > 0$ throughout the domain. The constrained implementation compares feasible controls rather than mechanically applying these formulas where their premises fail. At a binding box constraint, the appropriate one-sided derivative or complementary-slackness condition replaces an interior equality. The cardinal utility derivative is

$$\partial_u \left(\frac{c^{1-u}}{1-u} \right) = \frac{c^{1-u} \{1 - (1-u) \log c\}}{(1-u)^2}. \quad (42)$$

It need not have the sign suggested by a verbal interpretation of increased risk aversion. The full shadow value V_u incorporates this derivative, future opportunities, adjustment costs, covariance, and liquidation. The numerical policy is informative only relative to the declared primitives.

initial state and let

$$C^\pi = \mathbb{E}^\pi \int_0^\tau e^{-\rho t} \frac{\theta_t^2}{2} dt, \quad B^\pi = J_k^\pi + kC^\pi. \quad (43)$$

xed policy, B^π and C^π are independent of k : changing the coefficient does not the state law unless the policy is changed. In particular, the stopping time is l to be endogenous to the policy.

LEM 8—Adjustment costs and cumulative effort: *Suppose the feasible policy class primitives other than k are fixed, and optimizers exist at the coefficients being compared. $V(k) = \sup_\pi \{B^\pi - kC^\pi\}$ is decreasing and convex. If $k_2 > k_1$ and π_i is optimal at k_i ,*

$$C^{\pi_2} \leq C^{\pi_1}. \quad (44)$$

C^{π_i} is a subgradient of V at k_i . At differentiability points, all optimal effort levels equal For policies with objective losses at most e_1, e_2 at the respective coefficients,

$$C^{\pi_2} - C^{\pi_1} \leq \frac{e_1 + e_2}{k_2 - k_1}. \quad (45)$$

proof uses the two optimality inequalities and is given in Supplement S.3. The imate version is useful when numerical policies are used in an economic com-: the effort ordering can be assessed against a welfare-error budget. The result ot hold pointwise for $\theta(t, u, X)$ merely because (41) contains $1/k$. The shadow portfolio choice, consumption, and exit distribution all change when the problem ved.

theorem is an economic comparative static, not an identification result. Inferring data additionally requires an observation model for latent preferences, an ed moment map, and control of numerical error in that map. The solver supplies and value objects for such a model; a graph of the policy alone cannot establish cation. This distinction leaves the structural-estimation motivation intact while ing the additional objects it would require.

6. CONTRACT CHANGES AND CERTIFIED POLICY REUSE

6.1. Operating Benefits and the Outside Option

boundary contract permits a precise economic question: how do portfolio and nce choices respond when the benefit of remaining in the operating regime s relative to the outside option? Add a constant flow benefit m to the running ob- and a constant ℓ to the liquidation payoff, holding the diffusion, feasible controls, g rule, and consumption units fixed. These are changes in contract primitives. ed not be changes in the representation of a fixed preference ordering. policy p , let B_n^p be its expected baseline utility, including G but excluding the ent charge. In addition to its discounted effort C_n^p , define

$$A_n^p = \mathbb{E}^p \int_{t_n}^\tau e^{-\rho(v-t_n)} dv, \quad Z_n^p = \mathbb{E}^p e^{-\rho(\tau-t_n)}. \quad (46)$$

are taking expectations. Thus the same policy's payoff is

$$J_n^p(m, k, \ell) = B_n^p + mA_n^p - kC_n^p + \ell Z_n^p = \ell + B_n^p + dA_n^p - kC_n^p, \quad d = m - \rho\ell. \quad (47)$$

interpretation of d is the flow benefit of operation net of the annuity value of the liquidation increment. This is a comparison of two explicitly priced components of the contract. It does not require a preference-dependent normalization of consumption utility.

PROPOSITION 9—Contract contrast and economic moments: *Suppose the feasible policy class, transition law, and stopping rule do not depend directly on (m, k, ℓ) , and the terminal values are finite. Then*

$$V_n(m, k, \ell) = \ell + W_n(d, k), \quad W_n(d, k) = \sup_p \{B_n^p + dA_n^p - kC_n^p\}. \quad (48)$$

A point change $(m, \ell) \mapsto (m + \rho h, \ell + h)$ leaves the entire optimal-policy correspondence unchanged and adds h to value. The function W_n is convex in (d, k) , increasing in d , and decreasing in k . If p_i has value loss at most e_i at $q_i = (d_i, k_i)$, then

$$(d_2 - d_1)(A_n^{p_2} - A_n^{p_1}) - (k_2 - k_1)(C_n^{p_2} - C_n^{p_1}) \geq -(e_1 + e_2). \quad (49)$$

Points of differentiability, optimal duration and effort equal $\partial_d W_n$ and $-\partial_k W_n$, respectively.

PROOF: Equation (47) proves the first claim for each policy before optimization. The supremum of the displayed affine functions is convex; their slopes in d and k have the stated signs. Approximate optimality gives $J^{p_1}(q_1) \geq J^{p_2}(q_1) - e_1$ and $J^{p_2}(q_2) \geq J^{p_1}(q_2) - e_2$. Adding and cancelling the intercepts proves (49). The differentiability claim follows from the supporting affine function of any optimizer. *Q.E.D.*

Two implications sharpen the interpretation of the portfolio exercise. First, increasing running utility alone is not a harmless normalization: it changes d and therefore the reward for discounted operating duration. Increasing the terminal payoff by the corresponding annuity value leaves d unchanged and cannot cause a portfolio reversal. Second, observations of choices and state paths alone cannot separately identify m and ρ along this equivalence direction. This is an exact restriction on the optimal-policy correspondence, not a claim that d itself is identified. Identification of d still requires an observation model and economically informative variation. If the distribution of the termination time were independent of the policy, A^p would be common to all policies and even d would not change their rankings. Policy-dependent termination is therefore essential to the operating-benefit mechanism.

The supersolution argument for G in the baseline economy is not imposed on every proposed contract: adding (m, ℓ) changes its flow-generator comparison by d . The identities above concern payoff functionals and do not require a classical boundary condition for every counterfactual.

The coefficient-weighted charge is kC^p , whereas C^p is an unweighted quantity of adjustment effort. Neither Proposition 9 nor Theorem 8 makes kC^{p_k} monotone. We report both objects. We also distinguish a change in d from a change in correlation: the former changes the relative reward for remaining in operation, while the latter changes state covariance and need not preserve the same policy-value features.

next turn the economic decomposition into a computable decision rule. Consider a finite-horizon Markov model with positive continuation kernels. Its reward, net of stopped payments, is affine in $q = (d, k)$; its action sets and transition kernels are independent across q . A stored policy p_i has independently evaluated features (B^i, A^i, C^i) and a value

$$J_n^i(s; q) = B_n^i(s) + dA_n^i(s) - kC_n^i(s). \quad (50)$$

These factors can supply such policies, but their trained critics are not substituted for the evaluated features.

Let $q_1, \dots, q_M$ be anchor contracts. Suppose $\bar{V}_n^i$ bounds the optimal value above at q_i . Let $\sum_i \lambda_i q_i$, $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, define

$$L_n(s; q) = \max_i J_n^i(s; q), \quad U_n(s; q) = \sum_i \lambda_i \bar{V}_n^i(s). \quad (51)$$

Anchor rights may be selected to minimize the upper bound; in the one-dimensional case, adjacent anchors give a chord. The stored policy need not be optimal at its anchor for the lower bound to be valid. The upper bound, in contrast, is genuine anchor information: an evaluated feasible policy by itself is a lower bound, not an upper one.

THEOREM 10—Uniformly certified policy reuse: *Under the preceding hypotheses, $V_n^*(s; q) \leq U_n(s; q)$. Define a feasible switching policy by*

$$i_n(s; q) \in \arg \max_i J_n^i(s; q), \quad p_n^{\text{sw}}(s; q) = p_{i_n(s; q), n}(s). \quad (52)$$

every date and state,

$$0 \leq V_n^*(s; q) - J_n^{\text{sw}}(s; q) \leq U_n(s; q) - L_n(s; q). \quad (53)$$

If the right-hand side is at most ε on a parameter set, the same policy-construction rule is valid throughout that set. There is no additional horizon multiplier on this independently constructed value envelope.

PROOF: Every J^i is feasible, giving the lower bound. For any fixed policy p , affinity $J^p(q) = \sum_i \lambda_i J^p(q_i) \leq \sum_i \lambda_i \bar{V}^i = U(q)$. Taking the supremum proves the upper bound. For switching, the terminal equality is common to all stored policies. If $J_{n+1}^{\text{sw}} \geq 0$, positivity of the continuation kernel implies

$$J_n^{\text{sw}} = R_n^{p_{i_n}} + K_n^{p_{i_n}} J_{n+1}^{\text{sw}} \geq R_n^{p_{i_n}} + K_n^{p_{i_n}} J_{n+1}^{i_n} = J_n^{i_n} = L_n.$$

Forward induction and the upper bound finish the proof. Q.E.D.

Feasible-value maximum and switching arguments are established ideas in the literature; see Barreto et al. (2017), Chang (2021). The upper envelope is a convexification specialization of the policy-cache bound of Nemecek and Parr (2021). Our contribution here is explicit: the stopped-contract identity identifies the economically relevant parameter, the interval construction below certifies every parameter rather

portfolio conclusions. Neither successor features nor policy caching is claimed as a new architectural architecture.

The approximation theorem in Section 4 supplies one route to the required anchor over bounds. If a feasible anchor policy has certified loss e_i , then $J^i(q_i) + e_i$ is an admissible $\bar{V}^i$. For the finite model executed here, backward maximization over an explicit action menu provides the anchor values and an independent feature recursion checks them. The target is this declared finite model: its operator discrepancy is zero relative to itself, not relative to Brownian first passage or to a continuum of actions. The other approximation questions remain separate.

For a scalar interval $[a, b]$ at fixed k , write the two endpoint-policy lines as $b_a(s) + e_a(s)$ and $b_b(s) + da_b(s)$. Their maximum is a feasible lower bound. The upper chord uses this maximum is concave and piecewise affine in d . Its maximum occurs at an endpoint or at the intersection

$$d_{\times}(s) = \frac{b_b(s) - b_a(s)}{a_a(s) - a_b(s)}, \quad (54)$$

at that intersection is defined and lies in $[a, b]$. The program checks these locations at every date and state. It inserts an anchor at a worst certified parameter and repeats until the maximum gap is below the stated tolerance. A denominator close to zero is handled by two effectively parallel lines, with an explicit numerical tolerance. This is a finite approximation of a continuum bound, not a dense parameter grid relabeled as uniform coverage. Supplement S.7 gives the error version and the numerical slack convention.

6.3. Certifying Economic Decisions, Not Just Values

A welfare bound need not determine a portfolio sign when two actions have nearly equal values. We therefore use a separate separation calculation. Partition the feasible actions into a desired set A^+ and its complement A^- . For example, A^+ may contain strictly positive portfolio shares and A^- all nonpositive shares. Fix one action $a \in A^+$ and one stored continuation policy p_i . Its one-step value followed by that policy is $Q_n^i(s, a; d)$ in d :

$$Q_n^i(s, a; d) = R_n(s, a; d) + K_n^a J_{n+1}^i(d). \quad (55)$$

$\bar{Q}_n(s, b; d) = R_n(s, b; d) + K_n^b U_{n+1}(d)$, using the same upper chord throughout a compact interval.

PROPOSITION 11—Action separation over an interval: *If the same pair (a, p_i) satisfies*

$$\min_{d \in [a_0, b_0]} \left\{ Q_n^i(s, a; d) - \sup_{b \in A^-} \bar{Q}_n(s, b; d) \right\} > 0, \quad (56)$$

every optimal first action belongs to A^+ for every $d \in [a_0, b_0]$.

PROOF: For every fixed $b \in A^-$, the difference between the feasible tail value and upper continuation bound is affine on the interval. Positivity at both endpoints therefore implies positivity throughout. Taking the supremum over A^- preserves the strict separation, while $Q^i(s, a; d) \leq Q^*(s, a; d)$ and $Q^*(s, b; d) \leq \bar{Q}(s, b; d)$. No action in A^- can maximize the true finite-model criterion. Q.E.D.

that, selecting unrelated favorable endpoint actions would not prove (56). The implementation enumerates the feasible pairs and records the maximizing pair, both pointwise comparisons, and the resulting interval margin. An unresolved interval is resolved from a zero effect. This permits a sharp portfolio conclusion away from a region while retaining an explicit uncertainty region where the sign is not resolved.

6.4. The Exact Scalar-OLS Correspondence

Adaptive reward construction belongs to optimistic linear support (Alegre, Bazzani, and da Silva, 2022). The following observation identifies its exact relation to the full-policy implementation and prevents an implementation shortcut from being taken for a new policy-set principle.

PROPOSITION 12—Adjacent policies and the full policy envelope: *Fix k , and suppose d_i stores one policy that is optimal at every state and date at d_i . If d_i are adjacent anchors, then, pointwise in state and date,*

$$\max_i J^i(d) = \max\{J^a(d), J^b(d)\}, \quad d \in [a, b].$$

Consequently the full policy-line envelope and the adjacent-policy calculation have the same corner gap, including its possible corner location.

PROOF: Consider an earlier anchor $d_i < a$. Optimality implies $J^i(d_i) - J^a(d_i) \geq 0$ and $J^a(a) \leq 0$. This difference is affine, so its slope is nonpositive and $J^i(d) \leq J^a(d)$ for $d \geq a$. A later anchor $d_i > b$ is similarly dominated by J^b for $d \leq b$. There are no other anchors. Taking the maximum proves the assertion. Q.E.D.

This can be incorporated in a finite state description, so simultaneous statewise computation is not an obstruction beyond the reach of scalar OLS. It is nevertheless a nontrivial numerical objective from coverage at an initial distribution. We compare the two objectives using the same exact full-menu oracle and the same 10^{-3} target. The independent full-envelope calculation verifies Proposition 12; it is not reported as a separate trained implementation of SFOLS. Approximate anchor policies need not obey adjacent-policy identity, in which case one must retain all feasible policy values or the omitted lines.

7. POLICY REUSE WHEN TRANSITION LAWS CHANGE

Hard transfer holds the economic law of motion fixed. Changes in the frequency of a demand shock, the probability of a demand shock, or the survival of installed policy do not satisfy this restriction. Even with a fixed policy, their repeated effects on states need not be affine. This section supplies an upper correction for that arbitrariness and a feasible-policy representation with which to certify the resulting policy as factually optimal.

consider the finite-horizon additive problem, with a common finite state space and common feasible actions. At date n , write

$$T_{n,\lambda}^a v = R_{n,\lambda}^a + K_{n,\lambda}^a v, \quad (R_{n,\lambda}^a, K_{n,\lambda}^a) = (1 - \lambda)(R_{n,0}^a, K_{n,0}^a) + \lambda(R_{n,1}^a, K_{n,1}^a). \quad (57)$$

where K is a positive discounted, possibly killed kernel with row mass at most $\gamma_n \leq 1$. Payments are included in R ; the terminal function g is common. The parameter λ is known when the policy is selected and is held fixed across dates. Equation (57) can be interpreted as an independent regime draw at each date, not as advance information about future draws.

In contrast to reward-affine transfer, $\lambda \mapsto V_{n,\lambda}$ need not be convex. A two-date unconstrained chain that pays one only when the first regime is 1 and the second is 0 has value $(1 - \lambda)$. Both endpoint values are zero and the midpoint value is $1/4$. Interpolating endpoint optima therefore gives an invalid upper bound. The difficulty remains without a neural approximation, a numerical error, or policy switching.

Fix an interval $[a, b] \subseteq [0, 1]$, put $t = (\lambda - a)/(b - a)$, and let v^a, v^b be endpoint superfunctions:

$$v_n^j \geq \max_u T_{n,j}^u v_{n+1}^j, \quad v_N^j = g, \quad j \in \{a, b\}. \quad (58)$$

Exact endpoint optima are one admissible choice. Define, at every state and action,

$$D_n^u = (K_{n,a}^u - K_{n,b}^u)(v_{n+1}^b - v_{n+1}^a), \quad (59)$$

$$M_N = 0, \quad (60)$$

$$M_n(s) = \max \left\{ 0, \max_{u \in A_n(s)} [D_n^u(s) + \max\{(K_{n,a}^u M_{n+1})(s), (K_{n,b}^u M_{n+1})(s)\}] \right\}. \quad (61)$$

The inner maximum over the two endpoints is taken for the same action u . It does not admit a different control after observing the current regime.

THEOREM 13—A corrected transition-law chord: *Under (57)–(58),*

$$U_n(\lambda) = (1 - t)v_n^a + tv_n^b + t(1 - t)M_n \quad (62)$$

is an upper bound on $V_{n,\lambda}$ simultaneously at every state, every date, and every $\lambda \in [a, b]$. If $\|K_{n,a}^u - K_{n,b}^u\|_{\infty \rightarrow \infty} \leq L_{K,n}|b - a|$ and $\|v_{n+1}^b - v_{n+1}^a\|_{\infty} \leq L_{v,n+1}|b - a|$, then

$$\|M_n\|_{\infty} \leq (b - a)^2 \sum_{j=n}^{N-1} \Gamma_{n,j} L_{K,j} L_{v,j+1}, \quad \Gamma_{n,j} = \prod_{r=n}^{j-1} \gamma_r. \quad (63)$$

PROOF: Let $C_n = (1 - t)v_n^a + tv_n^b$. Affinity of rewards and kernels gives the exact identity

$$T_{n,\lambda}^u C_{n+1} - C_n = (1 - t)(T_{n,a}^u v_{n+1}^a - v_n^a) + t(T_{n,b}^u v_{n+1}^b - v_n^b) + t(1 - t)D_n^u.$$

The first two terms are nonpositive. Positivity and affinity of K imply $K_{n,\lambda}^u M_{n+1} \leq \max\{K_{n,a}^u M_{n+1}, K_{n,b}^u M_{n+1}\}$ pointwise. Consequently $T_{n,\lambda}^u U_{n+1} \leq U_n$ for every feasible u . The $U_N = g$, backward comparison proves the first assertion. Taking sup norms in (61) as $\|M_n\|_{\infty} \leq L_{K,n} L_{v,n+1} (b - a)^2 + \gamma_n \|M_{n+1}\|_{\infty}$; backward iteration gives (63). *Q.E.D.*

in continuation value. It vanishes when the kernels are fixed, recovering the transfer upper chord. It also vanishes when the relevant continuation functions change. These are economically distinct invariances. Equation (63) concerns the total upper correction, not the total loss of a policy library: a change in the optimal policy can still generate a first-order bank-coverage error. No general square-root complexity conclusion follows from this equation alone.

Endpoint neural approximation can be used without treating it as an exact optimizer. Suppose $\widehat{v}_N = g$ and certified one-sided residual bounds satisfy $\max_u T_{n,j}^u \widehat{v}_{n+1} - \widehat{v}_n \leq e_n^j$. Set $e_N^j = 0$ and $e_n^j = (r_n^j)_+ + \gamma_n e_{n+1}^j$. Then $v_n^j = \widehat{v}_n + e_n^j$ satisfies (58), by positivity and row-mass bound. Thus the theorem accommodates approximate neural endpoints and an explicit evaluation budget. A sampled training residual does not supply required r_n^j . The numerical implementation below uses exact finite-model endpoints and the transition-law correction, rather than endpoint estimation, is isolated.

7.2. Feasible Policy Values over the Entire Parameter Interval

Consider a deterministic Markov policy p that does not observe future regimes. With $N - n$ future regime labels, its value is a polynomial of degree at most h :

$$J_n^p(\lambda) = \sum_{j=0}^h \binom{h}{j} \lambda^j (1 - \lambda)^{h-j} F_{n,j}^p. \quad (64)$$

These coefficients have a direct conditional-expectation interpretation. Imagine drawing all h future regime labels before the experiment, but not revealing them to the agent. Conditional on exactly j labels being 1, the current label is 1 with probability j/h . Set $F_{N,0}^p = g$ and

$$F_{n,j}^p = \left(1 - \frac{j}{h}\right) \left[R_{n,0}^{p_n} + K_{n,0}^{p_n} F_{n+1,j}^p\right] + \frac{j}{h} \left[R_{n,1}^{p_n} + K_{n,1}^{p_n} F_{n+1,j-1}^p\right]. \quad (65)$$

The term with zero weight is omitted, so no nonexistent coefficient is used at $j = 0$ or $j = h$. The recursion remains valid with killing: labels after exit can be drawn without replacement, and the stopped payoff.

Neural representation permits an operational continuum certificate. Restrict the polynomial coefficients to a closed parameter cell by de Casteljau subdivision. Express (62) as a polynomial of degree h by degree elevation. For a library $\mathcal{P}$ and any cell I , let $U_{n,j}^I$ and $F_{n,j}^{p,I}$ be these coefficients and put

$$\epsilon_I = \max_{n,s} \left[\min_{p \in \mathcal{P}} \max_{0 \leq j \leq N-n} \{U_{n,j}^I(s) - F_{n,j}^{p,I}(s)\} \right]_+. \quad (66)$$

The lower bound of the minimum and maximum deliberately gives an upper bound rather than equality.

OLLARY 14—Transition-law policy reuse: *Choose at each state and date an index $\lambda \in \Lambda$ and use that policy's current action. The resulting feasible switching value $V_n^\lambda(s)$ satisfies*

$$0 \leq V_{n,\lambda}(s) - J_n^{p^\lambda}(s) \leq \epsilon_I, \quad \lambda \in I. \quad (67)$$

, and every date, without interpreting parameter samples as a uniform test.

PROOF: The Bernstein basis is nonnegative and sums to one. For each fixed policy, polynomial difference on I is bounded above by its largest coefficient difference. Taking a minimum over policies proves $U_n - \max_p J_n^p \leq \epsilon_1$. Backward induction and optimality give $J_n^{\text{sw}} \geq \max_p J_n^p$: after taking the selected policy's current action, subsequent switching cannot lower its continuation value. Theorem 13 completes the proof. *Q.E.D.*

The same lower representation can be paired with a different upper oracle. For example, disclose to a relaxed controller the number of remaining regime-1 draws, and disclose each realized label afterward. Replace the fixed action in (65) by a maximum over a *common* current action after combining the two weighted terms. Call the resulting vectors $W_{n,j}$. Then $\sum_j \binom{h}{j} \lambda^j (1 - \lambda)^{h-j} W_{n,j}$ is an upper value: every original policy is feasible in the enlarged information structure. This is an explicit, zero-penalty information relaxation in the sense of Brown, Smith, and Sun (2010), not a new general duality principle. Its triangular count recursion requires $h + 1$ coefficient vectors at a date with h periods left, rather than enumeration of all 2^h label histories. We report its information premium separately from the corrected-chord certificate; a loose information bound is evidence of a large policy loss.

Changing transitions has long been a subject of parametric and robust dynamic programming. Our result does not claim that such counterfactuals were previously possible. Its additional content relative to a reward-policy cache is the explicit discount-operator term in (61), its second-order upper-error account, and its combination of feasible policy polynomials and switching. These objects retain the same scalar parameter across the horizon rather than allowing an adversary to reset it freely at a node. In the application, the certificate supplies a new, fully specified uncertainty counterfactual for the original stopped preference economy.

7.3. The Price of Compressing an Upper Oracle

An upper value can be valid without being the sharpest available upper value. This distinction matters when an interval is to be accepted at a prescribed welfare tolerance. The relevant comparison holds the interval, feasible policies, and subdivision cells fixed. $I = [a, b]$, $t = (\lambda - a) / (b - a)$, and $h = N - n$. A localized count-informed recursion coefficients

$$W_{n,j}^I = \max_u \left\{ (1 - j/h) T_{n,a}^u W_{n+1,j}^I + (j/h) T_{n,b}^u W_{n+1,j-1}^I \right\}. \quad (68)$$

term with zero weight is omitted. At the terminal date the coefficient is g . The minimization uses the *same* action in the two branches. Conditional on the number of remaining draws from endpoint law b , a draw is from b with probability j/h . The relaxed controller learns the remaining count, whereas an admissible policy in the original economy can ignore it. Thus the associated polynomial is an upper value, not an employable anticipative policy.

For comparison, write the degree- h coefficients of the corrected chord as

$$U_{n,j}^I = (1 - j/h) v_n^a + (j/h) v_n^b + f_{h,j} M_n, \quad f_{h,j} = \frac{j(h-j)}{h(h-1)} \quad (h \geq 2). \quad (69)$$

nt law to be chosen afresh at every date and state:

$$B_N^I = g, \quad B_n^I = \max_{u,e \in \{a,b\}} T_{n,e}^u B_{n+1}^I. \quad (70)$$

t recursion is a finite-horizon, reward-valued rectangular relaxation. It discards
mon-parameter restriction more extensively than the count construction.

POSITION 15—Matched upper-oracle ordering: *Under the hypotheses of Theorem*
 $\leq U_{n,j}^I$ *for every state, date, and coefficient. Moreover, $W_{n,j}^I \leq B_n^I$. Positive Bernstein*
ion preserves both orderings. With identical feasible-policy coefficients and cells, the
count certificate is no greater than either comparator's certificate. No ordering between
 B^I *is required.*

thank the advisory referee for identifying the first ordering. Its economic-
implication is a tradeoff between upper precision and the work required
in it. The proof, including the endpoint and short-horizon cases, appears in
ment S.10. A two-period example makes the difference strict: the original value is
 $1 - t$, the chord coefficients are $(1, 1, 1)$, and the count coefficients are $(1, 1/2, 1)$.
 $1/2$, their upper values are 1 and $3/4$, respectively, against a true value of $1/2$.
correction compresses the count dependence. Its proof replaces a count-
convex combination of the two propagated corrections by their maximum
places endpoint action backups by endpoint supersolutions. It thereby trades pre-
fewer action-wide kernel applications. With H remaining dates and already
exact endpoint solutions, the implementation requires $4H - 6$ such applica-
the correction when $H \geq 2$, versus $H(H + 1) - 2$ for the count recursion with
neighboring continuations and shared endpoint coefficients. The rectangular
on requires $2H$. These are operation counts for the specified recursions, not
of elapsed times.

nonzero correction stores $(H - 1)S$ additional values for S states, against $H(H -$
terior count coefficients. Endpoint values are shared. The implementation
s one date's chord coefficients only when a certificate needs them. This saving
t remove the common feasible-policy bank: m policies require order mSH^2 value
ents and mSH policy indices. Action-wide temporary arrays and the transition
ntation also remain. A comparison of total work must therefore include anchor
ation, policy evaluation, local restriction, and subdivision, not only the upper
on.

onal bounds have a substantial verification literature. Parameter lifting in [Quat-](#)
[Dehnert, Jansen, Junges, and Katoen \(2016\)](#) relaxes repeated parameter depen-
nd refines regions; its two-toss example already exhibits the obstruction behind
. That work principally treats probabilistic specifications in parametric Markov
, whereas here the target is a finite-horizon reward optimum together with
missible policy lower value. Equation (70) is our explicitly specified regional
ator, not an implementation or timing claim about their verification software. It
datewise as well as statewise endpoint resets. The count construction retains a
n count of the endpoint draws but reveals that count to the optimizing controller.
policy Bernstein coefficients retain the original repeated-parameter dependence
t that information. The corrected chord retains its signed cross-operator term
ompressing the propagation of the correction. These are different relaxations of
ence, not alternative names for the same algorithm.

the second-order correction in Theorem 13 is not a second-order bound on the loss of endpoint-policy bank. A bound on the latter must allow optimal actions to change. For a finite target, let r_n bound the absolute reward, let $\beta_n \leq 1$ bound every kernel row β , and let

$$\rho_n = \max_u \|R_{n,1}^u - R_{n,0}^u\|_\infty, \quad \kappa_n = \max_u \|K_{n,1}^u - K_{n,0}^u\|_{\infty \rightarrow \infty}.$$

define nonnegative constants backward by

$$\begin{aligned} V_N &= \|g\|_\infty, & V_n &= r_n + \beta_n V_{n+1}, \\ L_N &= 0, & L_n &= \rho_n + \kappa_n V_{n+1} + \beta_n L_{n+1}, \\ Q_N &= 0, & Q_n &= \kappa_n L_{n+1} + \beta_n Q_{n+1}. \end{aligned} \tag{71}$$

These constants depend on primitives of the stored finite model, not on a smooth optimal-policy correspondence.

THEOREM 16—A total coefficient-error bound: *Suppose the two endpoint policies are optimal at every state and date. On an interval of width w , the corrected-chord certificate with these policies satisfies*

$$\epsilon_I \leq \max_{0 \leq n < N} \{2L_n w + \tfrac{1}{2} Q_n w^2\}. \tag{72}$$

The same upper estimate applies to the localized count certificate. Put $L_* = \max_n L_n$ and $Q_* = \max_n Q_n$. For $\epsilon > 0$, dyadic refinement terminates once

$$w \leq \min\{1, \epsilon / (4L_*), \sqrt{\epsilon / Q_*}\},$$

where a quotient by a zero constant imposes no restriction. Thus, for fixed finite-model constants, the worst-case number of scalar intervals is $O(\epsilon^{-1})$; an $O(\epsilon^{-1/2})$ policy-bank claim does not follow from the correction alone.

The proof bounds optimal values and all fixed-policy values by the same Lipschitz version, and then bounds their *coefficients*, not just sampled values. It is given in Supplement S.10. The result permits policy kinks. It does not remove dependence on the state/action target or furnish a high-dimensional parameter covering rate. Its practical use is to establish termination while allowing the realized coefficient certificate to be much tighter than a global Lipschitz estimate.

A useful allocation rule follows from Proposition 15. On each dyadic interval, first compute the compressed certificate. If it fails the target, compute the local count certificate before buying another anchor. Split only when that second certificate fails. For exact endpoint oracles and the same local endpoint-policy rule, this cascade never requests an anchor outside the tree requested by chord-only refinement: every interval accepted by the chord is also accepted by count, and every failed count interval would have failed the chord. Extra count work can reduce anchor demand, but need not reduce elapsed time. Our matched frontier below compares each upper method on common banks. The separately executed autonomous experiment reports chord, count, and cascade refinement from fresh policy caches; its timing is not inferred from the common-bank frontier.

continuity identity identifies which contract changes matter; it does not identify force of a portfolio reversal. To distinguish an operating-duration incentive from formation, fix an initial state and date and partition admissible policies by first risky position. Let $\mathcal{P}_+$ require a positive first position and $\mathcal{P}_-$ a nonpositive. Further actions remain unrestricted within the economic model. In the finite implementation the first position is held over the first rebalancing period, of length $\delta = 1/8$, of liquidation. These are nonempty finite policy sets. A continuous-time counterpart restricts the position over a positive-duration interval, not merely its value at the instant; Section 9 makes this distinction explicit. For that counterpart, replace $\mathcal{P}_\sigma$ by suprema unless attainment is established. The finite-policy conclusions concern the declared calendar.

$$V_\sigma(d, k) = \max_{p \in \mathcal{P}_\sigma} \{B_p + dA_p - kC_p\}, \quad \Delta(d, k) = W_+(d, k) - W_-(d, k), \quad \sigma \in \{+, -\}. \quad (73)$$

$\Delta > 0$ identifies a strictly positive optimal first position, while $\Delta < 0$ rules it out. $\Delta = 0$ is an indifference, not necessarily a unique threshold. The economic objects A_p are discounted duration and adjustment effort, not the first-period duration or effort kC_p .

PROPOSITION 17—Duration, adjustment effort, and a risk-switching locus: *For finite sets, W_+ and W_- are convex piecewise-affine functions of (d, k) . At a point where the switching feature vector in each class is locally constant,*

$$\Delta_d = A_+ - A_-, \quad \Delta_k = -(C_+ - C_-). \quad (74)$$

If $\Delta(d, k) = 0$ and $A_+ \neq A_-$ at such a point, the local switching locus has slope

$$\frac{dd^*(k)}{dk} = \frac{C_+ - C_-}{A_+ - A_-}. \quad (75)$$

Generally, on an interval of d at a fixed k , suppose every pair of optimizing policies, one in each class, satisfies $A_+ - A_- \geq \kappa > 0$. Then Δ is strictly increasing with slope at least κ in integral sense. Opposite endpoint signs imply exactly one crossing. If a computed difference has uniform error at most $e_+ + e_-$, any zero of that computed difference lies within $(e_+ + e_-)/\kappa$ of the true crossing.

PROOF: A maximum of finitely many affine functions is convex, Lipschitz on compact sets, and piecewise affine. On a cell with fixed optimizing feature vectors, differentiability gives (74); implicit differentiation gives (75). Across cells the value functions are piecewise continuous. Their derivatives almost everywhere are the features of active options, a finite-policy instance of the envelope argument in [Milgrom and Segal \(2002\)](#). Verifying the assumed derivative separation gives $\Delta(d_2, k) - \Delta(d_1, k) \geq \kappa(d_2 - d_1)$. Continuity gives existence and the strict inequality gives uniqueness. Finally, at an approximate zero the absolute true difference is at most $e_+ + e_-$; the same slope inequality bounds its distance from the true zero. Q.E.D.

positive-risk policies both keep the agent in operation longer and use more adjustment effort, a higher adjustment cost requires a larger operating benefit before the positive-risk class becomes preferable. If the effort ordering is reversed, the comparative advantage reverses. The aggregate theorem that optimal effort decreases with k does not undermine this ratio: it compares optimal choices across costs, whereas (75) compares constrained risk classes at the same cost. The duration-separation hypothesis must be established over the claimed interval; a bisection calculation or a derivative at one point does not establish it.

8.1. A Matched Preference-Formation Counterfactual

Let W_σ^{adj} retain the full adjustment set and let W_σ^0 restrict $\theta = 0$ at every state and date. State u remains stochastic, with the original volatility, covariance, operating region, and settlement rule. Thus the comparison removes deliberate preference formation, not preference risk, and changes no reward level or financial technology. Define its option value in each risk class by

$$\Omega_\sigma(d, k) = W_\sigma^{\text{adj}}(d, k) - W_\sigma^0(d), \quad \Omega_\sigma \geq 0. \quad (76)$$

The zero-adjustment model does not depend on k because its realized adjustment effort is zero.

PROPOSITION 18—Risk-specific option values: *At every contract,*

$$\Delta^{\text{adj}}(d, k) - \Delta^0(d) = \Omega_+(d, k) - \Omega_-(d, k). \quad (77)$$

In particular, preference formation reverses a nonpositive-risk ranking, $\Delta^0 < 0$, precisely when $\Omega_- - \Omega_+ > -\Delta^0$. It reverses a positive-risk ranking, $\Delta^0 > 0$, precisely when $\Omega_- - \Omega_+ > \Delta^0$. If Δ^0 has the same strict sign at two contracts but Δ^{adj} has opposite signs, the change in relative option values identifies the incremental force behind that reversal, rather than the generic continuity identity alone.

PROOF: Set inclusion gives nonnegativity in (76). Subtract the two definitions of Δ to obtain (77). The two strict inequalities are then necessary and sufficient for opposite signs. No interior first-order condition or portfolio-distance argument is required. *Q.E.D.*

The propositions suggest a focused experiment. We independently optimize both risk classes with and without deliberate adjustment, record (B, A, C) for the attaining policies, and compare their value differences. We also vary k while keeping the operating region, covariance, and settlement curvature fixed. The resulting brackets locate indifference by continuity and opposite signs, but are not reported as a globally unique switching threshold. The full first-action sign certificates from Section 6 retain their separate, stronger exclusion interpretation where applicable. Utility units are the same fixed cardinal units as in the original stopped economy; the ratio (75) translates adjustment-cost changes into operating-benefit changes without pretending to estimate effort from data.

option decomposition identifies the quantity that changes a risk ranking; a necessary condition is needed to predict its sign. Consider a two-stage specialization that uses the preference channel. A risk class $\sigma \in \{+, -\}$ delivers a positive consumption C_σ and an exogenously specified discounted operating duration A_σ . Before the option is realized, the agent chooses $\theta \in [-\bar{\theta}, \bar{\theta}]$ at cost $k\theta^2/2$. Set

$$F_\sigma(\theta) = \mathbb{E} \frac{C_\sigma^{1-u_0-\theta}}{1-u_0-\theta}, \quad B_\sigma = F_\sigma(0), \quad g_\sigma = F'_\sigma(0), \quad M_\sigma = \max_{|\theta| \leq \bar{\theta}} [-F''_\sigma(\theta)],$$

where $u_0 > 1 + \bar{\theta}$ and consumption has finite positive support. The class value is $B_\sigma + \Omega_\sigma(k)$, with $\Omega_\sigma = \max_{|\theta| \leq \bar{\theta}} \{F_\sigma(\theta) - B_\sigma - k\theta^2/2\}$. Unlike the optimized-policy results in (74), g_σ and M_σ are computed directly from the consumption lottery and the preference index.

PROPOSITION 19—Consumption exposure and the adjustment premium: *For $u > 1$,*

$$U_{uu}(c, u) = -\frac{c^{1-u}}{(u-1)^3} \{[1 + (u-1)\log c]^2 + 1\} < 0. \quad (78)$$

if $k + M_\sigma \leq \bar{\theta}$, then

$$\frac{g_\sigma^2}{2(k + M_\sigma)} \leq \Omega_\sigma(k) \leq \frac{g_\sigma^2}{2k}. \quad (79)$$

Furthermore, if $A_+ - A_- =: D_A > 0$ and

$$\underline{\Omega} := \frac{g_+^2}{2(k + M_+)} - \frac{g_-^2}{2k} > 0, \quad (80)$$

then consumption adjustment lowers the risk-indifference contract by at least $\underline{\Omega}/D_A$. Specifically, with $d = (B_+ - B_-)/D_A$, every $d \in (d^0 - \underline{\Omega}/D_A, d^0)$ strictly favors the nonpositive class without adjustment and the positive class with adjustment.

The argument uses global concavity in the preference index, not an interior first-order condition for an optimized portfolio. Low consumption affects the adjustment incentive through $U_{uu}(c, u) = c^{1-u}[1 + (u-1)\log c]/(u-1)^2$. A downside consumption exposure therefore increases the magnitude of the preference-adjustment benefit even when the adjusted utility is lower. Condition (80) compares that gain with curvature and risk; greater risk by itself is not sufficient. Supplement S.11 proves the bounds and shows M_σ can be bounded using endpoint preferences rather than an unverified preference grid.

In the declared family, take $u_0 = 2$, $\bar{\theta} = 0.2$, $C_- = 0.5$, and $C_+ = 0.05$ with probability p or otherwise, with $D_A = 0.5$. For every $p \in [0.06, 0.10]$ and $k \in [20, 80]$, the primitive data gives $\Omega_+ - \Omega_- > 0.00186$ and a threshold reduction exceeding 0.00372 in preference-benefit units. This is a continuum prediction from the primitive inequalities, the interpolation of a nine-point exercise. The finite support and preference interval imply $|g_+| \geq 1.4820$, $|g_-| \leq 0.6138$, and $M_+ \leq 17.342$; these conservative bounds imply the uniform result. The larger, point-specific lower bounds and the exactly optimized options are reported in the supplement. This specialization explains a channel; it

the economy is examined separately with decision-specific certificates.

8.3. Certifying a Risk Ranking over a Joint Counterfactual Region

welfare tolerance is not a tolerance for the sign of a small class-value difference. Therefore construct upper and lower values separately for each risk class and each investment regime. Fix $I_\lambda = [a, b]$ and $I_d = [d_l, d_h]$ and write $t = (\lambda - a)/(b - a)$ and $(d - d_l)/(d_h - d_l)$. For a fixed policy, its value is polynomial in t and affine in d . For a fixed λ , the optimal class value is convex in d , since the feasible policy set is unchanged and every policy value is affine.

Let $U_\sigma^l(t)$ and $U_\sigma^h(t)$ be valid transition-polynomial upper values at the two contracts, constructed using the constrained count or corrected-chord recursion. Then

$$U_\sigma(t, s) = (1 - s)U_\sigma^l(t) + sU_\sigma^h(t)$$

an upper value throughout the rectangle. Evaluate any feasible class-policy bank independently to obtain lower polynomials $L_{\sigma,p}(t, s)$. Denote their tensor Bernstein coefficients on a cell by $U_{\sigma,ij}$ and $L_{\sigma,p,ij}$.

PROPOSITION 20—Decision-specific tensor certificate: *Suppose the upper and lower polynomial constructions, including their coefficient operations, each have absolute error at most e . Throughout the closed cell,*

$$\begin{aligned} \underline{\Delta} &:= \max_{p \in B_+} \min_{i,j} \{L_{+,p,ij} - U_{-,ij}\} - 2e \leq \Delta, \\ \Delta &\leq \overline{\Delta} := \min_{p \in B_-} \max_{i,j} \{U_{+,ij} - L_{-,p,ij}\} + 2e. \end{aligned} \tag{81}$$

In particular, $\underline{\Delta}^{\text{adj}} > 0$ and $\overline{\Delta}^0 < 0$ certify opposite risk rankings under the same contract and risk-law parameter. They also imply $\Omega_+ - \Omega_- \geq \underline{\Delta}^{\text{adj}} - \overline{\Delta}^0 > 0$.

PROOF: A tensor Bernstein polynomial is a convex combination of its coefficients. A positive-class policy is a feasible lower value and each negative-class upper polynomial bounds that class optimum; their difference therefore bounds Δ from below. Taking a coefficient minimum, followed by a maximum over feasible policies, preserves the bound. Interchanging the classes gives the upper inequality. The two absolute error allowances add. Applying the option decomposition gives the final statement. *Q.E.D.*

This certificate does not require uniform duration separation, a regular active-policy or uniqueness of a threshold. Its object is a strict choice on a declared region. It uses the same transition parameter in all periods, permits later risky positions to change, and imposes the adjustment restriction at every date. In the application below, class-value tolerances are generated by these coefficient comparisons rather than borrowed from the 10^{-3} generic welfare result.

9. THE ECONOMIC DECISION AND ITS ROBUSTNESS

The object of a portfolio certificate is a decision under a specified contract, not an action attached to an unspecified limiting model. Three primitives are particularly

ogy that determines subsequent states, and the cardinal value assigned to utility in preferences. We make these primitives explicit and study their effects on the baseline experiment by changing the financial controls, adjustment technology, or liquidation rule of the baseline experiment.

9.1. A Rebalancing Contract with a Finite Settlement Lottery

The finite economy has eight rebalancing dates over one year. At each date the investor observes (u, X) , chooses (c, θ, π) , and holds that control until the next date or liquidation, whichever occurs first. The preference grid contains 33 tiers on $[1.2, 2.8]$; the wealth grid contains 49 balances on $[0.5, 2]$. The union of the two action meshes contains 65 elements. Three deposited state- and date-dependent feasible proposals complete the 1,568-action baseline menu. Removing those proposals from a lower-utility construction does not remove them from the target against which that policy is evaluated.

The contract is a primitive description of the transition experiment. Independently at every rebalancing date the law indicator chooses the correlation -0.25 with probability $1 - \lambda$ and 0.25 with probability λ . The probability λ is fixed for the entire contract; the indicator is not observed before the action. Conditional on the indicator, two independent symmetric normal distributions determine a proposed increment using the drift and volatility in (33)–(34). For an action a and $h = 1/8$ and signs (z_u, z_X) , that increment is

$$h \left((r + \pi(\mu - r))X - c \right) + \sqrt{h} \left(\pi X \sigma_X \{ \varrho z_u + \sqrt{1 - \varrho^2} z_X \} \right). \quad (82)$$

If the segment from the current state to the proposed state first meets the boundary of the grid at $\alpha < 1$, liquidation takes place at time αh , at that boundary point. Otherwise the arrival is settled on the four neighboring grid nodes by independent, uniform two-point lotteries in the two coordinates. Flow utility and the operating benefit accrue for αh , with the discount annuity $(1 - e^{-\rho \alpha h})/\rho$; liquidation pays the complete conventions, including arrival on a boundary node, are given in Appendix S.12.

The contract is a discrete rebalancing and randomized-settlement contract. The grid is part of the transition technology: it describes admissible recorded balances and preference levels. It is not an assertion that interpolation has no economic effect. It is an analytical convenience, not a calibration of an observed intermediary or an empirical finding that wealth is randomly rounded. The continuous diffusion in Section 5 remains the economic benchmark. Both specifications preserve consumption, costly information formation, risky investment, stochastic preferences, covariance, stopping, and liquidation. Their relationship is an approximation question rather than an identity. The posterior arrival Y settled to Z satisfies

$$\mathbb{E}[Z | Y] = Y, \quad \text{Cov}(Z) - \text{Cov}(Y) = \text{diag}(\mathbb{E}[(Y_i - l_i(Y))(r_i(Y) - Y_i)]). \quad (83)$$

The quantities l_i, r_i are adjacent grid nodes. Consequently, changing settlement granularity alters risk even when the rebalancing calendar and action menu are held fixed. The wealth-lottery perturbations below exploit this observation: they change an economically relevant primitive and reveal the movement of the original decision region. The contract is not used to assert an unproved diffusion limit.

Let $\mathcal{P}_+^\delta$ require the initial risky position to be positive throughout the first commitment interval of length δ , until liquidation, and let $\mathcal{P}_-^\delta$ require it to be nonpositive. In the experiment the control is constant within the first interval, so this is precisely the δ -action partition used by the solver. Later positions are unrestricted. In continuous time a constant initial position is one possible commitment contract; an adapted sign restriction over the same interval is a different admissible commitment contract. Either can be studied, but they must not be interchanged without changing the policy sets.

PROPOSITION 21—The duration of a risk classification: *For the diffusion in Section 5, changing a bounded measurable control at a single deterministic initial instant leaves its state and payoff unchanged, up to the usual indistinguishability convention. Hence a partition restricts only the sign of that isolated control value has equal class suprema. A strictly ordered lifetime class-value difference instead concerns a positive-duration restriction such as $\mathcal{P}_+^\delta$. When a time mesh is refined while the latter estimand is held fixed, the restriction must inure to cover the same interval δ .*

PROOF: Changing a bounded drift integrand on a singleton changes no Lebesgue integral. The difference of the corresponding stochastic integrals has quadratic variation equal to the integral of a bounded squared integrand over that singleton, which is zero. The resulting state and stopped payoff therefore coincide. Either initial sign can be assigned without changing the value. Holding the duration of a restriction fixed under refinement is then a condition on the experiment, not an attainment assumption. *Q.E.D.*

The economic horizon of the reported choice is thus one eighth of a year. This is distinct from the lifetime duration feature A_p in (73). A local Hamiltonian comparison in flow-value units and can also be informative, but it is not the lifetime commitment-value difference reported here.

9.3. Preferences over Preference Changes

Consumption risk aversion does not identify the cardinal payoff from changing the preference index. A useful explicit specification is

$$U_b(c, u) = \int_1^c x^{-u} dx + b(u) = \frac{c^{1-u} - 1}{1-u} + b(u). \quad (84)$$

Members have marginal consumption utility c^{-u} and relative risk aversion u . The reference consumption level is one in the fixed units of the economy; $b(u)$ is the payoff from being in preference state u at that reference consumption. It is a preference over preference states. It must be specified or, in an empirical application, separately disciplined. The baseline corresponds to $b_0(u) = 1/(1-u)$. Thus its cross-preference cardinal comparison is a substantive primitive rather than a consequence of the CRRA coefficient alone.

Consider first $b(u) = b_0(u) + a(u - u_0)$ in the two-stage family of Section 8.2. A common tilt a changes neither class's consumption lottery nor its fixed- u_0 payoff. Nevertheless, it changes the shadow benefit of adjustment in both classes. The following theorem result characterizes a nontrivial set on which the original economic mechanism survives.

statement bound 0.2, $C_- = 0.5$, $C_+ \in \{0.05, 0.8\}$ with probability $p \in [0.06, 0.10]$ on the outcome, and $k \in [20, 80]$. For every common tilt $a \in [-0.25, 0.25]$,

$$) - \Omega_-(a) \geq \frac{(1.482 - 0.25)^2 20 / (20 + 17.342) - (0.6138 + 0.25)^2}{160} > 0.0004173. \quad (85)$$

uration difference $D_A = 0.5$, the adjusted indifference contract is therefore at least 46 below the unadjusted one throughout this probability–cost–cardinal family.

proof in Supplement S.12 uses the same curvature restriction as the original tion and treats the common cost in both option bounds jointly. It is an analytic um statement, not an inference from a table of optimized examples. Its suffi- t radius is conservative. Conversely, at $(p, k) = (0.08, 40)$ the relative option is 837 at $a = 0$ but -0.01391977 at $a = 1$. The unadjusted indifference is 1.5 in both ne adjusted indifference changes from 1.40234327 to 1.52783953. The larger tilt de the robust family, and the reversal is retained as economically informative. de exposure and the cardinal value of preference changes jointly determine the ism.

9.4. A Dynamic Accounting and Reoptimization Experiment

dynamic economy does not inherit the two-stage direction mechanically. Even deliberate adjustment, its preference state is stochastic. We therefore recompute sk classes in both adjustment regimes under each change in primitives. In ear, an additive tilt need not leave the dynamic no-adjustment values unchanged. policy p , let F_p be discounted exposure to $(c^{1-u} - 1)/(1 - u)$; let B_p and H_p be responding exposures to $1/(1 - u)$ and $u - 2$; and retain discounted operat- ation A_p , quadratic adjustment effort C_p , and liquidation payoff G_p . Under a ization multiplier s , tilt a , and liquidation multiplier z , the exact policy value is

$$J_p(s, a, d, k, z) = F_p + sB_p + aH_p + dA_p - kC_p + zG_p. \quad (86)$$

eline is $(s, a, k, z) = (1, 0, 2, 1)$. All six features use the stopped transition protocol lude its endogenous exit distribution.

POSITION 23—Dynamic feature accounting with reoptimization: *Fix the finite on law and the class and adjustment restrictions. The optimized value is the maxi- the affine forms in (86). At a differentiability point its derivatives in (s, a, d, k, z) are $\lambda, -C, G$ for an optimizing policy. If p_0 is optimal at $q_0 = (s_0, a_0, d_0, k_0, z_0)$, then for*

$$W(q_1) - W(q_0) = J_{p_0}(q_1) - J_{p_0}(q_0) + \mathcal{R}(q_1; p_0), \quad \mathcal{R}(q_1; p_0) \geq 0. \quad (87)$$

ity holds separately for all four class–regime values. For the relative option, subtract nonpositive-class and no-adjustment comparisons with their appropriate signs; the g difference of reoptimization gains has no predetermined sign.

DF: There are finitely many deterministic Markov policies. The affine represen- ives the envelope statement. Add and subtract $J_{p_0}(q_1)$, use $W(q_0) = J_{p_0}(q_0)$, and ibility of p_0 at q_1 to obtain (87). Taking the four signed differences preserves the y, but not the sign of its remainder. Q.E.D.

by adjustment. It does not identify a causal contribution from data or decompose extracting primitives into invariant shares. For changes in the shock law we directly evaluate the baseline policy under the new law and again report the nonnegative from reoptimization. That calculation changes the feature distribution itself, rather than treating covariance as a reward coefficient. We also solve a zero-covariance law: setting $\lambda = 1/2$ would mix two nonzero-correlation shock laws and is not the same intervention.

The results reported below use the original $k = 2$ and all financial and preference constraints. They explain why a positive relative adjustment option can coexist with positive risk in both regimes, with nonpositive risk in both regimes, or with opposite choices. The separation is the economic content needed to connect a computed certificate to the mechanism of the full model.

9.5. Decision Error across Economic Targets

Let $[\ell_r, u_r]$ be a certified difference interval for a finite protocol in regime $r \in \{\text{adj}, 0\}$, including its numerical arithmetic allowance. If another target satisfies $|W_{\sigma}^{r, \text{target}} - W_{\sigma}^{\text{finite}}| \leq \eta_{\sigma}^r$, then

$$\Delta^{r, \text{target}} \in [\ell_r - \eta_+^r - \eta_-^r, u_r + \eta_+^r + \eta_-^r]. \quad (88)$$

It follows by subtracting the two class-value error intervals. It must not charge the already included arithmetic error a second time. For the historical rectangle, an additional common per-class error strictly below $2.6496827802505486 \times 10^{-5}$ would suffice for both original signs. That number is a required budget, not an estimate of the decision error.

Our robustness experiment instead uses explicitly specified finite settlement protocols and reoptimizes each. It retains all four corners and the center of the historical rectangle, reports its failures as well as its successes, brackets the movement of indifference contracts, and additionally certifies a common nested region. No unexplained constant is substituted into the approximation theorem to turn a finite-model statement into a decision claim. The approximation results and continuous-time economy are preserved; numerical evidence identifies exactly the decisions for which a further target-error budget is consequential.

10. PARTICIPATION, TERMINATION, AND PORTFOLIO PERMISSIONS

The operating obligation and the investment permissions are distinct economic primitives. An agent who can leave a costly activity need not value an adjustment technology as does an agent committed to continue. Likewise, a change in the permitted position need not have the same effect as a change in preferences. We embed the original operating economy in a participation contract, characterize the price of exiting its obligation, and identify the investment-permission margins that move the indifference loci. The original noncancellable experiment remains a member of this family.

10.1. An Operating Mandate with Participation and Limited Enforcement

At date zero, an agent may accept an operating mandate or take the outside payoff x_0 . Acceptance fixes the managed initial state $x_0 = (2, 1.25)$, the rebalancing calendar,

numeraire supports a signing grant q and an enforceable termination charge F . If a payment is credited to managed wealth X . Crediting the grant to that state changes consumption, transitions, and the experiment, and is not the contract considered here.

A mandate stipulates a minimum operating term mh , where $h = 1/8$ and $m \in \{1, 2, \dots, 8\}$. At an interior rebalancing date $n \geq m$, the agent may surrender for $G(x) - F$. At that date, operation is compulsory. At a resource or preference boundary the agent is discharged without a charge: the boundary records a contractually verifiable decision to continue within the permissible state domain. This discharge provision is not an elective surrender at an ordinary interior node. It explains why the underlying liquidation payoff does not imply the same right to terminate. The mandate $m = 8$ is the original noncancellable mandate, independently of F .

The charge is supported by externally enforceable guarantee capacity $F \leq E \leq \bar{E}$, where $\bar{E}$ is the available capacity limit. Let $C(E) \geq 0$ be the agent's nondecreasing cost of maintaining that capacity, measured in the separate numeraire. It may include a surety premium or a collateral carrying cost. It is independent of the within-period control conditional on the contracted capacity, and enters the participation constraint explicitly. No costless, unbounded borrowing against managed wealth is allowed. A fully collateralized interpretation is also possible if the collateral account and return are specified separately; its carrying cost cannot be silently omitted. The sections below first report $C(E) = 0$ so that the effect of the operating technology is isolated. A specified positive $C(E)$ adds to the required grant whenever participation

is required. These are assumptions of a theoretical procurement economy, not claims about a particular intermediary. They provide a contracting interpretation and a transparent participation test. They do not establish that this mandate, its settlement lottery, or its operating technology is optimal among all contracts. In particular, a principal must be willing to pay for the service; declaring the operation compulsory does not itself ensure participation.

Let $r \in \{\text{adj}, 0\}$ for the two adjustment regimes and $\sigma \in \{+, -\}$ for the initial adjustment class. The regime 0 prohibits deliberate adjustment, not stochastic movements in preferences. Let $W_\sigma^r(\lambda, d, F, m)$ denote the operating value before the signing grant and capacity cost. A deterministic Markov policy p specifies both operating actions and an elective surrender rule. Let τ be its first discharge, surrender, or terminal date. Let $\mathcal{A}_p$ for elective surrender before discharge or maturity, and define

$$\mathcal{A}_p = \mathbb{E}_p \left[\int_0^\tau e^{-\rho t} dt \right], \quad \mathcal{S}_p = \mathbb{E}_p [e^{-\rho \tau} \mathbf{1}_{\mathcal{E}_p}], \quad (89)$$

$$J_p(\lambda, d, F, m) = \mathcal{B}_p(\lambda, m) + d\mathcal{A}_p(\lambda, m) - F\mathcal{S}_p(\lambda, m). \quad (90)$$

The policy p contains consumption utility, preference-state utility, adjustment cost, and participation without the charge. In the finite protocol, the integral in (89) uses the within-period exit annuity. The surrender exposure is discounted incidence, not discounted exit probability. Fixing a policy fixes these features; reoptimizing a policy need not do so.

PROPOSITION 24—Participation, service procurement, and the surrender envelope: *For a fixed policy p , the function $J_p(\lambda, m)$ and a fixed feasible action technology, $W_\sigma^r = \max_{p \in \mathcal{P}_\sigma^r(m)} J_p$ is convex in (d, F) ,*

not satisfying participation is

$$q_r^*(E) = \left[G(x_0) - \max_{\sigma} W_{\sigma}^r + C(E) \right]_+. \quad (91)$$

Suppose a principal obtains flow b per unit of discounted operating duration and pays the grant, while termination charges are remitted to a separate enforcement institution. For a specified principal agent policy p_r^* with $\mathcal{A}_{p_r^*} > 0$, that contract is individually rational for both parties at minimum grant if and only if

$$b \geq \frac{q_r^*(E)}{\mathcal{A}_{p_r^*}}. \quad (92)$$

At a point with a unique exposed feature vector, the class-value derivatives are $W_d = \mathcal{A}_{p^*}$ and $W_F = -\mathcal{S}_{p^*}$. At a tie, the directional derivative is the maximum of $v_d \mathcal{A}_p - v_F \mathcal{S}_p$ over optimizing policies. These statements concern the specified contract, not an optimization over all mandates.

The proof is an affine-policy envelope argument, given in Supplement S.13. Equation (92) imposes a substantive participation restriction on the principal as well as on the agent. It compares compensation with delivered service rather than interpreting a low grant for an immediately abandoned activity as a procurement improvement. The displayed values are utility-numeraire quantities. They are not monetary willingness-to-pay estimates for an investor whose entire wealth follows CRRA utility. Define $\Delta^r = W_+^r - W_-^r$ and $\mathcal{O} = \Delta^{\text{adj}} - \Delta^0$. Where the feature derivatives exist,

$$\Delta_d^r = \mathcal{A}_+^r - \mathcal{A}_-^r, \quad \Delta_F^r = -(\mathcal{S}_+^r - \mathcal{S}_-^r). \quad (93)$$

As increasing a surrender charge need not increase either the relative adjustment or the positive-position advantage. The sign depends on which class uses surrender more. If all class-specific participation constraints bind, then

$$\mathcal{O} = (q_+^0 - q_+^{\text{adj}}) - (q_-^0 - q_-^{\text{adj}}), \quad (94)$$

with a common capacity cost and outside payoff. The relative option is the difference between the compensation savings from adjustment in the two position classes. Adjustment weakly lowers each class's required compensation; it need not lower it more for the positive class.

10.2. When Does a Finite Charge Implement Continued Operation?

Let V_n^r be the original noncancellable continuation value. For the new mandate the surrender at eligible interior nodes is

$$W_n^r(x) = \max \left\{ G(x) - F, \max_{a \in A_n^r(x)} \left[r_n^{\lambda,d}(x, a) + K_n^{\lambda}(x, a) W_{n+1}^r \right] \right\}. \quad (95)$$

Before date m only the second entry is available. Boundary nodes continue to pay G , terminal nodes pay G , and the date-zero operating action retains the original positive-position sign restriction. A charge for an early forced exit is not introduced.

Let R_n contain all live grid nodes reachable at date n from x_0 using any baseline position and either endpoint law, without voluntary surrender. This support set includes

neously for both regimes and for every $\lambda \in [0, 0.25]$.

POSITION 25—A statewise enforcement frontier: *For a fixed contract and $m \geq 1$,*

$$F_r^*(m; \lambda, d) = \max_{m \leq n < 8} \max_{x \in R_n \text{ interior}} [G(x) - V_n^r(x; \lambda, d)]_+, \quad (96)$$

empty maximum equal to zero. The voluntary-surrender and noncancellable values at every reachable continuation node if and only if $F \geq F_r^$. This condition implies of both initial class values. It is sufficient, but need not be necessary, for equality only initial state.*

Threshold is nonincreasing in m and in d . Moreover,

$$F_{\text{adj}}^*(m; \lambda, d) \leq F_0^*(m; \lambda, d). \quad (97)$$

ently, deliberate preference adjustment weakly substitutes for enforcement capacity in nting continued operation, even though its effect on the relative initial-position option e either sign. If $\underline{V}_n \leq V_n^0$ uniformly on a parameter region, replacing V^0 by $\underline{V}$ in (96) sufficient common charge for both regimes throughout that region.

necessity in this proposition is statewise: an eligible node with $G - F > V_n$ makes y at that node impossible. The sufficiency follows by backward induction. The n (97) follows from inclusion of feasible policies, not from an assumed sign of r-class option. This distinction matters in the numerical results. A low charge se the unadjusted positive class to use surrender disproportionately, reversing tive option even though adjustment still reduces the compensation required every class.

plementation of the regional frontier uses two feasible no-adjustment policies air all-date Bernstein values. Nonnegative duration makes the lower benefit nt $d = 0.4$ sufficient. On each of sixteen law subintervals, the minimum restricted nt is a policy lower value at each node. Taking the larger of the two policy values is still a lower value for V^0 . Subtracting the arithmetic allowance and izing the resulting liquidation deficit over R_n supplies the common charge. This ion does not require an optimizer at every law probability or a neural policy. It oids paying for enforcement at grid nodes that the initial mandate cannot reach. surrender provides the opposite endpoint of this frontier. If G is a supersolution y feasible one-step operating backup, backward induction gives $W_n = G$ at all er-eligible dates when $F = 0$. With the first interval compulsory, the agent then es only that interval of operation. With surrender additionally available at date e original risk classification is bypassed and the value is simply $G(x_0)$. Neither factual is an implementation of the noncancellable contract.

10.3. Investment Permissions and Their Shadow Values

mandate can separately restrict funded purchases and stock borrowing at the date. A long permission L limits risky holdings to LX_0 ; a short permission S orrowed risky holdings to SX_0 . A liquidity-reserve requirement can motivate the and a stock-loan or collateral line the latter. Their levels need not be symmetric.

estimates of market-wide restrictions. Both $\pi = L$ and $\pi = -S$ carry financial risk. We vary these permissions only at the first date, leaving all later menus, the 1/8-year decision horizon, and the transition protocol unchanged. To distinguish a permission set from a missed action-grid point, we allow every risky share in the relevant interval for each common consumption–adjustment pair. The three original first-date proposals are retained when feasible. Consumption and deliberate adjustment themselves remain their original finite pairs; the later target is still the complete 1,568-action menu. This is a precisely defined enlargement of initial financial permissions, not continuous relaxation or minimization of all three controls over the full horizon.

PROPOSITION 26—Exact first-date risky-share reduction: *Fix the optimized continuation V_1^r , a common pair (c, θ) , and initial state $x_0 = (2, 1.25)$. For $\pi \in [-0.6, 1]$, every proposed increment remains strictly inside the state domain. On this range, the first-date payoff $Q^r(c, \theta, \pi)$ under the finite settlement protocol is continuous and piecewise affine in π . Breakpoints are the risky shares at which a conditional arrival wealth coordinate crosses a grid node. Hence the nonpositive maximum over $[-S, 0]$ and the positive supremum over $(0, L]$ are obtained by checking endpoints and these breakpoints for each common (c, θ) together with feasible frozen proposals. The closure at zero is used only for the positive supremum; a reported positive maximizer must be strictly positive.*

This reduction replaces a dense risky-share scan by a finite exact search for the specified enlarged initial menu. The numerical implementation contains 121 common consumption–adjustment pairs and 2,689 candidate triples over the wider range. It does not convert finite consumption and adjustment choices into continuous ones. It does not convert the piecewise-affine settlement payoff into the diffusion’s local mean–variance formula.

In a smooth interpolation cell at a binding optimum, let

$$\mu_L^r = \partial_\pi Q^r(c_+^*, \theta_+^*, L), \quad \mu_S^r = -\partial_\pi Q^r(c_-^*, \theta_-^*, -S). \quad (98)$$

These are one-sided values of relaxing the long and short permissions. At a switch between optimizing actions the appropriate directional envelope replaces a single derivative. For initial-only nested permission sets, the positive class value is nondecreasing in L , and the nonpositive class value is nondecreasing in S , regardless of whether a differentiable multiplier exists.

At a regular indifference locus with $D_A^r = \mathcal{A}_+^r - \mathcal{A}_-^r \neq 0$, the envelope identities give

$$\frac{\partial d_r^*}{\partial F} = \frac{S_+^r - S_-^r}{D_A^r}, \quad \frac{\partial d_r^*}{\partial L} = -\frac{\mu_L^r}{D_A^r}, \quad \frac{\partial d_r^*}{\partial S} = \frac{\mu_S^r}{D_A^r}. \quad (99)$$

These are local statements under the exposed-feature and nonzero-slope conditions, global monotonicity assertions about a difference of convex functions in d . They separate three economic margins: duration values an operating benefit, surrender closure values a termination charge, and the first-date continuation gradient values a binding portfolio permission. Preference adjustment changes these objects through optimization. It is not a substitute for observing which permission actually binds.

ng surrender is compatible with the transition-law certificate. It is an additional with reward $G - F$ and zero live kernel. It therefore preserves affine dependency on the law probability, nonnegative substochastic kernels, and the finite-horizon continuation ordering. A fixed policy has the representation

$$J_p(\lambda, d, F) = \sum_{j=0}^8 b_{j,8}(\lambda) (B_{p,j} + dA_{p,j} - FS_{p,j}), \quad (100)$$

$b_{j,8}$ denotes the degree-eight Bernstein basis. Early surrender merely shortens effective polynomial degree; degree elevation retains the common representation. per chord and count recursions use the same expanded menu, including the state and state eligibility of surrender. Feasible lower policies retain their own choices when evaluated at another fee.

Each law interval, upper values at the four (d, F) corners give a tensor upper bound by convexity of the optimized value in these two reward parameters. Each policy lower value is affine in them. Restricted Bernstein coefficients then bound initial class differences uniformly on a three-parameter box. The maximum over policies is taken only after a bound valid for each individual policy has been checked. The full argument, including arithmetic, is given in Supplement S.13.

There are three separate targets. The mathematical settlement contract uses real arithmetic; the implemented certificate concerns its stored transition and reward arrays; diffusion or another settlement contract is a different economic target. Let $\epsilon_{\sigma,n}^r$ stand for transferring a class value between two specified finite-array targets. A net backward error account is

$$\epsilon_{\sigma,8}^r = \epsilon_G, \quad \epsilon_{\sigma,n}^r \leq \epsilon_{r,n} + \epsilon_{K,n} B_{n+1} + \beta_n \epsilon_{\sigma,n+1}^r, \quad (101)$$

Reward errors include surrender and boundary payments, ϵ_K is a row- ℓ^1 kernel bounding continuation magnitude, and β bounds row mass. The two class errors are added to the error in a difference certificate. The reported arithmetic allowance is the evaluation of the stored arrays, not an uncomputed ϵ_K for the exact constructor or diffusion approximation. Selected-control reconstruction checks consistency, does not estimate an interval enclosure of every transcendental operation. No target error is inferred from a small reconstruction discrepancy.

11. CORRESPONDENCE WITH REGIONAL PARAMETER METHODS

Appropriate comparison is not between a reward-control method and literature devoted to transition probabilities. [Quatmann, Dehnert, Jansen, Junges, and Katoen](#) define parametric Markov decision processes and reward properties. [Heck, Quatmann, Spel, Katoen, and Junges \(2025\)](#) develop dependency-sensitive parameter estimation for parametric Markov chains. The latter model class does not include the agent's choice in the same way, but its treatment of shared parameters is directly relevant. We find the correspondence at the level of the target and information structure.

The finite Bellman operator as

$$T_\lambda^\lambda(x) = \max_a \{r_n^\lambda(x, a) + K_n^\lambda(x, a)v\}, \quad (r_n^\lambda, K_n^\lambda) = (1 - \lambda)(r_n^0, K_n^0) + \lambda(r_n^1, K_n^1). \quad (102)$$

stochastic row. The absorbed state has zero subsequent reward; exit rewards already included in r_n^λ must not be paid twice. Set terminal rewards at date H . An action-independent reward can equivalently be assigned to an intermediate action state. These transformations produce a finite-horizon parametric reward MDP. Signed rewards can be accommodated directly in finite-horizon Bellman comparisons; when a nonnegative-reward encoding is desired, a date-specific common shift must also be paid on absorbed states so that its total is policy independent. These are model correspondences, not a priority claim or an assertion that two implementations are identical.

The distinctions are as follows. In the economic target, λ is common across dates and states, and the agent chooses the action before the endpoint-law draw. A rectangular relaxation permits endpoint choices to reset across state–date visits. It therefore has a dependency that a fixed- λ policy must obey. The count-information recursion counts the total number of one endpoint’s draws remaining, but does not reveal the next draw before the current action. Its maximization is over one common action for a conditional mixture, not separate actions chosen after observing which endpoint occurs. The extra count information yields an upper value for (102). Generalized counting and dependency-reducing transformations address related losses of parameter dependence; their pMC results should not be silently identified with this controlled-count recursion.

Our corrected chord is a further compression of the count bound. Its signed cross-rator term is

$$(K_n^a - K_n^b)(V_{n+1}^b - V_{n+1}^a), \quad (103)$$

followed by positive backward propagation of a statewise correction. It replaces an prior count triangle by one correction vector per nontrivial date, while preserving the upper bound and the coefficient ordering established in Supplement S.10. On the other side, actual feasible class policies are evaluated under the common parameter. Our Bernstein coefficients preserve the dependency rather than receiving the upper relaxation’s extra information. This is why an upper relaxation and a neural training dual cannot substitute for feasible-policy evaluation.

The contribution claimed here is this particular signed compression, its ordering and full policy-bank error analysis, and its measured use in a common-target economic decision certificate. The comparison does not claim that regional expected-reward control, dependency-sensitive abstraction, or policy reuse first becomes possible with D. The count and chord timing comparison below is also not a benchmark against the complete software systems of [Quatmann, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) and [Deck, Quatmann, Spel, Katoen, and Junges \(2025\)](#). It is an attributable comparison of the two mathematically matched upper components with the same economic target and lower evidence.

12. COMPUTATION AND INDEPENDENT POLICY EVALUATION

12.1. *The Stochastic Transition and Its Boundary*

For the portfolio economy we use four equiprobable weak-Euler branches. Independent signs $z_1, z_2 \in \{-1, 1\}$ generate

$$= \theta h + \sigma_u \sqrt{h} z_1, \quad \Delta X = [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} \{q z_1 + \sqrt{1 - q^2} z_2\}. \quad (104)$$

Grid	Dates	V_0	C_0	Seconds
17×25	4	-1.140548	0.009344	0.31
33×49	8	-1.136916	0.010335	2.07
65×97	16	-1.135019	0.010902	15.45

te (2, 1.25); $k = 2$. All rows optimize $5 \times 5 \times 7$ consumption, adjustment, and portfolio actions. C_0 is the integrated $\theta^2/2$. These are finite-approximation values, not certified continuous-time errors.

with variances and the cross covariance are present. On each branch we follow the path from s to $s + \Delta s$ until its first boundary intersection, at fraction $\alpha \in [0, 1]$ step. The branch contribution is

$$r_k(s, a) \frac{1 - e^{-\rho \alpha h}}{\rho} + e^{-\rho \alpha h} \{1_{\{\alpha=1\}} v_{n+1}(s + \Delta s) + 1_{\{\alpha < 1\}} G(s + \alpha \Delta s)\}. \quad (105)$$

A boundary state stops immediately and receives G . An interior step that hits a boundary exactly may equivalently use its boundary datum. No branch is returned from an interior state after an exit.

A polygonal stopped random walk is an approximation to Brownian stopping, not a Brownian first-passage simulation. The reference operator uses positive bilinear interpolation for live continuations. The neural training operator evaluates its frozen continuation network at those same physical successor locations. Policy evaluation currently uses a separate NumPy implementation with positive interpolation. These approximations are compared rather than assumed to be identical.

Operator tests use $1, u, X, u^2, X^2, uX$. At an interior test point, their discrepancies have corrected drift-squared Euler remainder and recover the complete covariance matrix. A state test verifies that opposite preference controls produce different conditional values even when the displacement is smaller than a grid cell. Interpolation preserves preference; nearest-node rounding can erase it. The differential actor graph, finite-actor graph, and immediate stopping boundary are also tested independently.

Table I reports a refinement sequence with $\max(\Delta u, \Delta X)^2/h$ decreasing from 0.04 to 0.001. This matters because interpolation error divided by the time step must be small in a generator approximation. The action mesh is held fixed in this sequence, so time refinement is assessed separately. These calculations diagnose discretization error; three meshes do not themselves constitute an error certificate for the continuous-time value.

12.2. Preference-Cost Comparisons

The cost panel solves the entire finite problem again at $k = 0.5, 2, 8$, using a 25×37 grid, eight dates, and $9 \times 9 \times 13$ controls. Consumption, preference adjustment, and portfolio share are all optimized. Discounted effort is evaluated under the resulting policy, including its effect on the exit time. Table II reports the initial state $(u, X) = (0.01, 0.01)$.

The discounted cumulative effort falls as k increases, consistently with Theorem 8. The effort and consumption coordinates at this initial state are at their bounds for the displayed coefficients. This is not evidence for the interior formulas in (41).

k	V_0	C_0	c_0	θ_0	π_0
0.5	-1.117524	0.014614	0.800	0.200	-0.500
2.0	-1.135835	0.009888	0.800	0.200	-0.500
8.0	-1.160291	0.001432	0.800	0.100	-0.500

row re-solves all dates and all three control coordinates on the same 25×37 grid, eight dates, and 9×13 action mesh. The initial state is $(2, 1.25)$. The comparison concerns cumulative effort, not a global pointwise policy ordering.

In particular, a short portfolio position in this calibration should not be described as an unconditional preference-hedging effect. The liquidation contract, the level of running utility, and the portfolio's effect on exit jointly determine the optimum. Occupation-weighted exit probabilities are computed by forward propagation of the interpolation-defined Markov chain; an unweighted fraction of exiting grid branches could measure a different object.

12.3. Neural Policies in the Preference Economy

The ordinary neural proposal has two tanh hidden layers of width 32 and three rounded outputs. The safeguarded critic has two tanh hidden layers of width 64. Its output is $G(s) + (1 - t)m(s)N_{\hat{c}}(s)$, where m vanishes on every face of the rectangle. The implementation uses separate L-BFGS fits, 2,048 uniformly sampled training states as seed, eight dates, and a feasible $5 \times 5 \times 7$ comparison set containing the actor's proposal. No grid reference values or optimal reference policies enter these training sets.

After training, the entire feedback sequence is frozen and evaluated on a 33×49 grid. The comparison reference uses the same number of dates and a finer $9 \times 9 \times 13$ action mesh. A separate $7 \times 7 \times 11$ deviation search uses the *evaluated policy continuation*, not the learned critic, and includes the incumbent action. Consequently, a reported gain is a feasible payoff improvement. It is not a norm of the difference between two actions and is not an upper bound over the continuum of actions.

Table III reports all three safeguarded seeds and retains a lower-budget, unsafeguarded pilot. The pilot differs in width and optimization budget, so its comparison is development diagnostic, not an isolated causal ablation. Table IV reports errors for full control vector. A small error at the initial state does not imply a small maximum error near the boundary, and a policy coordinate can change substantially near a switch. The critic, policy value, and feasible-deviation columns therefore remain distinct.

The hard boundary architecture attains the displayed boundary condition to numerical precision. It does not certify the interior value or the Brownian stopping approximation. The sizable worst-case diagnostics identify where additional computation is needed for a uniform guarantee; they are not suppressed by averaging across states and seeds. Conversely, the initial-state comparison and the re-solved cost panel provide economic information that is not captured by the largest training residual alone.

12.4. Repair and Contract Counterfactuals

The policy errors above can be repaired without treating the learned critic as a value function. We freeze the three neural policy arrays and form, at each date and state, the

Run	$V_R - J_0$	Critic RMSE	Critic max.	Gain max.	Seconds
Pilot 101	0.026720	0.066694	0.676974	1.340226	13.5
Safe 101	0.001018	0.047925	0.298592	0.228318	88.3
Safe 202	0.001366	0.044197	0.346920	0.210235	83.0
Safe 303	0.001177	0.052661	0.934824	0.228568	81.5

separately solved 33×49 , eight-date, $9 \times 9 \times 13$ reference at $(2, 1.25)$. Critic errors compare critic with its independently evaluated policy over the whole grid. Gain is the largest feasible improvement over all interior states and dates against the evaluated continuation. The pilot has a critic and training budget; it is not a matched ablation. Hard boundary discrepancies are evaluated only in the raw record.

TABLE IV

FULL-VECTOR POLICY DISCREPANCIES AGAINST THE MATCHED REFERENCE

Run	MAE c	MAE θ	MAE π	Max. c	Max. θ	Max. π
Pilot 101	0.0136	0.0678	0.3212	0.562	0.400	1.300
Safe 101	0.0047	0.0165	0.1065	0.562	0.400	1.300
Safe 202	0.0047	0.0157	0.0954	0.562	0.400	1.300
Safe 303	0.0048	0.0158	0.1085	0.562	0.400	1.300

absolute and maximum discrepancies include every interior common node and all eight dates. Dates have different economic units and are not added together or called a Hamiltonian gain. Both use their stated finite approximations; these are not errors against a continuous-time oracle.

of the $9 \times 9 \times 13$ reference mesh, the $7 \times 7 \times 11$ diagnostic mesh, and the three proposals. The common mesh contains 1565 distinct actions; the complete menu has 8 entries per node. Including the original diagnostic mesh is important: two distinct action meshes need not be nested merely because one has more points. Every tested profitable deviation is feasible in the repair calculation.

target is now an explicit eight-date, 33×49 killed Markov chain. A branch's interpolation weights are nonnegative; its stopped payoff, discounted duration, adjustment effort are immediate rewards. Backward maximization uses the *evaluation*-date optimal continuation, and a separate fixed-policy recursion evaluates the *repair*-date optimal continuation. In this finite model, the incumbent proposals are feasible and the repaired proposals weakly improve their values at every node. Table V gives the pre-repair error: the post-repair Bellman gain over the entire stated menu, using independently evaluated policy continuations, is 4.441×10^{-16} , below 10^{-10} .

Largest initial-fit gains occur near the low-risk-aversion and high-wealth portion of the rectangle. Their exact dates, states, and occupation masses are saved. Under both the initial state and the stated initial population, the single worst-gain nodes have zero occupation mass. This explains why a large all-state error can coexist with a much smaller initial welfare loss; it does not make the global error disappear. The population and the occupation-weighted gains are distinct calculations, and the latter are bounded as upper bounds on optimal-policy loss. The repair addresses all states rather than discarding the low-occupation ones.

With $d = 2$, the adaptive construction uses 47 anchors to cover every $d \in [0, 1]$. Its maximum two-policy chord gap is 0.00093070, below 10^{-3} at every date and state. The

Seed	Feasible gain	Initial loss	Population loss	All-node loss
101	0.228318	0.001058	0.000730	0.236122
202	0.215071	0.001406	0.000783	0.245815
303	0.228568	0.001217	0.000800	0.236109

losses are value differences in the same finite economy, relative to exhaustive backward optimization. The action menu is the union of the reference and original diagnostic meshes and three frozen neural proposals at each node. The initial state is $(2, 1.25)$; the population is uniform on grid nodes in $[1.6, 2.4] \times [1.6, 1.6]$. “All-node includes every date and state, not only occupied states. The repaired policy has zero remaining gain in the computed backups over this menu. The published diagnostic mesh is contained in the new menu.

TABLE VI
OPERATING-BENEFIT COUNTERFACTUALS

d	Value	Duration	c_0	θ_0	π_0	Sign value gap
0.00	-1.136507	0.887826	0.80	0.20	-0.50	-0.000876
0.25	-0.914548	0.887930	0.80	0.20	-0.50	-0.000277
0.50	-0.692266	0.890233	0.80	0.20	0.80	0.000299
0.75	-0.469703	0.890536	0.80	0.20	0.80	0.000869
1.00	-0.246682	0.894175	0.80	0.20	0.80	0.001281

Initial state $(u, X) = (2, 1.25)$, $k = 2$. Duration is A^p , not the probability of survival. The final column is the best value among nonnegative-share first actions minus the best value among nonpositive-share first actions, using the optimal finite-model continuation. A positive entry favors the positive-share group. These five direct solves are distinct from the continuum certificate, which uses interval envelopes and ordered feasible action–continuation pairs.

Every evaluation does not retrain an actor or solve a new control maximization at a counterfactual query. A direct comparison at 101 parameter values finds a maximum value loss of 2.409×10^{-4} ; this comparison validates the implementation but is not the source of the continuum claim. The latter follows from the complete breakpoint enumeration in Section 6.

Table VI reports five independently optimized initial-state counterfactuals. Applying Proposition 11 to the full menu certifies a negative optimal initial share on $[0, 0.361017]$ and a positive share on $[0.373654, 1]$, with inward-rounded displayed endpoints. The intervening band is unresolved by this certificate. The sign conclusion uses action-value iteration and therefore does not confuse a 10^{-3} global welfare tolerance with a 10^{-3} tolerance on an action coordinate.

The contract mechanism is economically specific. Raising the operating benefit reduces longer discounted operation and changes the portfolio from a short position to a long one in the displayed economy. Adding the corresponding annuity payment to liquidation leaves the full policy correspondence unchanged; an independently evaluated terminal-exposure recursion verifies the accounting identity. The point reversal also occurs when the shock correlation is set to zero or to 0.25. It is therefore not an unconditional sign prediction from negative preference correlation. The model does not claim that a unit operating benefit is empirically estimated or that portfolio response holds under every liquidation contract.

persist, while the portfolio at $d = 0.5$ changes from negative on the coarsest to positive on the two finer ones. Value levels also move with resolution. This is the location of a diffusion-model switch a separate approximation question; it does not invalidate the all-parameter statement for the specified finite target. The continuous-time operator and its error-discrepancy term remain in the analysis. In an actual 101-query workload, full direct solutions take 184.86 seconds, compared to 143 seconds for library selection and complete policy evaluation. Building the library and common kernel takes 87.47 seconds, including the interval-certificate calculation. These are incremental times given the same frozen proposals to both methods. The experiment measures a reusable family of complete policies and moments, rather than interpolating from a single network forward pass. The stored library contains state policy values; its construction does not establish a neural compression advantage in this low-dimensional economy.

12.5. Coupled Stochastic Resource Allocation

Examine a genuinely dynamic shared constraint, let $x_n \in \mathbb{R}^d$ denote productive coordinates and $a_n \in \mathbb{R}_+^d$ investment. The three-period model is

$$x_n = A_d x_{n-1} + a_n + e_{n+1}, \quad A_d = 0.8I + 0.04(I_{+1} + I_{-1}), \quad \sum_i a_{ni} \leq 0.25d. \quad (106)$$

I_{+1}, I_{-1} denote the adjacent off-diagonal matrices. The shock has four equally likely components, $e = z_1 \ell_1 + z_2 \ell_2$, where $\ell_{1i} = 0.06$ and ℓ_{2i} is evenly spaced from -0.04 to 0.04 . The covariance is rank two and generally has nonzero cross-stock entries. Heterogeneous stocks, neighbor spillovers, a common budget, and the aggregate cost couple the coordinates.

The objective is expected total cost,

$$C^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{n=0}^2 \left\{ q(x_n) + \frac{0.1}{d} \|a_n\|^2 \right\} + q(x_3) \right], \quad (107)$$

$$q(x) = \frac{\|x - \mathbf{1}\|^2}{d} + 0.2 \left(\frac{\mathbf{1}^\top (x - \mathbf{1})}{d} \right)^2.$$

In a finite stochastic economy, not a discretization claim about a different diffusion. Terminal shock paths are used when a policy is evaluated. The independent coordinate optimizes a nonanticipative scenario tree with 21 decision nodes per initial state. The objective is convex quadratic and its nodewise resource constraints are genuine constraints on investment.

The cost critic is

$$\hat{C}_n(x) = q(x) + b_n + \ell_n^\top (x - \mathbf{1}) + \sum_{j=1}^J w_{nj} [h_j^\top (x - \mathbf{1}) + b_j]_{+}^2, \quad w_{nj} \geq 0. \quad (108)$$

The hidden features include coordinate, aggregate, and random directions; $J = 12$. Output coefficients are fitted by constrained least squares to the separated targets. A width-32 tanh proposal network uses a slack-softmax feasibility

d	NBO upper loss	Proposal only	NBO sec.	Reference sec.	Passed
4	0.000645	0.01190	1.737	0.172	3/3
8	0.000461	0.09541	2.241	0.190	3/3
16	0.000567	0.06807	3.180	0.226	3/3

columns are worst held-out cost-loss upper bounds across three seeds and 32 initial states per seed. Proposal only removes deployment improvement from the same fitted actor weights. Timing is the median across seeds at a 10^{-3} cost-loss target; NBO includes fitting and full evaluation, and the reference solves the initial-state batch to its 10^{-3} gap target. A separate 10^{-7} reference gap is used for auditing, not for the timing comparison. The reference is a convex nonanticipative scenario tree, not a sparse grid.

o and is trained against the improved controls. At deployment, projected convex improvement uses the analytic critic gradient, checked against automatic differentiation, to reach the gap (27). Squared-ReLU features are suitable here because the finite-step operator needs a value and an action gradient, not a classical second-derivative residual. For each of $d = 4, 8, 16$ and seeds 101, 202, and 303, training uses 1,024 draws from $[0.25, 1.65]^d$. Evaluation uses 32 fresh initial states from $[0, 1.2]^d$ and their complete look-ahead trees. The reference is solved both to the common cost tolerance 10^{-3} for timing and to a gap below 10^{-7} for independent validation. Formula (28) gives a numerical upper bound on each tested policy’s cost loss. These are double-precision convex certificates; recorded feasibility and gradient discrepancies accompany them, rather than an assertion of interval-arithmetic proof.

Every displayed run meets the common cost-error target at its tested initial states. The shared constraint binds frequently, including at most or all initial states depending on dimension and seed. Thus the experiment tests a dynamically consequential constraint rather than an inactive constraint. The same trained proposal networks are evaluated without the deployment safeguard. This matched ablation, reported in the supplement, isolates the effect of the improvement calculation without changing actor weights or training data.

At this query batch and tolerance, the specialized convex reference is faster than training and evaluating the neural representation. The latter delivers a usable feedback approximation, but a crossover in amortized cost requires a separate measurement. The experiment establishes accurate constrained neural control for this model and a transparent cost–accuracy comparison; it does not establish superiority over a solver that exploits the model’s convex structure.

12.6. What the Actor and the Continuation Representation Contribute

The actor does not enter the resource experiment’s critic targets: the target controls are computed before actor distillation. A same-weight comparison must therefore separate the value of continuation approximation from the value of actor initialization. We reload all nine checkpoints and solve the same convex improvements from either actor output or zero. The largest complete-policy cost difference between these arms is 1.256×10^{-10} . Both attain the stated 10^{-3} cost tolerance on every original held-out state. Actor initialization can save iterations, but those savings do not by themselves imply that its training cost is amortized.

The nonneural comparator is a full convex quadratic, $q(x) + b + \ell^\top(x - \mathbf{1}) + (x - \mathbf{1})^\top H(x - \mathbf{1})$ with $H \succeq 0$. It is fitted by projected least squares on the same training-state

Actor-start loss	Zero-start loss	Quadratic loss range	Actor-zero difference
6.446×10^{-4}	6.446×10^{-4}	2.065×10^{-3} – 3.203×10^{-3}	1.256×10^{-10}
4.607×10^{-4}	4.607×10^{-4}	1.668×10^{-3} – 6.692×10^{-3}	1.104×10^{-10}
5.665×10^{-4}	5.665×10^{-4}	5.444×10^{-3} – 6.777×10^{-3}	2.679×10^{-11}

dimension includes seeds 101, 202, and 303 and the same 32 initial states per seed. The first two give the largest certified lifetime cost-loss upper bound across those seeds. The quadratic spans the three maximum losses of a separately trained full positive-semidefinite quadratic critic. The same feasible controls and full shock trees; a reference Frank–Wolfe gap below 10^{-7} is added measured cost difference. Actor and zero starts use identical frozen nonlinear critic weights.

TABLE IX

FRESH-STATE QUERY WORKLOADS AT A COMMON COST TOLERANCE

Queries	Max. loss bound	Training	Query	Reference	Cost ratio
32	6.024×10^{-4}	0.812	0.040	0.152	0.18
128	6.574×10^{-4}	0.812	0.119	0.420	0.45
512	6.574×10^{-4}	0.812	0.600	1.477	1.05
2048	6.754×10^{-4}	0.812	2.600	5.708	1.67

tion four, seed 101; held-out state seed 541021, uniform on $[0, 1.2]^4$. Larger workloads extend the fresh-state sequence. Times are seconds. Query time includes zero-start convex improvement and terminal paths per initial state. Reference time is for independent nonanticipative optimization to common 10^{-3} cost tolerance. The cost ratio is reference time divided by critic training plus query time. Reported maximum policy-loss bound is below 10^{-3} . Timings are medians of three evaluations; the tighter validation reference is not charged to either competing solver.

ation and used in its own backward recursion. Unlike a diagonal quadratic, it every cross-state term. Its projected-gradient stopping diagnostic and Hessian values are recorded. Table VIII shows that the richer squared-ReLU continuation attains the common welfare target whereas these quadratic fits do not. This is due for the representational flexibility used here, not a theorem that a neural continuation dominates all nonneural approximation classes.

comparison with the previous execution, we retain the fresh sequence of initial query volumes 32, 128, 512, and 2,048. No initial state is added to the training query includes the complete endogenous policy tree and its 64 terminal shock Table IX charges for critic training, zero-start online improvement, and policy tion, and compares that sum with the specialized scenario-tree solver at the 10^{-3} cost tolerance. The independent tighter reference remains solely an accuracy for both methods. At the largest workload the measured end-to-end cost ratio is with a maximum policy-loss bound of 6.754×10^{-4} . The intermediate workloads prevent that historical crossover against the original reference. They do not give the price the reusable quadratic representation used below.

historical crossover does not identify nonlinear continuation as the preferred the object. All these queries share a quadratic objective Hessian and an affine from initial state to its linear term. Section 12.9 gives the conventional solver that opportunity for reuse and changes the computational conclusion. The actor’s the distillation time, initialization time, optimizer iterations, and projected break-

measurements. The original table is preserved as an executed comparison, not noted into an advantage over the stronger reusable-QP baseline.

12.7. Smooth and High-Dimensional Controls

An additional stochastic LQR experiment uses correlated Gaussian shocks, neighbor coupling in the transition, and a rank-one cross-state cost at dimensions 4, 8, and 16. Its critic has quadratic features, its actor is linear, and its independent reference uses Jacobi recursion. This structured control checks backward evaluation, covariance matrix-vector products, actor coefficients, and full-policy values. It is not an experiment with a generic multilayer critic, nor a substitute for the constrained resource application.

Computational cost depends on the operator and representation. For a differential equation and rank- r loading B , the exact trace can be evaluated using r directional Hessian-vector products. A K -probe trace instead uses K such contractions after forming the probe vectors. Their total costs include network differentiation, loading multiplication, Hessian-vector products, optimization, sampling, and the iterations needed to achieve an error target. Per iteration differentiation does not establish a polynomial solution-complexity theorem, but a threshold such as dimension 50 follows from that calculation. The semigroup experiments report their own costs without attributing them to unexecuted stochastic gradient acceleration.

12.8. Same-Oracle Optimistic Support and Neural-Action Attribution

Table X compares two coverage objectives with an identical exact Bellman oracle, on menu, state space, and tolerance. The initial-distribution version concentrates all weight on $(u, X) = (2, 1.25)$ at date zero. The statewise version takes the maximum gap across all 1,617 states and all eight decision dates. Both start from $d = 0, 1$, insert a worst-case starting corner, and use independently evaluated policy features. We report common kernel construction separately; bank memory counts policies and features, excluding shared transition arrays. The initial-distribution target needs only three anchors. Its

TABLE X
EXACT-ORACLE SCALAR OPTIMISTIC SUPPORT

Coverage objective	Anchors	Seconds	MiB	Global bound
Initial state/date	3	5.82463	1.14732	0.0529688
All states/dates	47	91.2708	17.9746	0.000930698

Notes: Same exact full-menu oracle and 10^{-3} target. The initial-state bound is 0.000410746. Global means over infinite states, dates, and $d \in [0, 1]$. Common kernel setup costs 2.78081 seconds and is excluded from preparation columns. MiB counts policies and features, not shared kernels or upper buffers. The independent full-line/adjacent-line discrepancy is 4.44089e-16.

The larger all-state/date bound demonstrates why that guarantee cannot be silently dropped for a different population or a later date. The statewise calculation uses 47 anchors. An independent calculation with every policy line reproduces its bound and verifies adjacent-policy identity to floating-point tolerance. This is an executed comparison of guarantee scope and the scalar-OLS correspondence, not evidence that a new name obscures the oracle count.

the 1,565-action common mesh. The upper information continues to come from the original full, neural-inclusive target. Table XI reports the anchor losses and the sum over every contract interval. The mesh-only bank remains within 10^{-3} of the target for all $d \in [0, 1]$, states, and dates. This is not a comparison against an restricted optimum, and it is not an autonomous mesh-only anchor-discovery experiment: the 47 locations and the full-target upper oracle are shared. The neural

TABLE XI

REMOVING NEURAL ACTIONS FROM THE FEASIBLE POLICY BANK

Feasible lower policies	Anchors	Anchor loss	Interval bound
Full menu	47	0	0.000930698
Common mesh only	47	3.87125e-05	0.000930698

Both rows use the original neural-inclusive optimal upper target at the same anchors. The second row moves all three neural candidates only from the lower policies. The interval bound covers all states, and $d \in [0, 1]$. Mesh-policy construction costs 88.0996 seconds; this is not an autonomous anchor-discovery time.

have a small, nonzero value contribution at some nodes, but are unnecessary for clearance. Exact dynamic programming constructs the repaired bank; independent valuation supplies its features; the upper oracle supplies the welfare benchmark; switching supplies deployment. The initial trained proposals and their profitable options remain reported. The transition-law theorem is an additional guarantee applicable to either neural or mesh policies; it does not require reassigning these roles to learning.

12.9. A Reusable Structural Solver for the Resource Economy

the 21 nonanticipative node controls into $z \in \mathbb{R}^{21d}$. For every initial state x , the model has cost

$$C_x(z) = \frac{1}{2}z^T H z + (F x + f_0)^T z + c(x), \quad z_h \geq 0, \quad \mathbf{1}^T z_h \leq 0.25d. \quad (109)$$

Matrices H, F and vector f_0 are assembled once from the affine state paths. We use controls by their node probabilities, compute the resulting Hessian's largest value once, and apply accelerated projected minimization. The scaling is constant at a node, so the node's feasible-set projection is unchanged. Each query starts at controls. The computation uses no future-shock information, stored query answer, query warm start.

Comparison follows the structural counterexample in the fifth-round advisory with an explicit independent assembly in the revision source. We check the reduced costs and gradients against the original full-tree routine. Every query is tested on all 64 terminal paths, and its stopping gap is recomputed from the full tree gradient, not just the assembled matrix. Table XII uses the same fresh states and 10^{-3} target as the earlier workload. Setup is included for the QP; only retraining, zero-start improvement, and full policy evaluation are included in the earned arm. Query timings are medians of three serial single-thread runs. The

Queries	Learned total	QP total	QP query	QP max gap
32	0.884792	0.00759194	0.00434819	0.000633261
128	0.964309	0.014003	0.0107592	0.000664162
512	1.57165	0.0405753	0.0373315	0.000664162
2048	3.66748	0.143248	0.140004	0.000677426

es: Times are seconds, single-thread, query medians of three. The learned total includes critic retraining and full policy evaluation; QP total includes structural setup and independent full-tree gradient/gap evaluation. All queries start at zero controls. The largest learned policy-loss upper bound is 0.000675399. original reference timings and all timing samples remain in the JSON record.

able QP is faster in these workloads, including online time alone. The complete-gaps certify its costs at the same tolerance. This is a substantive change in the computational conclusion, not a revision of the historical elapsed times. A fitted PSD-dratic *continuation approximation* and the exact parametric quadratic *program* are erent comparators. The recorded failure of the former does not imply a failure of the er. Consequently this resource model supports controlled neural approximation and onponent attribution, but supplies no empirical basis for choosing learning over the able structural solver. The additional transferable guarantee studied next concerns onlinear stopped economy with changed transition laws, not a retiming of this dratic example.

12.10. A Continuum of Shock-Law Counterfactuals

et law 0 be the original one-step quadrature with shock correlation -0.25 , and law e otherwise identical quadrature with correlation 0.25 . At every date, law 1 occurs n probability λ , independently across dates and without being known before that e's control. The state transition, exit payment, and discounted continuation are the esponding mixtures. The state grid, consumption and portfolio bounds, adjustment u, terminal payoff, and three frozen neural proposals remain unchanged. This is e change in the frequency of positive versus negative co-movement regimes in the e economy. It is not the claim that interpolating two quadratures equals discretizing y intermediate Brownian correlation.

or each of the contract contrasts $d = 0, 0.5, 1$, we solve endpoint problems and ptively subdivide the transition-parameter interval using Theorem 13. The endpoint cies are evaluated by (65). Eight closed Bernstein cells within each anchor interval nd its full policy-value gap; no target problem is solved at those cell endpoints ely to label a grid uniform. Table XIII reports the maximum certificate over every e, date, and $\lambda \in [0, 1]$. The exact parameter remains common across the horizon. e certificate is distinct from the older reward interval: the displayed d values are e fixed contracts, not a new joint continuum claim over (d, λ) . The count column his retained table uses one global interval and only the three feasible policies $= 0, 0.5, 1$. The corrected column uses adaptive localization and 17–19 anchor cies. Both columns are valid, but their difference confounds the upper construction, ization, and lower-policy coverage; it is not evidence of a sharpness advantage for e correction. Section 12.12 supplies the matched comparison and the work account. position 15 establishes that, on the same intervals and policy bank, localized count eakly sharper.

d	Anchors	Corrected bound	Count bound	Prep. seconds
0	19	0.000904573	0.0127756	52.5879
0.5	18	0.000899829	0.0123306	58.5677
1	17	0.000941412	0.00928855	53.1281

Each row covers every $\lambda \in [0, 1]$, all 1,617 states, and eight decision dates at the displayed fixed d . Corrected preparation includes endpoint solves, policy-coefficient construction, and interval propagation, excluding the common two-kernel build (5.43095 seconds) and subsequent independent evaluation. The count bound uses its separate informed upper oracle and three feasible anchor policies. It is asserted that it does not meet the 10^{-3} target. No joint (d, λ) -continuum or diffusion guarantee is asserted.

For each final anchor interval is independently replayed at its endpoints and midpoint to verify the complete backward optimization of the same mixture model. Fixed-policy evaluation of terminal values are also checked against a direct selected-kernel recursion. Small models enumerate all regime sequences, and two-date models enumerate all deterministic Markov policy. These tests distinguish a correct polynomial identity from an accidentally favorable main-model result. The uncorrected-chord negative has endpoint values zero and midpoint value $1/4$; some valid correction or informed upper oracle is therefore necessary. This negative control does not establish the necessity of this particular correction among valid constructions.

Calculations use double precision. The transition arrays are positive, the prescribed bounds are exhaustively checked, and recorded arithmetic discrepancies are bounded with explicit tolerances. These are finite-model numerical certificates, not a floating-point rounding interval arithmetic or a diffusion discretization guarantee. A CSR representation accelerates multiplication by the identical transition rows; its values are checked against the inherited array implementation, while fixed-policy evaluation uses a separate arithmetic path.

12.11. *The Preference-Formation Contribution to Risk Choice*

[XIV](#) evaluates the risk-class comparison in Section 8. Both arms retain preferences, the original covariance, the same operating boundary and settlement, and the same consumption and portfolio menus. The restricted arm sets deliberate adjustment at *all* states and dates. The difference $\Omega_+ - \Omega_-$ is computed from separately evaluated and independently evaluated risk-class policies, not from a first-order condition imposed on a binding control. At $d = 0.5$, the adjustment-enabled economy favors the first risky position, whereas the otherwise identical economy without deliberate adjustment favors the nonpositive class. The same disagreement occurs at $d = 0.375$ and $d = 0.4$. Thus the relative preference-formation option is large enough to reverse the ranking under an unchanged stopping contract. At $d = 1$, both economies favor the positive class: the operating-contract incentive can produce a reversal without deliberate preference formation, but the formation option changes where that reversal occurs. This separates a generic duration incentive from a preference-specific ranking option (77). Neither fixed consumption at a bound nor movement between portfolios alone supplies this identification.

[XV](#) holds the financial and stopping primitives fixed and varies the adjustment-coefficient. It reports opposite-sign brackets and the attaining policies' duration

d	Δ^{adj}	Δ^0	$\Omega_+ - \Omega_-$
0	-0.000875905	-0.00133454	0.000458639
0.25	-0.000276562	-0.000730716	0.000454155
0.35	-4.62573e-05	-0.000489849	0.000443592
0.365	-1.17117e-05	-0.000453738	0.000442026
0.375	1.13188e-05	-0.000429663	0.000440982
0.4	6.88949e-05	-0.000369477	0.000438372
0.5	0.000299199	-0.000135684	0.000434883
1	0.00128126	0.000950467	0.000330789

es: Positive Δ favors a positive first risky position; negative Δ favors the nonpositive class. Both classes minimize every subsequent action. The restricted arm sets deliberate adjustment to zero at all states and es but retains preference shocks and every financial and stopping primitive. Each reported value is ependently evaluated in the same cardinal utility units.

effort differences. The ratio in (75) is a local regular-branch prediction in cardinal v-utility units per unit of adjustment cost, not a global derivative through every pol-kink. We do not claim uniqueness from a bisection bracket; uniqueness requires the form duration-separation condition of Proposition 17. The no-adjustment bracket arately shows which part of the risk response remains when the formation option is oved. Increasing the adjustment-cost coefficient moves the observed opposite-sign

TABLE XV
RISK INDIFFERENCE AND THE DURATION-EFFORT COMPARISON

k / restriction	Opposite-sign bracket	$A_+ - A_-$	$C_+ - C_-$	Ratio
0.5	[0.34687805, 0.34689331]	0.00230304	4.6285e-05	0.0200973
2	[0.37007141, 0.37008667]	0.00230304	2.12312e-05	0.00921875
8	[0.48182678, 0.48184204]	0.00230304	7.90303e-05	0.0343156
No adjustment	[0.55891418, 0.55892944]	0.00230304	0	0

es: Each bracket has opposite risk-advantage signs at its endpoints and width at most 2×10^{-5} . roves existence of a crossing, not global uniqueness. Features and their ratio are evaluated at the ppoint. The ratio predicts a regular local branch slope only under the stated nonzero-duration and erentiability conditions. A zero-adjustment policy has zero adjustment effort. Bracket endpoints and r signed values are stored, rather than inferred from rounded table entries.

cket toward higher duration compensation. The no-adjustment bracket lies further ne right. These results accord with the positive effort–duration ratios of the attain-class policies, subject to the local regularity conditions. The duration–effort and on-value propositions organize these optimized-policy comparisons. The primitive dition in Proposition 19 supplies a separate predictive result, and Proposition 20 plies the class-specific transition-law certificate used below. Changing the operating ndary or settlement curvature would define further economic contracts, rather n innocuous numerical robustness checks. Their effects are not identified by the sent matched restriction. The original boundary calculation, correlation panel, and t numerical refinements remain available and are not silently replaced by the new eriment.

Now hold fixed what differed in the retained global comparison. For every one of the 1 original anchor intervals, the corrected chord, localized count recursion, and the regular upper recursion use the same two endpoint policies and the same eight Bernstein cells. Every anchor is freshly solved on the full 1,617-state, eight-action target, and its policy coefficients are independently reevaluated. The values and coefficients reproduce their R6 records to the deposited numerical precision. The three full-bank rows therefore reproduce the sixth-round matched comparison while adding a common-machine work account. Two additional $d = 0.5$ retain every second or fourth anchor and include the right endpoint.

TABLE XVI
MATCHED UPPER-ORACLE PRECISION AND WORK

d	Anchors	Upper oracle	Bound	Upper sec.	Total sec.
0	19	Compressed	0.000904573	27.02	53.34
		Local count	0.000733846	95.12	121.40
		Rectangular	0.0187608	21.62	47.90
0.5	18	Compressed	0.000899829	25.46	51.01
		Local count	0.000685103	88.91	114.43
		Rectangular	0.0171499	20.01	45.52
0.5	6	Compressed	0.00698458	7.34	19.26
		Local count	0.00338132	25.61	37.52
		Rectangular	0.0427253	5.75	17.66
0.5	10	Compressed	0.00327336	13.35	29.79
		Local count	0.0020246	46.32	62.74
		Rectangular	0.0255638	10.35	26.77
1	17	Compressed	0.000941412	23.64	47.04
		Local count	0.000435745	81.90	105.27
		Rectangular	0.0150829	18.24	41.60

All states, dates, and $\lambda \in [0,1]$; identical two-policy lower banks and eight cells within each interval. The 19-, 18-, and 17-anchor banks are the original R6 banks; the 6- and 10-anchor banks are the $d = 0.5$ bank. Total time includes common kernel setup, anchor solves, policy evaluation, constraint restriction, upper construction, and certification. Shared stages are attributed equally. One single-thread pass; diagnostic replay is excluded equally. Kernel-call and persistent-array counts, interval bound, and stage times are in the replication records.

The localized count certificate is sharper in every full-bank row, as Proposition 15 states. The compressed construction buys smaller upper-oracle work rather than a loss. At eight dates its 26 action-wide kernel applications per interval compare to 30 for the cached count recursion and 16 for the rectangular relaxation. The constraint correction needs seven additional state vectors, compared with 28 interior vectors; neither number includes the shared endpoint values or the uncompressed policy bank. The rectangular relaxation is cheaper in kernel work but loses much precision here by allowing repeated resets of the regime probability. Thus the comparison separates a validity result, a sharpness result, and a cost result.

The elapsed-time columns include kernel construction, the required anchor solves, policy evaluation, local coefficient restriction, upper construction, and subdivision. All stages are measured once and attributed equally. The compact implementation

removed from the timing. These are one-pass serial single-thread measurements, medians of independently rebuilt machines. The operation counts provide a reproducible work comparison independent of elapsed-time noise. No new neural training earned-speed advantage is inferred from them.

Common banks do not determine how many anchors an independently refined method would request. Table XVII therefore starts three procedures at $[0, 1]$ with empty arm caches, fixes $d = 0.5$, and uses the same left-first dyadic subdivision and 10^{-3} set. The chord-only and count-only arms use their own certificates to decide whether to split. The cascade tries count only after the chord fails. Every rejected parent interval, fully solved anchor, lower evaluation, and cell test is included in that arm's cost. This is an autonomous refinement experiment, unlike the coarsened-bank rows of Table XVI. The chord-only procedure requests 18 anchors; local count requests 11; the cascade

TABLE XVII
AUTONOMOUS REFINEMENT FROM EMPTY POLICY CACHES

Rule	Anchors	Final bound	Kernel calls	Total sec.
Chord	18	0.000899829	858	73.41
Count	11	0.000924471	1330	115.38
Cascade	11	0.000924471	1614	127.73

es: Full target, $d = 0.5$, tolerance 10^{-3} . Each arm starts at $[0, 1]$ and includes all rejected parent intervals. Kernel calls are action-wide upper-oracle applications, not anchor-solve calls. Total time includes both. Cascade tries local count only after compressed certification fails. All arms use identical dyadic split order, endpoint policies, and eight certificate cells.

requests 11. Thus the cascade avoids 7 endpoint optimizations relative to chord-only refinement at the same certified tolerance. Its elapsed time is 127.73 seconds, against 71 and 115.38 for the two single-oracle procedures, respectively. The measured costs, rather than the anchor reduction alone, determine the computational comparison. The cascade's anchor set is checked to be a subset of the chord-only set, as the nesting argument predicts. It need not be the fastest procedure: buying a sharper certificate also assumes work. These finite-model results support a choice among upper-information targets; they do not establish a dimension-free neural advantage or the optimality of any particular refinement strategy.

12.13. A Certified Preference-Adjustment Effect on a Region

he economic certificate holds the operating boundary, settlement function, consumption menu, financial technology, and stochastic preference shocks fixed. At $(X) = (2, 1.25)$ and $k = 2$, consider the whole rectangle

$$(\lambda, d) \in [0, 0.25] \times [0.4, 0.45].$$

parameter λ is the same per-period probability of the positive-co-movement regime used above, not a new reward coefficient. Each risk class is solved at the four corners with and without deliberate adjustment. Their feasible policies are then evaluated as transition polynomials with affine contract coefficients. Localized upper values at the contract endpoints are interpolated in d using convexity of

rectangle.

TABLE XVIII
A JOINT-REGION CERTIFICATE FOR THE ADJUSTMENT EFFECT

Adjustment regime	Lower Δ	Upper Δ	Max. class gap
Adjustment available	6.88873×10^{-5}	0.000324537	7.5953×10^{-9}
No deliberate adjustment	-0.000369485	-5.29937×10^{-5}	2.19355×10^{-6}

initial state $(u, X) = (2, 1.25)$, $k = 2$, full finite target, entire $(\lambda, d) \in [0, 0.25] \times [0.4, 0.45]$. Bounds the arithmetic allowance. Maximum class gap is the larger of the two class-specific upper-minus-lower certificates in that regime. The per-class arithmetic allowance is 3.79765×10^{-9} ; the lower on the relative adjustment option is 0.000121881. Eight corner optimizations produce sixteen class policies; eight count constructions certify the rectangle. No generic 10^{-3} welfare tolerance is these signs.

positive first-risk class is strictly preferred throughout the region when adjustment is available, and the nonpositive class is strictly preferred throughout the same when it is unavailable. Both signs survive the separate numerical allowance. The difference bounds the relative adjustment option from below, so the conclusion is an incremental preference-formation effect under unchanged financial and other primitives. It is not the assertion that adjustment is necessary for every risk-induced risk reversal. The earlier $d = 1$ comparison, where both regimes select the risk, remains in the paper.

The calculation uses the transition representation to certify the economically relevant region. The upper-information and feasible-policy errors are measured separately for four class values; they are not assigned the generic 10^{-3} welfare tolerance. The coefficients and a coefficient-only replay accompany the policy bank. Supplement gives the round-to-nearest arithmetic account relative to the stored finite model. Independent corner replay is an implementation check, not a replacement for the inequality. The result concerns this initial state and rectangle, not all initial uniqueness of a global switching threshold, or a continuous-time discretization.

13. ECONOMIC ROBUSTNESS AND THE COST OF A DECISION CERTIFICATE

The preceding experiments establish the operator and transfer comparisons on their own targets. We now hold the economic decision itself fixed when comparing the components, and then vary the economic primitives with both adjustment and reoptimized. Every historical comparison remains in the preceding tables. The final evidence neither substitutes a generic welfare tolerance for a sign margin nor moves the original region when its interpretation is challenged by a perturbed technology.

13.1. *The Original Region and Its Moving Boundary*

XIX reproduces the original full-menu corner and its movement under finer settlement. The change from 49 to 97 nodes switches the no-adjustment ranking corner; the adjusted ranking remains positive. The positive relative adjustment

Wealth nodes	Menu	Δ^0	Δ^{adj}	Relative option
49	Common	-5.3001×10^{-5}	3.2453×10^{-4}	3.7753×10^{-4}
49	Full	-5.3001×10^{-5}	3.2453×10^{-4}	3.7753×10^{-4}
97	Common	6.1263×10^{-5}	4.2641×10^{-4}	3.6515×10^{-4}
97	Full	6.1263×10^{-5}	4.2641×10^{-4}	3.6515×10^{-4}
145	Common	9.7943×10^{-5}	4.6009×10^{-4}	3.6215×10^{-4}
145	Full	9.7943×10^{-5}	4.5954×10^{-4}	3.6160×10^{-4}

rows retain 33 preference tiers, eight dates, and $(\lambda, d) = (0.25, 0.45)$. Common uses the identical 65-action union. Full also uses the three unchanged proposals at original nodes and their specified wealth interpolation at new nodes. Both classes and both adjustment regimes are reoptimized. The original region is not silently replaced by a favorable subset.

on therefore survives a perturbation under which adjustment ceases to be necessary positive risk. This distinction is not a failure of the finite-array inequality. It identifies part of the economic conclusion that depends on the settlement lottery. The common-menu and prolonged-full experiments give the same qualitative result. Their menus are not silently equated: interpolation of frozen proposals at new nodes can controls beyond the common mesh. The common-menu arm isolates the wealth-ery change without that additional policy-menu change. All four original corners the center are retained for every grid and both menus, with 120 class values and cted-policy replays. At the center, the no-adjustment difference remains negative on hree wealth grids. Thus the evidence maps the movement of the boundary rather asserting that the opposite-choice phenomenon disappears everywhere.

TABLE XX
LOCAL INDIFFERENCE BRACKETS WITH THE CALENDAR HELD FIXED

Wealth nodes	λ	Adjusted contract bracket	No-adjustment bracket
49	0	[0.370068, 0.370117]	[0.558887, 0.558936]
49	0.125	[0.339551, 0.339600]	[0.515869, 0.515918]
49	0.25	[0.309082, 0.309131]	[0.472998, 0.473047]
97	0	[0.320166, 0.320215]	[0.513477, 0.513525]
97	0.125	[0.288135, 0.288184]	[0.467773, 0.467822]
97	0.25	[0.256055, 0.256104]	[0.422266, 0.422314]
145	0	[0.302783, 0.302832]	[0.497852, 0.497900]
145	0.125	[0.270313, 0.270361]	[0.451367, 0.451416]
145	0.25	[0.237646, 0.237695]	[0.405078, 0.405127]

l brackets use the full prolonged menu. Bisection begins on $[0.2, 0.6]$ and retains every evaluated ference. Opposite signs and continuity locate a crossing; global uniqueness is not inferred. Changing e wealth lottery shifts both boundaries without changing the one-eighth-year commitment interval.

he indifference brackets in Table XX explain that movement on the contract scale. n regime boundaries shift as the wealth lottery is changed, and the interval between n remains economically nonempty at the inspected mixture weights. The full solver e records all bisection evaluations. These are local brackets, not a proof of global notonicity of the difference of two convex value functions.

Wealth nodes	Lower Δ^{adj}	Upper Δ^0	Both signs certified
49	6.8695×10^{-5}	-2.0913×10^{-4}	Yes
97	1.7639×10^{-4}	-9.4581×10^{-5}	Yes
145	2.1143×10^{-4}	-5.7401×10^{-5}	Yes

ne closed region is $[0, 0.125] \times [0.4, 0.425]$. Bounds shown use the full-menu bank and corrected Count and mesh-only lower-bank checks are also retained. This is a continuum certificate in for each of three specified finite protocols, not a continuum claim over every wealth grid or the g diffusion. The region was selected after inspecting the earlier robustness results; it was not stered.

XXI supplies an additional regional statement. It uses one common nested le and separately certifies its two signs under each of the three full-menu finite ls. The old rectangle and its failed corner are retained in Table XIX; the nested s not presented as an unannounced replacement. Its validity over intermediate ts and mixture probabilities comes from tensor coefficient bounds, not from a few parameter points. The discrete set of settlement protocols is itself not ed to a continuum of all possible discretizations.

13.2. Which Primitive Supports the Dynamic Option?

TABLE XXII

DYNAMIC MECHANISMS AT THE ORIGINAL REGION CENTER

Intervention	$10^4 \Delta^{\text{adj}}$	$10^4 \Delta^0$	10^4 relative option
Baseline	1.9674	-2.0933	4.0607
Tilt $a = -0.25$	1.5564	-1.3521	2.9085
Tilt $a = 0.25$	2.3858	-2.8501	5.2359
Tilt $a = 1$	3.8022	-5.1768	8.9790
Normalization $s = 0$	9.1981	9.1981	0.0000
Normalization $s = 0.5$	11.6551	10.5243	1.1309
Normalization $s = 1.5$	-8.9524	-15.8240	6.8716
Liquidation $z = 0$	-20.0244	-24.7728	4.7484
Liquidation $z = 2$	21.1887	17.8547	3.3340
Zero-covariance law	4.1345	0.9752	3.1593

for the named intervention, parameters are the original $(\lambda, d, k) = (0.125, 0.425, 2)$. Every entry nizes both risk classes in both adjustment regimes. The normalization multiplier changes $b_0(u) = u)$, not marginal consumption utility. Values shown as zero at $s = 0$ are numerically negligible, asserted exact zero theorem. These model counterfactuals are not empirically identified causal

dynamic counterfactuals distinguish the reference-state payoff from ordinary version. At the center, the baseline relative adjustment option is approximately 10^{-4} . Removing $b_0(u)$ while preserving marginal consumption utility makes this ce numerically negligible and leaves positive risk in both regimes. Increasing its tier to 1.5 instead gives a larger positive relative option while making both risk

economic primitive of the dynamic example, not an innocuous normalization. The common tilt illustrates why the two-stage result is not a calibration of the full economy. A tilt of one reverses the relative option in the two-stage example, whereas in the dynamic economy it increases the positive relative option at the inspected target. In the latter experiment the no-adjustment value changes because preferences continue to fluctuate, and consumption, continuation, and liquidation opportunities are endogenous. The uniform primitive theorem remains valid on its stated family; the dynamic calculation supplies the separate evidence needed for this economy. Liquidation and covariance also move the choice boundary. Eliminating the liquidation payoff makes both rankings nonpositive while leaving a positive relative adjustment option. Doubling that payoff makes both rankings positive. The zero-variance law gives positive risk in both regimes. These interventions retain the controls, deliberate-adjustment restriction, and positive-duration decision definition. They reveal interactions in the full model rather than crediting every change in risk to preference formation alone.

TABLE XXIII

OPTIMIZED-MOMENT ACCOUNTING FOR THE RELATIVE ADJUSTMENT OPTION

Component	Contribution in baseline utility units
Consumption exposure	-5.1603×10^{-6}
Cardinal preference-state payoff	5.0928×10^{-4}
Tilt exposure at baseline	0
Operating duration	-1.3726×10^{-5}
Adjustment effort	-4.9862×10^{-5}
Liquidation payoff	-3.4460×10^{-5}

The sum is the baseline relative option. Each moment is evaluated at its own class-regime optimizer. This accounting does not assign invariant causal shares; policy changes and primitive interactions are reported separately through reoptimization.

The optimized-moment account in Table XXIII puts the baseline magnitudes in common cardinal units. The preference-state payoff contributes about 5.093×10^{-4} in the relative option, partly offset by consumption exposure, duration, effort, and liquidation terms. This is an exact accounting across four separately optimized policies, a causal identification theorem. The deposited counterfactuals additionally evaluate the original policies under each new primitive and report their reoptimization gains. They make clear which changes are valuation of a fixed exposure and which require altered behavior.

13.3. Attributable Computation at the Economic Accuracy Target

Table XXIV compares streaming chord and count implementations on the original 8-action target and the original rectangle. Both use the same full-menu upper bounds and independently evaluated class policies. Both certify the two risk rankings. The count coefficients are weakly tighter, but the chord requires less upper construction risk; in this experiment its tighter competitor does not change the accepted economic conclusion. The total includes model construction, frozen-proposal loading, endpoint minimization, lower evaluation, the arithmetic audit, and independent selected-control

Upper	Lower bank	Lower Δ^{adj}	Upper Δ^0	Upper sec.	Total sec.
Thord	Full	6.8695×10^{-5}	-5.2801×10^{-5}	2.02	10.60
Count	Full	6.8695×10^{-5}	-5.2801×10^{-5}	7.89	16.47
Thord	Mesh only	6.8695×10^{-5}	-5.2801×10^{-5}	2.02	13.79

target is the original 1,568-action economy on $[0, 0.25] \times [0.4, 0.45]$. Every arm is charged for proposal loading, full-menu upper-anchor optimization, feasible-policy evaluation, the economic audit, and independent replay. The mesh-only bank additionally pays for its own lower-construction. Common setup is measured once and charged identically; these are serial single-measurements, not independent-machine medians. The per-class arithmetic allowance is 10^{-7} .

TABLE XXV

INITIAL OPTIMIZERS BESIDE THE ORIGINAL DECISION CERTIFICATE

Regime	Class	Value	c	θ	π
Adjustment	+	-0.759029726	0.8	0.2	0.8
Adjustment	−	-0.759226466	0.8	0.2	-0.5
No adjustment	+	-0.794487049	0.8	0.0	0.8
No adjustment	−	-0.794277722	0.8	0.0	-0.5

center $(\lambda, d) = (0.125, 0.425)$, noncancellable mandate, complete finite menu. Both the long permission 0.8 and the short permission 0.5 bind. Consumption is also at its upper limit; deliberate consent is at its upper limit when permitted. The nonpositive position is risky, and the no-adjustment retains stochastic preferences.

The gain is therefore attached to the economic decision tolerance rather than an added 10^{-3} welfare target.

The mesh-only lower-bank arm is evaluated against the same full-menu upper target and also certifies both signs. Its separately charged construction is a diagnostic of the target, not a claim to be the most efficient non-neural solver. No network is trained for the new arms: the three historical proposals define part of their common, already certified target. Their loading and extension are charged in construction; historical target training is a sunk input reported in the retained earlier experiments, not silently added as a new training speedup. Adding an identical historical construction charge to the arms changes total percentages, but not the measured difference between these components.

The new arm addresses the lower-bank storage issue directly. For the baseline focal task, sixteen policies require 827,904 bytes of control indices, and their retained optimal-duration coefficients require 2,304 bytes. The kernel and reward payload is 69,952 bytes. These payloads exclude common temporary arrays and process overhead; the evidence records process peak resident memory separately, without adding a shared process peak to one upper arm. The largest memory object remains the position representation, not the upper coefficient correction. The streamed focal task does not claim to preserve simultaneous access to all dates' and states' lower components at the same storage cost.

The new dense audit checks 60 models, 240 class recursions, and 720 parameter-point comparisons, including first-action and no-adjustment restrictions. Full-model policies are replayed through the original selected-control transition formula. All executed

Contract	Regime	Duration	Grant q^*	$b_{\min}$
$F = 0.00, m = 1$	Both	0.124688	0.110570	0.886775
$F = 0.70, m = 1$	Adjustment	0.684865	0.775123	1.131790
$F = 0.70, m = 1$	No adjustment	0.436339	0.789425	1.809203
$F = 0.85, m = 1$	Adjustment	0.890233	0.781344	0.877685
$F = 0.85, m = 1$	No adjustment	0.887898	0.816592	0.919691
$m = 8$	Adjustment	0.890233	0.781344	0.877685
$m = 8$	No adjustment	0.887898	0.816592	0.919691

agent chooses the better initial class in each regime. Grants and charges use a separate additive utility measure, not managed wealth. Here $C(E) = 0$; for positive capacity cost add $C(E)$ to the grant and divide the added amount by duration for the change in $b_{\min}$. The principal obtains the reported flow only if the operation is profitable. These are not empirical monetary compensating variations.

Tables, upper and lower coefficients, source hashes, and stage timings accompany the manuscript. The bounds concern the specified finite protocols and their documented stochastic model. No empirical replay discrepancy is used as a substitute for the additional target-error control in (88).

14. THE PRICE OF COMMITMENT AND THE LOCATION OF THE RISK BOUNDARY

The following experiments change surrender eligibility or first-date investment permissions while preserving the original economic experiment as a reported comparison. Operating classes are reoptimized under the relevant changed primitive. The source programs use the hash-pinned transition constructor, complete later menus, and frozen proposal files of the preceding experiment. No network is retrained, no historical scientific input is replaced, and no settlement-grid result is silently reassigned to a different set.

14.1. Participation Is Quantitatively Material

At the original center $(\lambda, d) = (0.125, 0.425)$, immediate liquidation pays $G(x_0) = 23143551$. The noncancellable adjusted optimum is -0.7590297258 , whereas the adjustment optimum is -0.7942777218 . Thus the minimum signing grants, before capacity carrying cost, are 0.7813440809 and 0.8165920769. Adjustment reduces the compensation required to undertake the mandate by 0.0352479960. The signing grant is not a wealth injection into the portfolio problem; its separate-account assumption is essential to these numbers.

Table XXVI also relates the grant to the amount of service actually delivered. With a cancellable operation, the principal's break-even service flow is lower under adjustment. The calculation does not impose that benefit on the principal: below the played threshold, the specified service procurement is not mutually acceptable at minimum grant. Free surrender after the first interval requires less compensation because it delivers much less operation. It is not a cheaper way of obtaining the same service. For a positive carrying cost $C(E)$, add $C(E)$ to the grant and $C(E)/\mathcal{A}_{p^*}$ to the break-even service flow.

A concrete positive-cost procurement example illustrates the participation margin. If the available capacity is at least 0.85, use $F = E = 0.85$, let $C(E) = 0.01$ at that capacity,

m	F	$10^4 \Delta^{\text{adj}}$	$10^4 \Delta^0$	$10^4 \mathcal{O}$
1	0.00	5.1294	5.1294	0.0000
1	0.40	5.1294	5.1294	0.0000
1	0.70	43.0565	77.1751	-34.1186
1	0.80	1.9674	-1.9186	3.8860
1	0.85	1.9674	-2.0933	4.0607
2	0.00	5.4927	5.4008	0.0919
4	0.00	5.9646	5.9646	0.0000
6	0.00	22.4187	23.0450	-0.6263
7	0.00	25.6088	26.0814	-0.4726
8	0.00	1.9674	-2.0933	4.0607

center, full finite operating menu, four reoptimized class values per row. The first interval compulsory. $m = 8$ is the original mandate and has no eligible surrender date. Absolute values monotone in F and m in the required directions; the displayed differences need not be. All 19 center options remain in the numerical deposit.

the principal's net service flow be $b = 0.90$. With adjustment the minimum grant 3441 and the principal's surplus is 0.0098659. Without adjustment the minimum 0.8265921 and the principal's surplus is -0.0274839 . Thus the adjusted mandate is financially acceptable while the same service-price and capacity terms cannot support the unadjusted mandate. This is a constructive theoretical example in the stated free-account economy, not an empirical calibration or an optimality claim about the mandate menu.

14.2. A Finite Surrender Charge and a Nonmonotone Relative Option

With a zero charge with the first interval compulsory, all four optimized policies survive at the first eligible date. The positive and nonpositive values are respectively 2558999182 and -0.0887688376826 in both regimes. The relative adjustment is zero. The positive initial action is $(0.8, 0, 0.8)$; the nonpositive action is $(-1/15)$ in the original finite menu. That menu does not contain zero risky assets and the negative position is not described as risk-free. An independent date-zero calculation instead gives $G(x_0)$, as predicted by the supersolution argument. The free-surrender result is not inferred from a few optimized rows alone. At all dates, the all-action endpoint check at $d = 0.45$ has largest interior residual 4966453 for the operating backup of G . Affinity in λ and nonnegative duration pass the check to the original rectangle. The boundary residual is zero. The conclusion benchmark's previously established supersolution inequality has the same economic implication for its own free-stopping problem; this is not a convergence argument between the two models.

At the intermediate charge $F = 0.70$, the relative option is *negative*, approximately -341186. Adjustment still improves each individual class value. What changes is the class in which it is most valuable. Discounted surrender exposures are 0.248753 and 0.7 in the adjusted positive and nonpositive classes, compared with 0.560607 and 0.5 without adjustment. The two regimes therefore value the termination margin differently. These exposures enter (93); they are not a causal decomposition obtained by fixing the optimized policy fixed while claiming it remains optimal everywhere.

m	Term (years)	Reachable-state bound	All-node bound
1	0.125	0.834860	2.530979
2	0.250	0.764293	2.231418
3	0.375	0.681582	1.912037
4	0.500	0.585751	1.570892
5	0.625	0.474158	1.206311
6	0.750	0.342991	0.877726
7	0.875	0.187395	0.492947
8	1.000	0.000000	0.000000

form over $(\lambda, d) \in [0, 0.25] \times [0.4, 0.45]$ and both adjustment regimes, on the original finite menu. Bounds are rounded upward to six decimals after the 10^{-7} allowance. A capacity E at least the stated bound implements the original mandatory continuation values with charge $F = E$, provided $E \leq \bar{E}$. These are sufficient statewise bounds, not the smallest initial-state fees or an optimal contract-design solution.

The charge 0.80 restores opposite rankings at the center, but not the original uniform conclusion. The positive no-adjustment policy still uses the surrender option, with bounded incidence 0.0689951. A regional certificate at that charge fails to make upper class difference negative. At $(\lambda, d) = (0, 0.4)$, direct reoptimization gives a positive no-adjustment difference 0.00139238. The deposit retains this adverse corner as well as the failed certificate. Raising the charge to 0.85 restores the original values and bounds at the center. The uniform result below establishes more than that pointwise derivation.

Minimum operating terms provide another way of changing the obligation. With surrender after two, four, six, or seven intervals, both classes continue to prefer the relative position at the center in both regimes. The relative option is not monotone in minimum term: for example, it is approximately -0.0000473 at seven intervals, and comes 0.0004061 under the original eight-interval mandate. Absolute class values lose the required monotonicity in the fee and the minimum term. Their differences do not inherit it.

14.3. An Enforcement–Commitment Frontier on the Original Region

The reachable supports contain 30 nodes after one interval and 735 after seven intervals. Applying Proposition 25 with the two feasible no-adjustment lower policies gives the sufficient regional charges in Table XXVIII. For surrender available after the first interval, the computed upper bound is 0.8348592806; the displayed sufficient charge is rounded upward. Using every grid node instead would require a sufficient bound above 2.53. Reachability therefore matters economically: the guarantee need cover the actual mandate's possible continuation states, not unrelated initial conditions. These are upper bounds on the statewise enforcement requirement, not estimates of the smallest fee preserving just the initial class ranking. They apply to the original finite menu on target and the whole rectangle $[0, 0.25] \times [0.4, 0.45]$. They do not automatically cover the enlarged continuous first-date permission menus studied below. Larger minimum terms reduce the sufficient guarantee capacity. Adjustment weakly reduces the exact statewise requirement by (97); the supplement also reports the separately derived center thresholds in both regimes.

Contract	$10^4 \Delta^{\text{adj}}$ bound	$10^4 \Delta^0$ bound	Both signs
$m = 8$	[0.6869, 3.2473]	[-3.6968, -0.5280]	Yes
$F = 0.80, m = 1$	[0.6869, 3.2473]	[-3.6968, 13.9258]	No
$F \in [0.85, 0.90], m = 1$	[0.6869, 3.2473]	[-3.6968, -0.5280]	Yes

uses the original rectangle $[0, 0.25] \times [0.4, 0.45]$, both adjustment regimes, and four law subcells. and count are executed separately and give the same displayed bounds. Every unscaled difference 2×10^{-7} arithmetic padding relative to the stored arrays. The fee interval uses a joint fee–benefit–efficient bound. The failed $F = 0.80$ row and its adverse reoptimized corner are retained.

TABLE XXX
THE TWO REGIME-SPECIFIC PERMISSION BOUNDARIES

λ	L_{adj}^*	L_0^*	S_{adj}^*	S_0^*
0.000	0.792922	0.816033	0.506281	0.486275
0.125	0.789009	0.810784	0.509481	0.490704
0.250	0.785115	0.805683	0.512575	0.495063

$= 0.425$, with no voluntary surrender. The L roots hold $S = 0.5$; the S roots hold $L = 0.8$. Only the risky-share permission changes. Search is exact over the breakpoint reduction conditional on 121 consumption–adjustment pairs and feasible original proposals. Reported roots are floating-point enclosures, not interval enclosures or continuously optimized consumption and preference-adjustment

XXIX certifies the original opposite-position comparison for every

$$(\lambda, d, F) \in [0, 0.25] \times [0.4, 0.45] \times [0.85, 0.90], \quad m = 1. \quad (110)$$

reamed upper constructions pass with the stated arithmetic padding. Thus the conclusion is implemented by a finite, explicitly priced enforcement capacity surrender right, not only by formally forbidding exit. The capacity remains a relative economic restriction: participation requires its carrying cost to be funded, free and intermediate-charge outcomes remain different. The same original rather than a replacement favorable rectangle, appears in all three rows of the

14.4. Both Investment-Permission Boundaries Move

breakpoint search reproduces the original finite-menu class values at all nine (λ, d) contracts to numerical precision. The maximum over continuous first-order shares, conditional on each finite consumption–adjustment pair, therefore reveal an omitted within-box improvement in these comparisons. The original roots are at both position limits, as shown beside the original decision table. The preference-hedging formula remains an interior characterization, not a formula for constrained optimizers.

XXX places both switching permissions on the same scale. At the center and $F = 0.5$, the adjusted class changes sign at $L \simeq 0.7890095$, and the no-adjustment $L \simeq 0.8107837$. The baseline $L = 0.8$ lies between them. Increasing it to 0.82

λ	L	S	d_{adj}^*	d_0^*
0.000	0.80	0.50	0.370085	0.558915
0.000	0.82	0.50	0.176862	0.386619
0.000	0.80	0.51	0.457404	0.636790
0.125	0.80	0.50	0.339574	0.515873
0.125	0.82	0.50	0.139861	0.334568
0.125	0.80	0.51	0.429666	0.597689
0.250	0.80	0.50	0.309087	0.473014
0.250	0.82	0.50	0.102936	0.282746
0.250	0.80	0.51	0.401952	0.558753

voluntary surrender. Later-date policies are reoptimized at every trial benefit on the unchanged full nu. All reported local crossings have positive optimizing duration differences. Local numerical brackets feature values are deposited; they do not establish global uniqueness or a directed-rounding root closure. The original permissions and both adverse first-date extensions are shown at every inspected probability.

ses both, so both regimes select a positive position. Holding $L = 0.8$ instead, the corresponding short-permission switches are $S \simeq 0.5094806$ and $S \simeq 0.4907037$. Increasing from 0.5 to 0.51 crosses the adjusted boundary, so both regimes select a nonpositive position. The relative adjustment option remains positive in these two particular permission interventions. Its sign alone therefore does not determine whether adjustment changes the selected position class.

at the baseline center, the long-permission shadow values are approximately 0.79009 with adjustment and 0.0194114 without adjustment. The short-permission shadow values are 0.0200460 and 0.0225172. These quantities explain the sensitivity in the units per unit of risky-share permission. The exact interpolation-cell derivatives agree with central finite differences to less than 10^{-10} in the reported nine-contract audit. One- or two-percentage-point change in permissions can be large relative to an initial s -value margin near 2×10^{-4} .

the operating-benefit frontiers in Table XXXI are recomputed with continuation policies reoptimized as d changes. They are not obtained by extrapolating a fixed policy's duration. At $\lambda = 0.125$, the original local crossing pair is approximately (0.339574, 0.515873). With the long permission 0.82, it becomes (0.139861, 0.334568); with the short permission 0.51, it becomes (0.429666, 0.597689). Thus the same central benefit 0.425 can lie between, above, or below the two boundaries. The movement is consistent with the duration and permission-price identities in (99). The numerical brackets identify local crossings; global uniqueness in d and directed-rounding closures of the roots are not inferred.

the economic content of the example is consequently a priced interaction between reference adjustment, operating commitment, and constrained exposure. Adjustment reduce the compensation and enforcement required to operate while changing which binding permission is preferred. The sign of that change depends on the surrender margin and on the location of the two permission-sensitive value surfaces. The original cardinal-tilt theorem, the settlement-granularity evidence, and the same-set computational comparisons remain necessary complementary statements; none substitutes for these contract margins.

Model	Learned m	Learned π	Policy max. error	Relative A error
Merton	0.04124996	0.74999998	3.94e-08	9.13e-13
Epstein–Zin	0.04550010	0.30000000	1.03e-07	6.89e-12

efficient $A = \exp(q)$ and bounded actor parameters are learned by separated updates. Analytical ratios are used only to compute the displayed errors. The sign condition and normalized HJB are tested by. These runs use homothetic features rather than an unrestricted neural value class.

15. RECURSIVE UTILITY, TEMPORAL SELVES, AND DYNAMIC GAMES

15.1. Merton and Epstein–Zin Policy Evaluation

A smooth portfolio benchmark uses stationary infinite-horizon CRRA utility with $\psi = 1$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$. With $c = mX$ and no terminal bequest, analytical ratios are

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2} = 0.75, \quad m^* = \frac{\rho + (\gamma - 1)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\}}{\gamma} = 0.04125. \quad (111)$$

These are stationary quantities, not finite-horizon policies with an arbitrary terminal condition. Under the displayed optimum, the discounted negative wealth moment in the verification argument decays at rate $m^* > 0$.

For recursive utility, put $a = 1 - 1/\psi$ and $b = 1 - \gamma$. The difference-form Epstein–Zin problem with $c > 0$, $bV > 0$ is

$$f_{EZ}(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (112)$$

The arithmetic case requires a limit rather than substitution at $a = 0$. In the numerical case $\gamma = 5$, $\psi = 1.5$, with the same asset and discount parameters, homothetic ratios give

$$\pi^* = 0.3, \quad m^* = \psi\rho + (1 - \psi)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\} = 0.0455. \quad (113)$$

For $c = AX^b/b$, the coefficient is $A = (\rho(m^*)^{a-1})^{b/a}$. Consumption utility is increasing on the declared domain. A negative value when $b < 0$ is required by that domain and the sign error.

To execute fixed-policy evaluation for the unknown coefficient A and separate policy-differentiation updates for bounded consumption and portfolio parameters, the coefficient is represented by $A = \exp(q)$; it and the controls are learned, not set to targets. The analytical ratios are used afterward for validation. Table XXXII compares these homothetic-feature solves from the ordinary neural networks used for homogenous preferences.

Element S.4 derives the policy equations, the positive branch of the evaluation problem, and a verification class with the required transversality. For the displayed optimum, the linearized discount rate is 0.1115, the relevant wealth moment decays at rate 0.066, and their difference is $0.0455 > 0$. A separate finite-horizon architecture imposes $bV > 0$ and an exact terminal sign by a transformed output. It is a sign test, not an additional solution of a finite-horizon Epstein–Zin problem.

β	c_0/w	c_1/w	c_2/w	c_3/w	Eval. residual
0.7	0.2827991	0.3401848	0.4312698	0.5978885	1.78e-15
1.0	0.2163112	0.2651939	0.3467520	0.5099987	1.78e-15

consumption dates and terminal log wealth, $\Delta = 1$, $\rho = 0.04$. Continuation is evaluated without multiplying by β again. Independent bounded one-shot optimizations against these frozen continuation es give no gain exceeding 10^{-9} on the tested wealth levels.

ne role of these calculations is to test recursive value dependence and the evaluation ph. Neither a low residual nor the positivity transform is by itself a proof of recursive-ty existence for every policy. Equally, the fact that an argmax has no analytic nula does not define a fully nonlinear PDE. Semilinearity concerns dependence on highest-order derivatives. Control of diffusion can make the HJB fully nonlinear, reas a difficult drift-control maximization with a fixed diffusion need not do so. The DE literature also contains value representations rather than exclusively Pontryagin ints (Han, Jentzen, and E, 2018); these methods should not be excluded by that nction.

15.2. Sophisticated Temporal Selves

et the state obey $s_{n+1} = F(s_n, c_n, \varepsilon_{n+1})$ and let $\delta = e^{-\rho\Delta}$. A sophisticated current self uates a one-period deviation using $\beta\delta$ on ordinary continuation utility. The equilib- n system is

$$c_n^*(s) \in \arg \max_{c \in \mathcal{C}(s)} \{u(c)\Delta + \beta\delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (114)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))], \quad V_N = G. \quad (115)$$

factor β belongs in the current self's improvement criterion but not in each step of ordinary continuation evaluation. Repeating it in (115) instead defines a different ount recursion. We use the same equations in the main text, supplement, and lementation.

he executed example has four consumption dates, terminal log wealth, no inter- $w_{n+1} = w_n - c_n$, $0 < c_n < w_n$, and $\Delta = 1$. With $V_n(w) = A_n \log w + B_n$ and $A_N = 1$, kward induction gives

$$\frac{c_n}{w_n} = \frac{1}{1 + \beta\delta A_{n+1}}, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log z_n + \delta \{A_{n+1} \log(1 - z_n) + B_{n+1}\}. \quad (116)$$

ct concavity establishes the one-shot optimum. We compare it with independent nded scalar optimization and evaluate the ordinary recursion at four wealth levels. onsumption columns are ratios, not levels. The no-present-bias case $\beta = 1$ recovers onential discounting.

two-consumption-date example with terminal log wealth and $\delta = 1$ makes the nction transparent: the sophisticated first ratio is $1/(1 + 2\beta)$, whereas the erroneous metrically repeated-beta recursion gives $1/(1 + \beta + \beta^2)$. They equal 0.416667 and 6621 at $\beta = 0.7$. The discrepancy is economic, not a change of notation.

layer i , evaluation freezes the policy profile, and improvement maximizes only i 's criterion with rivals' controls detached. A dynamic equilibrium must also account for unilateral changes in the player's entire continuation policy. We therefore compute a one-period Hamiltonian gap from dynamic exploitability,

$$\text{Exploit}_i(s) = \sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}; s) - J_i(\pi_i, \pi_{-i}; s). \quad (117)$$

Capacity-regression has two firms, binary capacities $K_i \in \{0, 1\}$, and demand $D \in \{1, 1.3\}$ with transition matrix $\begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$. Output is $q_i = K_i/3$, and revenue is $\kappa(q_i - q_j)$. Investment $a_i \in [0, 1]$ is the probability of capacity one next period, conditional on which the firms' capacity draws are independent. Its cost is $0.4a_i^2/2$. This is the cost of maintaining capacity; it is not a free reset. The horizon is four dates, discount factor is 0.95, and terminal value is $0.05K_i$.

At every date and all eight states, stage best responses are clipped affine functions of the rival's investment. Their composed contraction is checked, and iteration is started from three initial profiles. After the equilibrium profile is computed, a separate hard best-response program freezes the rival at every date and optimizes the own firm's continuation. The original output compared the final, date-zero best-response value. We now compare the best-response value with the equilibrium value at every step of the backward recursion; the all-date gain is below the stated 10^{-10} tolerance. The saved policies and values cover the complete finite state space.

Capacity-regression makes next capacity independent of current capacity, conditional on the chosen investment. Its early policy consequently has a closed form. For the demand transition matrix, the first three dates satisfy

$$a_n(D) = \frac{\delta\{\mathbb{E}[D' | D]/3 - 1/9\}}{\kappa + \delta/9}, \quad a_3 = \frac{0.05\delta}{\kappa}. \quad (118)$$

We use this case as a regression test and add economically persistent capital. With $\kappa = 1$ and $s = 0.8$, the new twelve-date transition is

$$\Pr(K_{i,n+1} = 1 | K_{i,n}, a_{i,n}) = sK_{i,n} + (1 - sK_{i,n})a_{i,n}. \quad (119)$$

Capacity-regression creates capacity when none is installed and offsets the risk of losing an installed unit. Current capacity now changes the marginal return to investment. All low-profit, cost, demand-transition, and terminal-value primitives are held fixed. One-period best responses remain clipped affine functions of the rival's action. Implementation enumerates all nine pairs of binding and interior constraints, rather than supposing interiority. The maximum composed best-response slope product is 0.8, below one. Table XXXIV shows dependence on installed capacity, the rival's capacity, demand, and the remaining horizon. A separate full best-response recursion computes the maximum unilateral dynamic gain of 4.441×10^{-16} over all 192 date-player-state combinations. Deliberately replacing one late-date action by a nonoptimal investment yields a large detected gain at a recorded date and state, checking that the diagnostic does not inadvertently reduce to date zero.

Capacity-regression is an independent dynamic-game reference, not a trained high-dimensional neural network. A separate automatic-differentiation test verifies that the own-action gradient is present and the rival gradient is absent. The static Cournot comparison,

K_1	K_2	$a_{1,0}(D=1)$	$a_{1,0}(D=1.3)$	$a_{1,10}(D=1)$	$a_{1,11}(D=1)$
0	0	0.700239	0.816869	0.521484	0.118750
0	1	0.665094	0.812634	0.442261	0.118750
1	0	0.141975	0.163600	0.108478	0.023750
1	1	0.133116	0.162538	0.088661	0.023750

elve investment dates and capacity survival $s = 0.8$. Capacity-one probability is $sK_i + (1 - sK_i)a_i$. Investment therefore depends on installed capacity and on continuation opportunities. The largest independently computed full unilateral deviation gain over all 192 date–player–state combinations is 41×10^{-16} .

n Nash quantities $1/3$ and joint-profit quantities $1/4$, remains a diagnostic for distinction between equilibrium and collusion, not a substitute for the dynamic check. Supplement S.5 sets out the equilibrium operator, resource normalization for per markets, and a conditional small-player approximation bound. Symmetry of normalization is not used as an equilibrium certificate.

he portfolio and recursive-utility benchmarks build on [Merton \(1971\)](#), [Epstein Zin \(1989\)](#), and [Duffie and Epstein \(1992\)](#). The normalization and admissible comparison class used here are specified in Supplement S.4.

16. CONCLUSION

neural Bellman Operators separate proposal, evaluation, improvement, and certification so that a computational policy can be used in an economic counterfactual. The substantive question is not whether a method contains a neural network, but whether representation and its error account support the change in primitives being studied. For rewards with fixed transitions, the policy-feature and optimistic-support geometry is established. We retain its complete finite-model implementation, action-signatures, and repaired policies, while identifying precisely the part shared with scalar optimistic linear support. For changing transition laws, an endpoint chord alone is invalid. The cross-operator correction in Theorem 13, feasible Bernstein policy values, statewise switching provide a continuum welfare guarantee that incorporates missing effect. Approximate neural endpoints can enter through certified residual resolutions rather than assumed exact evaluation.

the stopped preference economy, operating benefits and liquidation levels reduce contract contrast. A risk-specific duration–effort ratio then characterizes regular movements of an indifference locus, and a matched restriction on deliberate preference adjustment identifies the option-value contribution to the ranking of risky positions. Primitive consumption-exposure condition supplies a positive risk-specific option and over a probability–cost family. In the unchanged finite stopped economy, tensor Bernstein upper and lower values certify opposite risk rankings throughout a joint track–transition region. This connects the reusable transition representation to the economic decision itself. The numerical threshold brackets retain their separate direct–indirect interpretation; uniqueness and a diffusion interpretation are not inferred from a direct regional sign certificate.

he matched count comparison determines what the transition correction buys. It is not the sharpest upper oracle: localization with the same policy bank favors count

and that allows policy kinks. The common-bank frontier and the autonomous experiment separately report the price of precision and the demand for new solutions. Lower-policy storage and exhaustive finite-action work remain in that category.

The resource comparison establishes the importance of reusable structure on both sides of a computational comparison. A nonlinear convex critic improves on the fitted value function continuation in the recorded experiments, but an exact reusable parametric program is faster than learned deployment for the tested short quadratic problems. Neither that fact nor the mesh-only policy-bank ablation invalidates the common-bank certificate. They determine what can be attributed to learning and what should be attributed to policy evaluation, structural optimization, and parameter reuse. The new economic analysis makes the decision horizon and cross-preference cardinality explicit. A first-risk choice concerns a positive-duration commitment, not a choice changed at a single instant. The primitive option result survives a nontrivial tilt family, while full-model counterfactuals show that its dynamic magnitude and effect on risk choice depend on preference-state payoffs, liquidation, duration, and variance. Wealth-lottery perturbations move the original indifference boundary; failures remain alongside an additional common-region certificate. These are not distinctions between economic propositions, not substitutes for the target-control needed to transfer a sign to the diffusion benchmark.

The original finite decision target, streaming chord and count implementations of the same risk comparison with different work. The experiment charges common action and validation costs, retains a mesh-only lower-policy comparator, and separates kernel, policy, and coefficient storage separately. It therefore links compression of economic accuracy requirement without converting a component saving into a cost about neural necessity or total process memory.

Participation and enforcement provide a further economic interpretation. A separately financed mandate is acceptable only if its signing grant covers the agent's outside cost and capacity cost, and if the principal values the delivered operating service sufficiently. At the original center, deliberate adjustment reduces the minimum grant by 0.03525 utility-numeraire units. A statewide enforcement frontier shows that adjustment weakly substitutes for guarantee capacity. On the original parameter rectangle, a charge between 0.85 and 0.90 implements the original opposite-position result when surrender is available after the first interval. Free surrender eliminates the option at the inspected initial contracts, while an intermediate charge reverses the adverse outcomes remain part of the same contracting analysis.

Actual financial choices bind both the long and short permissions. An exact risky-share breakpoint calculation and the reoptimized benefit frontiers show that modest permission changes move both regime boundaries past the original central point. The permission shadow values and discounted surrender exposures identify the regions behind that movement. They explain why a preference-adjustment permits participation value, and its effect on the chosen position class are different from the economic claims. The error account likewise distinguishes evaluation of the stored result from a real-arithmetic constructor or another economic target.

The integrated dynamic result establishes when preference adjustment changes the optimal choice in a contract with an economically active counterfactual surrender option. On the certified joint domain, downward adjustment is dominated at every reachable live state, while the adjusted and unadjusted regimes strictly prefer

tion and the relative adjustment value. A low-preference alternative initial state at a different discharge incentive and substantial downward-adjustment value; the of does not erase that distinction. The two-stage primitive theorem remains an dependent mechanism, not an explanation imposed on the stochastic economy.

ricing both enforcement technologies turns conditional implementability into an institutional comparison. The minimum grant and direct term cost select a full compulsory term, an intermediate term, or a short financial implementation under explicitly ed primitives. When a full term is free, its dominance is recovered. The active-ender region and the inactive implementation family answer different questions ut the same continuation economy; neither is presented as a substitute for the other. joint certificate therefore links numerical precision to the direction of adjustment, ranking of positions, and the scope of an operating contract.

he continuous-time formulation, finite transition targets, neural proposals, recur-ferences, temporal selves, and persistent dynamic competition remain distinct s of the framework. Keeping their assumptions and evidence explicit permits new representations and counterfactuals without changing the economic meaning of the orted welfare guarantee.

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The new Section S.15 proves the all-response service enclosure, joint procurement selection, signed opportunity-exposure identities, and the witness-reconstruction arguments. This supplement gives complete proofs for Neural Bellman Operators and its economic applications. The new dynamic section proves the signed transport criterion on feasible reachable states, decomposes the adjustment option into current and future continuation gains, and establishes the joint law–benefit–fee–permission certificate. It distinguishes active counterfactual surrender from implementation with an inactive stop option and derives the minimum procurement cost when both compulsory terms and financial capacity are priced. The replication account records coefficient arrays, lower-policy identities, support sets, numerical error allowances, and executed adverse interventions. All earlier operator, approximation, preference, transition-law, regional certification, participation, permission, recursive-utility, equilibrium, and computational proofs remain included. Statements concerning stored-array evaluation are separate from transfer to a different constructor or diffusion target.

S.1. OPERATOR PROOFS AND QUANTITATIVE APPROXIMATION

S.1.1. *Exact Policy Improvement*

We use the assumptions of the main paper’s exact separated-operator proposition. In particular, $I(p)(t, s)$ is a selector maximizing $\mathcal{H}^a[E(p)](t, s)$, not a stationary point of a parameterized actor objective. Continuity of E is in the norm controlling the value, time derivative, first spatial derivatives, second spatial derivatives, and the stated boundary values. The admissible compact policy class is closed under I . These are assumptions in an exact boundary-value problem, not universal properties of neural policy classes.

PROOF OF EXACT SEPARATED-OPERATOR CONSISTENCY: Write $V_n = E(p_n)$ and $V_{n,t} = I(p_n)$. The evaluation equation gives $V_{n,t} + \mathcal{H}^{p_n}[V_n] = 0$. Pointwise maximization lies

$$V_{n,t} + \mathcal{H}^{p_{n+1}}[V_n] \geq 0.$$

assumed comparison and verification inequalities, with the same boundary operator for both policies, imply $V_{n+1} \geq V_n$. Precompactness of the evaluation image implies boundedness and hence a pointwise limit $\bar{V}$. Every subsequential limit in the declared strong topology has this same value component. Its derivative components are the derivatives of that value, because uniform convergence of the functions and their derivatives preserves differentiation. Consequently all strong accumulation points coincide and $V_n \rightarrow \bar{V}$ strongly.

Let $p_{n_j} \rightarrow p$ be a policy subsequence, which exists by compactness. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity of improvement at p and continuity of evaluation

$$E(I(p)) = \lim_j E(I(p_{n_j})) = \lim_j V_{n_j+1} = \bar{V}.$$

$$0 = \bar{V}_t + \mathcal{H}^{I(p)}[\bar{V}] = \bar{V}_t + \sup_a \mathcal{H}^a[\bar{V}].$$

oundary traces also pass to the limit. Comparison identifies $\bar{V}$ with the optimal and verification identifies p as optimal because $E(p) = \bar{V}$. This argument does require $p = I(p)$ or convergence of the entire policy sequence. In the presence of maximizing actions, different optimal policies may share the same limiting value. *Q.E.D.*

This proof is an application of exact policy-improvement reasoning; it is not a claim of optimality over [Jacka and Mijatović \(2015\)](#). The assumption of a compact strongly regular transition image can fail at corners or under degenerate diffusions. The stopped process is therefore evaluated by its specified positive transition approximation which has asserted to satisfy every classical hypothesis globally.

S.1.2. Finite-Horizon Value-Loss Bound

For simplicity, put $\Gamma_{n,n} = 1$ and $\Gamma_{n,j} = \prod_{\ell=n}^{j-1} \gamma_\ell$. Both $\mathcal{T}_n^a$ and its policy version are monotone and γ_n -Lipschitz in the sup norm. The implemented operator is $\tilde{\mathcal{T}}_n^a$. The three additive error bounds at $\hat{v}_{n+1}$ are

$$\begin{aligned} \epsilon_n &\geq \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \delta_n &\geq \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\hat{p}_n} \hat{v}_{n+1} \right\|_\infty, \\ \kappa_n &\geq \sup_a \left\| (\tilde{\mathcal{T}}_n^a - \mathcal{T}_n^a) \hat{v}_{n+1} \right\|_\infty. \end{aligned}$$

$$= \sup_a \tilde{\mathcal{T}}_n^a, \text{ and } \|\hat{v}_N - G\|_\infty \leq \eta.$$

PROOF OF THE MAIN VALUE-LOSS THEOREM: Let $D_n = \left\| \hat{v}_n - J_n^{\hat{p}} \right\|_\infty$. Inserting the implemented and target fixed-policy backups gives

$$D_n \leq \epsilon_n + \kappa_n + \gamma_n D_{n+1}.$$

$= \|\hat{v}_n - V_n^*\|_\infty$. The supremum inequality $|\sup_a f_a - \sup_a g_a| \leq \sup_a |f_a - g_a|$ and the negative improvement gap give

$$R_n \leq \epsilon_n + \delta_n + \kappa_n + \gamma_n R_{n+1}.$$

By induction, with $D_N, R_N \leq \eta$, proves

$$\begin{aligned} D_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta, \\ R_n &\leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \delta_j + \kappa_j) + \Gamma_{n,N} \eta. \end{aligned}$$

and with coefficients $2\epsilon_j + \delta_j + 2\kappa_j$ and terminal coefficient 2η .

Q.E.D.

the theorem is finite-horizon. It permits a recursive operator with $\gamma_n > 1$ but then adds amplification explicitly through $\Gamma_{n,j}$. It does not imply contraction for an arbitrary Epstein–Zin aggregator. Boundary killing can be included in the transition kernel by treating the boundary payoff as a known reward component; only continuation values contribute to γ_n .

S.1.3. Differential Residual and Boundary Error

Let v be a $C^{1,2}$ candidate for the additive, stopped, discounted problem. Define $\hat{v} = v_t + \ell^a + \mathcal{L}^a v - \rho v$. Suppose $|R^{\hat{p}}| \leq \epsilon(t)$, $\sup_a R^a - R^{\hat{p}} \leq \delta(t)$, and the payoff discrepancy at terminal time or exit is at most η . Assume the localized Itô formula passes expectations by uniform integrability.

For every admissible p , stopped Itô integration gives

$$J^p(t, s) - v(t, s) = \mathbb{E}_{t,s}^p \left[\int_t^\tau e^{-\rho(r-t)} R^{p_r}(r, S_r) dr + e^{-\rho(\tau-t)} (G - v)(\tau, S_\tau) \right].$$

An upper bound for an arbitrary policy is $\eta + \int_t^T e^{-\rho(r-t)} (\epsilon(r) + \delta(r)) dr$. The absolute discrepancy for $\hat{p}$ is at most $\eta + \int_t^T e^{-\rho(r-t)} \epsilon(r) dr$. Taking a supremum over p and adding gives the differential corollary. The argument is conditional on uniform residual and boundary bounds. Replacing these bounds by sampled root mean squared losses is not a valid proof step.

S.1.4. Feasible Candidates, Convex Gaps, and Reference Bounds

If $Q(s, a)$ is L_a -Lipschitz in a and a feasible set $\mathcal{C}(s)$ is an h_a -net of the admissible actions, then a maximizer over $\mathcal{C}(s)$ is within $L_a h_a$ of the full supremum. Adding a marginal proposal to $\mathcal{C}(s)$ cannot reduce its achieved backup value. Neither statement bounds the evaluation error of the critic that generated Q .

For differentiable convex F on a compact convex set A , its tangent inequality implies

$$F(a) - \inf_{b \in A} F(b) \leq \nabla F(a)^\top a - \min_{b \in A} \nabla F(a)^\top b =: g_F(a).$$

For $A = \{a \geq 0 : \sum_i a_i \leq B\}$, the minimum is $B \min_i (0, \min_i \partial_i F(a))$. The same formula applies to a scenario tree with separate feasible actions at each information node: use the node-specific linear minimization gaps, using the gradient of the complete probability-weighted objective. This gradient must include all descendant costs.

If a feasible reference has objective C_R and gap G_R , then $C_R - G_R \leq C^* \leq C_R$. Any other feasible policy, evaluated on the same tree, therefore satisfies

$$C_\pi - C_R \leq C_\pi - C^* \leq C_\pi - C_R + G_R.$$

The program evaluates both objectives by all 64 terminal branches. It does not insert a neural critic as a substitute for policy cost. The reported interval is based on ordinary double-precision evaluation of these inequalities; it is not an interval-arithmetic enclosure of rounding error.

is μ -strongly concave and the approximate continuation is within e of the one, an action whose approximate improvement gap is at most δ obeys

$$\|\hat{a} - a^*\|^2 \leq \frac{2}{\mu}(\delta + 2\gamma e).$$

the value of each action changes by at most γe , and strong concavity bounds the true suboptimality below by $\mu \|\hat{a} - a^*\|^2 / 2$. This is not applied to a noncon-
 DU maximand. In the convex resource problem the immediate quadratic cost
 s strong convexity $0.2/d$ for minimization. Its deterioration with dimension is
 d in the conditioning discussion rather than hidden in a dimension-free speed

S.2. COUNTEREXAMPLES, TRACE ESTIMATION, AND DIFFERENTIATION

S.2.1. *Why a Joint Penalty Does Not Define the Economic Solution*

Consider $-V' + V = 0$ on $[0, 1]$, $V(1) = 1$. Its solution is $V^*(t) = e^{t-1}$. The historical
 site objective was of the form

$$L(V) = \int_0^1 \{V(t) + [-V'(t) + V(t)]^2\} dt.$$

A possible perturbation $V_\varepsilon(t) = V^*(t) - \varepsilon(1 - t)$ preserves the terminal condition but
 dual $-\varepsilon(2 - t)$. Consequently

$$L(V_\varepsilon) - L(V^*) = -\frac{\varepsilon}{2} + \frac{7\varepsilon^2}{3} < 0 \quad \text{for } 0 < \varepsilon < \frac{3}{14}.$$

The exact economic solution is not even a local minimizer of this penalty. Separate
 blocks, objectives, and verification conditions are substantive changes, not a
 of notation for that joint loss.

S.2.2. *A Stable Actor Stationary Point Need Not Be Optimal*

on $[-2, 2]$, let $H(a) = -(a^2 - 1)^2 + 0.2a$. Its stationary equation is $4a^3 - 4a -$
 . There are three real roots. At the negative local maximizer, approximately
 9943532, $H''(a) = 4 - 12a^2 < 0$, so gradient ascent has a locally asymptotically
 equilibrium. The positive maximizer is approximately 1.0241203002 and has
 payoff by approximately 0.3998745872. Both are isolated. The example embeds
 finite-horizon control problem with flow $H(a)$ and exact evaluation $(T - t)H(a)$.
 esequently, even exact fast evaluation, diminishing separated step sizes, bounded
 , and zero noise do not turn isolated actor stationary points into policy-
 cement fixed points. Under a suitable stochastic-approximation theorem the
 d conclusion concerns the limiting invariant or stationary set. Global optimality
 nally requires an argument excluding suboptimal attractors. Convex improve-
 n the resource example supplies such an argument for each surrogate action
 n; it does not automatically eliminate critic approximation error. Constant-step
 as no convergence theorem in this paper, and the executed neural runs use the
 ly recorded L-BFGS and constrained least-squares routines instead.

for the exit-time problem $dX_t = a_t dt$, $|a_t| \leq 1$, with flow -1 and zero payoff on $\{0, 1\}$, the value is $V(x) = |x| - 1$. The viscosity equation is $1 - |V'| = 0$. The smooth solutions

$$W_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2}$$

satisfy the boundary data, while their squared strong residual under the uniform distribution tends to zero:

$$\frac{1}{2} \int_{-1}^1 \left(1 - \frac{|x|}{\sqrt{x^2 + \varepsilon^2}}\right)^2 dx \sim \left(2 - \frac{\pi}{2}\right) \varepsilon.$$

Nevertheless $W_\varepsilon \rightarrow 1 - |x|$, which has the wrong sign and is not the desired viscosity solution. The asymptotic expression follows by $x = \varepsilon y$ and integration of the resulting integrable function on $[0, \infty)$. The code also evaluates the finite- ε integral by independent adaptive quadrature.

Monotonicity, stability, and consistency with an appropriate comparison principle are central to viscosity approximation; see [Barles and Souganidis \(1991\)](#). Smooth network approximation of a function and its derivatives, as in [Hornik, Stinchcombe, and White \(1990\)](#), concerns a smooth target and does not approximate a derivative at a kink or face a viscosity-selection argument. Historical kink and residual arrays remain available as archival material, not as evidence that an unregularized strong-residual selects the correct solution.

S.2.4. Probe Budget and the Squared-Loss Estimand

Let $A = B^\top \nabla^2 v B$ be symmetric, z_k independent standard Gaussian vectors, and $q_k = z_k^\top A z_k$. Gaussian fourth moments give $\mathbb{E} q_k = \text{tr } A$ and $\text{Var}(q_k) = 2 \|A\|_F^2$. For deterministic a and $\bar{q}_K = K^{-1} \sum_k q_k$,

$$\mathbb{E} \left(a - \frac{1}{2} \bar{q}_K \right)^2 = \left(a - \frac{1}{2} \text{tr } A \right)^2 + \frac{\|A\|_F^2}{2K}.$$

In contrast $K^{-1} \sum_k (a - q_k/2)^2$ has excess expectation $\|A\|_F^2/2$, independent of K . For regression matrix $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\|A\|_F^2 = 15$, $a = 1.25$, and the exact-trace loss is 1.5625. The estimation excess is $7.5/K$.

An unbiased alternative for $K \geq 2$ is the off-diagonal statistic

$$U_K = \frac{(\sum_k r_k)^2 - \sum_k r_k^2}{K(K-1)}, \quad r_k = a - q_k/2.$$

Independence implies $\mathbb{E} U_K = (\mathbb{E} r_k)^2$. Under an integrable derivative envelope, differentiating its expectation yields the gradient of the exact squared residual. The realized statistic can be negative; unbiasedness neither makes it a nonnegative sample loss nor guarantees optimizer stability. If the action or parameters are selected using the probe probes, the fixed-parameter identity is not an adaptive generalization theorem. Independent held-out probes are needed to interpret such selection.

K	Correct mean	Population	MC s.e.	Old mean	Debiased mean
1	8.91810	9.06250	0.22234	8.91810	–
4	3.47199	3.43750	0.04758	9.17095	1.57233
16	2.03146	2.03125	0.01492	9.06287	1.56270
64	1.68023	1.67969	0.00640	9.06132	1.56307

dependent Gaussian batches per row; the exact-trace squared residual is 1.5625. Correct mean probes before squaring; old mean averages individual squared residuals. The debiased off-diagonal requires at least two probes. Its standard errors and both variance estimands are retained in the word.

row averages 20,000 independent batches. The Monte Carlo standard error is to the repeated-batch mean, not to a single quadratic form or its within-batch error. The numerical test checks the known expectation within six Monte Carlo standard errors and preserves the wrong mean-of-squares statistic as a negative control. It does not assign value or policy errors to this algebraic experiment.

S.2.5. Executable Computational-Graph Tests

test suite forms a quadratic critic and a control-dependent diffusion loading, computes its automatic Hessian with the covariance, and checks both parameter blocks. Actor update, critic coefficients receive no gradient but the action-dependent critic does. In the critic update, the policy output is detached while the critic coefficients receive gradients. A finite-transition actor derivative is also checked against differences in all three controls. Freezing critic parameters does not detach the state input: the continuation gradient with respect to the action-induced state is computed.

A separate test compares the resource critic’s explicit gradient with PyTorch automatic differentiation. The convex squared-ReLU feature architecture is a transparent special case of the input-convex approach of [Amos, Xu, and Kolter \(2017\)](#). Exact directional derivatives or Hessian-vector products, following [Pearlmutter \(1994\)](#), can avoid computing a dense Hessian when the diffusion has low rank. The complete costs depend on network size, diffusion multiplication, sample count, probe count, and iteration iterations.

S.3. THE STOPPED PREFERENCE-FORMATION ECONOMY

S.3.1. Admissibility and the Boundary Contract

anterior state is $(u, X) \in (1.2, 2.8) \times (0.5, 2)$, with the controls and coefficients as in the main paper. The process is stopped at the first exit or $T = 1$, whichever comes first. It receives $G(u, X) = -0.02(u - 2)^2 + 0.1 \log X$ at that time. A start on the boundary is settled immediately. This is a finite-horizon contract on an open operating domain; it is not a claim that the unstopped diffusion remains in a rectangle.

With bounded progressively measurable controls, the wealth SDE with coefficients in wealth and the additive preference SDE have a well-defined solution up to exit. At exit, wealth and consumption are bounded away from zero. The flow, terminal wealth, and quadratic adjustment cost are bounded on the stopped domain. Thus the

ness.

the original state-constraint interpretation cannot hold with these coefficients. Preference volatility is 0.05 at both endpoints, and the maximal wealth drift at the lower boundary is $(0.02 + 0.8 \times 0.06)0.5 - 0.05 = -0.016$. At zero portfolio volatility it is negative. Neither fact is repaired by projecting an Euler endpoint back onto the angle. The stopped contract retains all interior coefficients and makes the change in boundary economics explicit.

S.3.2. A Boundary-Compatible Payoff

The choice of G has a useful analytic check. Its derivatives are $G_u = -0.04(u - 2)$, $G_X = -0.04$, $G_X = 0.1/X$, $G_{XX} = -0.1/X^2$, and $G_{uX} = 0$. For every admissible action every $k \geq 0$,

$$\begin{aligned} \ell^a + \mathcal{L}^a G - \rho G &\leq \max_{u \in [1.2, 2.8]} \frac{0.8^{1-u}}{1-u} + 0.0064 + 0.0043 - 0.00005 + 0.0032846 \\ &< -0.8162. \end{aligned}$$

The consumption bound follows because utility is increasing in consumption. Maximization over u at $c = 0.8$ gives the upper endpoint; it can also be checked by the explicit derivative. The 0.0043 term bounds $0.1[0.02 + 0.06\pi - c/X]$ by using $\pi \leq 0.8$, $\pi \geq 0.025$. The omitted portfolio diffusion term is nonpositive, and so are adjustment costs. The last term bounds $-\rho G$ uniformly.

The stopped Itô integration therefore implies $J^p \leq G$ for every admissible policy. To obtain boundary continuity from below, fix for example a constant admissible nonzero portfolio share. Its two-dimensional covariance is uniformly positive definite on the closed rectangle because $X \geq 0.5$ and $|\varrho| < 1$. The boundary is regular for this fixed diffusion, and its expected exit time tends to zero as the start tends to a boundary point. Bounded rewards and continuity of G imply that this policy's stopped payoff tends to G there. Since its payoff is bounded above by the optimal value and the latter is bounded above by G , the value has the declared boundary limit. This argument establishes boundary compatibility; it does not assert global classical differentiability at corners.

S.3.3. Transition, Covariance, and First Exit

For step h , form four equiprobable sign pairs $(z_1, z_2) \in \{-1, 1\}^2$. Let

$$\begin{aligned} d_u &= \theta h + \sigma_u \sqrt{h} z_1, \\ d_X &= [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} [\varrho z_1 + \sqrt{1 - \varrho^2} z_2]. \end{aligned}$$

From an interior state s , α is the first fraction in $[0, 1]$ at which the line $s + \alpha d$ meets a face, or one if it stays inside. An exit branch has payoff

$$\frac{1 - e^{-\rho \alpha h}}{\rho} \ell(s, a) + e^{-\rho \alpha h} G(s + \alpha d).$$

At a boundary start the backup is exactly $G(s)$. In particular there is no nonzero contribution from a boundary node. The reference evaluator uses nonnegative bilinear transition weights that sum to one on surviving branches. The signs reproduce conditional first moments and the covariance matrix of the diffusion. Polynomial tests on $1, u, X, u^2, X^2, uX$ distinguish covariance terms and drift-product terms of order h . At the test state and action the covariance is

$$\begin{pmatrix} 0.0025 & -0.002 \\ -0.002 & 0.0256 \end{pmatrix}.$$

Root-mean-square generator discrepancy decreases by a factor of ten when h decreases from 10^{-3} and again to 10^{-4} . Preference controls of -0.2 and 0.2 give different mean values even when both displacements are smaller than a spatial grid cell. No nearest-neighbor map erases the control.

Line-exit rule is an approximation to diffusion killing, not exact Brownian bridge simulation. Before-correction overshoots, branch-exit frequencies, occupied exit probabilities and post-exit payoffs are distinct recorded objects. A reflection-regulator field is applicable and is represented by null with that explanation. For the joint state–time dependent, the squared largest mesh width divided by h decreases as $0.04, 0.02, 0.01$; the control refinements are also retained. These observations support consistency of the statistics but do not supply a numerical bound for every κ_n in the main theorem.

S.3.4. Economic Comparative Statics

For each admissible policy let $B(p)$ denote cost-free benefit. Define discounted cumulative effort by $C(p) = \mathbb{E} \int_0^{\tau_p} e^{-\rho t} \theta_t^2 / 2 dt$. The stopping time is policy dependent and directly dependent on k for a fixed policy. Thus $V(k) = \sup_p [B(p) - kC(p)]$ is concave and convex. For optimizers p_1, p_2 at $k_1 < k_2$, revealed optimality gives

$$\begin{aligned} B(p_1) - k_1 C(p_1) &\geq B(p_2) - k_1 C(p_2), \\ B(p_2) - k_2 C(p_2) &\geq B(p_1) - k_2 C(p_1). \end{aligned}$$

The first implies $(k_2 - k_1)[C(p_2) - C(p_1)] \leq 0$. Also $-C(p)$ is a subgradient of V at any p which p is optimal, and $V'(k) = -C(p)$ wherever V is differentiable. For policies with value losses at most e_1, e_2 , the same inequalities with these errors imply

$$C(p_2) - C(p_1) \leq \frac{e_1 + e_2}{k_2 - k_1}.$$

The result applies to the continuous stopped model and separately to each exactly computed positive finite approximation. A finite-grid comparison cannot set $e_i = 0$ for the continuous problem merely because its discrete maximization was exhaustive. Reported effort is computed both backward and by forwarding the induced state probabilities. The latter multiplies live branch mass by bilinear weights and accumulates expected cost through the line-exit fraction. This is an occupation calculation for the transition-defined finite Markov chain. It provides an economically interpretable average rather than an unweighted mean of control magnitudes. A reduction of this state does not imply a pointwise reduction of θ , consumption, or portfolio share. The result enables a comparative static identify k from observations without a specified data set and an injective moment map.

S.4.1. Stationary Merton Benchmark

Write $b = 1 - \gamma$, $V(X) = AX^b/b$, $c = mX$, and

$$g(m, \pi) = r + (\mu - r)\pi - m + \frac{1}{2}(b - 1)\sigma^2\pi^2.$$

exponential utility discount ρ , the fixed-policy stationary equation is

$$A = \frac{m^b}{\rho - bg(m, \pi)}$$

the positive branch $\rho - bg > 0$. Optimization gives $\pi^* = (\mu - r)/(\gamma\sigma^2)$ and

$$m^* = \frac{\rho - (1 - \gamma)[r + (\mu - r)^2/(2\gamma\sigma^2)]}{\gamma}.$$

$(\rho, \gamma, \mu, r, \sigma) = (0.04, 2, 0.08, 0.02, 0.2)$ the targets are $(m^*, \pi^*) = (0.04125, 0.75)$. Wealth remains positive under this proportional policy. Its discounted terminal candidate expectation decays at rate $\rho - bg(m^*, \pi^*) = m^* > 0$, verifying its own transversal-condition. The usual verification comparison is made over policies for which the expected local martingale and discounted terminal inequalities pass to the limit. The executed feature solver treats $q = \log A$, the consumption ratio, and the portfolio ratio as unknowns. It alternates the fixed-policy scalar evaluation root with an automatic-differentiation actor optimization, without feeding the analytical targets into updates. Positivity is imposed by $A = e^q$. This is a learned homothetic feature model; ratios alone would not be a neural-function benchmark on an unrestricted state space. The no-short regression at $\mu = 0.01$ instead has unconstrained share -0.125 and constrained share zero at every positive wealth. It is not a wealth-dependent switching problem.

S.4.2. Epstein–Zin Driver and the Admissible Comparison Class

Use $a = 1 - 1/\psi$, $b = 1 - \gamma$, with $a, b \neq 0$, $c > 0$ and $bV > 0$. The driver is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}.$$

Log limits must be taken separately when a denominator vanishes. For a homothetic candidate $V = AX^b/b$, dividing the stationary HJB by AX^b gives

$$\frac{\rho}{a} \{m^a A^{-a/b} - 1\} + g(m, \pi) = 0.$$

As a fixed proportional policy has a unique positive homothetic coefficient whenever $ag/\rho > 0$. Its consumption first-order condition is $\rho m^{a-1} A^{-a/b} = 1$, and the portfolio first-order condition gives the same Merton share with risk aversion γ . Eliminating A yields

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2},$$

$$m = \psi\rho + (1 - \psi) \left[r + \frac{1}{2\gamma\sigma^2} \right],$$

$$A^* = (\rho(m^*)^{a-1})^{b/a}.$$

$(\gamma, \psi, \mu, r, \sigma) = (0.04, 5, 1.5, 0.08, 0.02, 0.2)$ this gives $m^* = 0.0455$ and $\pi^* = 0.3$. The sized Hamiltonian is strictly concave in the consumption ratio and portfolio in the specified positive domain. The selected ratios lie inside the implemented bounds.

precision about infinite horizon, the verification class consists of policies for the policy-wise recursive equation exists on the branch $bY > 0$, its localized comparison argument is integrable, and the comparison-weighted terminal remainder is. On compact subsets of $c > 0, bY > 0$ the driver is locally Lipschitz, but this does not imply a global Lipschitz constant on the entire half-line. One may first stop at compact localization sets, apply the mean-value linearization in the value argument, then impose uniform integrability and the weighted terminal limit to remove the approximation. With $\Gamma_t = \exp(\int_0^t f_V(c_s, \zeta_s) ds)$ for the intermediate value ζ_s , the required remainder is of the form $\mathbb{E}[\Gamma_T |V(X_T) - Y_T^p|] \rightarrow 0$. The candidate HJB inequality and comparison yield $Y_0^p \leq V(X_0)$, with equality for the maximizing policy. The selected homothetic policy itself, $f_V = -0.1115$, while the growth exponent of $b\gamma = 0.066$. The candidate comparison-weighted value therefore decays at rate $-0.066 = 0.0455 > 0$. The numerical record evaluates the fixed-policy equation to get errors, and the analytic calculation verifies this decay. This specifies a genuine recursive benchmark with an explicit admissible verification class; it does not assert that all bounded controls satisfy the terminal condition automatically. The sign-enforcing architecture is actually executed: $V = e^q X^b / b$, hence $bV = e^q X^b > 0$. The separate finite-horizon architecture test uses

$$V_{\xi}(t, X) = \frac{1}{b} \exp\{\log[bG(X)] + (T - t)N_{\xi}(t, X)\}$$

> 0 . It verifies the terminal identity and sign on test points. It is a domain test, evidence that an additional finite-horizon recursive problem was trained. These solutions follow the policy-wise recursive-utility formulation of [Duffie and Epstein](#) rather than replacing recursion by a pointwise evaluation at $c = 1, V = -1$.

S.4.3. A Structured Stochastic Dynamic Control Check

$\in \{4, 8, 16\}$, the additional six-period LQR problem uses

$$A = 0.82I + 0.07(J + J^\top), \quad B = 0.2I, \quad Q = I + 0.31\mathbf{1}^\top/d, \quad R = 0.15I,$$

Gaussian shock loading $\Sigma = 0.08(I + 0.31\mathbf{1}^\top/d)$, with terminal matrix Q . Here J is on its upper adjacent diagonal. The linear neural actor and quadratic feature are trained by policy evaluation and improvement. The independent reference is the finite-horizon Riccati recursion, including the noise contribution to the value estimate. A second recursion evaluates the learned fixed policy rather than using its critic as a performance score.

The experiment has cross-state dynamics and non-diagonal stochastic covariance and binding control constraints. It is retained as a structured dynamic regression. The main shared-budget resource experiment supplies the separately executed baseline case; neither is relabeled as an adaptive sparse-grid comparison.

d	Actor coeff. error	Cost excess max.	Neural sec.	Riccati sec.
4	4.25e-06	3.03e-11	0.0571	0.00037
8	2.25e-06	2.14e-11	0.0724	0.00028
16	4.30e-06	3.84e-11	0.1330	0.00033

periods, linear actors and quadratic feature critics, independent Riccati reference, and separate frozen- y evaluation. Cost excess is normalized per dimension in the implementation. This structured untrained regression is not the main binding-resource experiment and supplies no generic neural speedantage.

S.5. TEMPORAL SELVES AND DYNAMIC COMPETITION

S.5.1. *Sophisticated Consumption–Savings*

Let one period have length Δ , $\delta = e^{-\rho\Delta}$, and terminal payoff $G(w) = \log w$. The acting self chooses $c \in (0, w)$ to maximize $\Delta \log c + \beta \delta V_{n+1}(w - c)$, whereas ordinary continuation evaluates the chosen policy as

$$V_n(w) = \Delta \log c_n(w) + \delta V_{n+1}(w - c_n(w)).$$

$V_{n+1} = A_{n+1} \log w + B_{n+1}$, strict concavity gives

$$z_n := \frac{c_n(w)}{w} = \frac{\Delta}{\Delta + \beta \delta A_{n+1}},$$

$$A_n = \Delta + \delta A_{n+1},$$

$$B_n = \Delta \log z_n + \delta A_{n+1} \log(1 - z_n) + \delta B_{n+1}.$$

Starting with $A_N = 1, B_N = 0$ yields an exact sophisticated equilibrium. The coefficient is independent of β , but the policy and the constant are not. The reference test uses $\rho = 0.04$, four consumption dates, $\rho = 0.04$, and $\beta \in \{0.7, 1\}$. At four wealth levels it separately solves the bounded one-shot consumption problem and checks the evaluation identity with the frozen continuation coefficients.

For two consumption dates, $\delta = 1$, the correct first-date ratio is $1/(1 + 2\beta)$. Reusing acting self's maximand as its stored continuation would instead yield $1/(1 + \beta + \beta^2)$. For $\beta = 0.7$ the difference is 0.4166667 versus 0.4566210. The $\beta = 1$ check alone cannot reveal this error. These checks implement the written temporal game rather than an ordinary geometric-discount control problem. They do not identify the finite-state reference as a trained neural solution.

S.5.2. *A Stochastic Capacity Duopoly*

There are four decision dates. Demand is $D \in \{1, 1.3\}$ with probability 0.8 of staying at its current value. Each firm's installed capacity $K_i \in \{0, 1\}$ determines current output $q_i = K_i/3$. Current operating profit is $q_i(D - q_i - q_j)$. Firm i chooses a maintenance renewal intensity $a_i \in [0, 1]$ at cost $0.4a_i^2/2$. Independently across firms conditional on current actions, next-period capacity is one with probability a_i and zero otherwise. The discount factor is 0.95 and terminal value is $0.05K_i$. There are eight Markov states and four dates, so complete state enumeration is possible.

response is therefore the projection onto $[0, 1]$ of its discounted marginal contribution gain divided by 0.4. The joint map is affine before projection, with slope defined by the cross difference of the four next-capacity continuation values. The algorithm checks the product of the two own best-response slopes, iterates from multiple profiles, and compares their limits. This finite example has a contraction bound of 0.4. It does not establish uniqueness in arbitrary dynamic games.

Independent backward best-response recursion freezes the rival's entire state-dependent policy at every subsequent date and allows the tested firm to deviate at all dates. The difference from the profile value is dynamic exploitability. This is stronger than testing against a single current Hamiltonian against an inconsistent continuation. The report computes the maximum over all dates, states, and players, with a numerical tolerance of 10^{-6} . A separate automatic-differentiation test confirms that a player's loss does not differentiate the rival's parameters.

A static special case $q_i(1 - q_i - q_j)$ remains a useful negative control. Symmetric output is $1/3$, whereas joint-profit output is $1/4$. At the latter profile, changing a player's quantity to its own best response is profitable. A symmetric or jointly optimal algorithm is therefore not a substitute for a Nash calculation. The dynamic capacity problem is a finite-state equilibrium computation; no competitive limit or general neural-network convergence is inferred from it.

S.5.3. Market-Size Normalization and a Conditional Small-Player Bound

To obtain the economically relevant many-firm extension without conflating different scales, consider per-firm normalized output y_i and actual output $q_i = y_i/N$, with constant demand $P = a - b\bar{y}$, $\bar{y} = N^{-1} \sum_i y_i$. Multiplying each firm's monetary payoff by N gives normalized flow $y_i P - C(y_i)$ under consistently scaled costs. Its direct strategic derivative is $-by_i/N$. Investment, depreciation, and state transitions must be scaled and retained as well; this current-flow derivative by itself does not establish the existence of dynamic equilibria.

A precise conditional implication is available. Suppose that, against a fixed rival profile, the finite- N player's transition operators and a price-taking comparison operator are monotone and γ_n -Lipschitz, have the same terminal payoff, and differ uniformly by at most η_n/N on all policies and continuation values used in the comparison. Then the N -player optimal for the price-taking problem loses at most

$$\frac{2}{N} \sum_{j=n}^{N-1} \Gamma_{n,j} \eta_j$$

relative to the finite- N best response. To prove it, backward iteration bounds the difference of the two optimal values by $N^{-1} \sum_j \Gamma_{n,j} \eta_j$, and bounds the two evaluations of a fixed policy by the same amount. Their sum gives the result. Uniform Lipschitz bounds of aggregate-state effects is one route to the operator hypothesis, but it must be verified in a specified model. This proposition concerns a unilateral deviation bound, not the selection or convergence of a sequence of equilibria. It is a theoretical extension of the model and is not recorded as an executed large- N experiment.

This section retains the foundational neural experiments and their full checkpoint provenance. The fifth-revision additions and independent rechecks are described in S.8.

S.6.1. Source and Evidence Identities

The authoritative R4 source record consists of `ECTA_R4.tex`, `SUPP_R4.tex`, their explicitly included sections, and `replication/r4/`. The base review is pinned at commit `79a7d84`. The replication manifest records the reachable source commit, full SHA-256 digests of the source files, package versions, and output identities. Source files are committed before their authoritative execution. A later result commit can add generated outputs without changing that source anchor.

Execution status, evidence class, and numerical-tolerance status are separate fields. A completed program is not automatically a solved control problem. The coupled source runs have a specified held-out cost-loss target and independent reference gaps. The NDU comparison reports discrepancies and does not assign a passed uniform evidence certificate. Algebraic checks, domain tests, finite-state equilibria, and structured numerical solvers each have their own evidence class. An unavailable continuous-time error bound or reflection regulator is null with an explanation, not a numerical zero. Scientific digests exclude elapsed time and hardware measurements. File digests still identify complete files, including timing records. Floating-point algorithms, package versions, and BLAS implementations can change last digits across platforms. The source saved weights permit direct policy replay independently of retraining. Timings include neural fitting and full evaluation for the stated initial-state batch; they do not rely on a universal hardware comparison or unmeasured amortization advantage.

S.6.2. Architectures and Optimization

For the stopped preference model the safeguarded neural run uses separate two-hidden-layer tanh actors of width 32 and critics of width 64 at each of eight decision steps. The actor has three bounded output coordinates. The critic writes the value as G times remaining time times a product vanishing on all spatial faces times its network output. The training sample contains 2,048 interior states. L-BFGS caps are 160 actor and 1,000 critic iterations with strong-Wolfe line search; actual function evaluations and iteration counts are recorded when produced by the optimizer. The implementation never trains on the independent grid reference’s values or policies.

The unsafeguarded pilot uses width-32 critics and 1,024 training states with a smaller critic budget. It is retained as a development baseline, not used as a matched causal ablation. The resource experiment supplies a matched ablation: the same fitted actor weights are evaluated with and without pointwise improvement at deployment. It holds training, initial states, shocks, and model primitives fixed.

For resources the convex critic is

$$q(x) + \ell^\top(x - \mathbf{1}) + c_0 + \sum_{j=1}^{6d+102} \alpha_j [\max\{w_j^\top(x - \mathbf{1}) + b_j, 0\}]^2, \quad \alpha_j \geq 0.$$

Directions consist of signed coordinate axes, signed aggregate directions, and 32 seeded Gaussian directions, each at offsets $-0.5, 0, 0.5$. The coefficients are fitted by bounded

d	Seed	NBO upper loss	Proposal only	Reference gap	Binding at 0
4	101	6.45e-04	1.19e-02	8.32e-08	0.812
4	202	4.88e-04	8.14e-03	7.40e-08	0.938
4	303	5.23e-04	6.31e-03	7.02e-08	0.938
8	101	4.29e-04	9.54e-02	9.79e-08	0.938
8	202	4.61e-04	3.28e-02	9.97e-08	1.000
8	303	2.03e-04	3.89e-02	9.75e-08	1.000
16	101	5.41e-04	6.73e-02	9.54e-08	1.000
16	202	5.67e-04	6.81e-02	9.63e-08	1.000
16	303	5.24e-04	6.65e-02	6.02e-08	1.000

uses the stored 32 initial states and every one of the 64 terminal shock branches. The reference gap is the complete scenario-tree linear minimization gap at the tighter audit tolerance. Binding is the fraction of actions exhausting the common budget. All declared dimension–seed combinations are included.

uses squares with a 10^{-5} ridge term. The actor has one width-32 tanh layer and a softmax output enforcing the common budget. Its 160-iteration L-BFGS fit distills the greedy actions. The convex pointwise improvement uses an explicit gradient-tangent upper bound and checks its linear minimization gap rather than equating a local point with a maximum. All three seeds, 101, 202, and 303, are included at dimension.

An independent scenario tree contains one root action, four next-date actions, and four last-date actions for each tested initial state. Its objective sums all state and control costs and the 64 terminal costs with their probabilities. An independent backward pass computes the gradient. Reference optimization stops at a common-accuracy of 10^{-3} for timing and is separately continued to 10^{-7} for policy-cost auditing. Participation is enforced by one action per information node, not one freely chosen action per complete shock path.

S.6.3. *Reproduction Order*

The repository README gives executable commands for the following sequence: generate temporal, trace, utility, and game tests; stochastic grid refinements and re-solved cost panels; the retained unsafeguarded neural pilot; all three safeguarded runs; nine resource runs; independent frozen-policy and occupation analysis; computational-graph and checkpoint-replay tests; table generation; source/output test generation; and LaTeX builds. The table generator reads the deposited JSON data, so manually entered historical arrays cannot silently become current numerical data.

The source snapshot also retains the previous manuscripts, supplements, reports, error diagnostics, and arrays unchanged. The preservation map locates their sub-topics in the new manuscript or this supplement. Retention is not endorsement of incorrect claims. The development log records the standard-library module-collision discovered during implementation, the corrected execution, and the difference of the pilot. Failed commands are not relabeled as successful scientific experiments.

or NDU, a reference value difference compares two specified finite approximations evaluates a neural policy against a specified finite reference. The actor discrepancy reports consumption, preference adjustment, and portfolio separately because they are different units. A feasible Bellman gain is measured in continuation-value units and uses an independently evaluated frozen continuation. It is a lower bound on the minimum profitable one-step deviation over a richer action set, not an upper bound on deviations.

or resources, the full stochastic tree gives the evaluated policy cost. The reference agent gap bounds its unknown optimum on that same tree, resulting in the interval $[-1, 1]$. The held-out initial states are explicitly stored, so the bound is not extrapolated to all states. The Merton and Epstein–Zin policy errors compare learned homothetic coefficients with analytical solutions, while LQR errors compare structured learned policies with an independent Riccati solution. Temporal and game exploitability use separate best-response calculations against the actual continuation being evaluated. The standard errors concern a repeated-batch scalar loss and are not policy errors.

S.7. STOPPED CONTRACTS AND VALUE-ENCLOSED POLICY REUSE

S.7.1. Discounted Duration and Terminal Exposure

For any realized stopping time $\tau \leq T$, direct integration gives

$$\rho \int_t^\tau e^{-\rho(v-t)} dv + e^{-\rho(\tau-t)} = 1.$$

This equality holds before conditioning and before optimizing. In the positive finite model, each branch satisfies the same identity with its actual live duration αh . Thus the independently evaluated exposure recursion is

$$Z_n^p(s) = z_n(s, p_n(s)) + K_n^{p_n} Z_{n+1}^p(s), \quad Z_N^p = 1,$$

where z_n is the discounted probability weight of settlement during the current branch. K_n^p contains only live discounted interpolation weights. Duration satisfies $A_n^p = \rho A_n^p + K_n^p A_{n+1}^p$, $A_N^p = 0$. The row identity $\rho a_n^p + z_n^p + K_n^p \mathbf{1} = 1$ proves $\rho A_n^p + Z_n^p = 1$ by induction. This includes immediate settlement for an initial boundary node.

The program does not define Z as $1 - \rho A$ and then test the resulting tautology. It equates Z with its own killed-payment recursion and checks the identity for randomly selected contracts and stored-policy switches. It then compares the original four-feature payoff $B + mA - kC + \ell Z$ with the reduced payoff $\ell + B + (m - \rho\ell)A - kC$. The largest discrepancy and the source hash are recorded. For state paths generated by an optimality selection, all changes $(m, \ell) \mapsto (m + \rho h, \ell + h)$ preserve the set of feasible optimal paths; this is the precise sense in which choices alone do not identify the two levels separately.

The monotonicity result concerns a common transition and feasible-policy class. Changing the shock correlation, the control bounds, or the stopping mechanism does not satisfy that condition. Nor does a different parameter-specific action menu necessarily preserve convexity of the optimized finite value in the original parameters. We therefore freeze the complete union menu over the contract interval before building the binary.

feasible anchor policy p_i have certified loss $\hat{c}_n^i(s)$ at its anchor q_i . Exact evaluation of the valid upper bound $\bar{V}_n^i = J_n^i(q_i) + c_n^i$. If evaluation instead supplies an interval, the valid anchor upper bound is $\bar{J}_n^i(q_i) + c_n^i$. The empirical maximum of sampled residuals cannot be substituted for either ingredient.

Similarly, an error version of switching can be stated without conflating approximate features with exact evaluation. Suppose $\|\hat{J}_n^i - J_n^i\|_\infty \leq e_n$ uniformly over stored anchors at the query, and choose the policy index maximizing $\hat{J}_n^i$. Let $L_n = \max_i J_n^i$ and $L_n - J_n^{\text{sw}})_+ \|\infty$. If continuation kernels have row sums at most γ_n , then

$$D_n \leq 2e_n + \gamma_n D_{n+1}, \quad D_n \leq 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j \quad (\text{S.1})$$

Common exact terminal evaluation. To prove the first inequality, the selected anchor has exact value at least $L_n - 2e_n$. The next switching value is at least the selected anchor value minus D_{n+1} ; positivity gives the result. Hence

$$V_n^* - J_n^{\text{sw}} \leq U_n - L_n + 2 \sum_{j=n}^{N-1} \Gamma_{n,j} e_j.$$

The exact-feature theorem in the main paper is the $e_n = 0$ case. The executed finite approximation uses this case up to recorded floating-point errors; it does not assign zero to the neural critic. This distinction allows neural approximation to provide anchors while the certificate uses separately evaluated policies.

S.7.3. A Finite Certificate for an Entire Interval

Adjacent anchors $a < b$, two feasible endpoint-policy lines are $b_1 + da_1$ and $b_2 + da_2$, intersecting at every date and state. With an affine upper chord $u_0 + du_1$, define

$$g(d) = u_0 + du_1 - \max\{b_1 + da_1, b_2 + da_2\}.$$

On either side of the intersection of the two lines, g is affine. Its maximum on $[a, b]$ therefore occurs at a , at b , or at their intersection if present. Parallel lines require distinct endpoints. Taking the maximum over the finite set of dates and states gives a complete bound without discretizing the contract interval. Enlarging the policy library may improve the maximum of feasible policy values; using just the endpoint pair defines an interval is conservative and reduces certification cost.

The adaptive construction begins with contracts zero and one. At every iteration it updates the complete bound, selects a worst interval, and solves a new anchor at the center attaining that interval's largest gap. Each new policy is independently certified. The complete anchor sequence, worst gaps, and parameter locations are tracked, including the stopping criterion. The implementation has a maximum-anchor limit that raises an error rather than publishing an uncertified library when the process is not reached.

All computations use double precision. The pairwise-intersection calculation treats near-slope differences below 10^{-14} as parallel. The omitted variation of a difference between lines on the unit contract interval is at most 10^{-14} ; the verification checks

both quantities. This is a numerical certificate based on explicit finite inequalities, an interval-arithmetic proof of every floating-point operation. Independent policy versions, kernel comparisons, and direct query solves test the implementation of the qualities.

S.7.4. Whole-Interval Action Separation

At a fixed date and state, choose one feasible first action a and one library continuation. Both its reward and its continuation value are affine in the contract. For each competing first action b , the upper-chord continuation gives another affine function. Their difference is positive at both interval endpoints, it is positive at every convex combination of the endpoints. Taking a supremum over all competing actions then gives the main action-separation proposition.

The implementation maximizes the minimum of the two endpoint margins over candidate pairs (a, p_i) . This ordering of maximization and minimization is essential. Taking the minimum of two independently maximized endpoint margins would permit different favorable witness at each endpoint and would not establish a single affine lower bound on the interval. Saved output includes the selected library index, the action, both sign margins, and the reported status of each interval. The displayed negative-region endpoint is rounded downward and the positive-region endpoint upward; neither rounded interval extends beyond its certificate.

S.7.5. Relation to Existing Policy-Transfer Results

The feature decomposition and the policy-improvement logic build on [Barreto et al. \(2017\)](#), [Chang \(2021\)](#). The anchor upper bound is the convex-combination case of [Necek and Parr \(2021\)](#), whose policy-cache method already addresses whether an existing policy is sufficiently accurate for a new reward function. Our construction uses that bound on the economically reduced stopped-contract parameter, checks every interval breakpoint and every finite state-date node, and applies a separate interval-wide action-separation inequality. The independently verified finite-economy results and the stopping-contract interpretation are the substantive objects reported here. We do not claim that the underlying supremum-of-affine-functions argument, policy caching, or actor-critic separation was first introduced in this paper.

S.8. ADDITIONAL COUNTERFACTUAL AND EQUILIBRIUM COMPUTATIONS

S.8.1. Joint Refinement and Contract Sensitivity

The complete refinement panel is reported in Table [S.4](#). Unlike refining time on a fixed interpolation grid, these calculations change state spacing, time spacing, and the time mesh together. They retain the consumption and preference controls as active optimization coordinates. The coarse, intermediate, and fine initial values at $d = 0$ are different; the portfolio at the midpoint is also sensitive to the first refinement. The point contract comparison is more stable than the estimated switching location. All results remain evaluations of declared finite economies rather than upper and lower bounds on the stopped diffusion.

The correlation panel in the machine-readable output independently resolves the intermediate economy at correlations -0.25 , 0 , and 0.25 . For each, the initial share

State grid	Dates	Action grid	d	Value	Duration	π_0
17×25	4	$5 \times 5 \times 7$	0.0	-1.140548	0.884705	-0.50
17×25	4	$5 \times 5 \times 7$	0.5	-0.698148	0.885220	-0.50
17×25	4	$5 \times 5 \times 7$	1.0	-0.253215	0.891936	0.80
33×49	8	$7 \times 7 \times 11$	0.0	-1.136633	0.887830	-0.50
33×49	8	$7 \times 7 \times 11$	0.5	-0.692395	0.890237	0.80
33×49	8	$7 \times 7 \times 11$	1.0	-0.246813	0.894453	0.80
49×73	16	$9 \times 9 \times 13$	0.0	-1.133946	0.888358	-0.50
49×73	16	$9 \times 9 \times 13$	0.5	-0.689493	0.890188	0.80
49×73	16	$9 \times 9 \times 13$	1.0	-0.243752	0.894848	0.80

controls are reoptimized at every resolution. These refinements use the displayed meshes without a frozen neural proposals. The endpoint reversal survives, whereas the midpoint changes sign in the first and second resolution. This table measures sensitivity of finite economies, not a certified for the continuous stopped diffusion.

TABLE S.5

ADJUSTMENT QUANTITY AND COEFFICIENT-WEIGHTED EXPENDITURE

k	C^{p_k}	kC^{p_k}	Value
0.5	0.014614	0.007307	-1.117524
2.0	0.009888	0.019777	-1.135835
8.0	0.001432	0.011455	-1.160291

complete finite problem is solved separately at each coefficient on a 25×37 grid with eight dates 9×13 actions. The unweighted quantity is decreasing; the coefficient-weighted charge is not. g is included in both quantities.

s from -0.5 at $d = 0$ to 0.8 at $d = 1$. We do not transfer the feature bank across changes in transition law. Duration, values, and all controls are stored for each solve. The normalization comparison, by contrast, changes only the reward π and retains the same transition and policy class.

S.5 distinguishes discounted adjustment effort from its coefficient-weighted. The same decomposition is evaluated at the original cost-panel grid and with original complete control vector. The theorem applies to the first quantity; the coefficient produces the largest weighted charge in this panel.

S.8.2. Matched Actor-Free and Quadratic Comparisons

ine actor-free comparisons reload the deposited nonlinear critic weights and the initial states. Both actor-start and zero-start algorithms use the same projected improvement method, gradient, Lipschitz step bound, and stopping tolerance. l policy is then evaluated through all 64 terminal shock paths. Saved output es actor initialization, optimization time, iterations at each date, complete life-est, and the reference gap. The original proposal-only ablation remains in S.6; ng the safeguard and removing the actor are different experiments. late training costs, critic-only backward fits use the same training-state draws ture construction as the original nonlinear critic. Actor distillation is timed

d	Seed	Actor train	Actor query	Zero query	Projected break-even
4	101	1.483	0.0414	0.0424	48512
4	202	0.741	0.0432	0.0451	12544
4	303	0.742	0.0436	0.0429	None
8	101	0.808	0.0607	0.0626	13568
8	202	0.797	0.0637	0.0713	3392
8	303	0.794	0.0489	0.0512	11488
16	101	0.878	0.1004	0.1035	8864
16	202	0.879	0.1174	0.1326	1856
16	303	0.875	0.1051	0.1079	9888

onds per 32-state complete-tree query batch. Actor training is the distillation fit alone; common target instruction and critic training are not charged twice. The last column is $32 \lceil T_{\text{actor}} / (t_{\text{zero}} - t_{\text{actor}}) \rceil$ when measured saving is positive. It is a linear fixed-batch projection, not an observed crossover. “None” means that actor initialization was not faster in that timing comparison. Optimizer iteration counts and amortized initialization times are saved in the machine-readable results.

separately against the same improved-action targets. Table S.6 reports the complete nine-timing panel. A positive measured online saving is converted into a clearly labeled fixed-batch break-even projection. The separately executed fresh-state workloads in the main paper, not this projection, support the observed amortization comparison. The quadratic comparator minimizes squared fitting error over

$$q(x) + b + \ell^T z + z^T H z, \quad z = x - \mathbf{1}, \quad H \succeq 0.$$

regression is convex in (b, ℓ, H) . Its gradient is computed from the full quadratic Hessian matrix. Projected accelerated gradient leaves (b, ℓ) unrestricted and projects the Hessian matrix onto the positive-semidefinite cone by clipping negative eigenvalues. A small ridge term stabilizes the design. The reported projected-gradient norm is below the declared 10^{-8} stopping threshold in the executed fits. At deployment the analytic gradient is $\nabla q(x) + \ell + 2Hz$, and its Hessian bound is the largest eigenvalue of $\nabla^2 q + 2H$. The common investment projection and full scenario-tree cost calculation are unchanged. This comparison tests a genuinely coupled convex quadratic, rather than a separable or unoptimized baseline.

The fresh-state workload charges only for the computations used by each competing method. The reusable policy’s cost is critic training plus zero-start online improvement plus full policy evaluation. The specialized solver’s cost is a complete nonanticipative reference optimization at tolerance 10^{-3} . A reference solved more tightly to gap below 100% supplies the accuracy upper bounds for both; it is not charged asymmetrically as the cost of one solver’s online cost. Hardware, dependencies, seeds, and individual times accompany the outputs. A three-period, rank-two-shock experiment cannot establish dimension-independent performance, even when its measured query reuse is useful.

S.8.3. Persistent Capacity and All-Date Dynamic Deviations

Let $b_i = 1 - sK_i$, so that next-period capacity probability is $sK_i + b_i a_i$. Write $W_i(k'_1, k'_2)$ as player i ’s continuation value after averaging over next demand. For player one the

$$a_1 = \text{clip} \left(\frac{\delta b_1}{\kappa} \{W_1(1,0) - W_1(0,0) + (sK_2 + b_2a_2)\Delta W_1\}, 0, 1 \right), \quad (\text{S.2})$$

$\Delta W_1 = W_1(1,1) - W_1(1,0) - W_1(0,1) + W_1(0,0)$. The corresponding formula for player two. The action criterion is strictly concave because the continuation is in the player’s own probability and the action cost has curvature $\kappa > 0$. Enumerate whether each clipped best response is at zero, interior, or one gives nine active-set combinations. Feasible candidates are checked against both original best-response functions.

Product of the two affine slopes bounds the Lipschitz constant of the composed response; clipping is nonexpansive. Its recorded maximum below one therefore has a unique one-period fixed point for every continuation game. Backward induction supplies a finite-horizon Markov-perfect profile. This argument does not require current-capacity independence and does not rely on convergence from a single iteration.

For independent dynamic verification, let $J_{i,n}$ be the value of the frozen profile and $J_{i,n}$ the full unilateral best-response value

$$B_{i,n}(s) = \max_{a_i} \{r_i(s, a_i, p_{-i,n}(s)) + \delta \mathbb{E}[B_{i,n+1}(s') \mid s, a_i, p_{-i,n}(s)]\}.$$

At each backward date, the code stores $B_{i,n}(s) - J_{i,n}(s)$ for both players and all eight dates. It reports the largest positive gain, the signed maximum and minimum, and the date, player, capacities, and demand. The historical code had placed the comparison after the date loop, which verified only date zero. The corrected scope is added for both the renewal regression and the persistent economy.

For negative control, the final-period investment of player one at capacities (1,1) and demand is replaced by one. The independent routine recomputes the frozen-profile values and all best responses. It detects a gain exceeding 0.1, at a recorded late-date. This test would not be passed by a routine that merely reports convergence of the equilibrium solver. The reset case $s = 0$, four dates, reproduces its early closed-form results and terminal investment to numerical precision. Both examples remain exact date references, not neural-equilibrium training results.

S.8.4. Executable Files and Preservation

The fifth-revision package is in `replication/r5`. It imports the unchanged fourth-revision transition, resource primitives, and saved neural checkpoints. The contract definitions are in `contracts.py` and `economic_robustness.py`. The remaining experiments are in `resource_ablation.py` and `persistent_game.py`. The generator reads executed JSON; the validation script replays policy features, non-identity identities, kernel comparisons, and certificate conditions. The build records the build and output hashes and preserves the exact review-base commit.

For other papers, theorem proofs, raw negative controls, referee reports, and neural checkpoints remain in their original paths. The new manuscript carries the original mathematical and economic extensions forward and adds the contract-reuse and current-capacity material. The accompanying response identifies where each issue in the fourth-revision report is treated. Build completion and numerical verification do not require journal submission or editorial approval.

S.9.1. Coefficient Construction and Interval Bounds

For a fixed policy, pre-draw the h remaining iid regime labels. Conditional on their count j , labels are exchangeable, the first label is 1 with probability j/h , and the remaining count is $j - 1$ or j . Applying the same count-blind policy to these conditional counts gives the main coefficient recursion. Averaging over the binomial count distribution proves the polynomial representation. This is an evaluation identity, not a policy improved to condition its action on the count.

For the separate upper calculation the controller is allowed to observe the remaining count, and past labels are revealed after their transitions. Every original policy can ignore this extra information. Backward maximization of the weighted common-action criterion therefore gives an upper coefficient vector. Averaging over counts preserves the upper inequality. Optimizing the regime-0 and regime-1 actions separately *before* averaging their weighted mean would instead disclose the current label and define a different relaxation. The implementation takes the maximum after the mean.

For $h \geq 2$, the degree- h Bernstein coefficients of the corrected upper value on a local interval are

$$U_{n,j} = \left(1 - \frac{j}{h}\right) v_n^a + \frac{j}{h} v_n^b + \frac{j(h-j)}{h(h-1)} M_n, \quad j = 0, \dots, h.$$

For $h = 1$ the correction is zero because the common terminal payoff has zero endpoint difference; at $h = 0$ the only coefficient is g . Degree elevation thus represents the quadratic upper function exactly. De Casteljau subdivision restricts both upper and lower coefficients to the same closed cell before taking their coefficient differences. The method is conservative because $\sup_t \min_p f_p(t) \leq \min_p \sup_t f_p(t)$. The minimum is taken separately for each state and date, not before the statewise maximum.

The stopping condition uses the maximum of these cell bounds, not the largest loss observed at replay parameters. The replay additionally checks the whole value vector, dominance by the upper function, and improvement from switching relative to the best fixed policy. The script stores each anchor policy, its independently evaluated polynomial coefficients, each interval's upper coefficients, cell bounds, and validation statistics. The count-information and corrected-chord outputs remain separate so that the loose information bound cannot be substituted for the stopping certificate.

S.9.2. A Reusable QP with the Original Information Structure

At a tree node, write its state as $x_h = M_h x + S_h z + b_h$. Let $Q = I/d + 0.2\mathbf{1}\mathbf{1}^\top/d^2$ and p_h be its probability. Expansion of all stage and terminal costs gives

$$F = 2 \sum_h p_h S_h^\top Q S_h + 2 \text{diag}_h(0.1 p_h I/d), \quad F = 2 \sum_h p_h S_h^\top Q M_h, \quad f_0 = 2 \sum_h p_h S_h^\top Q (b_h - \mathbf{1}).$$

The control diagonal includes decision nodes only. The sums over state costs include terminal nodes. Let D repeat each decision node's probability across its d controls. A Lipschitz constant for the transformed gradient is the largest eigenvalue of $D^{-1/2} H D^{-1/2}$. An accelerated step is a nodewise simplex projection of $z - D^{-1} \nabla C_x(z)/L$. Every ten iterations we check the complete convex tangent gap; the final gap is independently calculated by the original tree-gradient routine. The reported query time includes that

ve totals; the tighter deposited reference enters only the accuracy comparison. assembly is a new source implementation of the structural comparison described fifth-round report, not an invention of parametric quadratic programming. l-start workloads use the same saved initial states. No array of known query ns initializes a new query. Complete costs, gaps, timing samples, feasibility and gradient comparisons are retained in the execution record.

S.9.3. Risk-Class Evaluation and Scope

given (d, k) , first solve the original finite problem backward. At the initial optimize the current action separately over positive and nonpositive portfolio ns using the same optimal continuation. Evaluate each resulting complete policy ndently to obtain (B, A, C) . This computes first-risk-class values, not policies ned to keep the same sign forever. Repeat the entire backward solve with $\theta = 0$ y state and date for the no-deliberate-adjustment model. Preference volatility s present. The two risk-specific option values are nonnegative by set inclusion, air difference is checked against the independently formed difference of risk ges.

tion starts only when the risk advantage has opposite endpoint signs. The records both signs, the bracket, the active class features, and the ratio of their nd duration differences. It does not assign a global uniqueness flag. A duration-ver bound valid throughout an interval would justify the main uniqueness he present numerical brackets do not assert that additional hypothesis. The o-sign exclusions in the inherited reward experiment therefore retain their independent status.

S.9.4. Evidence and Reproduction

current source record is `ECTA_R6.tex`, `SUPP_R6.tex`, the sixth-revision sec- ectory, and `replication/r6/`. Sections S.1–S.8 preserve earlier substantive specifications, and evidence with their historical identities. The new manifest e reachable source commit and hashes the executable sources, immutable input oints, generated records, and tables. The public execution log distinguishes preparation, experiments, validation, and document build.

support comparison uses `bank_comparison.py`. Transition calculations ansport.py and `kernel_certificate.py`. The remaining programs are himism.py, `structural_qp.py`, and the independent small-model checks in tests.py. Numerical libraries are single-threaded for timing. Two developmen- sition-certificate runs were stopped to eliminate repeated interval calculations n to replace array multiplication by an equivalent CSR multiplication. The final e the cached intervals and verified CSR rows. This is an execution optimization, ange in the state, action, transition, or welfare target. The all-state scalar-OLS ison retains the inherited common oracle implementation for both objectives.

revised source does not reclassify the historical neural fits, exact grid repairs, or ames. In particular, removal of neural proposals still meets the original reward-elfare target, and the reusable QP still supplies the relevant stronger resource ator. A completed build or a passing finite-model validation does not establish acceptance, broad neural superiority, or a diffusion approximation theorem.

This section supplies the additional proofs for the matched comparison and the re-refinement result. All statements concern the same finite state/action target and a near affine mixture of positive substochastic operators. Approximation of a diffusion is a separate question.

S.10.1. Localized Count Dominance

On an interval $[a, b]$, let $h = N - n$, and suppress its superscript. Write $C_{n,j} = (1 - j/h)v_n^a + (j/h)v_n^b$ and $f_{h,j} = j(h - j)/(h(h - 1))$ for $h \geq 2$. The corrected upper coefficients are $U_{n,j} = C_{n,j} + f_{h,j}M_n$. At $h = 0$, both constructions equal g . At $h = 1$, the correction is zero because both endpoint terminal functions are g , and the pure endpoint human backups are no greater than their endpoint supersolutions. Suppose $W_{n+1,j} \leq U_{n+1,j}$. Positivity permits substitution of U for W in the count version. For an interior count, expanding the part containing C gives

$$\begin{aligned} (1 - j/h)T_{n,a}^u v_{n+1}^a + (j/h)T_{n,b}^u v_{n+1}^b \\ + f_{h,j}(K_{n,a}^u - K_{n,b}^u)(v_{n+1}^b - v_{n+1}^a) \leq C_{n,j} + f_{h,j}D_n^u. \end{aligned}$$

This identity includes the reward terms, whose weights sum to one. For $h > 2$, the propagated correction is

$$f_{h,j} \left\{ \frac{h - j - 1}{h - 2} K_{n,a}^u M_{n+1} + \frac{j - 1}{h - 2} K_{n,b}^u M_{n+1} \right\}.$$

These two weights are nonnegative and sum to one. Keeping the current action u fixed, the entire expression is at most

$$C_{n,j} + f_{h,j}\{D_n^u + \max(K_{n,a}^u M_{n+1}, K_{n,b}^u M_{n+1})\} \leq C_{n,j} + f_{h,j}M_n.$$

For $h = 2$ the next correction is zero and the same conclusion follows from D_n^u . Counts U and h use the endpoint supersolution inequalities directly. Maximization over common action completes induction. This is the comparison identified in the R6 advisory report, including its same-action and short-horizon qualifications.

For the rectangular comparison, induction gives $W_{n+1,j} \leq B_{n+1}$. Each count backup is a convex combination of two quantities no greater than $\max_{u,e} T_{n,e}^u B_{n+1} = B_n$, so $W_{n+1,j} \leq B_n$. Each original policy can ignore count information, proving that the count polynomial itself is an upper value. Positive polynomial evaluation and positive deiteljau subdivision preserve coefficient dominance. Subtracting the same lower coefficients, taking coefficient maxima, policy minima, positive parts, and state/date maxima preserves the ordering of the certificates. These statements do not establish that the count is faster.

S.10.2. Operation Counts and Storage

With exact endpoint values shared, count coefficients $j = 0, h$ need no additional human solve. At remaining horizon $h \geq 2$, all h next-date coefficients require one multiplication by each base kernel. Neighboring results can be cached while looping through j . The total is $2 \sum_{h=2}^H h = H(H + 1) - 2$ action-wide kernel applications. The

ver, totaling $4H - 6$ for $H \geq 2$; it is zero for $H = 1$. The rectangular recursion has $4H - 6$ applications per date. Reward combinations and maxima are additional operations. The production code's counts are checked by an instrumented dense matrix at horizons 1, 2, 3, 4, 6, 8, 12, 16.

At each date, the correction stores $H - 1$ nonzero state vectors; known zero vectors are retained in its compact representation. The count method has $\sum_{h=1}^H (h - 1) = H(H - 1)/2$ interior coefficient vectors. Endpoint values are common inputs. The correction bank for one feasible policy contains $\sum_{h=0}^H (h + 1)$ state vectors; across m policies this is $O(mSH^2)$, in addition to $O(mSH)$ policy indices. The implementation uses the chord one date at a time during certification, and charges that expansion to the correction. These persistent-array counts exclude action-wide work buffers for the sparse kernel, and are not claims about process peak resident memory. If a policy has at most z stored successors and A actions, an action-wide application costs $O(zA)$. Naïve repeated de Casteljau restriction of m_l local policy arrays to B cells costs $O(m_l B^3)$; it can dominate when action optimization is cheap.

S.10.3. Proof of the Total Coefficient-Error Bound

Let V_n, L_n, Q_n be the primitive recursions in the main text. By backward induction, the fixed-policy value and every optimal value has absolute magnitude at most V_n . The bounded continuation f with $\|f\|_\infty \leq V_{n+1}$,

$$\|T_{n,\lambda}^u f - T_{n,\lambda'}^u f\|_\infty \leq |\lambda - \lambda'|(\rho_n + \kappa_n V_{n+1}).$$

Monotonicity and the row-mass bound give a Lipschitz constant L_n for both optimal and all fixed-policy values, whether or not the maximizing action changes. The efficient bound is needed in addition to this pointwise result. Let p^a be an a -optimal Markov policy and let $F_{n,j}^{p^a,I}$ be its Bernstein coefficients on $I = [a, b]$. For $n < N$, its conditional-count recursion is a convex combination of endpoint- a and endpoint- b backups of next-date coefficients. Relative to the pure- a policy value, induced by the preceding inequality give

$$\|F_{n,j}^{p^a,I} - J_n^{p^a,a}\|_\infty \leq \beta_n L_{n+1} w + (\rho_n + \kappa_n V_{n+1}) w = L_n w.$$

The induction bounds the absolute coefficient magnitude by V_n . It is valid even when conditional counts change the regime probabilities: the current mixture weights are between zero and one, and the induction bound is uniform over the remaining

Exact endpoint optimization gives $J_n^{p^a,a} = v_n^a$ at every state and date.

$\|v_n^b - v_n^a\|_\infty \leq L_n w$, while the signed cross term obeys $\|D_n^u\|_\infty \leq \kappa_n L_{n+1} w^2$. The induction recursion therefore implies $\|M_n\|_\infty \leq Q_n w^2$. For $h \geq 2$, $0 \leq f_{h,j} \leq 1/2$. Conse-

$$U_{n,j}^I - F_{n,j}^{p^a,I} \leq 2L_n w + \frac{1}{2} Q_n w^2.$$

At the terminal date the difference is zero. Selecting one feasible policy already establishes the claimed bound; taking the better policy can only tighten it. Subdivision is positive and cannot invalidate the bound. Count dominance transfers the bound to the local count construction.

A complete dyadic grid first reaching width w_* has fewer than $2/w_*$ intervals and most one more anchor. This proves finite termination and the worst-case $O(\epsilon^{-1})$ state-
 at for fixed primitive constants. The result does not require second-order smoothness
 optimal values. For nonexact endpoints, their independently certified one-sided
 dual and policy-evaluation errors must be added before this termination rule is
 d.

S.10.4. Scope of the Matched Frontier

ne full experiment reoptimizes every original R6 transition anchor on the unchanged
 et and independently reevaluates its policy coefficients. It uses the same endpoint-
 cy pair and eight closed cells on every compared interval. For $d = 0.5$, it also
 pares common coarser banks retaining every second or fourth original anchor,
 ays including 1. These locations are inherited computational design choices, not
 ly timed autonomous anchor searches. Setup, anchor optimization, policy evalua-
 , local coefficient restriction, upper construction, and cell certification are charged
 ach method's reconstructed total. Shared work is measured once and attributed
 atically. Numerical ordering checks and replay diagnostics are validation overhead,
 part of any arm's deployment workload. The results are one complete serial, single-
 ad pass per row on one machine; no cross-machine historical time is used as a
 aparator. The raw stage times and every interval bound are deposited.

S.11. PRIMITIVE ADJUSTMENT BOUNDS AND DECISION-CERTIFICATE REPLICATION

S.11.1. The Primitive Option Bound

or $z = u - 1 > 0$ and $a = \log c$,

$$U(c, u) = -\frac{e^{-za}}{z}, \quad U_u(c, u) = \frac{e^{-za}(1 + za)}{z^2}.$$

ifferentiating again gives the strictly negative second derivative in the main text. For
 e positive consumption support the expectation can be differentiated term by term.
 or's integral formula and concavity imply, for $|\theta| \leq \bar{\theta}$,

$$g_\sigma \theta - \frac{1}{2} M_\sigma \theta^2 \leq F_\sigma(\theta) - F_\sigma(0) \leq g_\sigma \theta.$$

tracting $k\theta^2/2$, evaluating the lower quadratic at $\theta = g_\sigma/(k + M_\sigma)$ when this is
 ible, and maximizing the upper quadratic on the whole line give the option bounds.
 upper bound remains valid if the actual optimum is constrained. Subtracting the
 ative-class upper bound from the positive-class lower bound gives the relative-
 on inequality.

nce $D_A = A_+ - A_- > 0$ is a primitive constant in this specialization,

$$\Delta^0(d) = B_+ - B_- + dD_A, \quad \Delta^{\text{adj}}(d) = \Delta^0(d) + \Omega_+ - \Omega_-.$$

n functions have the same strictly positive slope. Their unique indifference contracts
 erefore differ by $(\Omega_+ - \Omega_-)/D_A$. This proves the stated threshold shift and the
 rvening interval of opposite rankings. Uniqueness here follows from the explicitly
 stant duration difference in the two-stage model; it is not carried over to the full
 ped portfolio model.

ature does not have to be estimated by sampling preference indices. In the
n above,

$$-U_{uu}(c, 1+z) = \frac{e^{-za}[(1+za)^2 + 1]}{z^3} = \int_a^\infty t^2 e^{-zt} dt.$$

nd derivative with respect to z is $\int_a^\infty t^4 e^{-zt} dt \geq 0$. The integral is finite for every
also when $a < 0$. Hence the negative second derivative of expected utility is
in z , and its maximum on the preference interval is attained at an endpoint. It
e in the probability of a two-point consumption lottery. Its maximum over the
probability/preference rectangle can therefore be bounded at its four vertices.
 $u_0 = 2$, $\bar{\theta} = 0.2$, $C_- = 0.5$, and $C_+ \in \{0.05, 0.8\}$, the positive-class shadow is affine
ative for every $p \in [0.06, 0.10]$. Its absolute value lies between 1.4820 and 3.1180;
bsolute negative-class shadow is below 0.6138; and $M_+ < 17.342$. Feasibility of the
ound trial follows from $3.1180/20 < 0.2$. For every $k \in [20, 80]$,

$$\Omega_+ - \Omega_- \geq \frac{1.4820^2}{2(80 + 17.342)} - \frac{0.6138^2}{2 \cdot 20} > 0.00186.$$

nd is conservative because it takes adverse cost values separately in its positive
gative terms. Division by $D_A = 0.5$ gives a reduction greater than 0.00372. This
a statement on the entire probability/cost rectangle, not on the finite set of
ed optimizations.

TABLE S.7

PRIMITIVE PREDICTIONS AND OPTIMIZED ADJUSTMENT OPTIONS

p	k	Lower option	Actual option	Min. shift	Test d
0.06	20	0.0261621	0.0311074	0.0523241	0.723838
0.06	40	0.016882	0.0185198	0.0337641	0.733118
0.06	80	0.00973227	0.0102147	0.0194645	0.740268
0.08	20	0.0681238	0.0811337	0.136248	1.43188
0.08	40	0.0441692	0.0488284	0.0883384	1.45583
0.08	80	0.0257477	0.0271885	0.0514953	1.47425
0.10	20	0.12072	0.147117	0.24144	2.12928
0.10	40	0.0800377	0.0900327	0.160075	2.16996
0.10	80	0.0475675	0.0507768	0.095135	2.20243

two-stage CRRA family: $u_0 = 2$, $|\theta| \leq 0.2$, $C_- = 0.5$, $C_+ \in \{0.05, 0.8\}$, and $A_+ - A_- = 0.5$. Option
 $\Omega_+ - \Omega_-$. The primitive lower bound determines the test contract, at which every row has
 Δ^{adj} . The continuous-family proof covers all $p \in [0.06, 0.10]$ and $k \in [20, 80]$, not just these rows.
re in fixed cardinal utility and operating-benefit units.

able compares the primitive lower bound with a one-dimensional strictly con-
optimization in the same cardinal utility units. For each row the test contract is
1 by $d = d^0 - \underline{\Omega}/(2D_A)$, before using optimized option values. It therefore tests
ction from the bound rather than selecting a contract after locating a numerical
l. The consumption values, probabilities, and duration difference specify an ana-
family, not estimated parameters. In particular, the eight-date stopped economy
its original state dynamics and is not replaced by this specialization.

the risk certificate uses all 1,617 states, all eight decision dates, and the original 8-action target, comprising the common union mesh and the three frozen neural proposals. The declared initial state is $(u, X) = (2, 1.25)$, the adjustment cost is $k = 2$, contract interval is $[0.4, 0.45]$, and the one-step shock-law mixture probability lies in $[0, 0.25]$. This region was selected during model exploration; it is not a preregistered or exhaustive characterization of all reversals.

For each adjustment regime and each of the four parameter corners, one backward optimization yields both first-risk classes: dates after the first use the regime’s common feasible menu, and the two first-date maxima enforce positive and nonpositive risky positions separately. This is eight corner optimizations, yielding sixteen feasible class policies. The no-adjustment menu imposes zero deliberate adjustment at every date and date but retains preference shocks. Later portfolio signs remain unrestricted. Endpoint values feed eight localized count constructions, one for each contract endpoint, contract interval, and adjustment regime. Each corner policy is independently evaluated at reward contrasts zero and one, giving its Bernstein intercept and duration coefficients; this covers arbitrary intermediate d without a new policy solve.

Convexity of the optimal class value in d validates linear interpolation of its upper envelope, not of its optimal policies. Feasible-policy values are affine in d exactly. The resulting degree-eight-by-degree-one tensor coefficients certify the entire rectangle. Subdivision is available if a sign interval contains zero; acceptance of the original rectangle in the deposited execution means no further subdivision is required. Pointwise replay at the corners checks implementations but is not the source of the continuum sign claim. All upper and lower tensor coefficients, policy indices, and policy-feature arrays are deposited, with a coefficient-only replay program that does not call an optimizer.

S.11.4. Numerical Error Account

The exact-arithmetic theorem is stated separately from its floating-point implementation. The application uses an a priori round-to-nearest allowance relative to the *stored kernel and reward arrays*. It is not a claim of directed-rounding interval arithmetic, but it does not include errors in choosing the state/action grid, constructing the quadratic model, or approximating a diffusion. Those omissions do not change which finite set the certificate concerns.

Every stored continuation row is nonnegative with at most sixteen pre-aggregation entries. Let $\hat{\beta}$ be the largest computed row sum plus 10^{-12} . The program verifies $\hat{\beta} \leq 1 + 2 \cdot 10^{-12}$. This margin dominates summation error for sixteen nonnegative entries of this size. Let R be twice the largest sum of the absolute stored base reward, continuation reward at $|d| \leq 1$, and effort reward at $k = 2$, across both common kernels and every neural-proposal kernel. The use of $|d| \leq 1$ covers the intermediate feature evaluation at $d = 1$, not merely the reported interval ending at 0.45. Let $G = 2\|g\|_\infty$ and

$$V = (G + HR) \max(1, \hat{\beta})^H, \quad q = 4096(H + 1), \quad \gamma_q = \frac{q\mathbf{u}}{1 - q\mathbf{u}}, \quad \mathbf{u} = 2^{-53}.$$

per-class allowance used in the calculation is

$$e = 2\gamma_q(1 + 2V + R) \max(1, \hat{\beta})^H. \quad (\text{S.3})$$

at most sixteen products and fifteen additions; constructing its sparse row at most sixteen additions of original entries. Adding the three reward components, mixing endpoint laws, and mixing conditional-count branches takes fewer than additional scalar operations per path per date. An endpoint action maximum is expansive in the sup norm: its error is bounded by the largest candidate error, multiplied by the number of actions. Thus an endpoint solve, a selected-policy endpoint recursion, or a count recursion has fewer than $128H$ serial scalar operations per contributing path. Computing intercept/duration features from values at t and $t+1$ adds one subtraction and one multiply-add per coefficient, with absolute error bounded by $2V$. A degree- H restriction uses at most two triangular deconvolutions per path, with fewer than $6(H+1)^2$ scalar operations per retained coefficient even if counted without sharing. For $H=8$, these counts, subsequent differences, and coefficient extrema are dominated by $4096(H+1)$. Positive mixing and row-mass propagation propagate the absolute bound by at most the displayed mass factor. The factors R, G and outside γ_q cover input rounding margins and the addition of separately computed paths. The formula is an absolute-error allowance; signed reward cancellations are not analyzed with a relative-error claim. No overflow or underflow occurs in the displayed value range.

The same allowance is charged separately to the upper and feasible lower computation. A class-value gap therefore includes $2e$, and a class-difference sign interval includes $2e$. The relative-option conclusion adds the already enlarged difference bounds. The information-relaxation and policy-bank errors are controlled by their respective coefficient gaps, not absorbed into an unexplained floating-point tolerance. Independent selected-kernel replay checks the actual arithmetic discrepancy, but small replay errors are not substituted for the a priori allowance. Supplementary records the value and reward magnitudes, row-mass bound, operation budget, and absolute allowance.

S.11.5. *Reproduction and Preservation*

Run `bash replication/r7/run_all.sh` from the repository root, then compile `R7.tex` and `SUPP_R7.tex`. The driver executes the dense independent checks, the primitive family, the joint-region certificate, the full matched upper comparison, and the autonomous dyadic refinement. The full matched frontier reports one measured pass; the autonomous comparison starts each arm with a fresh policy cache and includes the retained parent intervals. All retained R1–R6 numerical files are inputs or historical records, not silently regenerated by the R7 driver. A byte-level preservation check covers every inherited R6 file except the explicitly advanced revision index. A source directory, environment record, complete stdout/stderr logs, generated tables, and redacted manuscript copies accompany the revision.

S.12. DECISION PRIMITIVES, DYNAMIC MECHANISMS, AND STREAMING REPLICATION

S.12.1. *The Finite Transition Protocol*

The stochastic state at a decision date is a node of the declared preference–wealth controls belong to the union of the meshes $(9, 9, 13)$ and $(7, 7, 11)$ in consumption, investment, and risky share, after removing exact duplicate actions. There are 1,565 actions, including 185 with zero deliberate adjustment. Three frozen feasible

only the deposited proposals. On the 97- and 145-node grids we extend their three control coordinates linearly in wealth, separately at each preference node and date. The convex control box makes the extension feasible. The common-menu experiment includes all three proposals at every date, keeping precisely the same 1,565 controls on every grid. It never invokes the historical shortcut that also removed one of the meshes. The conditional one-step law has four equiprobable sign pairs. Its candidate arrival is the current state plus the increment in the main text. The fraction α is the smallest positive fraction at which the straight segment meets the boundary, truncated above at $\alpha = 1$. For a current boundary node $\alpha = 0$ and liquidation is immediate. A strictly earlier arrival $\alpha < 1$, pays the discounted liquidation value at the actual hit point and contributes live transition mass. An arrival at $\alpha = 1$ is settled on neighboring nodes. If the arrival is at a boundary node, the next decision has zero operating duration and immediately triggers liquidation; this has the same calendar-time settlement value. At the terminal date all surviving nodes pay G . These conventions reproduce the inherited transition structure; no reflected state, wealth injection, or additional independent exit draw is introduced.

For an interior arrival coordinate $y \in [l, r]$, the settlement probabilities at l and r are $(r - y)/(r - l)$ and $(y - l)/(r - l)$. Their mean is y and their variance is $(y - l)(r - y)$. Conditional coordinate independence makes the conditional covariance diagonal. Applying the law of total covariance proves the variance identity in the main text. The identity applies to the live settlement lottery; early-liquidation payoffs remain evaluated at their actual hit points. It supplies an economic reason for examining regularity rather than dismissing its effects as arithmetic noise.

S.12.2. Proof of the Uniform Cardinal-Tilt Bound

Write $F_\sigma(\theta) = \mathbb{E}[U(C_\sigma, u_0 + \theta)]$. The original curvature proof gives $-M_\sigma \leq F''_\sigma \leq 0$ on the adjustment interval. Adding $a\theta$ leaves the curvature unchanged and replaces $F'_\sigma(0)$ by $g_\sigma + a$. Taylor's integral formula gives

$$(g_\sigma + a)\theta - \frac{1}{2}(k + M_\sigma)\theta^2 \leq F_\sigma(\theta) - F_\sigma(0) + a\theta - \frac{1}{2}k\theta^2 \leq (g_\sigma + a)\theta - \frac{1}{2}k\theta^2.$$

The positive-class lower quadratic is evaluated at $\theta = (g_+ + a)/(k + M_+)$. The negative-class upper quadratic is maximized without imposing the adjustment bound; relaxing the bound preserves the upper inequality. With $A = 0.25$, the rounded constants from (1) imply

$$|g_+ + a| \geq 1.482 - A, \quad |g_- + a| \leq 0.6138 + A, \quad M_+ < 17.342.$$

The trial is feasible because $(3.118 + A)/20 < 0.2$. Thus, putting $P = (1.482 - A)^2$, $Q = (0.6138 + A)^2$, and $M = 17.342$,

$$\Omega_+(a) - \Omega_-(a) \geq \frac{1}{2k} \left\{ P \frac{k}{k + M} - N \right\}.$$

The expression in braces increases with k . At $k = 20$ it is strictly positive. It follows that the last display is bounded below by

$$\frac{P \cdot 20 / (20 + M) - N}{2 \cdot 80} = 0.000417380273806974 \dots$$

probability and preference ranges follows from the affine probability dependence of the point-curvature argument in S.11. The proof therefore does not rely on the 45 checks supplied with this revision. The positive root of $P20/(20 + M) - N = A$ variable, is about 0.2718421; it is a sufficient-radius boundary for this bound, sharp economic threshold at every (p, k) .

unadjusted class difference is affine in the contract with slope $D_A = 0.5$. Tilting $(-u_0)$ leaves both unadjusted two-stage payoffs unchanged. The unique two-difference therefore shifts by $-(\Omega_+(a) - \Omega_-(a))/D_A$. This proves the uniform shift. At $a = 1$ the sign can reverse, as the retained counterexample shows. The baseline invariance nor this threshold direction is imposed on the dynamic play, where u continues to move when deliberate adjustment is prohibited.

S.12.3. *Dynamic Feature Moments and the Transition-Law Experiment*

Six features in the main text are evaluated by six separate selected-policy back-recursions. Flow features include the actual discount annuity on each branch. Liquidation feature includes both early exit and terminal liquidation. The reward valuation is exact for the declared finite law. For every newly optimized class a second evaluation reconstructs the candidate increment and its interpolation directly from the selected controls, rather than reading the stored sparse kernel. The feature-weighted objective is then compared with both the Bellman value and that replay at every state and date.

Changes in (s, a, d, k, z) , the fixed-policy effect is a linear combination of und feature moments. The reoptimization remainder follows by adding and subtracting the value of the baseline optimizer under the new reward. For a kernel change, denote the old policy value and $\tilde{V}^p$ its value under the new kernel and rewards. The difference obeys the exact recursion

$$\tilde{V}_n^p - V_n^p = \tilde{r}_n^p - r_n^p + (\tilde{K}_n^p - K_n^p)V_{n+1}^p + \tilde{K}_n^p(\tilde{V}_{n+1}^p - V_{n+1}^p),$$

the appropriate terminal difference. Backward substitution gives the fixed-policy effect. Adding the separately reoptimized value adds a nonnegative gain to each class. Taking a four-class contrast need not preserve that sign. The zero-covariance experiment sets both conditional laws' correlation to zero and rebuilds their law; it does not identify that law with an equal mixture of the two baseline endpoint

The mechanism program executes 36 specification-contract comparisons, each with 144 class values, for 144 class solves. It preserves the original $k = 2$, the eight-date horizon, and the complete action menu. The retained interventions comprise common law tilts, the reference-state normalization multiplier, the liquidation multiplier, the rating-benefit contrast, the mixture weight, and the separate zero-covariance model select readable comparisons; the JSON retains every run, six moments per direct-replay discrepancies, baseline-policy counterfactuals, and reoptimization results. No fitted data, identified causal shares, or two-stage calibration is inferred from computations.

S.12.4. *Streaming the Upper and Feasible Lower Recursions*

The count recursion at a date uses only the coefficient array at the immediately preceding date. Its endpoint coefficients are shared with the endpoint optimal values.

for their common continuation has been computed. The same observation applies to corrected chord: later-date corrections are common to the two first-action classes, only the initial maximum is split. Thus sharing those continuations changes neither admissible actions nor information. The one-period case has only its two endpoint coefficients and is handled directly.

For a degree- h policy value, the coefficient at count j uses only counts j and $j - 1$ at next date. We therefore retain two adjacent date arrays and discard each completed h -date array. At the initial date, only the declared initial state's coefficients are defined. Two passes, at operating contrasts zero and one, yield the policy's intercept and duration coefficients. Once evaluated, a policy occupies HN_S integer control indices and $2(H + 1)$ floating-point focal coefficients. An m -policy bank therefore need not store $m(H + 1)(H + 2)N_S/2$ value coefficients for this focal-state task. The rolling temporary coefficient storage is $O(HN_S)$; policy storage remains $O(mHN_S)$. These savings do not give all-state, all-date policy-value access for free. The inherited all-state certificate and its storage account remain separate results.

For a common bank, replacing an upper coefficient array by a smaller valid one cannot change the certificate. The exact coefficientwise ordering from S.10 is unchanged by rolling storage or by splitting the first-action mask after common continuation. The independent dense program tests the paired implementation for both adjustment times and both signs, including a one-date horizon. It implements the conditional-mixture and admissible actions separately from the production routine. Its 60 domain models give 240 class recursions and 720 point comparisons. Such checks test implementation; the ordering theorem and induction establish the general result. The sparse transition implementation aggregates the same at-most-sixteen pre-regression entries per row as the inherited positive interpolation constructor. For the wealth grids it retains disjoint CSR blocks rather than allocating a second full concatenated matrix. This storage change preserves rows and rewards. Each block is checked against a separately computed weighted-gather product before use. The common and appended-action matrices, reward arrays, temporary action arrays, policy values, and coefficient arrays have distinct roles; their costs are not conflated with the number of upper correction vectors.

S.12.5. Arithmetic and Target Error

The revised certificate uses a per-class allowance of 10^{-7} , deliberately larger than the historical count allowance. As in S.11, its scope is floating-point evaluation relative to stored finite transition and reward arrays, not errors in constructing a diffusion approximation. The code verifies nonnegative rows, bounded row mass, finite rewards, correct selected-policy replay, and coefficient replay. Supplementary evidence records arithmetic assumptions and the construction checks. The allowance is not a directed-bounding interval enclosure of transcendental evaluations in the transition constructor. For completeness, let β be the largest stored row mass enlarged by 10^{-12} , let R be the largest sum of absolute reward components over $|d| \leq 1$, and let $G = 2\|g\|_\infty$. $B_H = G$, $E_H = M_H = D_H = 0$, and $\gamma_j = ju/(1 - ju)$ for $u = 2^{-53}$. The audit uses

$$B_n = R + \beta B_{n+1},$$

$$E_n = \beta E_{n+1} + \gamma_{4096} \{1 + R + 4\beta(B_{n+1} + E_{n+1})\},$$

$$M_n = 4w\beta B_{n+1} + \beta M_{n+1},$$

$$+ \gamma_{4096} \{1 + 8w\beta(B_{n+1} + E_{n+1}) + 4\beta(M_{n+1} + D_{n+1})\}.$$

is the mixture interval width, and the last two quantities are zero at the last date, where the endpoint continuation values coincide. B bounds exact endpoint values, E bounds endpoint and selected-coefficient errors, M bounds correction values, and D bounds their errors. The term $4w\beta E_{n+1}$ charges the propagated difference of two imperfect endpoints. Nonnegative mixing, nonexpansiveness of maxima, absolute dot-product error bound justify the recursions. Each sparse row has sixteen original entries; aggregation, reward assembly, law mixing, coefficient and correction construction have substantially fewer than 4,096 serial scalar operations per date. Signed cancellation is bounded in absolute magnitude, not by a error in the final value.

Upper coefficient error is bounded by $E_0 + D_0/2 + \gamma_{32}\{1 + 2(B_0 + E_0) + M_0 + D_0\}$. For lower coefficients we use $3E_0 + \gamma_{16}(1 + 4B_0)$, enlarged by $\gamma_{4096(H+1)}(1 + 4B_0)$ for restriction passes. A final $\gamma_{32}\{1 + 4(B_0 + M_0/2)\}$ covers coefficient differences at the end. The audit requires the larger resulting per-class bound to be below 10^{-7} for a certificate is accepted. An action maximum introduces no factor equal to the number of actions. These deliberately loose counts cover each contributing arithmetic operation and do not replace a signed intermediate by its small realized final value.

A conservative absolute arithmetic budget for the new recursions is provided by the routine deposited with the revision. It includes endpoint-value error propagated through the signed cross-operator correction, rather than treating that correction as a separate error. The coefficient inequalities are enlarged separately on their upper and lower bounds. In particular, each difference bound includes 2×10^{-7} . The 10^{-3} generic welfare bound is not used to certify the much smaller decision margins. An additional error for the economic target must enter the separate class-value transfer inequality in the next section. Observed wealth-grid movement is evidence about the importance of that component, not an upper bound on it.

S.12.6. *Reproduction and Version Boundary*

Full revision entry points are `ECTA_R8.tex` and `SUPP_R8.tex`. Run `bash reproduction/r8/run_all.sh`, followed by the two documented `latexmk` commands. The readable source, generated tables, complete outputs, execution logs, reports to the latest report, and preservation manifest are deposited together. The source files record hashes, and the workflow records the exact source commit. The historical main paper, all earlier appendices, numerical countercomparisons, and referee comments are retained. New interpretation and additional evidence are incorporated into the complete manuscript, not supplied only in an author-response file.

S.13. PARTICIPATION, SURRENDER, AND PERMISSION FRONTIERS

This section proves the additional contracting results and specifies the numerical details of their implementation. The finite state space, bounded rewards, eight-interval action, first-action classification, and frozen later menus are those of S.12. A surrender is available only at eligible interior nodes. Its reward is $G - F$ and its live action on row is zero. Boundary discharge continues to pay G . The first interval remains active in every reported position-class experiment.

for a deterministic Markov policy, augment the original action labels by a surrender label. Add an absorbing cemetery state if needed to represent killed probability mass. Reaching the surrender label, pay $G - F$ once and receive zero thereafter. The altered terminal payoff is G . Backward substitution in this policy recursion, or direct summation over the finite path space, gives

$$J_p = \mathcal{B}_p + d\mathcal{A}_p - F\mathcal{S}_p.$$

operating duration is nonnegative and bounded by $(1 - e^{-\rho})/\rho$. An elective surrender occurs at most once, so $0 \leq \mathcal{S}_p \leq 1$. In particular, the fee exposure cannot be exceeded by the expected number of visits to surrender-eligible states.

There are finitely many deterministic Markov policies in the original action target. They attain the unrestricted and sign-restricted optima; randomization does not improve a finite-horizon expected payoff. Each policy payoff is affine in (d, F) , so their minimum is convex, nondecreasing in d , and nonincreasing in F . If $F' \geq F$, every ad policy satisfies $-(F' - F) \leq J_p(F') - J_p(F) \leq 0$. Taking policy maxima on the appropriate side gives the 1-Lipschitz property. Increasing the minimum term removes surrender policies, so it weakly lowers each class value. None of these orders implies an order for a difference of class values.

For completeness, consider a direction $v = (v_d, v_F)$ at a contract $z = (d, F)$. A finite set of nonoptimizing policies has a strictly positive value gap from the maximum, unless it is empty. For sufficiently small positive steps, only policies optimizing at z can determine the directional derivative. The derivative is therefore

$$D_v W(z) = \max_{p \in \arg \max J(z)} \{v_d \mathcal{A}_p - v_F \mathcal{S}_p\}.$$

This proves the exposed-feature statement, including the case of several policies with the same feature vector. It is the standard finite envelope logic used in the main paper, and a new general envelope theorem.

The external numeraire makes total agent utility $q - C(E) + \max_\sigma W_\sigma^*$. Comparing this quantity with $G(x_0)$ and imposing $q \geq 0$ gives the stated participation grant. The managed initial state is unchanged, so the grant does not enter the Bellman recursion. While the principal receives flow b while the mandate operates, its expected payoff at the minimum grant is $b\mathcal{A}_{p_r^*} - q_r^*(E)$. Termination charges go to the separate enforcement institution. Dividing by strictly positive duration gives the procurement condition. If the agent optimum is not unique, the contract must specify an admissible selection or the relevant worst-case duration; the proposition uses a specified selection. It does not assume that all payoff-equivalent policies deliver identical service.

Both regime policy sets use the same financial opportunities and outside payoff. No-adjustment policies are feasible in the adjustment regime. Therefore $W_\sigma^{\text{adj}} \geq W_\sigma^0$ and $q_\sigma^{\text{adj}} \leq q_\sigma^0$ at a common capacity cost. Where all class-specific participation gaps are positive, subtracting the two participation equations within each class gives $q_\sigma^0 - q_\sigma^{\text{adj}} = W_\sigma^0 - W_\sigma^{\text{adj}}$. The four-class contrast yields the price interpretation of the relative option. The sign of that contrast is not implied by its two component inequalities.

The separate-account assumption is economically material. A model in which all payments enter managed CRRA wealth would require a new state transition, consumption response, and participation calculation. Similarly, collateral opportunity costs

than the ex ante $C(E)$ used here. These are alternative economic targets, not details that can be inserted without reoptimization.

S.13.2. The Statewise Enforcement Frontier

R_n be the live support reachable using the union of endpoint kernels and all feasible actions. Nonnegative kernel entries imply support closure: if $x \in R_n$ and a row from x gives positive mass to y , then $y \in R_{n+1}$. Elective surrender can follow any paths, but cannot introduce a new live state. The union may be larger than the support of any particular optimized policy; this is intentional.

Choose $F \geq F_r^*(m; \lambda, d)$. At maturity the optional and compulsory values both equal F because they coincide at all reachable nodes at date $n + 1$. By support closure their operating backups coincide at any reachable node at date n . At an eligible interior node, $F \leq V_n^r$ by the definition of F_r^* ; taking the maximum with surrender cannot raise the value. At an ineligible node the operating backup is the only option, and at a boundary node the two options agree by construction. Backward induction proves statewise equality, and hence equality of both sign-restricted initial values. If instead $F < F_r^*$, some eligible interior node has $G - F > V_n^r$, at which its optional value is strictly higher. This proves the necessary direction for equality at every reachable node. Such a node cannot affect either initial optimum, which is why the threshold is not described as the smallest initial-sign-preserving fee.

Increasing m removes terms from the maximum. Increasing d weakly raises V_n^r at every state by nonnegative policy duration, so it weakly reduces the maximum. The conclusion gives $V_n^{\text{adj}} \geq V_n^0$, and hence $F_{\text{adj}}^* \leq F_0^*$. Thus adjustment weakly reduces the capacity needed to implement the same mandatory operating values. When C is increasing, an available smaller implementing capacity can also weakly reduce its operation cost. This claim concerns implementation capacity, not monotonicity of social risk preference.

Regional calculation uses the two no-adjustment noncancellable policies optimized at $(\lambda, d) = (0, 0.4)$ and $(0.25, 0.4)$. Their later-date policies are feasible regardless of the initial class used to store the policy. For each policy and each date $n \geq 1$, compute the Stein coefficients in the *physical* mixture probability λ . Restrict those coefficients to sixteen subintervals of $[0, 0.25]$. Their coefficient minimum is a uniform policy value at every state on that subinterval. Take the larger of the two policy lower bounds and subtract 10^{-7} before comparing with G . Since duration is nonnegative, the upper bound also applies for all $d \in [0.4, 0.45]$. It is below V^0 and therefore below the regime values. Maximizing the resulting liquidation deficit over eligible reachable states gives the sufficient regional frontier.

The resulting bound 0.8348592805606 for $m = 1$ includes the stated arithmetic allowance. The upward-rounded value 0.834860 is sufficient relative to the stored arrays. Maximizing deficit in this bound occurs at date one, node $(1.95, 1.34375)$, on the first subinterval. Restricting to the actual initial state alone would not justify ignoring the continuation constraint. Conversely, including every unrelated grid node raises the sufficient bound to 2.5309784144. Neither bound is a claim about a continuum of admissible managed states.

S.8 gives the direct-solver statewise thresholds at the original center. They use the reachable support and the separately optimized mandatory continuations. In comparison, the one-interval thresholds are approximately 0.7767542 with adjustment and

m	Adjustment	No adjustment
1	0.7767542	0.8135332
2	0.7161762	0.7458682
3	0.6442995	0.6661886
4	0.5592377	0.5734029
5	0.4575809	0.4648691
6	0.3346276	0.3367796
7	0.1842712	0.1842768
8	0.0000000	0.0000000

ect evaluations of $\max_{n \geq m, x \in R_n} [G(x) - V_n^*(x)]_+$ at $(\lambda, d) = (0.125, 0.425)$. Both use the same all-action shable support. The initial point alone can require a smaller charge than statewide equality. These at calculations are distinct from the upward-rounded sufficient regional bounds in the main paper.

35332 without adjustment. They illustrate the policy-inclusion order. The sufficient onal bounds in the main text additionally cover all contracts in the rectangle and feasible-policy lower values; the two tables answer different questions.

G is an operating supersolution at every date, the zero-charge recursion has $W_n = \kappa(G, \mathcal{T}_n G) = G$ at the last eligible decision date. Repeating the argument proves ality at all eligible dates. Earlier mandatory dates retain their operating recursion. ate zero, admitting free surrender yields $G(x_0)$ as well. In the finite implementation, ry operating action is checked at both law endpoints and the upper benefit endpoint. e the one-step expression is affine in the law weight and its duration coefficient is negative, these endpoint inequalities cover the entire original rectangle. The large erved slack is a supersolution check for the stored arrays, not an interval proof for uncomputed transition constructor.

S.13.3. Risky-Share Breakpoints and Permission Derivatives

x an initial (c, θ) pair. For each law and sign realization the proposed preference rdinate is $2 + h\theta + \sqrt{h}\sigma_u z_u$, independent of π . The wealth coordinate has the form

$$y_X(\pi) = a_c + b_{q,z}\pi, \quad a_c = 1.25 + h(r1.25 - c),$$

$$b_{q,z} = h(\mu - r)1.25 + \sqrt{h}\sigma_X 1.25(qz_u + \sqrt{1 - q^2}z_X).$$

the declared constants, $|y_u - 2| < 0.043$ and $1.054 < y_X < 1.365$ throughout the er risky-share domain, all common (c, θ) pairs, and both law endpoints. These con- rative bounds lie strictly inside $[1.2, 2.8] \times [0.5, 2]$. The whole line segment therefore s interior, the exit fraction is one, and the flow annuity is independent of π .

onditional on a shock, the preference interpolation weights are constant as π varies. veen any two wealth-node crossings, each wealth interpolation weight is affine in y_X , ce affine in π . The bilinear interpolant of any continuation vector is consequently ne in π there. Averaging shocks and the law mixture preserves affinity on the mon partition. The flow reward is independent of π , so the first-date action value ontinuous and piecewise affine. A maximum over a compact interval is attained at ndpoint or a breakpoint. A positive open interval uses its closure only to compute

find a positive maximizer. The original finite frozen proposals are compared only when feasible.
 In interpolation cell, write $I_X V_1(y)$ for the slope of the interpolant in wealth. Continuation gives

$$\partial_\pi Q(c, \theta, \pi) = \sum_{\varrho} \Pr(\varrho) \frac{1}{4} \sum_z e^{-\rho h} I_X V_1(y_{\varrho, z}(\pi)) b_{\varrho, z}.$$

There is no within-period flow derivative in this expression. If the winning pair is fixed and the relevant optimum remains at L or $-S$, the value of relaxing the decision is the corresponding endpoint derivative. At a breakpoint or an action tie, choose the appropriate one-sided piecewise derivative or directional envelope. Because later-date permissions do not change V_1 , no later-menu derivative is being omitted. Changing d , unlike changing only a first-date permission, changes the optimized continuation. The finite envelope theorem nevertheless gives the derivative of each value as its optimizing discounted duration. Differentiating $\Delta^r(\lambda, d_r^*, F, L, S) = 0$ at a regular crossing then gives the three frontier formulas in the main text. If $D_A^r = 0$, if the vector is not exposed, or if a control switch destroys differentiability, these are not asserted. The deposited benefit-root calculations reoptimize continuation by trial benefit and retain local endpoint signs. Their floating-point root brackets and directed-rounding interval enclosures and do not prove that no other crossing

S.13.4. *The Joint Fee–Benefit–Law Certificate*

The stopping policy evaluated under affine endpoint kernels has a polynomial of degree at most the remaining horizon in λ . This follows directly from the recursion: each live step multiplies by one affine mixture, whereas a surrender row has a zero kernel. Padding with zero future rewards and a cemetery state permits degree reduction to the common horizon. Applying the same recursion to the base reward, the terminal reward, and unit surrender charge yields the three focal coefficient vectors $\vec{\delta}_p$.

In the (d, F) corner, the signed correction and the count recursion use the operating policy and surrender actions together. Their proofs in S.10 apply because the added action has a bounded affine reward and a nonnegative zero kernel. The last-decision endpoint calculations agree at the terminal vector, and the initial sign mask is applied only to the terminal values. The count information and the signed correction are not granted a different surrender date or a different charge from the lower policies.

For a fixed law probability, the optimized value is a maximum of affine functions of (d, F) . Its value inside a rectangle is therefore bounded above by the bilinear convex combination of its values at the four corners. Replacing those corner values by valid polynomial uppers preserves the inequality. A fixed policy value is affine in (d, F) and equals the corresponding corner interpolation. Restrict both the upper and policy values to a law subinterval and express them in a common Bernstein degree. For a policy p and nonpositive upper U_- , the minimum of all coefficients of $J_p - U_-$, using the reward-parameter corner indices, bounds that difference everywhere in (d, F) . Taking the largest such bound across positive policies remains valid. The upper

ties. Taking the worst cell gives the regional statement. No policy is selected using information unavailable in its actual evaluation. The deposit constructs the full four-corner fee–benefit bank at the two law endpoints for both adjustment regimes and both initial classes. The $F \in [0.85, 0.90]$ certificate uses four law subcells, while retaining the original full menu plus the eligible surrender option. The favorable signs are not inferred by extrapolating the pointwise no-surrender observation at the center. The reachable enforcement bound supplies an additional, separate sufficient explanation for their equality with the noncancellable signs. The 0.80 certificate fails; direct reoptimization at $(\lambda, d) = (0, 0.4)$ gives a positive no-statement difference 0.0013923796303, so this failure is not merely an uninformative lower bound.

S.13.5. Arithmetic, Target Transfer, and Executed Validation

The arithmetic assumptions are those in S.12: stored reward arrays and nonnegative position arrays are evaluated in round-to-nearest binary64 arithmetic. Rewards can have either sign; nonnegativity applies to transition weights and duration, not to utility. The surrender reward is included in the absolute reward bound and its row mass is zero. Let $\gamma_j = j2^{-53} / (1 - j2^{-53})$. The R9 audit reuses the B, E, M, D backward versions in S.12 with row-mass bound $\beta = 0.9950124791936824$, absolute reward bound $R = 27.2130993863$, and law width $1/4$. The bound covers $|d| \leq 1$ and $0 \leq F \leq 1$. For the additional three-feature lower representation, the audit uses

$$L_{\text{err}} = 5E_0 + \gamma_{4096(8+1)}(1 + 8B_0),$$

the same upper error as in S.12, enlarged finally by $\gamma_{64}(1 + 4B_0 + 2M_0)$ for coefficient subtraction and extrema. The factor five charges the subtraction of the base error in constructing the duration and surrender features and their recombination, instead of treating those features as exact. The restriction budget dominates both details. The teljau passes. A sparse row has at most sixteen entries before aggregation; adding a surrender label does not increase a dot-product length. Maxima are nonexpansive and introduce no factor equal to the number of feasible actions. The resulting derived class bound is 1.4305784×10^{-8} , below the deliberately larger accepted allowance. Every displayed difference interval is enlarged by 2×10^{-7} .

The stored-array target is not silently identified with a real-arithmetic transition constructor. For two finite targets with a common action correspondence, subtract the Bellman backups, add and subtract one kernel applied to the other's continuation, and use $|\max_a q_a - \max_a \tilde{q}_a| \leq \max_a |q_a - \tilde{q}_a|$. An absolute reward discrepancy ϵ_r , ℓ^1 discrepancy ϵ_K , continuation magnitude bound B , and row-mass bound β give $\epsilon_{r,n} + \epsilon_{K,n}B_{n+1} + \beta_n\epsilon_{n+1}$. The terminal discrepancy initializes this recursion. Surrender, forced-discharge, and boundary payments must enter the reward or terminal discrepancy. Different action sets require a separate action-approximation bound or a correspondence with a priced improvement error. A diffusion additionally requires its own consistency and approximation argument. The arithmetic allowance already charged to the stored-array certificate is not counted a second time as a measured constructor error.

The new independent validation has two components. Twelve random two-state, one-interval models are checked by enumeration of 81 deterministic continuation policies per initial class and parameter point. There are 120 class–point comparisons.

10^{-16} ; count never exceeds the signed chord beyond the declared numerical tolerance. A separate full-economy check evaluates 80 class-parameter comparisons across both regimes, and five mixture probabilities. Selected-control reconstruction of the stored sparse rewards and kernels differs from the optimized value by at most 4.45×10^{-16} . Fixed-policy polynomial replay differs by at most 4.45×10^{-16} . Final checks cover reward-parameter convexity, action eligibility, and the distinctness of the free-surrender contract. These are implementation checks, not substitutes for preceding proofs.

The complete economic run retains 19 center surrender/term specifications, 20 further contract rows, their four class policies and moments, sixteen all-action supersolution checks, the eight sufficient enforcement bounds, nine two-regime permission comparisons, and eighteen reoptimized benefit roots. Three regional fee specifications are checked with both streamed upper constructions. All 398 protected historical or original source entries checked by the execution manifest remain unchanged. The action target is the continuous first-date risky-share envelope conditional on finite option-adjustment pairs; the fee certificate target is the original finite menu plus a fee. The deposit does not claim a joint theorem simultaneously enlarging those different action targets.

S.13.6. *Reproduction*

The complete manuscript entry points are `ECTA_R9.tex` and `SUPP_R9.tex`. From the repository root, run the following commands in order:

```
on replication/r9/run.py -mode all
on replication/r9/validate.py
on replication/r9/tables.py
```

The source inventory, historical hash checks, generated JSON, policy and coefficient execution transcript, manuscript build log, and point-by-point response are bundled together. R9 adds new readable source files and new evidence directories. The original R8 manuscript, supplement, all earlier economic applications and adverse comparisons, and the latest advisory report remain available at their original paths.

14. DYNAMIC TRANSPORT, JOINT CONTRACTS, AND PRICED IMPLEMENTATION

This section supplies the proofs and numerical specifications for the section entitled “Dynamic Adjustment and the Price of Commitment” in the main paper. Main-paper results in this independently compiled supplement are identified by their proposition number. The substantive definitions are repeated here to avoid dependence on cross-section references.

S.14.1. *Proof of the Reachable-State Transport Proposition*

Let $V_i^\uparrow$ be the value with nonnegative deliberate drift, including the same elective stop as the unrestricted problem. For a negative action a , let a^0 be its feasible zero-drift counterpart. Given enclosures $\ell \leq V_{n+1}^\uparrow \leq h$, write $V_{n+1}^\uparrow = v + z$, where $v = (h + \ell)/2$ and $z \leq e = (h - \ell)/2$. Entrywise positivity gives

$$(K^a - K^{a^0})V_{n+1}^\uparrow \leq (K^a - K^{a^0})v + |K^a - K^{a^0}|e.$$

ts. Taking the convex combination of these endpoint upper bounds bounds the difference under every mixture law. Maximizing that affine upper expression over the two endpoints is valid. This is slightly more conservative than evaluating the absolute value of the mixed signed kernel itself, since $|(1 - \lambda)D_0 + \lambda D_1| \leq (1 - \lambda)|D_0| + \lambda|D_1|$. The charged evaluation allowance then yields $Q^a(V_{n+1}^\uparrow) < Q^{a^0}(V_{n+1}^\uparrow)$. At terminal time the unrestricted and restricted values agree. Suppose they agree at date $n + 1$ on the common reachable set. Closure of the set under all feasible live transitions permits replacement of unrestricted continuation by restricted continuation every date- n backup. Every negative-drift backup is strictly dominated by a feasible zero-drift backup. All remaining operating and stop actions are available in the upward problem. Hence the two maxima agree. Backward induction proves equality on every reachable set, including either initial financial class: the counterpart has the same risky share, so it respects the class restriction. Boundary nodes have the common discharge rate and require no strict operating-action inequality. The argument uses all feasible processors, not only successors of the ultimately chosen policy.

For completeness, the continuation enclosures used in the computation follow from separate backward comparison. Start with the common terminal array. At each later date, evaluate the lower continuation at the lower benefit and higher fee and minimize over the law endpoints before maximizing over upward-feasible actions. This is a lower bound, because it permits an adverse law choice for each action. Evaluate the upper continuation at the higher benefit and lower fee and maximize over both law endpoints over actions. This is an upper bound. Add outward error padding after each stage. Further recursion assumes that the optimal policy is constant on the box.

At the first date, the restricted objective is piecewise affine in the risky share for each consumption–drift pair. The difference between the negative-drift action and its counterpart is piecewise affine on the union of their breakpoint sets. Its maximum over every admissible permission interval is therefore covered by the union of these sets and the largest-permission endpoints. A separately proposed action needs its own counterpart or direct domination proof. The deposited frozen proposals all have negative drift on the reachable live states; this statement is checked rather than assumed.

S.14.2. Continuation Incentives and the Dynamic Class Differential

On a branch on which the exit fraction α is independent of deliberate preference drift, arrival preference changes by $\alpha h \theta$, while consumption and the financial transition are fixed. The fundamental theorem of calculus for absolutely continuous, piecewise differentiable continuation gives the transport integral in the main paper. Integrating quadratic flow cost gives $k\theta^2 \mathbb{E}[(1 - e^{-\rho \alpha h})/\rho]/2$. Immediate preference discharge requires a different branch calculation and is not covered by this smooth-transport interpretation; the signed-row test still applies.

An exact decomposition clarifies the dynamic comparison. Suppose the selected first financial controls (c_σ, π_σ) agree between the adjusted and unadjusted class optima, checked at the reported central contracts. Write $a_\sigma = (c_\sigma, \theta_\sigma, \pi_\sigma)$ and $a_\sigma^0 = (c_\sigma, 0, \pi_\sigma)$.

$$W_\sigma^{\text{adj}} - W_\sigma^0 = Q^{a_\sigma}(V_1^{\text{adj}}) - Q^{a_\sigma^0}(V_1^{\text{adj}}) + K^{a_\sigma^0}(V_1^{\text{adj}} - V_1^0). \quad (\text{S.4})$$

initial action. Subtracting the two classes yields the reported relative local and terms. If the financial controls differ, the additional term $Q^{a_0}_v(V_1^0) - W_\sigma^0$ must be added; it is not generally zero. The replication code fails rather than silently ignoring that term when its equality premise fails.

The boundary-state diagnostic, continuation is further written as $V_1 = G + (V_1 - G)$ which separates the liquidation benchmark from the liability of remaining in operation. The code independently reconstructs noncost flow, adjustment cost, immediate discharge, live liquidation-benchmark exposure, and continuation excess. The reference-boundary mass is obtained by evaluating the boundary indicator on the stipulated first-date settlement lottery. It is discounted incidence, not an expected probability and not a continuous-time hitting probability. The large minus-upward value difference is a reoptimized all-date comparison; the one-step difference is not substituted for that full difference.

The primitive two-stage result and its common-cardinal-tilt robustness remain separate theorems. No step of this proof substitutes their cost coefficients or adjustment coefficients for those of the eight-date economy.

S.14.3. *Proof of the Joint-Region Certificate*

A deterministic feasible stopping policy for the dates after the first, and a feasible immediate action. Affine dependence of rewards and kernels on the repeatedly drawn law gives a degree-at-most-eight Bernstein polynomial in λ by backward conditioning on the number of regime-1 draws. Conditional on the policy, the coefficients are determined by d and F . The first-date coefficient is obtained by applying its selected first-date coefficients to the seven-period continuation coefficients. Early stopping shortens effective horizon but does not alter the degree-elevated representation.

At every benefit–fee corner, compute endpoint optima for the largest permissions. The lower corrected chord, or the count-information recursion with the same information bounds each class value over the law interval. Convexity of optimized value in law bounds an interior reward point by the multilinear interpolation of these upper values. Both methods allow the same elective stop action at the same law. The first-date upper backup optimizes over the breakpoint menu corresponding to the largest permissions; the later upper backups use exactly the original full action

for lower values, retain feasible policies optimized at the smallest permissions. They are feasible at every permission pair in the box. Fix one such policy across all reward corners before taking a regional bound. Let $L_{c,j}^p$ be its restricted Bernstein coefficient at reward corner c and law coefficient j , and let $U_{c,j}^\sigma$ be an upper coefficient for a permission class. Nonnegativity and unit sum of the tensor interpolation weights

$$J_p - W_\sigma \geq \min_{c,j} \{L_{c,j}^p - U_{c,j}^\sigma\}.$$

the maximum of this bound over feasible lower policies is valid. Reversing the argument applies an upper difference bound. Importantly, taking a separate best lower policy at each coefficient or reward corner before the minimum is not justified and is not true. The policy identity is held fixed through the complete corner-and-coefficient interpolation.

me. For active surrender, apply it to the positive zero-adjustment class with surrender and to that same class with all eight intervals compulsory. The first-date permission schemes are treated in the same conservative manner. The resulting bounds include 10^{-7} for each difference. Their displayed decimal bounds in the main paper are rounded outward.

A policy has zero discounted elective-surrender incidence, its operating behavior payoff are feasible in the noncancellable economy. It cannot exceed the latter class supremum. A strictly positive lower bound on the value gain from permitting surrender therefore implies that any policy closer to the surrender value than that bound must have positive elective-surrender incidence. This conclusion remains valid for a class defined by a supremum over strictly positive shares. The closure at zero is used only in upper search; every retained lower policy has a strictly positive initial share in class

The proof is a continuum argument, not an inference from the six interior validation tracts. Those direct reoptimizations test the implementation against the derived bounds. The transport certificate supplies the independent statewise directional state-ent. Neither a point replay nor a class-value bound alone would supply it.

S.14.4. *Proof of the Priced Implementation Proposition*

For regime r , let V_n^r denote the noncancellable continuation and let $\mathcal{R}_n$ be the feasible-transport tube. A fee F implements equality of surrender and noncancellable values on eligible reachable nodes precisely when

$$F \geq F_r^*(m) := \max_{m \leq n < 8} \max_{x \in \mathcal{R}_n \text{ live}} [G(x) - V_n^r(x)]_+.$$

The empty maximum at $m = 8$ is zero. This is a statewise threshold, not a necessary threshold for matching only the initial value. The earlier proof by backward induction relies without change to the enlarged first-date menu because the reachable tube now contains its successors. Policy inclusion gives $V_n^{\text{adj}} \geq V_n^0$ and hence $F_{\text{adj}}^*(m) \leq F_0^*(m)$.

Conditional on implementing the selected noncancellable allocation, W^r and delivered service A_r do not depend on the chosen instrument. For a given term, nondecreasing capacity cost makes the smallest sufficient capacity optimal. Participation requires $q \geq G(x_0) - W^r + C(E)$ and $q \geq 0$; hence the minimum grant is the stated positive part. The principal pays that grant and direct term cost. Minimizing over the grant terms proves the procurement formula, and comparison with bA_r proves the break-even condition. At the threshold, continued operation must be the specified no-response selection. Requiring strict stopping incentives instead gives the infimum fees decrease to the threshold; the code separately checks strictly higher fees and reports them as implementation tests, not as the exact continuous optimum.

When $z = G(x_0) - W^r \geq 0$, the grant increment from capacity is exactly $C(E)$. For arbitrary z , monotonicity and unit Lipschitz continuity of $[\cdot]_+$ give $0 \leq [z + C(E)]_+ - z \leq C(E)$. Comparing this increment with saved direct term cost proves the general argument comparison, including the nonbinding-participation case. With $D(m) = (m - 1)/8$ and binding participation, each term's cost is affine in κ . The lower envelope selects the economically preferred institution; crossings of these affine functions, not free-term omission, determine its changes.

enforcement costs. The principal's fees are zero on the implemented path, so no fee revenue finances the grant. For active contracts outside this implementation any different fee recipient or principal revenue assumption must be entered manually.

S.14.5. *Arithmetic, Targets, and Reproduction*

source fixes 1,617 continuation states, the inherited 1,568 later operating actions, states, and the stored first-date breakpoint rows. The first rows are nonnegative and no larger than the inherited discount bound. The stopped reward includes the boundary payoff and the elective fee. A full array manifest records data types, and SHA-256 identities for the primitive reward, duration, effort, CSR, and the arrays generated by the pinned sources.

Gamma-bound calculation in the R9 arithmetic audit is rerun at the new interval and fee bound. Its per-class bound is smaller than the charged 10^{-7} . The first-checkup has at most sixteen pre-coalescence transition entries per row. A signed piece of two such rows has at most thirty-two; its absolute-value transport bound is a further 2×10^{-7} per comparison. The interval recursion is padded by 10^{-7} per date. These pads exceed the round-to-nearest operation bound for the stored states and bounded eight-step continuation values. Positive interpolation weights at extrema do not amplify a uniform one-sided value error beyond the stated row-recursion. The support test uses the nonzero entries of these stored nonnegative

There is an additional distinction at the first date. The certified operator is the piecewise-affine interpolation of its deposited knot rows for each finite consumption-ir. Direct evaluation of the original transition formula at selected caps is checked against this representation; agreement is an implementation test, not an enclosure of the arithmetic constructor. Rebuilding with another environment requires checking the stored array identities or treating the rebuilt arrays as a separately identified set. No constructor or diffusion enclosure is inferred from last-bit agreement.

The supplied runner executes the nine directional replays, the alternative-state witness dynamic decomposition, all twelve joint fee-permission interventions, both the upper methods, the full reachable-state transport test, all eight enforcement alternatives, selected-control reconstructions, and interior-region validations. It saves the coefficient arrays and lower policies, not only their final extrema. The common bill, construction, and validation are charged to each matched upper-method load. Kernel payload is reported separately from process peak memory, which is measured. Historical training and full historical timing sweeps are preserved but described as newly rerun.

The proof establishes the indicated finite-array economic result. Transferring initial inequalities to another target requires the additional value-error account stated in the main paper. Transferring directional dominance requires a compatible statewise support and reachability account. These are different mathematical obligations.

S.15. SERVICE, PROCUREMENT, EXPOSURE, AND WITNESS RECONSTRUCTION

This section supplies the additional arguments used by the R12 main paper. Sections 4 above are retained in full. Their references to earlier main-paper statements

ces with the R12 labels below refer to the new main paper. The numerical target in this section is the canonical R12 array system; identities of historical targets are not ntly reassigned.

S.15.1. Service bounds without differentiability

x an item j , an adjustment regime, and a feasible policy π . The definition of operat-
duration gives

$$J_j(\pi; d + h) = J_j(\pi; d) + hA_j(\pi), \quad J_j(\pi; d - h) = J_j(\pi; d) - hA_j(\pi). \quad (\text{S.5})$$

se equalities hold with surrender, forced discharge, deliberate adjustment, and
hastic kernels. The benefit parameter changes only the reward coefficient, not a
nel, feasible set, capacity cost, term cost, or stopping charge. This is why the two
urbed problems can identify service. Varying a primitive that also changes the
sition or stopping technology would not justify the same argument.

ROOF OF PROPOSITION 1: Let π be η -optimal at d . Optimality of the perturbed
e and near-optimality at the center imply

$$W_j(d + h) \geq J_j(\pi; d + h) \geq W_j(d) - \eta + hA_j(\pi), \quad (\text{S.6})$$

$$W_j(d - h) \geq J_j(\pi; d - h) \geq W_j(d) - \eta - hA_j(\pi). \quad (\text{S.7})$$

first inequality yields $hA_j(\pi) \leq W_j(d + h) - W_j(d) + \eta$; the second yields $hA_j(\pi) \geq$
 $W_j(d) - W_j(d - h) - \eta$. Substituting the adverse endpoints of the three certified inter-
proves (7). Intersecting with $[0, \bar{A}]$ preserves validity. Randomization preserves
two affine identities after expectation, and the same inequalities follow. No limit-
derivative or stable optimizer has been assumed. If the positive-class supremum
ot attained, every existing feasible near-optimal policy still obeys the displayed
qualities; the proof does not use an exact maximizing selector. *Q.E.D.*

he exact value function is convex and nondecreasing in d , as a supremum of affine
ctions with nonnegative slopes. This observation helps interpret the secants but is
an additional assumption needed in the proof. At a kink, distinct optimal policies
y have different service. The finite interval covers that set of service responses; it is
a standard error for a unique derivative.

he numerical queries have error allowance $\epsilon = 10^{-7}$, so the implemented endpoints
 $\ell_t = \widehat{W}(d + t) - \epsilon$ and $u_t = \widehat{W}(d + t) + \epsilon$. The response tolerance is $\eta = 10^{-8}$ and
0.01. Thus each untruncated service secant charges $(2\epsilon + \eta)/h = 2.1 \times 10^{-5}$, in
dition to economic curvature or policy switching between the queried benefits. The
cker reconstructs the values at all three benefits; it does not read a favorable service
rval from the generator's summary and call that a proof.

S.15.2. Participation and robust comparison of discrete instruments

ROOF OF THEOREM 2: For a η -optimal response to item j , operating payoff is at
t $W_j(d) - \eta \geq \ell_0 - \eta$. Under payment $\bar{q}_j$ its participation payoff is therefore at least

$$\ell_0 - \eta + [G(x_0) - \ell_0 + C(E) + \eta]_+ - C(E) \geq G(x_0). \quad (\text{S.8})$$

participating response under nonnegative payment q has operating payoff no less than $W_j(d) \leq u_0$. Therefore $q \geq [G(x_0) - u_0 + C(E)]_+ = \underline{q}_j$.

osition 1 and $b \geq 0$ now imply that the selected item's surplus under its execution payment is at least $\underline{\Pi}_j$. Every participating near-optimal response to every item, by a feasible payment, yields surplus at most $\overline{\Pi}_j$. In particular, these upper bounds rival both the smallest potentially feasible payment and the most favorable possible feasible service; their joint attainability is not required for a valid upper bound. Comparison against those bounds proves the first assertion. The no-procurement regime has value zero and is explicitly included.

To prove the uniform assertion, fix a candidate and a competitor. Their lower-minus-upper difference is affine in (b, κ) because all service and payment interval endpoints are linear in (b, κ) and $D_\kappa(m)$ is affine. Each point of the price rectangle is a convex combination of its vertex values; the affine difference is the same convex combination of its vertex values. Positivity at every vertex implies positivity throughout. The selected candidate's lower difference relative to zero has the same property. There are finitely many competitors, so the argument applies simultaneously to all of them. Q.E.D.

The theorem identifies the optimal discrete instrument, not the exact minimum payment within that item. Its executable grant includes numerical and response constraints. If an application requires strict participation with an additional grant $\delta > 0$, the proof works whenever δ is smaller than the minimum certified separation. If best responses are imposed, they form a subset of the certified response class. No reporting or truth-telling constraint is implied by the complete-information menu. The fee recipient matters. In this environment the fee is paid to an external agent in the private sector; no term FH is added to purchaser revenue. A different institutional arrangement changes the purchaser's objective and requires a new comparison. Capacity costs are paid in the separate numeraire, while term enforcement cost is paid directly by the purchaser. Payments do not alter managed wealth, so the service bounds obtained by setting the grant remain applicable after the grant is quoted.

S.15.3. Verification of the procurement theorem

In each regime, the menu has $2 \cdot 6 \cdot 8 = 96$ items. The generator and independent checker solve the continuation problem for two regimes, six capacities, eight terms and eight benefits, or 288 parameter queries. Each query supplies both initial class values. The checker independently optimizes every later-date action backup on every state, verifies that the supplied later policy is feasible and value-attaining to its numerical tolerance, and returns the first action and its value, and compares the entire stored continuation-value with not only the initial scalar.

First-date action pairs are finite and the risky-share operator is affine between the knots. Its maximum on a closed permission interval is at a knot or permission endpoint. All permission endpoints used in this numerical theorem are canonical knots. For every admissible consumption–adjustment pair; the checker tests this property by comparing the rounded requested permission. The positive class is treated by its closure under optimization. Feasible strictly positive first actions are retained in the chosen set of contracts and the regional lower witnesses.

Three reconstructed value intervals yield service and payment intervals through the derived formulas. At each of the four (b, κ) vertices and in each regime, the checker

zero. It also checks that the same candidate wins at all four vertices. The exact generated gaps and input identities are deposited in `independent_validation.json`; forward-displayed bounds in the main tables are computed from that record. This finite arithmetic certificate, the analytic error allowance, and Theorem 2 establish Theorem 3. Inter-point reoptimization alone would not establish the rectangle. Nominal policy moments give a separate accounting check. With the stored selected policy, the generator evaluates (B, A, H) by policy-specific recursions and checks $W = dA - FH$. These moments explain the chosen operating allocation but are not substituted for the all-response service enclosure in the decision certificate. The response variance in the theorem concerns the economic agent's response class; the theorem does not infer that every numerical maximizing policy is itself 10^{-8} -optimal merely because a floating-point argmax was returned.

S.15.4. Class-specific exposure and its tail representation

PROOF OF PROPOSITION 4: Add and subtract $Q^{\text{adj}}(a_\sigma^0)$ and $Q^0(a_\sigma^0)$ from $W_\sigma^{\text{adj}} - W_\sigma^0$. Since a_σ is an adjusted optimum,

$$\Omega_\sigma = Q^{\text{adj}}(a_\sigma) - Q^{\text{adj}}(a_\sigma^0) + Q^{\text{adj}}(a_\sigma^0) - Q^0(a_\sigma^0) + Q^0(a_\sigma^0) - W_\sigma^0 \quad (\text{S.9})$$

$$= g_\sigma + K_\sigma O_1 + B_\sigma. \quad (\text{S.10})$$

The reward of the fixed zero-drift action is unchanged between the two regimes, so the middle difference is precisely $K_\sigma O_1$. Feasibility of that action for the zero-adjustment regime gives $B_\sigma \leq 0$. Subtracting the two class identities proves (14). Decomposing a signed row as $D^+ - D^-$ and using positivity separately on each part proves (15). *Q.E.D.*

For completeness, let ordered state values be $o_1, \dots, o_M$ and let D_j be a signed kernel difference with total mass zero. Define $T_i = \sum_{j=i}^M D_j$. Since $o_j = o_1 + \sum_{i=2}^j (o_i - o_{i-1})$, rearranging finite sums gives

$$\sum_{j=1}^M D_j o_j = o_1 \sum_{j=1}^M D_j + \sum_{i=2}^M (o_i - o_{i-1}) \sum_{j=i}^M D_j = \sum_{i=2}^M T_i (o_i - o_{i-1}). \quad (\text{S.11})$$

This proves the sufficient opportunity-tail condition in the main paper. Without equal masses the first term must be retained. Substochastic live kernels can be augmented by a zero-option cemetery state to obtain equal total mass; the position of that zero option in an asserted monotone ordering must be valid. This transformation does not add a large reward, which is already accounted for in the current payoff.

$\underline{\delta}_i \leq o_i - o_{i-1} \leq \bar{\delta}_i$, a lower bound is $\sum_i (T_i^+ \underline{\delta}_i - T_i^- \bar{\delta}_i)$, with an analogous upper bound. Consequently a sufficient class reversal is obtained when a lower bound on relative local gain, signed future exposure, and financial-replacement difference exceeds $-\Delta^0$. This is a transferable restriction on the mechanism. The application does not assert that primitive wealth ordering alone proves the required ordering of future adjustment values.

At the reported point solutions the zero-drift financial counterpart is also a zero-adjustment optimizer, and both financial-replacement terms reconstruct as zero to numerical tolerance. Their presence in the proposition remains necessary. A different

authorize their removal from the general formula. The signed exposure is by wealth bins in the deposited diagnostic record; negative and positive exposures can be much larger in absolute value than their net difference. This is an economic reason to retain signed transport rather than replace it by unrelated base value errors.

S.15.5. Regional lower witnesses and upper recursions

fixed feasible policy, write a value at date $n + 1$ in degree $H - 1$ Bernstein form common law parameter. Affineness of the one-period reward and kernel in that order gives the degree- H coefficient recursion

$$\beta_{n,i}^{\pi} = \left(1 - \frac{i}{H}\right) \{R_{n,0}^{\pi} + K_{n,0}^{\pi} \beta_{n+1,i}^{\pi}\} + \frac{i}{H} \{R_{n,1}^{\pi} + K_{n,1}^{\pi} \beta_{n+1,i-1}^{\pi}\}, \quad (\text{S.12})$$

omitting endpoint terms omitted. Reward coefficients multiply the degree-elevation in the same way as continuation coefficients. A stopping action has zero live-kernel and current payoff $G - F$. The terminal condition is the stored terminal value the checker implements this selected-policy recursion directly on the primitive including the first-date action.

low subinterval, de Casteljau restriction transforms the global coefficients with- changing the polynomial. At any fixed reward corner, nonnegativity and partition of the Bernstein basis imply that the minimum coefficient is a lower bound and the maximum coefficient an upper bound. A policy feasible under the smaller permission remains feasible at every larger permission in the box. Its reward/fee difference is affine while that policy remains fixed. Thus the minimum over all corner values is a valid uniform lower bound for that single policy, and the maximum over the policy bank may be taken only *after* this minimum. The proof never chooses a single policy for each coefficient.

difference of two optimized values, let L_j^{π} be the restricted coefficient vector of the policy for target j and let U_k be upper coefficients for target k . Then

$$\max_{\pi} \min_{\text{corners}, i} (L_{j,i}^{\pi} - U_{k,i}) - 2\epsilon \leq W_j - W_k \leq \min_{\pi} \max_{\text{corners}, i} (U_{j,i} - L_{k,i}^{\pi}) + 2\epsilon. \quad (\text{S.13})$$

minima/maxima across policies need not use the same policy, but each policy is held fixed within its corner-and-coefficient operation. This proves the reduction for the adjusted spread, unadjusted spread, and surrender gain. The independent first reconstructs the lower and upper coefficients from primitive arrays and repeats this reduction; it does not accept their roles on trust.

count-information and signed-compression induction is retained in Supplement the new implementation the endpoint optimizations, count interior maximization and signed majorant propagation are separately recoded. Each is tested over the state space and all relevant reward corners. The claimed regional scope is one law cell and the displayed permission/reward box; selected-policy checks at interior points are not used as a substitute.

downward-drift elimination, the independently constructed support starts with the first-date transition rows and repeatedly includes every successor of every later row. The upward-only continuation intervals use adverse benefit/fee

and in the main paper then dominates every omitted negative-drift action by its viable zero-drift counterpart. Backward induction from terminal values gives equality of all and upward-only optima on each reachable set. Date- and state-specific frozen comparisons are checked for unpaired negative drifts. This recreates the valid R11 reasoning of the separately identified R12 target.

S.15.6. *Immutable inputs and the scope of independent checking*

The canonical manifest names every logical input archive, every physical archive part, every byte hash, and every array's shape, dtype and content hash. The first-date operator of all later operators are loadable without constructor calls. The archive format allows for nonobject numeric arrays. The loader checks array inventory, finiteness, sorted row indices, nonnegative transitions, and the stated subprobability mass bound. Witness identities refer to the expected canonical-manifest hash, so changing a manifest tag with its arrays does not pass the original witness check.

The explicit new-target command refuses an existing target directory. The generation command similarly refuses an existing witness-output directory. Reproduction reads committed canonical inputs by default. A separately identified target can be generated to another directory, but its result is not silently attributed to the original target. The final R11 expected manifest and all inherited results remain intact. The R12 comparison file records primitive differences without treating small floating-point differences as an operator-transfer proof.

The independent verifier uses the deposited identities as the initial anchor, then applies semantic tests to the mathematical witnesses. Its region reducer rejects the tree's destroyed upper array even when the original file hash is not consulted. A sparse disposable-file test rejects that mutation by archive identity. Other fixtures test feasible policies, negative transition weights, excess row mass, altered continuation closures, stale favorable summaries, and an incorrect canonical identity. These are final execution tests; no original witness or remote historical file is modified.

The numerical tolerance for comparing two binary64 reconstructions is not itself the economic error bound. The analytic arithmetic audit supplies a per-value bound, and the other advertised allowance is charged in the economic intervals. The audit explicitly includes construction and diffusion error. Agreement between two implementations supporting evidence against coding mistakes, not a theorem that the underlying non-arithmetic construction is identical to another platform's constructor. Source comparison target identity, witness identity, verification result, and final deposit commit are ordered as different objects.

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